



$\text{\LaTeX}$ M1 Notes I & II

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Github: <https://github.com/Baudelaireee/Notebook>

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Part I

M1 Notes I

1 9/28 Notes for "Calculus on Manifolds" by Spivak

Problem. Can we derive the explicit formula from an equation with several variables? Here are two examples :

$$x^2 + y^2 - 1 = 0 \quad (1)$$

and

$$e^{xy} + \sin y + x^2 - 2 = 0 \quad (2)$$

In the first example, it is clearly that we can express y as a function of x in a certain interval. But in the second example, it seems that we can draw a picture to visualize the implicit curve.

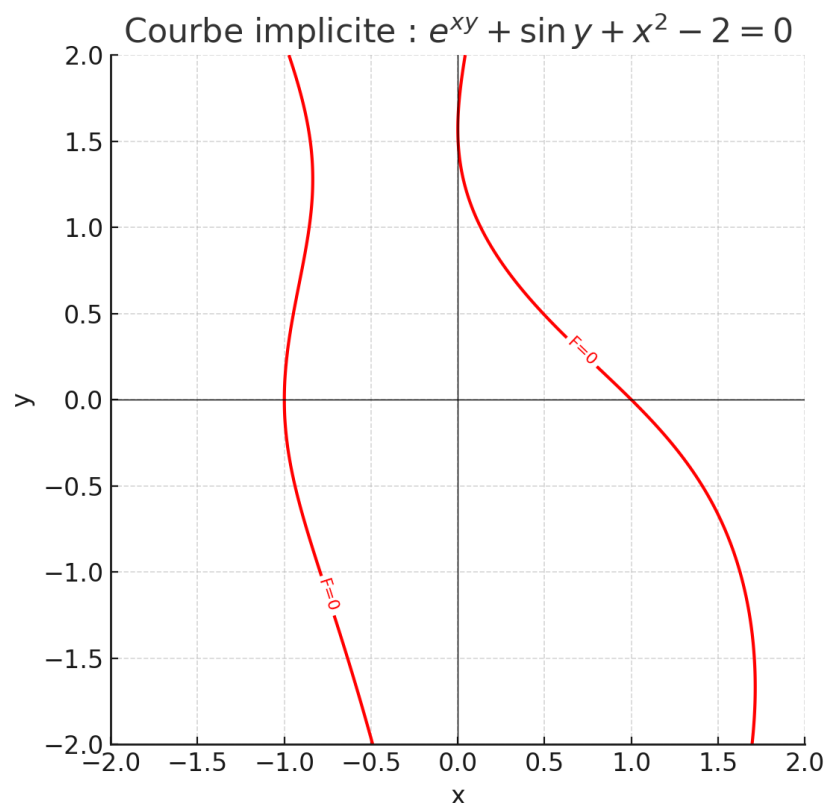


Figure 1: $e^{xy} + \sin y + x^2 - 2 = 0$

The curve here seems very smooth, so for example, at the point $(x_0, 0)$ in the curve we

can express y as a function of x in the neighborhood, but it does not mean that we can write the function explicitly.

For thinking about this problem, we assume here we have a multivariable function $F(x, y)$ and we want to solve the equation $F(x, y) = 0$. Like equation (1), we can express it as $F(x, y) = x^2 + y^2 - 1$. Here we assume F is a smooth function, then we can get the total differential of F

$$df = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy$$

If we assume the existence of the implicit function $y = f(x)$, then along the curve ($G(x) = F(x, f(x)) = 0$) we can get

$$0 = G'(x) = \frac{\partial F}{\partial x}(x, f(x)) + \frac{\partial F}{\partial y}(x, f(x)) \cdot f'(x)$$

hence we can get the derivative of the function f can be denoted by

$$f'(x) = -\frac{\frac{\partial F}{\partial x}(x, f(x))}{\frac{\partial F}{\partial y}(x, f(x))}$$

let us check some necessary condition here:

1. We try to restrict the function to be vanishing at some point, that means in the voisinage of the point, there exists something like a curve $F(x, y) = 0$ such that we can do the total differential like above, so here we need two conditions: locally, $F(a, b) = 0$ for some point (a, b) and a enough smooth function F .
2. We need to make sure the formula for $f'(x)$ makes sense, so locally $\frac{\partial F}{\partial y}(a, b) \neq 0$.

Actually, under these hypotheses, we can prove the existence of the implicit function $y = f(x)$ in the voisinage of the point (a, b) such that $F(x, f(x)) = 0$, which completes the implicit function theorem in the case of two variables.

Theorem 1.1 (Implicit Function Theorem in $\mathbb{R}^2$). Let F be a C^1 function defined on an open set containing the point (a, b) . Assume that $F(a, b) = 0$ and $\frac{\partial F}{\partial y}(a, b) \neq 0$. Then there exists an open interval I containing a , an open interval J containing b , and a unique C^1 function $f : I \rightarrow J$ such that for all $x \in I$, $F(x, f(x)) = 0$. Moreover, the derivative of f is given by

$$f'(x) = -\frac{\frac{\partial F}{\partial x}(x, f(x))}{\frac{\partial F}{\partial y}(x, f(x))}$$

Proof. here we just give a proof of the existence of the implicit function f , the propoerties have benn proved as above. The idea here is to construct a locally diffeomorphism and then use the inverse function theorem.

We take the map $\Phi(x, y) = (x, F(x, y))$, then we can compute the Jacobian matrix of Φ at the point (a, b)

$$D\Phi(x, y) = \begin{pmatrix} 1 & 0 \\ \frac{\partial F}{\partial x}(x, y) & \frac{\partial F}{\partial y}(x, y) \end{pmatrix}$$

The condition $\frac{\partial F}{\partial y}(a, b) \neq 0$ implies $\det D\Phi(a, b) \neq 0$ the Jacobian matrix is invertible, hence by the inverse function theorem, there exists an open set W containing (a, b) and an open set W' containing $\Phi(a, b) = (a, 0)$ such that $\Phi : W \rightarrow W'$ is a C^1 diffeomorphism.

Finally we can construct the implicit function: the inverse $\Phi^{-1}(x, y) = (x, g(x, y))$ for some C^1 smooth function g in W' , so we define the implicit function on the domain $D = \{x \in \mathbb{R} | (x, 0) \in W'\}$ (**RMQ:** here D is the intersection of W' and the x -axis, notice that at least $(a, 0) \in W'$, so D is not empty and can be seen as an open interval containing a)

$$f(x) = g(x, 0)$$

then

$$\Phi(x, f(x)) = \Phi(x, g(x, 0)) = \Phi \circ \Phi^{-1}(x, 0) = (x, 0)$$

by the definition of Φ we can conclude that $F(x, f(x)) = 0$ □

Let us back to the two examples above, if we apply the implicit function theorem here, we can get a clear ODE for the implicit function $y = f(x)$, that is an equivalent view of the implicit function theorem.

Example 1.1. See the equation (2), we want to find the express of y as a function of x , so see the whole equation as a curve $F(x, y) = e^{xy} + \sin y + x^2 - 2$, then by the implicit function theorem, we can conclude a new relationship between x and y :

$$y' = f'(x) = -\frac{F_x(x, y)}{F_y(x, y)} = -\frac{ye^{xy} + 2x}{xe^{xy} + \cos y}$$

so finding the explicit formula of y is equivalent to solving above ODE to get a explicit solution. The information here is given by the differential structure of the curve, and notice that the ODE contains the original information about the curve.

However, sometimes we can find the explicit formula of the implicit function, for example, in equation (1), we can express y as a function of x explicitly:

$$y = \pm\sqrt{1 - x^2}$$

and the corresponding ODE here is

$$y' = -\frac{x}{y}$$

It's a separable ODE, we can solve it easily to get the explicit formula of the $y = f(x)$.

2 CA: Some examples and technics

Example 2.1. We consider two algebraic number fields $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{5})$. and we can study the algebraic integer ring in two fields, then we can find that

$$K = \mathbb{Q}(\sqrt{2}), \quad \mathcal{O}_K = \mathbb{Z}[\sqrt{2}]$$

and

$$K = \mathbb{Q}(\sqrt{5}), \quad \mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{5}}{2}\right]$$

here $X^2 + X + 1$ gives a monic polynomial with integer coefficients such that $\frac{1+\sqrt{5}}{2}$ is a root, that shows that $\frac{1+\sqrt{5}}{2}$ is an algebraic integer, so $\mathbb{Z}[\sqrt{5}] \subsetneq \mathcal{O}_K$.

It's a classic example about ring of algebraic integers, we can say more about quadratic fields

Theorem 2.1. Let d be an integer without square factors, and let $K = \mathbb{Q}(\sqrt{d})$ be a quadratic field. Then the ring of integers $\mathcal{O}_K$ is given by

$$\mathcal{O}_K = \begin{cases} \mathbb{Z}[\sqrt{d}] & \text{if } d \equiv 2, 3 \pmod{4}, \\ \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right] & \text{if } d \equiv 1 \pmod{4}. \end{cases}$$

Proof. One proof elementary is to use p-adic valuation, here is the outline:

- Prove the **main lemma**: $a + b\sqrt{d}$ is an algebraic integer if and only if $2a \in \mathbb{Z}$ and $a^2 - db^2 \in \mathbb{Z}$.
- Verify that if $a + b\sqrt{d}$ is an algebraic integer, then $v_p(a) \geq 0$ and $v_p(b) \geq 0$ for any odd prime p . (Notice that here is not sufficient to say $a, b \in \mathbb{Z}$)
- Consider the case of 2-adic valuation we can get the condition $v_2(a) \geq -1$ and $v_2(b) \geq -1$.
- Prove that $v_2(a) = -1$ and $v_2(b) = -1$ if and only if $d \equiv 1 \pmod{4}$, otherwise $v_2(a) \geq 0$ and $v_2(b) \geq 0$, then we can conclude the result.

□

Notice that it is not so easy to prove the similar result for cubic fields or higher degree fields, the concept of **integral basis** is useful here.

operation of ideals: a useful technique to compute the isomorphism.

Proposition 2.2. Let M be a A -module, I be an ideal of A and N be a submodule of M , then we have the isomorphism:

$$I(M/N) \cong (IM + N)/N$$

In particular, when $M = A$ and $N = J$ is a ideal, then we have

$$I(A/J) \cong (I + J)/J$$

Proof. we consider the natural map $\pi : M \rightarrow M/N$. We can verify that $I(M/N)$ is a submodule of M/N , then by correspondence theorem, there exists a submodule in M :

$$\begin{aligned}\pi^{-1}(I(M/N)) &= \pi^{-1}\left\{\sum_{\text{finite}} a_k \cdot (m_k + N) \mid a_k \in I, m_k \in M\right\} \\ &= \pi^{-1}\left\{\sum_{\text{finite}} a_k m_k + N \mid a_k \in I, m_k \in M\right\} \\ &= IM + N\end{aligned}$$

Then again by the image of natural map we can match

$$I(M/N) = (IM + N)/N$$

□

Proposition 2.3. If $I \subset R$ as a ideal, $a \in I$ and $b \notin I$, then

$$a + b \notin I$$

Proof. If $a + b \in I$, then $b = (a + b) - a \in I$, which is a contradiction. □

3 Fundamental group and Covering Space

3.1 Homotopy and π_1 functor

”deforming” is a topological concept, it refers to the process of continuously transformation.

Definition 3.1. Let $f, g : X \rightarrow Y$ be two continuous maps between topological spaces X , we say that f is **homotopic** to g if there exists a continuous map $H : X \times I \rightarrow Y$ such that

$$H(x, 0) = f(x), \quad H(x, 1) = g(x)$$

and we denote it by $f \simeq_H g$.

Furthermore, if there exists a subset $A \subset X$ such that for any $a \in A$, the homotopy $H(a, -) : I \rightarrow Y$ is constant, then we say that f is **homotopic relative to A** to g , and we denote it by $f \simeq_{H,A} g$.

In particular, a map $f : X \rightarrow Y$ is (relative) **nullhomotopic** if it is relative homotopic to a constant map.

One point is that relative homotopy share the most properties with homotopy, the proof is similar, just checking the condition on A . Here are some basic properties of homotopy:

Proposition 3.1. Let $X, Y, Z \in \mathbf{Top}$

(1) (relative) homotopy defines **an equivalence relation** on $\mathcal{C}(X, Y)$.

homotopy behaves well in composition:

(2) If $f, g : X \rightarrow Y$ and $h : Y \rightarrow Z$, then (relative) homotopy

$$f \simeq_{H,A} g \implies h \circ f \simeq_{F,A} h \circ g$$

with $F(x, t) = h(H(x, t))$.

(3) If $f, g : X \rightarrow Y$ and $h : Z \rightarrow X$, then (relative) homotopy

$$f \simeq_{H,A} g \implies f \circ h \simeq_{F,h^{-1}(A)} g \circ h$$

with $F(z, t) = H(h(z), t)$.

(3) If $f, g : X \rightarrow Y$ and $f', g' : Y \rightarrow Z$, then (relative) homotopy

$$f \simeq_{H,A} g, \quad f' \simeq_{H',A'} g' \implies f' \circ f \simeq_{F,A} g' \circ g$$

with

$$F(x, t) = \begin{cases} f'(H(x, 2t)) & t \in [0, \frac{1}{2}] \\ H'(g(x), 2t - 1) & t \in [\frac{1}{2}, 1] \end{cases}$$

The relative homotopy holds when $f(A) \subset A'$ and $g(A) \subset A'$.

Proof. (1) the proof of reflexive and symmetric property is not difficult, but the proof of transitive properties needs a technical lemma, which is very important in algebraic topology.

Lemma (Gluing Lemma).

Let X be a topological space with closed subsets A and B such that $X = A \cup B$. If $f : A \rightarrow Y$ and $g : B \rightarrow Y$ are continuous maps such that $f(x) = g(x)$ for all $x \in A \cap B$, then there exists a unique continuous map $h : X \rightarrow Y$ such that $h|_A = f$ and $h|_B = g$.

with this lemma we can prove that if $f \simeq_F g$ and $g \simeq_H h$, then we can construct a homotopy $H : X \times I \rightarrow Y$

$$H(x, t) = \begin{cases} F(x, 2t) & t \in [0, \frac{1}{2}] \\ H(x, 2t - 1) & t \in [\frac{1}{2}, 1] \end{cases}$$

then it finishes the proof. For the relative case, let $F(x, -)$ and $H(x, -)$ be constant for any $x \in A$, then $F(x, 1) = g(x)$ and $H(x, 0) = g(x)$ show that $F(x, -) = G(x, -) = H(x, -)$.

(2)(3)(4) are direct verifications. □

Remark 3.1. Category **Top** is too big and rough that we cannot do more, with above properties, we can define the quotient category

$$\mathbf{hTop} \mid \begin{array}{l} \text{objects: topological space } X \\ \text{morphisms: homotopy class } [f] \in \mathcal{C}(X, Y) / \sim \end{array}$$

similarly if we consider the relative homotopy, it also defines a quotient category of pair topological space $\mathbf{Top}^2$, where the object is couple (X, A) with $A \subset X$ as a subspace, and the morphism between (X, A) and (Y, B) is a continuous map $f : X \rightarrow Y$ such that

$$f(A) \subset B, \text{ i.e. a commutative diagram: } \begin{array}{ccc} X & \xrightarrow{f} & Y \\ i_A \uparrow & & \uparrow i_B \\ A & \xrightarrow{f|_A} & B \end{array} \quad \text{Then the correspondence}$$

quotient category is denoted by

$$\mathbf{hTop}^2 \mid \begin{array}{l} \text{objects: } (X, A) \in \mathbf{Top}^2 \\ \text{morphisms: } [f] \in \mathcal{C}(X, Y) / \sim_{relA} \end{array}$$

In above two categories, the composition of morphism is well-defined by the **composition of homotopy**, i.e. we can define

$$[f] \circ [g] := [f \circ g]$$

for any $g : X \rightarrow Y$ and $f : Y \rightarrow Z$. And the identity morphism of a object X is $[id_X]$.

In category, we define two objects are isomorphic if there exists two morphisms between them such that their composition is identity morphism, in above two categories, for any two topological spaces X and Y , they are isomorphic if there exists two classes $[f] \in [X, Y]$ and $[g] \in [Y, X]$ such that

$$[f] \circ [g] = [id_Y] \quad \text{and} \quad [g] \circ [f] = [id_X]$$

which isomorphism is very important in algebraic topology, hence it deserves a name:

Definition 3.2. Let X and Y be two topological spaces, we say that they are **homotopy equivalent** if there exists two continuous maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that their compositions are homotopic to identity maps:

$$f \circ g \simeq id_Y \quad \text{and} \quad g \circ f \simeq id_X$$

The following object of the section is to define a functor from $\mathbf{Top}$ to other categories, here we see a simple example to $\mathbf{Set}$ category, the detail can be found in Rotman's book (GTM 119).

Definition 3.3. $\pi_0 : \mathbf{Top} \rightarrow \mathbf{Set}$ is a functor defined as following:

- On objects: for any topological space X , $\pi_0(X) := X / \sim_c$, where $\sim_c$ is the equivalence relation defined by path-connectness, i.e. $X / \sim_c$ is the set of path-connected components of X .

- On morphisms: for any continuous map $f : X \rightarrow Y$, we define

$$\pi_0(f) : \pi_0(X) \rightarrow \pi_0(Y), \quad [x] \mapsto [f(x)]$$

which maps a component to another component under image, which is ensured by the continuity.

should pay attention that homeomorphic condition implies homotopy equivalence, but not vice versa, for example

$\mathbb{S}^1$ and $\mathbb{R}^2 - \{0\}$ are homotopy equivalent but not homeomorphic.

Here π_0 gives a simple way to distinguish the spaces by counting the number of path-connected components, and we can prove the following proposition:

Proposition 3.2. If $f, g : X \rightarrow Y$ are homotopic, then $\pi_0(f) = \pi_0(g)$. In particular, If X, Y are homotopy equivalent, then there exists a bijection between $\pi_0(X)$ and $\pi_0(Y)$, i.e. they have same cardinality.

Proof. Let $H : X \times I \rightarrow Y$ be a homotopy from f to g , for any $x \in X$, we define a path $\gamma : I \rightarrow Y$ by

$$\gamma(t) = H(x, t)$$

then $\gamma(0) = f(x)$ and $\gamma(1) = g(x)$, hence $f(x)$ and $g(x)$ are in the same path-connected component, which implies that $\pi_0(f)([x]) = [f(x)] = [g(x)] = \pi_0(g)([x])$.

For the second part, if X and Y are homotopy equivalent, then there exists two continuous maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that

$$f \circ g \simeq id_Y, \quad g \circ f \simeq id_X$$

then by the first part we have

$$\pi_0(f) \circ \pi_0(g) = \pi_0(f \circ g) = \pi_0(id_Y) = id_{\pi_0(Y)}$$

and similarly

$$\pi_0(g) \circ \pi_0(f) = id_{\pi_0(X)}$$

hence $\pi_0(f)$ and $\pi_0(g)$ are bijections between $\pi_0(X)$ and $\pi_0(Y)$. □

This simple construction shows the following concept:

(1) If a functor $F : \mathbf{Top} \rightarrow \mathbf{C}$ satisfies that for any two homotopic maps $f, g : X \rightarrow Y$, $F(f) = F(g)$, then it is called **homotopic invariant**, then the functor can induce a well-defined functor on **hTop**. Here π_0 is clearly a homotopic invariant, and we can induce a functor $\bar{\pi}_0 : \mathbf{hTop} \rightarrow \mathbf{Set}$ by $\bar{\pi}_0([f]) = \pi_0(f)$.

(2) It often happens that two spaces are not homeomorphic, but they have the similar shape (i.e. homotopy equivalent), usually some topological properties just depends on the the shape of spaces, hence homotopy equivalence will be a good condition to use.

(3) Actually π_0 is so limited that we can not talk more about the detail of the spaces, one reason is that **Set** is also too rough, we can not get more information from it.

From complex analysis and differential geometry, some information can be reflected by loops, i.e. closed paths. for example we consider a such application:

$$w : C(\mathbb{C}^*, 1) \rightarrow \mathbb{C}, \quad \gamma \mapsto \int_{\gamma} \frac{dz}{z}$$

where $C(\mathbb{C}^*, 1)$ denotes the set of closed paths in $\mathbb{C}^*$ starting from 1, with the basic knowledge we can know that all possible values of w is of the form $2\pi in$ for some integer n . Which requires us to classify all path closed paths in $\mathbb{C}^*$, and we can do a very rough partition here without proof, it is very intuitive:

$$C(\mathbb{C}^*, 1) = \sqcup_{k \in \mathbb{Z}} C_k$$

where C_0 denotes the set of closed paths which does not enclose the zero, while C_k for $k \neq 0$ denotes the set of closed paths which enclose the zero k times, moreover we make convention that the positive direction is counter-clockwise, hence which allows us to define a winding number for any closed path in $\mathbb{C}^*$:

$$W : C(\mathbb{C}^*, 1) / \sim \rightarrow \mathbb{Z}, \quad [\gamma] \mapsto \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z}$$

Furthermore we want to conclude a homomorphism to additive group, because we clearly know that if we concatenate two closed paths γ_1 and γ_2 , then the winding number of the new path is the sum of two original winding numbers:

$$W([\gamma_1 * \gamma_2]) = W([\gamma_1]) + W([\gamma_2])$$

hence it gives a group isomorphism. Indeed, the naive partition above is just realtion of homotopy, if we take two homotopy closed path γ_1, γ_2 , then there exists a homotopy H such that $\gamma_1 \simeq_{H, \{1\}} \gamma_2$, we can see that

$$W([\gamma_1]) - W([\gamma_2]) = \frac{1}{2\pi i} \int_{\gamma_1 - \gamma_2} \frac{dz}{z} = \frac{1}{2\pi i} \int_S \frac{dz}{z} = 0$$

the surface S is enclosed by γ_1 and $-\gamma_2$, and they will not enclose the zero by homotopy, hence Stoke's theorem (or Cauchy's theorem) shows that the integral is zero since it is a closed form on S . It is a inspiration to the fundamental group theory, and it is a really classic and important example, Hence we should pay our attention to the closed paths of a spaces, and define a group via homotopy.

Definition 3.4. For any two path $f, g : I \rightarrow X$ of a topological space with same end-points, they are **path homotopic** if they are relative homotopic with respect to $\{0, 1\}$, i.e. there exists a continous map $H : I \times I \rightarrow X$ such that

$$H(s, 0) = f(s), \quad H(s, 1) = g(s), \quad H(0, t) = f(0) = g(0), \quad H(1, t) = f(1) = g(1)$$

for any $s, t \in I$.

As a special case of realtive homotopy, it defines an equivalence reation on $\mathcal{C}(I, X)$, i.e. the set of all paths in X . If g is a path starting from the endpoint of path f , i.e. $f(1) = g(0)$, we can connect it to get a new path, we can formilize it

$$(f * g)(t) := \begin{cases} f(2t), & t \in [0, 1/2] \\ g(2t - 1), & t \in [1/2, 1] \end{cases}$$

It is the ideal way to define the algebra on the set of path homotopy classes, and in fact it works not bad:

Proposition 3.3. Let X be a topological space and $x_0, y_0 \in X$, let $C(X; x_0, y_0)$ be the set of all paths from x_0 to y_0 and $\sim$ be the path homotopy relation, then $C(X; x_0, y_0)/\sim$ forms a groupoid under the operation of concatenation of paths:

$$[f] * [g] := [f * g]$$

Proof. It needs to check the definition:

(1) operation is well-defined: if $f \simeq f'$ and $g \simeq g'$, then

$$[f] * [g] = [f * g] \stackrel{?}{=} [f' * g'] = [f'] * [g']$$

(2) associativity: for any three paths f, g, h with proper endpoints, we can verify that $([f] * [g]) * [h] = [f] * ([g] * [h])$ by direct computation. □

In the system, there does not exists a real identity elemnt, but we can actually find right-identity and left-identity. We define two special paths here:

$$e_{x_0}(t) = x_0, \quad e_{y_0}(t) = y_0$$

for any $t \in I$, then for any path $f \in C(X; x_0, y_0)$, we can verify that

$$[e_{x_0}] * [f] = [f], \quad [f] * [e_{y_0}] = [f]$$

although the two constant paths are exactly not in $C(X; x_0, y_0)$, in particular when $x_0 = y_0$, they are the same path, and it is in $C(X; x_0, x_0)$, hence it will be the identity element of the groupoid, hence we can talk more if we consider about loops(The algebraic reason):

Proposition 3.4. Let X be a topological space and $x_0 \in X$, let $C(X; x_0)$ be the set of all loops based at x_0 and $\sim$ be the path homotopy relation, then $(C(X; x_0)/\sim, *, [e_{x_0}])$ forms a group under the operation of concatenation of paths.

Proof. Based on the previous proposition, we only need to check:

(1) $[e_{x_0}]$ is exactly the identity element.

(2) for any path f , it is natural to reverse the path:

$$\bar{f} : I \rightarrow X, \quad t \mapsto f(1 - t)$$

then we can verify that its class is the inverse: $[f] * [\bar{f}] = [e_{x_0}]$ □

This group is called the **fundamental group** of X based at x_0 , and it is denoted by $\pi_1(X, x_0)$, to consider the functoriality here, we consider a special case of pair topological category

$$\mathbf{Top}_* \left| \begin{array}{l} \text{objects: topological space with base point } (X, x_0) \\ \text{morphisms: continous map with } f(x_0) = y_0 \end{array} \right.$$

Notice that $f, g : (X, x_0) \rightarrow (Y, y_0)$ is homotopy if there exists a continous map $H : X \times I \rightarrow Y$ such that $f \simeq_{H, \{x_0\}} g$, i.e. relative homotopy with respect to the base point, it is induced by $\mathbf{hTop}^2$!

Definition 3.5. $\pi_1 : \mathbf{Top}_* \rightarrow \mathbf{Grp}$ is a covariant functor as following:

-on objects: fundamental group $\pi(X, x_0) = C(X; x_0) / \sim$

-on morphisms: For any continous map $f : (X, x_0) \rightarrow (Y, y_0)$, we define

$$\pi_1(f) : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0), \quad [\gamma] \mapsto [f \circ \gamma]$$

usually we denote $\pi_1(f)$ by f_* .

Here are some basic properties of fundamental group functor, it can be left to verify:

1. induce map f_* is well-defined as a group homomorphism.
2. **functoriality**: for any two continous maps $f : (X, x_0) \rightarrow (Y, y_0)$ and $g : (Y, y_0) \rightarrow (Z, z_0)$, then

$$(g \circ f)_* = g_* \circ f_*$$

3. Let $a \in X$ be a based point, and X_a is the path-connected component of X containing a , then the inclusion map $i : X_a \rightarrow X$ induces an isomorphism

$$i_* : \pi_1(X_a, a) \rightarrow \pi_1(X, a)$$

4. If $f : I \rightarrow X$ is a path connecting two based points x_0 and x_1 , then it induces an isomorphism

$$f_{\#} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1), \quad [\gamma] \mapsto [f * \gamma * \bar{f}]$$

In particular, if X is path-connected, then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$ for any two based points $x_0, x_1 \in X$.

5. product keeps fundamental group: for any two based topological spaces (X, x_0) and (Y, y_0) , then

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

Finally, we conclude the homotopy invariant property of fundamental group functor, it is really important that which allows us to identify the fundamental group of homotopy equivalent spaces.

Theorem 3.5. If $f, g : (X, x_0) \rightarrow (Y, y_0)$ are homotopic continuous maps, then $f_* = g_*$. In particular, if (X, x_0) and (Y, y_0) are homotopy equivalent, then their fundamental groups are isomorphic.

Proof. Let $f \simeq_H g$, then for any $[\gamma] \in \pi_1(X, x_0)$, we define a homotopy

$$F : I \times I \rightarrow Y, \quad (s, t) \mapsto H(\gamma(s), t)$$

It is continuous by composition, and $F(s, 0) = f \circ \gamma(s)$ and $F(s, 1) = g \circ \gamma(s)$, moreover $F(0, t) = H(x_0, t) = y_0$ and $F(1, t) = H(x_0, t) = y_0$, hence it shows that $f \circ \gamma \simeq g \circ \gamma$, i.e. $f_*([\gamma]) = g_*([\gamma])$. For the second part, if (X, x_0) and (Y, y_0) are homotopy equivalent, then there exists two continuous maps $f : (X, x_0) \rightarrow (Y, y_0)$ and $g : (Y, y_0) \rightarrow (X, x_0)$ such that

$$f \circ g \simeq id_{(Y, y_0)}, \quad g \circ f \simeq id_{(X, x_0)}$$

then by the first part and functoriality we have

$$f_* \circ g_* = (f \circ g)_* = (id_{(Y, y_0)})_* = id_{\pi_1(Y, y_0)}$$

and similarly $g_* \circ f_* = id_{\pi_1(X, x_0)}$, QED. □

Remark 3.2. The theorem shows that fundamental group is a homotopy invariant, hence it induces a well-defined functor $\bar{\pi}_1$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathbf{Top}_* & \xrightarrow{\pi_1} & \mathbf{Grp} \\ \downarrow \gamma & \nearrow \bar{\pi}_1 & \\ \mathbf{hTop}_* & & \end{array}$$

i.e there exists a unique factorization of π_1 through $\mathbf{hTop}_*$, that means functor π_1 depends indeed on the homotopy type of spaces, and what we just compress the original category, the construction of fundamental have two steps in general: (1) Firstly we forget the specific topological structure of spaces, and just record the shape of spaces and classify them by homotopy equivalence, which gives the homotopy category $\mathbf{hTop}_*$.

(2) In each homotopy type, we choose a representation to study, and we study them in a certain viewpoint: look at the loops based at a point, actually it will reflect 1-dimensional information of the space, in other words, we forget some other higher-dimensional information of the space.

3.2 Simple cases

The calculation of fundamental group is not easy in general, and the fundamental group induce a definition of a special connected space:

Definition 3.6. A topological space X is **simply connected** if it is path-connected and its fundamental group is trivial, i.e. for any based point $x_0 \in X$, $\pi_1(X, x_0) = \{[e_{x_0}]\}$.

In complex analysis, simply connected spaces plays a important role and many good properties holds under this condition, and roughly speaking, simply connected spaces in 2-dimension is just a space without holes. It is easy to prove that $\mathbb{R}^n$ is simply connected for any $n \geq 1$, and more generally, we can conclude that:

Proposition 3.6. Any convex subset X of $\mathbb{R}^n$ is simply connected.

Proof. For any two points $x_0, x_1 \in X$, the line segment connecting them is contained in X by convexity, hence X is path-connected. For any loop $\gamma \in C(X; x_0)$, we can define a homotopy

$$H : I \times I \rightarrow X, \quad (s, t) \mapsto (1 - t)\gamma(s) + tx_0$$

then clearly $H(s, 0) = \gamma(s)$ and $H(s, 1) = x_0$, moreover $H(0, t) = x_0$ and $H(1, t) = x_0$, hence it shows that $\gamma \simeq_{H, \{0,1\}} e_{x_0}$, which implies that $\pi_1(X, x_0)$ is trivial. $\square$

As we have mentioned before, we construct a group isomorphism by winding integration:

$$C(\mathbb{C}^*, 1) / \sim \cong \mathbb{Z}$$

under the operation of concatenation, and we prove that same homotopy class implies the same winding number, hence realtion of homotopy is just the subrealtion here, and in fact that this realtion is exactly same with the path homotopy realtion, which means that

$$\pi_1(\mathbb{C}^*) = \mathbb{Z}$$

however, it is not easy to prove that same winding number implies path homotopy, by comparison with specific integration and analysis, homotopy is a more abstract concept, some new ideas and tools are needed. Here we admit the result of fundamental group of $\mathbb{C}^*$, and we will see it will be a key to caculate many fundamental groups later.

Claim 3.7.

$$\pi_1(\mathbb{S}^1) \cong \mathbb{Z}$$

The reason is that $\mathbb{C}^*$ is homotopy equivalent to $\mathbb{S}^1$, but more geometrically we realize that the fundamental group depends on the shape of spaces, and $\mathbb{S}^1$ can viewed as a "smallest" model same as $\mathbb{C}^*$, more precisely we define a special case of homotopy equivalent:

Definition 3.7. Let X be a topological space and $i : A \hookrightarrow X$ be a embedding map.

-If there exists a continous map $r : X \rightarrow A$ such that $r \circ i = id_A$, then A is called a **retract** of X , and r is called a **retraction**.

-Furthermore if $i \circ r \simeq id_X$ gives a homotopy, then A is called a **deformation retract** of X , and r is called a **deformation retraction**.

By the definition, we can **verify** that deformation retract is a special case of homotopy equivalent, hence it induces a general isomorphism of fundamental groups:

Claim 3.8.

$$\pi_1(\mathbb{S}^n) \cong \pi_1(\mathbb{R}^{n+1} - \{0\})$$

and similarly we can give some other examples:

$$\pi(A, a) \cong \mathbb{Z}$$

where A is the annulus $\{z \in \mathbb{C} : r < |z| < R\}$ for some $0 < r < R$, and $a \in A$. Pay attention that **retract is not necessary deformation retract**, for example the unit circle $\mathbb{S}^1$ is a retract of the disk $D = \{z \in \mathbb{C} : |z| \leq 1\}$ by the retraction, but they are not same homotopy type since $\pi_1(D) = \{0\}$ as a convex set.

We should also notice a particular case of deformation retract: the whole space can be contracted to just one point, i.e. the case of $A = \{\text{pt}\}$, it is indeed possible, for example we consider $\mathbb{R}^n$ and define a deformation retraction

$$r : \mathbb{R}^n \rightarrow \{0\}, \quad x \mapsto 0$$

with the inclusion $i : \{0\} \rightarrow \mathbb{R}^n$ by $i(0) = 0$, then we can verify

$$r \circ i(0) = 0, \quad i \circ r(x) = 0$$

and we can find a homotopy

$$H : \mathbb{R}^n \times I \rightarrow \mathbb{R}^n, \quad (x, t) \mapsto (1 - t)x$$

which satisfies

$$H(x, 0) = x, \quad H(x, 1) = 0$$

hence it shows that $\mathbb{R}^n$ is deformation retract to a point, which implies the fundamental group is just the trivial group $\pi_1(\{\text{pt}\})$. In general, we can conclude the special spaces:

Definition 3.8. Let X be a topological space, X is **contractible** if it can be deformation retracted to a point, equivalently(**verify!**), the identity map id_X is nullhomotopic.

Pay attention that

contractible spaces is a special case of simply connected spaces.

It is not true vice versa, a non-trivial example is $\mathbb{S}^2$ whose fundamental group is trivial, but it is not contractible. However, the calculation of fundamental group needs some discussion, what we can do here is just to see how a path in sphere can be contracted to a point, draw some pictures we can know the continuous transformation of a loop can be done by passing the pole, it gives us a intuition that the fundamental group is trivial.

To better interpretation the idea, we introduce a special construction of contractible space:

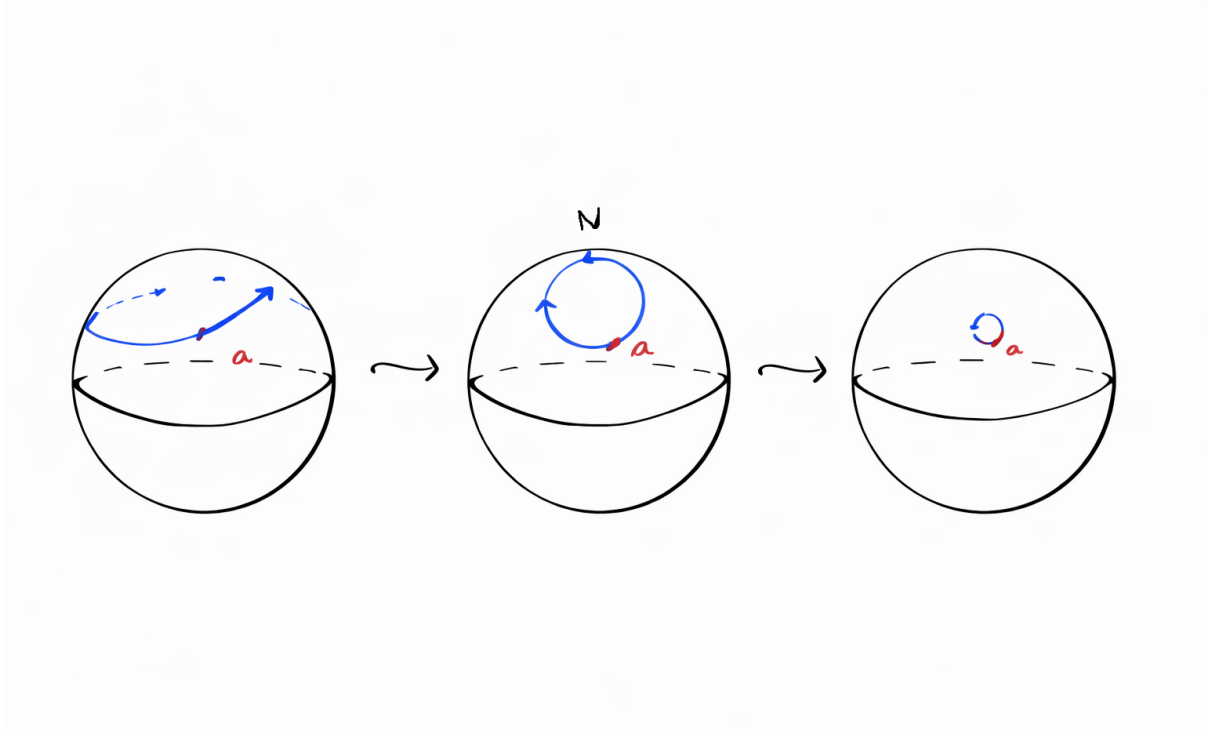


Figure 2: Contract a loop in $\mathbb{S}^2$ to a point by passing the pole

Definition 3.9. Let X be a topological space, the **cone** of X is defined by

$$C(X) := (X \times I) / (X \times \{1\})$$

i.e. we identify all points in $X \times \{1\}$ to a single point, which is called the **cone point**. Similarly, the suspension of X is defined by

$$S(X) := (X \times I) / (X \times \{0\}, X \times \{1\})$$

i.e. we identify all points in $X \times \{0\}$ to a single point, and all points in $X \times \{1\}$ to another single point, which are called the **suspension points**.

with this construction, we can identify $\mathbb{S}^1$ by homeomorphism

$$\mathbb{S}^2 \cong S(\mathbb{S}^1) \cong C(\mathbb{S}^1) \sqcup_{\mathbb{S}^1} C(\mathbb{S}^1)$$

the last construction is to glue two copies along the $\mathbb{S}^1$, hence the question is reduced to consider when the cone and suspension are contractible

Proposition 3.9. For any topological space X ,

- The cone $C(X)$ is contractible.
- If X is contractible, then the suspension $S(X)$ is also contractible.

- If X is path-connected, then the suspension $S(X)$ is also path-connected with $\pi_1(S(X), a) = 0$

Proof. □

However, we can not talk more about whether $\mathbb{S}^2$ is contractible or not, If $S(X)$ is contractible, we can not conclude that X is contractible or not, and the example is very messy, hence it is an ideal way to handle the problem. Anyway, the contractible property of $\mathbb{S}^n$ is equivalent to the brouwer fixed-point theorem, and someother important results in topology, it reflects the difficulty of the problem without algebraic topology tools.

Claim 3.10. $\mathbb{S}^n$ is not contractible, in particular $n \geq 2$

$$\pi_1(\mathbb{S}^n) \cong \{0\}$$

Theorem 3.11. Let $f : \mathbb{S}^1 \rightarrow X$ be a continous map, then the following statements are equivalent:

- (1) f is nullhomotopic.
- (2) f can be extended to a continous map $\tilde{f} : D^2 \rightarrow X$ such that $\tilde{f}|_{\mathbb{S}^1} = f$.
- (3) The induced homomorphism $f_* : \pi_1(\mathbb{S}^1) \rightarrow \pi_1(X)$ is trivial.

Proof. □

Remark 3.3. It is a techinc and useful result, and generally we can think of fundamental group from the view of $\mathbb{S}^1$. For any closed path $\gamma : I \rightarrow X$ with $\gamma(0) = \gamma(1)$, we can identify it as a map from $\mathbb{S}^1$ to X by the following diagram

$$\begin{array}{ccc} I & \xrightarrow{\gamma} & X \\ \downarrow \sim & \nearrow f & \\ \mathbb{S}^1 & & \end{array}$$

Then It makes sense to consider Hom functor $\text{Hom}_{\mathbf{Top}_*}(\mathbb{S}^1, -)$ which admits object same with $C(X; x_0)$, and use the factorization and Yoneda lemma we can get clearly the representation of fundamental group functor:

$$\pi_1(-) \cong \text{Hom}_{\mathbf{hTop}_*}(\mathbb{S}^1, -)$$

It is the category construction of fundamental group functor, here if we use $\mathbb{S}^n$ to replace $\mathbb{S}^1$, then we can get the general homotopy group π_n .

3.3 Van-Kampen Theorem

For a Torus, by classic homeomorphism we have $T^2 \cong \mathbb{S}^1 \times \mathbb{S}^1$, hence by product property of fundamental group, we have

$$\pi_1(T, a) \cong \pi_1(\mathbb{S}^1, a_1) \times \pi_1(\mathbb{S}^1, a_2) \cong \mathbb{Z} \times \mathbb{Z}$$

for any based point $a = (a_1, a_2) \in T^2$. Now if we remove a point from the torus, then will the fundamental group change? What's the new fundamental group? To see the homotopy type of $T^2 - \{pt\}$, we can consider the definition of torus by quotient

$$T^2 = [0, 1]^2 / \sim$$

with relation $(0, y) \sim (1, y)$ and $(x, 0) \sim (x, 1)$ for any $x, y \in [0, 1]$, then we can see that removing a point from the torus is just the quotient space of the square removing a point inside it

$$\begin{array}{ccc} [0, 1]^2 - \{p\} & \xrightarrow{r} & \partial[0, 1]^2 \\ \pi \downarrow & & \downarrow \pi \\ T^2 - \{pt\} & \xrightarrow{\bar{r}} & \partial[0, 1]^2 / \sim \end{array}$$

Here r is the deformation retraction from the square removing a point to its boundary, and quotient relation $\sim$ do not affect the deformation by definition, hence the induced map by UPQ-TOP $\bar{r}$ is exactly a deformation retraction, hence we can conclude the homotopy equivalence.

The object we get is actually a wedge sum of two circles, formally

Definition 3.10. Let (X, x_0) and (Y, y_0) be object of $\mathbf{Top}_*$, the **wedge sum** of X and Y is defined by

$$X \vee Y := (X \sqcup Y) / (x_0 \sim y_0)$$

i.e. we glue the two spaces at their base points. and we define the point $[x_0] = [y_0]$ in $X \vee Y$ as the base point

From the view of category, wedge sum is just the coproduct in $\mathbf{Top}_*$, and we (should) know that coproduct in $\mathbf{Grp}$ is the free product, hence it is natural to guess that the fundamental group of wedge sum is the free product of fundamental groups, and in fact it is true.

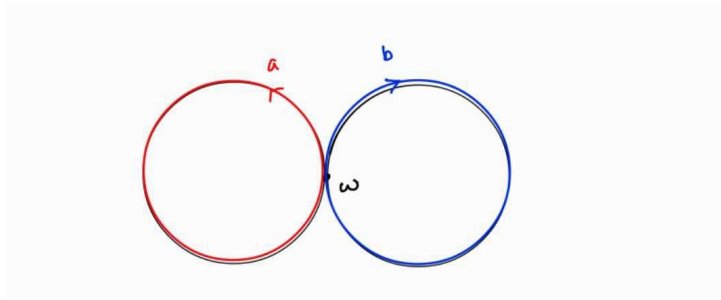


Figure 3: Wedge sum of two circles $\mathbb{S}^1 \vee \mathbb{S}^1$

let A and B be two copies of $\mathbb{S}^1$, and consider the wedge sum with the base point at w , and we suppose that a and b be two paths based at w going around A and B respectively once (see the figure), then clearly $[a]$ and $[b]$ are the generators of $\pi_1(A, w)$ and $\pi_1(B, w)$ respectively, but by conactenation of paths $a*b$ seems not be homotopic to $b*a$ (although it is not so clear to see it), hence the fundamental group seems and be generated by $\langle [a], [b] \rangle$, it is the free product group with rank 2.

Claim 3.12.

$$\pi_1(\mathbb{S}^1 \vee \mathbb{S}^1) \cong \pi_1(\mathbb{S}^1) * \pi_1(\mathbb{S}^1) \cong \mathbb{Z} * \mathbb{Z}$$

In generally, π_1 is not a **left-adjoint functor**, hence it does not preserve coprouducts, but in some certain case we can see that it does, and in particular preserve push out.

Conversely, we admit the result of fundamental group here, and then we think about the fundamental group of torus in another way, although we have known that $\pi_1(T^2) \cong \mathbb{Z} \times \mathbb{Z}$. we take small open disc U in T^2 , and set $D \subsetneq U$ as a smaller closed disc, and then we set $V = T^2 - D$ to be another open set such that $U \cap V$ is homotopy equivalent to $\mathbb{S}^1$:

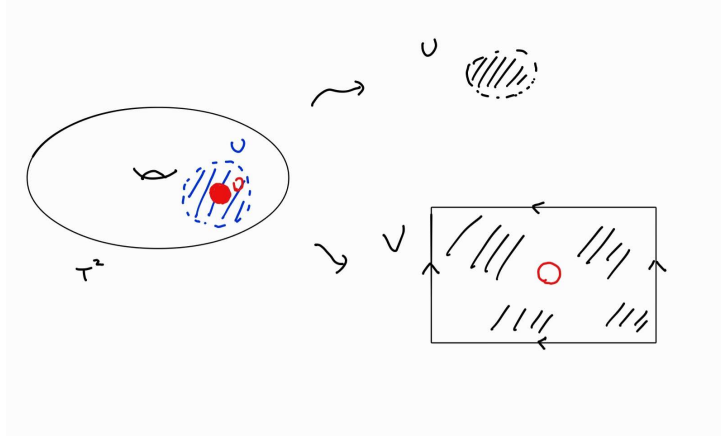


Figure 4: Open cover of T^2 by U and V

V can be seen as a deformation retraction of $T^2 - \{\text{pt}\}$, and U is simply connected (in view of manifold, it is homeomorphic to $\mathbb{R}^2$), hence we get their fundamental groups by taking a based point $a \in U \cap V$:

$$\pi_1(U, a) \cong \{0\}, \quad \pi_1(V, a) \cong \mathbb{Z} * \mathbb{Z}, \quad \pi_1(U \cap V, a) \cong \mathbb{Z}$$

In the view of topology, T^2 can be viewed as the identification space of U and V along their intersection, i.e. we have the following diagram:

$$\begin{array}{ccc} U \cap V & \xrightarrow{i} & U \\ j \downarrow & & \downarrow p \\ V & \xrightarrow{q} & T^2 \end{array}$$

To explicitly give the induced maps, we set the generator of $\pi_1(U \cap V, a) \cong \mathbb{Z}$ to be $[\gamma]$, and the generators of $\pi_1(V, a) \cong \mathbb{Z} * \mathbb{Z}$ to be $[\alpha]$ and $[\beta]$, then we can prove that induced maps of j are $j_*([\gamma]) = [\alpha][\beta][\alpha]^{-1}[\beta]^{-1}$, although it is intuitive to see, but the proof is not trivial, we sketch the outline here:

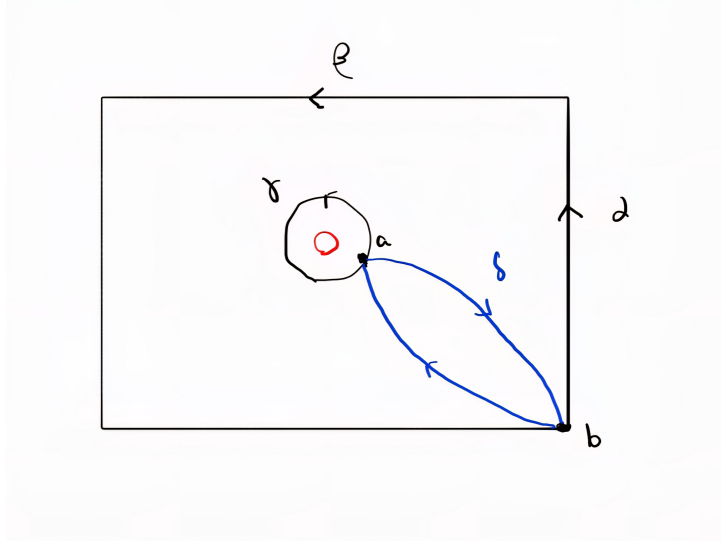


Figure 5: the referred paths

(1) Choosing a point b of the vertex $[0, 1]^2$, and let $\alpha\beta\alpha^{-1}\beta^{-1}$ be a path of V based at b , then prove γ and $\alpha\beta\alpha^{-1}\beta^{-1}$ is (free) homotopy in V .

(2) the path connectness ensures to choose a path l in U connecting a and b , and prove that the loop is nullhomotopic in V .

(3) By concatenation, $\delta\alpha\beta\alpha^{-1}\beta^{-1}\bar{\delta}$ gives a loop based at a , and then prove that it is path homotopic to γ in V , and then we can conclude that

$$[\gamma] = [\delta][\alpha\beta\alpha^{-1}\beta^{-1}][\bar{\delta}] = [\alpha\beta\alpha^{-1}\beta^{-1}]$$

By the **universal property of coproduct (free product of groups)**, p_* and q_* induce a unique homomorphism

$$\varphi : \pi_1(V) * \pi_1(U) \rightarrow \pi_1(T^2)$$

such that $\varphi \circ k = p_*$ and $\varphi \circ l = q_*$, where k and l are the inclusion maps of $\pi_1(V)$ and $\pi_1(U)$ into the free product respectively, and then the functor π_1 gives the following commutative diagram:

$$\begin{array}{ccc} \pi_1(U \cap V) & \xrightarrow{i_*} & \pi_1(U) \\ \downarrow j_* & & \downarrow k \\ \pi_1(V) & \xrightarrow{l} & \pi_1(U) * \pi_1(V) \\ & \searrow q_* & \downarrow \varphi \\ & & \pi_1(T^2) \end{array}$$

(Note: In the original image, there is an additional curved arrow from $\pi_1(U)$ to $\pi_1(T^2)$ labeled p_* .)

Then we should analysis φ such that we can get the structure of $\pi_1(T^2)$. Indeed, $\pi_1(U \cap V)$ is trivial, hence k and p_* just $0 \mapsto 0$, hence we just need to consider j_* and commute diagram $\varphi \circ l = q_*$. Notice that $\pi_1(V) = \pi_1(U \cap V) = \mathbb{Z} * \mathbb{Z}$, hence $l = id$ and $\varphi = q_*$.

Consider inclusion $q \circ j$, and γ is nullhomotopice in T^2 by embedding, hence $q_* \circ j_*([\gamma]) = [e_a]$, which implies that

$$q_*([\alpha][\beta][\alpha]^{-1}[\beta]^{-1}) = [e_a]$$

Notice that $q_*([\alpha])$ and $q_*([\beta])$ is not trivial and generate $\text{Im } q_*$, then $\text{Im } q_*$ must be abelian. q_* is surjective because any path in T^2 can be countinously deformed to a new path which deoes not pass U , hence by first isomorphism theorem we can conclude that

$$\pi_1(T^2) = \langle q_*(\alpha), q_*(\beta) \rangle / \langle q_*([\alpha][\beta][\alpha]^{-1}[\beta]^{-1}) = [e_a] \rangle$$

actually it is just $\mathbb{Z}^2$ and can be seen as the pushout of the diagram:

$$\{0\} \longleftarrow \mathbb{Z} \longrightarrow \mathbb{Z} * \mathbb{Z}$$

Theorem 3.13 (Van-Kampen Theorem).

Let X be a topological space, U and V be two open subsets covering X , if U , V and $U \cap V$ is path-connected, then for any $x_0 \in U \cap V$, we have

$$\pi_1(X, x_0) = \pi_1(U, x_0) *_{\pi_1(U \cap V, x_0)} \pi_1(V, x_0)$$

i.e. the natural map

$$\varphi : \pi_1(U, x_0) * \pi_1(V, x_0) \rightarrow \pi_1(X, x_0)$$

is surjective with kernel the normal subgroup generated by $i_*([\gamma])j_*([\gamma])^{-1}$, where $[\gamma]$ are the genertors of $\pi_1(U \cap V, x_0)$, and i_* and j_* are the induced map by inclusion

$$i : U \cap V \rightarrow U, \quad j : U \cap V \rightarrow V$$

Proof.

□

Remark 3.4. Consider some speical case of Van-kampen:

1. If $U \cap V$ is just a point, then it is the case of wedge sum, hence

$$\pi_1(U \vee V, x_0) \cong \pi_1(U, x_0) * \pi_1(V, x_0)$$

with x_0 being the wedge point.

2. More generally, if $U \cap V$ is simply connected, then the amalgamation product is just the free product, i.e.

$$\pi_1(X, x_0) \cong \pi_1(U, x_0) * \pi_1(V, x_0)$$

3. If U is simply connected, then the normal subgroup is only generated by $j_*([\gamma])$, hence

$$\pi_1(X, x_0) \cong \pi_1(V, x_0) / \langle j_*([\gamma]) \rangle$$

In above example, torus T^2 is just the case.

3.4 Covering Spaces

In the previous section, we claim that the fundamental group of $\mathbb{S}^1$ is $\mathbb{Z}$, we will prove it in this section.

Theorem 3.14. $\pi_1(\mathbb{S}^1) \cong \mathbb{Z}$

In this example, what we do is to lift the paths of $\mathbb{S}^1$ to $\mathbb{R}$, and then the paths with the different winding numbers will show the difference and motivates us to conclude the homotopy.

Definition 3.11. The covering space of a topological space X is a topological space E together with a continuous surjective map $p : E \rightarrow X$ satisfying:

For any $x \in X$, there exists an open neighborhood U of x such that $p^{-1}(U)$ is a disjoint union of open sets in E

$$p^{-1}(U) = \sqcup_{i \in I} U_i$$

and for each U_i , the restriction $p|_{U_i}$ gives a homeomorphism from U_i to U .

Usually we use the notation (E, p) or just a map $p : E \rightarrow X$ to denote a covering space, and we call E the **covering space** of X with **covering map** p . For any $x \in X$, the preimage $p^{-1}(\{x\})$ is called the **fiber** of x , and we denote it by F_x .

The covering space have the following topological property:

- (a) covering map is open and a local homeomorphism.
- (b) If $Y \subset X$ is a subspace, then the preimage $p^{-1}(Y)$ gives a covering space of Y with the covering map $p|_{p^{-1}(Y)}$.
- (c) X is Hausdorff if and only if E is also Hausdorff.
- (d) If X is path-connected, then all fiber have the same cardinality, hence we can define the **degree** of covering space as the cardinality of fiber.
- (e) If E is compact, (path) connected, then X is also compact, (path) connected.

Similar with $\mathbb{S}^1$, each path in X can be lifted to a path in E , formally in covering space we define a lifting of a map $f : Y \rightarrow X$ as a map $\tilde{f} : Y \rightarrow E$ such that $p \circ \tilde{f} = f$:

$$\begin{array}{ccc} & & E \\ & \nearrow \tilde{f} & \downarrow p \\ Y & \xrightarrow{f} & X \end{array}$$

Then we can conclude the a general lifting property for covering spaces, they are always be seen as a universal properties.

Proposition 3.15 (lifting criterion).

Let $p : E \rightarrow X$ be a covering space, Y is a path-connected and locally path-connected topological space, and $p(e_0) = x_0$. Then for any continuous map $f : (Y, y_0) \rightarrow (X, x_0)$, the **unique lifting** $\tilde{f} : (Y, y_0) \rightarrow (E, e_0)$ exists if and only if

$$f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(E, e_0))$$

Proof. □

Remark 3.5. This proposition shows that in the certain case, the lifting map to the covering space is unique up to fibre of the based point, and this criterion covers many case of lifting:

1. In particular, if $Y = [0, 1]$ (or generally simply connected space), then the lifting always exists and is unique, and it is the lifting of paths:

$$\begin{array}{ccc} & & E \\ & \nearrow \tilde{\gamma} & \downarrow p \\ I & \xrightarrow{\gamma} & X \end{array}$$

For any path $\gamma : I \rightarrow X$ with $\gamma(0) = x_0$, there exists a unique lifting path $\tilde{\gamma} : I \rightarrow E$ such that $\tilde{\gamma}(0) = e_0$

2. If $Y = I \times I$, then we can consider the lifting of path-homotopy, For any homotopy $H : I \times I \rightarrow X$ with $H(-, 0) = f$, if $\tilde{f}$ is a lifting of f , there exists a unique lifting homotopy $\tilde{H} : I \times I \rightarrow E$ such that $\tilde{H}(-, 0) = \tilde{f}$.
3. If Y is path-connected and locally path-connected, and then we can consider the general lifting of homotopy, For any homotopy $H : Y \times I \rightarrow X$ with $H(-, 0) = f$, if $\tilde{f}$ is a lifting of f , there exists a unique lifting homotopy $\tilde{H} : Y \times I \rightarrow E$ such that $\tilde{H}(-, 0) = \tilde{f}$

$$\begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & E \\ \downarrow i & \nearrow \tilde{H} & \downarrow p \\ Y \times I & \xrightarrow{H} & X \end{array}$$

The remark 2 is the special case of 3, the proof is not difficult: set $f(y_0) = x_0$ and fibre $p(e_0) = x_0$, then homotopy $H(y_0, 0) = f(y_0) = x_0$, then there exists a unique lifting $\tilde{H}$ with $\tilde{H}(y_0, 0) = \tilde{f}(y_0) = e_0$, then we need to verify that $\tilde{H}(-, 0) = \tilde{f}$ exactly, indeed $\tilde{H}(-, 0) : Y \rightarrow E$ is a lifting of $H(-, 0) = f$, hence by the uniqueness of lifting we can conclude the result.

Corollary 3.16. Let $p : E \rightarrow X$ be a covering space with $p(e_0) = x_0$, and γ and γ' be two paths in X starting at x_0 , if they are path homotopic, then their unique liftings at e_0 has the same endpoint and are path homotopic.

Proof. □

In particular, if γ is a loop based at x_0 , then its lifting $\tilde{\gamma}$ must end at the fiber of x_0 , it is ensured by $p \circ \tilde{\gamma} = \gamma$. Hence if we consider two homotopic loops based at x_0 , then their liftings at e_0 must end at the same fiber point, and hence we can get a natural map

$$\phi_{e_0} : \pi_1(X, x_0) \rightarrow F_{x_0}, \quad [\gamma] \mapsto \tilde{\gamma}(1)$$

where $\tilde{\gamma}$ is the unique lifting of γ starting at e_0 , and we call it the **lifting correspondence** at e_0 . Should pay attention that this map just depends on the point of fiber, and hence if we fix a homotopy class of loops γ based at x_0 , then we can get a natural endomorphism in the sense of **Set**

$$T_{[\gamma]} : F_{x_0} \rightarrow F_{x_0}, \quad e_0 \mapsto \phi_{e_0}([\gamma])$$

It motivates us to consider the group action of $\pi_1(X, x_0)$ on the fiber F_{x_0} .

Theorem 3.17. Let $p : E \rightarrow X$ be a covering space with $x_0 \in X$, then the right action $\pi_1(X, x_0) \curvearrowright F_{x_0}$ defined by

$$e_0 \cdot [\gamma] := \phi_{e_0}(\gamma) = \tilde{\gamma}(1)$$

is well-defined, and for any $e_0 \in F_{x_0}$ the orbit of e_0 is exactly the image of lifting correspondence ϕ_{e_0} :

$$\text{Orb}(e_0) = \text{Im}(\phi_{e_0})$$

then we have the following conclusions:

- If covering space is path-connected, then the action is transitive.
- If covering space is simply connected, then the action is free and transitive.

Proof. The outline of the proof is here:

(1) By the definition of group action, we just need to verify that for any class $[\gamma]$, endomorphism $T_{[\gamma]}$ is bijective, to prove that we just need to prove

$$T_{[\gamma]}^{-1} = T_{[\bar{\gamma}]}$$

(2) Next, check the definition of orbit, then the proof of transitive is equivalent to the proof that lifting correspondence is surjective.

(3) Finally, the freeness is equivalent to the injectivity of lifting correspondence. □

Lifting correspondence gives a new proof of $\pi_1(\mathbb{S}^1) \cong \mathbb{Z}$:

Proof. $p : \mathbb{R} \rightarrow \mathbb{S}^1, \quad t \mapsto e^{2\pi i t}$ is a covering map, and we set the base point $x_0 = 1$, and then we can get the fiber is $F_{x_0} = \mathbb{Z}$. The real line is simply connected, hence the lifting correspondence at 0 is a bijection

$$\phi : \pi_1(\mathbb{S}^1, 1) \rightarrow \mathbb{Z}, \quad [\gamma] \mapsto \tilde{\gamma}(1)$$

and we prove that it is a homomorphism to additive group $\mathbb{Z}$, then we can conclude the isomorphism. Indeed, let γ and δ be two loops based at 1, and their liftings at $0 \in \mathbb{R}$ are $\tilde{\gamma}$ and $\tilde{\delta}$ respectively, and we set $\tilde{\gamma}(1) = m$ and $\tilde{\delta}(1) = n$. Notice the peridoic

$$p(x + n) = x$$

hence we set a new path of $\mathbb{R}$ by $f(t) = m + \tilde{\delta}(t)$, then $f(0) = \tilde{\gamma}(1)$ allows us to get conactenation $\tilde{\gamma} * f$, and then we can verify that

$$\tilde{\gamma} * f(0) = 0, \quad p(\tilde{\gamma} * f) = \gamma * \delta$$

hence $\tilde{\gamma} * f$ is the unique lifting of $\gamma * \delta$ at 0, and then we have

$$\phi([\gamma * \delta]) = \tilde{\gamma} * f(1) = m + n = \phi([\gamma]) + \phi([\delta])$$

□

The fundamental group of covering space has a deep relation with the fundamental group of base space, and in many case we can find that the fundamental group of covering space is easier to comput.

Proposition 3.18. Let $p : E \rightarrow X$ be a covering space with $p(e_0) = x_0$, then

-covering map p induces an **injective** homomorphism

$$p_* : \pi_1(E, e_0) \rightarrow \pi_1(X, x_0), \quad [\gamma] \mapsto [p \circ \gamma]$$

- $\text{Im } p_*$ is the **stabilizer** of e_0 under the action $\pi_1(X, x_0) \curvearrowright F_{x_0}$.

- In particular, if E is path-connected and the degree of covering is n , then

$$n = [\pi_1(X, x_0) : \text{Im } p_*]$$

Proof.

□

Proof. Let $\delta = p \circ \gamma$ for some loop γ based at e_0 , then $\delta(1) = \delta(0) = x_0$ shows that γ is the lifting of δ . And we set $\delta' = p \circ \gamma$, if $[\delta] = [\delta']$ in $\pi_1(X, x_0)$, then there exists a path homotopy $\delta \simeq_H \delta'$, notice that $H(-, 0) = \delta$ has a lifting γ and $H(-, 1) = \delta'$ has a lifting γ' , so there exists a unique lifting homotopy $\tilde{H}$ such that $\tilde{H}(-, 0) = \gamma$ and $\tilde{H}(-, 1) = \gamma'$. For any $s \in I$, we have

$$p \circ \tilde{H}(0, s) = H(0, s) = x_0, \quad p \circ \tilde{H}(1, s) = H(1, s) = x_0$$

hence $\tilde{H}(0, -)$ and $\tilde{H}(1, -)$ are two lifting path of constant path c_{x_0} starting at e_0 , by the uniqueness of lifting path, they are c_{e_0} , which shows that $\tilde{H}$ is a path-homotopy.

For any $[\gamma] \in \pi_1(X, x_0)$, $e_0 \cdot [\gamma] = e_0$ means that the lifting of γ at e_0 is a loop, hence there exists $[\tilde{\gamma}] \in \pi_1(E, e_0)$ such that $p \circ \tilde{\gamma} = \gamma$, hence $\text{Stab}(e_0) \subset \text{Im } p_*$. Conversely, for any $p_*(\delta) \in \text{Im } p_*$, δ is the lifting of $p \circ \delta$ at e_0 , and as a loop we have

$$e_0 \cdot [p \circ \delta] = \delta(1) = e_0$$

It is clearly that $n = |F_{x_0}|$ and apply orbit-stabilizer theorem we can conclude

$$|\text{Orb}(e_0)| = [\pi_1(X, x_0) : \text{Stab}(e_0)] = [\pi_1(X, x_0) : \text{Im } p_*]$$

the path-connectness of E ensures that the action is transitive, hence $|\text{Orb}(e_0)| = |F_{x_0}| = n$. $\square$

Corollary 3.19.

$$\pi_1(\mathbb{RP}, 1) \cong \mathbb{Z}/2\mathbb{Z}$$

Proof. There are the things need to verify:

- (1) $\mathbb{S}^2$ is the covering space of $\mathbb{RP}^2$ of degree 2.
- (2) $\mathbb{S}^2$ is simply connected so that $\text{Im } p_* = 0$.
- (3) The fibre of $1 \in \mathbb{RP}^2$ has two points $F_1 = \{1, -1\}$.
- (4) $2 = [\pi_1(\mathbb{RP}^2, 1) : \{0\}]$ shows that $\pi_1(\mathbb{RP}^2, 1) \cong \mathbb{Z}/2\mathbb{Z}$ by the group theory.

$\square$

3.5 The classification of covering spaces

This section is to conclude the correspondence between covering spaces and group actions of fundamental group, which is one type of geometry-algebra correspondence.

Definition 3.12. Let X be a topological space, then the category of covering spaces over X , denoted by $\mathbf{Cov}_X$ with data:

- objects: covering spaces (E, p) or $p : E \rightarrow X$.

- morphisms: for two covering spaces (E, p) and (F, q) , a morphism $f : E \rightarrow F$ is a continuous map such that $q \circ f = p$, can be seen as the commutative diagram:

$$\begin{array}{ccc} E & \xrightarrow{f} & F \\ & \searrow p & \downarrow q \\ & & X \end{array}$$

In particular, f is just the lifting of p to F , hence some requirement of connectness can ensure the uniqueness of morphism, and by lifting criterion, comparison the image of the induced map can ensure the existence of morphisms.

Similarly, we should consider the category of group actions of fundamental group, generally we set a group G and define the category G -space or G -set:

Definition 3.13. The category of G -sets, denoted by $\mathbf{Set}^G$ with data:

-objects: a **G-set** is defined as (X, G) , where $G \curvearrowright X$ gives a right group action, and denote it by $x \cdot g$.

-morphisms: For any two G -set (X, G) and (Y, G) , a **G-map** is defined as $f : X \rightarrow Y$ such that for any $x \in X$ and $g \in G$,

$$f(x \cdot_X g) = f(x) \cdot_Y g$$

where $\cdot_X$ and $\cdot_Y$ are the group actions of G on X and Y respectively.

Pay attention that the definition is based on the right group action, because we define the action of fundamental group on the fiber as a right action, generally left action can also work.

With these preparations, the proposition 3.18 can be rewritten as a functor between two subcategories:

Proposition 3.20. Let X be a **path-connected** topological space with $x_0 \in X$, then there exists a faithful functor $F : \mathbf{Cov}_X \rightarrow \mathbf{Set}^{\pi_1(X, x_0)}$ with data:

- objects: $(E, p) \mapsto (p^{-1}(x_0), \pi_1(X, x_0))$ with the right action defined as above (lifting loops).

- morphisms: for any morphism $f : (E, p) \rightarrow (E', p')$, define

$$F(f) : p^{-1}(x_0) \rightarrow p'^{-1}(x_0), \quad e_0 \mapsto f(e_0)$$

i.e. $F(f) = f|_{p^{-1}(x_0)}$.

Proof. □

Corollary 3.21. Under above condition, we define $\mathbf{finiCov}_X$ be the subcategory of $\mathbf{Cov}_X$ consisting of finite covering spaces, and $\mathbf{finiSet}^{\pi_1(X, x_0)}$ be a subcategory of $\mathbf{Set}^{\pi_1(X, x_0)}$ consisting of finite $\pi_1(X, x_0)$ -sets, then the restriction of functor F gives a faithful functor

$$F : \mathbf{finiCov}_X \rightarrow \mathbf{finiSet}^{\pi_1(X, x_0)}$$

In the proof of $\pi_1(\mathbb{RP}^2, 1) = \mathbb{Z}/(2)$, the condition of simply-connectness of covering space $\mathbb{S}^2$ is a lucky case, generally the covering space may not be simply connected although their fundamantal group is smaller than the base space.

Example 3.1.

Definition 3.14. Let X be a topological space, it is called **semilocally simply connected** if for any $x \in X$, there exists an open neighborhood U of x such that the induced map

$$i_* : \pi_1(U, x) \rightarrow \pi_1(X, x)$$

By the definition, we define a subcategory of topological spaces category:

$$\mathbf{BonTop} \subset \mathbf{Top}$$

by requiring the objects to be path-connected, locally path-connected and semilocally simply connected topological spaces, this condition ensures the existence of simply connected covering space:

Proposition 3.22. Let $(X, x_0) \in \mathbf{BonTop}_*$, then there exists a covering space $(\tilde{X}, \tilde{p})$ with $\tilde{p}(\tilde{x}_0) = x_0$, such that $\tilde{X}$ is simply connected.

Proof.

□

Conversly, we want to reconstruct the covering space from the fundamantal group of base space, in the sense of **BonTop** we can get a trival covering space and a simply connected covering space, it is two extreme of the subgroup of fundamental group, and we can generalize it to any subgroup:

Theorem 3.23 (classification of covering spaces).

Let $(X, x_0) \in \mathbf{BonTop}_*$, and $H \leq \pi_1(X, x_0)$ be a subgroup, then there exists a covering space (E, p) with $p(e_0) = x_0$, such that the induced map

$$p_* : \pi_1(E, e_0) \rightarrow \pi_1(X, x_0)$$

has image H .

-Essentially, the functor $F : \mathbf{Cov}_X \xrightarrow{\sim} \mathbf{Set}^{\pi_1(X, x_0)}$ is an equivalence of categories.

Proof.

□

In particular, each path-connected covering space implies a transtive action of fundamental group, conversely, each transitive group action can reconstruct a path-connected covering space: ...

according to that we get the correspondence

$$\{\text{path-connected covering space}\} \longleftrightarrow \{\text{transitive action}\}$$

3.6 Deck transformations and normal covering spaces

This section is to refine the coorespondence,

020

Definition 3.15. Let $p : E \rightarrow X$ be a covering space, then a **deck transformation** (or automorphism) is a homeomorphism $f : E \rightarrow E$ such that $p \circ f = p$.

$$\begin{array}{ccc} E & \xrightarrow{f} & E \\ & \searrow p & \downarrow p \\ & & X \end{array}$$

In the above definition:

- Equivalently in the sense of category, a deck transformation is an automorphism of object (E, p) in category $\mathbf{Cov}_X$.
- All the deck transformations of covering space $p : E \rightarrow X$ form a group under composition, denoted by $\text{Deck}(E/X)$ or $\text{Aut}(E/X)$.
- The condition of hemomorphism can be replaced by continous bijection, since the covering map is open.

The deck transformation group has a natural action on the covering space, and especially on the fiber F_{x_0} for some $x_0 \in X$:

$$\text{Deck}(E/X) \curvearrowright F_{x_0}, \quad e_0 \cdot f = f(e_0)$$

In particular E can be seen as the disjoint union of fibers, and we can consider to construct the covering space by the action of fundamental group: Take x_0 and x_1 be two distinct points of X , we assume X is **path-connected**, then there exists a path α connecting x_0 and x_1 such that

$$\pi_1(X, x_0) \xrightarrow{\sim} \pi_1(X, x_1), \quad [\gamma_0] \mapsto [\alpha * \gamma * \bar{\alpha}]$$

By the group action of $\pi_1(X, x_0)$ and $\pi_1(X, x_1)$ on the fiber F_{x_0} and F_{x_1} respectively, we can get two bijections between fibers:

$$f_{[\gamma_0]} : F_{x_0} \rightarrow F_{x_0}, \quad e_0 \mapsto e_0 \cdot [\gamma]$$

and

$$f_{[\gamma_1]} : F_{x_1} \rightarrow F_{x_1}, \quad e_1 \mapsto e_1 \cdot [\gamma_1]$$

where $[\gamma_0] \in \pi_1(X, x_0)$ and $[\gamma_1] \in \pi_1(X, x_1)$. Notice that for any $[\alpha] \in \pi_1(X, x_0)$, with this method we can try to gluing the application of the fibers to give a bijection

$$F : \pi_1(X, x_0) \rightarrow \text{Deck}(E/X), \quad [\gamma] \mapsto f_\gamma$$

while the problem is that is the map well-defined?

Lemma 3.24. Let $p : (E, e_0) \rightarrow (X, x_0)$ be a covering space, and E is path-connected and locally path-connected, then the following statements are equivalent:

- (1) $\text{Deck}(E/X)$ acts **transitively** on the fiber F_{x_0} .
- (2) $\varphi : \pi_1(X, x_0) \rightarrow \text{Deck}(E/X)$ is surjective group homomorphism.
- (3) $\text{Im } p_*$ is a **normal** subgroup of $\pi_1(X, x_0)$.

If a covering space satisfies one of the above conditions, then it is called a **normal covering space** or **galois covering space**.

Proof. □

Remark 3.6. In particular, galois covering space gives a short exact sequence

$$1 \rightarrow \pi_1(E, e_0) \xrightarrow{p_*} \pi_1(X, x_0) \xrightarrow{\varphi} \text{Deck}(E/X) \rightarrow 1$$

It encodes the relation between fundamental group of covering space and base space, via deck transformation group, in particular if the covering space is simply connected, then we have an isomorphism

$$\pi_1(X, x_0) \cong \text{Deck}(E/X)$$

Hence the calculation of fundamental group can be transformed to the calculation of deck transformation group, which will be easier in many cases, hence the existence of simply connected covering space motivates us to consider the use of method in **BonTop**.

As we have discussed above, normal covering space allows us to construct a global automorphism along the fundamental group action on fibers, and the automorphism group can act on the covering space freely. However, in many cases we find that a topological space can be denoted as the orbit space of some group action, i.e.

$$G \curvearrowright X \implies X/G := \{G \cdot a \mid a \in X\}$$

In particular, the galois covering space shows that deck transformation group acts on the fiber transitively, and the fiber is always discrete, hence the transformation group should be discrete too, which motivates us to consider the discrete action.

Lemma 3.25. Let $G \curvearrowright X$ be a **discrete** group action on a topological space X , if the action is free and proper, then the quotient map

$$p : X \rightarrow X/G, \quad x \mapsto G \cdot x$$

is a covering map, i.e. X is a covering space of X/G .

Proposition 3.26. Let $p : E \rightarrow X$ be a covering space, if E is path-connected and locally path-connected, the covering space is normal if and only if there exists a discrete

group G acting on E freely and properly such that $X \cong E/G$ homeomorphic and the following diagram commutes:

$$\begin{array}{ccc} & E & \\ q \swarrow & & \searrow p \\ E/G & \xrightarrow{\sim} & X \end{array}$$

In particular,

4 Some Results by Algebraic Topology

4.1 The fundamental theorem of algebra

It is a famous theorem in algebra, but the proof can appear in many branches of mathematics via some basic ideas. Here a proof by algebraic topology is given here.

“Tous les théorèmes fondamentaux de l’algèbre reposent sur la compacité.”

— Jean-Pierre Serre, *“Cours d’arithmétique”*

Theorem 4.1. A polynomial equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

with complex coefficients has at least one root in $\mathbb{C}$.

Proof. The outline of the proof is as follows:

(1) Prove that the continuous map $f : \mathbb{S}^1 \rightarrow \mathbb{C} - \{0\}$ defined by $f(z) = z^n$ is not nullhomotopic.

(2) Prove the special case of the theorem when $|a_{n-1}| + \cdots + |a_1| + |a_0| < 1$

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

(3) Prove the general case by (2).

Here are the details:

(1) Notice that $\mathbb{S}^1$ can be embedded in $\mathbb{C} - \{0\}$ by the inclusion map i , and similarly we can define a map $g : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ by $g(z) = z^n$, then we can write f as the composition

$$f = i \circ g$$

then by the functoriality of fundamental group, we have

$$f_* = i_* \circ g_*$$

notice that i_* is injective since $\mathbb{S}^1$ can be seen as a retract of $\mathbb{C} - \{0\}$, and for the generated loop $[a]$ of $\pi_1(\mathbb{S}^1)$, we can calculate that $g_*([a]) = [g(a)] = [a^n]$, hence we can explicitly write the formula of $g_* : \mathbb{Z} \rightarrow \mathbb{Z}$ by $x \mapsto nx$, clearly it is injective.

(2) We prove the special case by contradiction, suppose that the polynomial has no root in $\mathbb{C}$, hence we can define a continuous map $p : D \rightarrow \mathbb{C} - \{0\}$ by

$$p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0 = 0$$

where D is the unit disk in $\mathbb{C}$, similarly if we restrict p to the boundary $\mathbb{S}^1$, i.e. we define $h = p|_{\mathbb{S}^1}$, then by functoriality

$$h_* = p_* \circ j_*$$

where j is the inclusion from $\mathbb{S}^1$ to D . Notice that j_* is the trivial map since $\pi_1(D)$ is trivial, hence h_* is also the trivial map, i.e. h is nullhomotopic.

However, in (1) we have shown that f is not nullhomotopic, then a homotopy from f and h can lead to the contradiction, we define $H : \mathbb{S}^1 \times I \rightarrow \mathbb{C} - \{0\}$ by

$$H(z, t) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_1z + a_0)$$

then clearly $H(-, 0) = f$ and $H(-, 1) = h$.

(3) For the general case, we make a substitute $z = cy$ for some $c > 0$, then

$$y^n + \frac{a_{n-1}}{c}y^{n-1} + \cdots + \frac{a_1}{c^{n-1}}y + \frac{a_0}{c^n} = 0$$

if c is proper, i.e.

$$\left|\frac{a_{n-1}}{c}\right| + \left|\frac{a_{n-2}}{c^2}\right| + \cdots + \left|\frac{a_1}{c^{n-1}}\right| + \left|\frac{a_0}{c^n}\right| < 1$$

then we can conclude the existence of root, and we can make a estimate to find that

$$c > \frac{\max\{|a_{n-1}|, \dots, |a_1|, |a_0|\}}{n}$$

□

4.2 Brouwer fixed point theorem

The Brouwer Fixed Point Theorem is a fundamental result in topology asserting that a continuous self-map of a compact convex set in finite-dimensional Euclidean space must have a fixed point. It captures a purely topological phenomenon: a ball (or simplex) cannot be continuously mapped into itself without leaving at least one point fixed.

Theorem 4.2 (Brouwer). Let D^n be the closed unit disc of $\mathbb{R}^n$, then any continuous map $f : D^n \rightarrow D^n$ has at least one fixed point, i.e. there exists $x \in D^n$ such that $f(x) = x$.

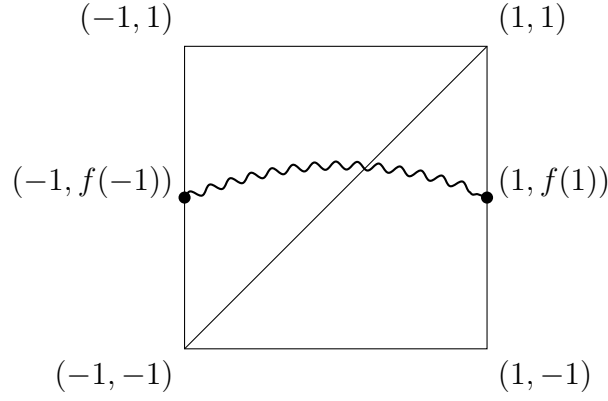
Remark 4.1. The case of $n = 1$ is easy to prove by analytic methods; geometrically $D^1 = [-1, 1]$ implies that we can observe the graph of f . Hence the question is equivalent to finding the intersection of the graph and the line $y = x$. A simple but instructive topological argument is to consider the following two **open** sets:

$$A = \{(x, f(x)) \mid f(x) > x\}, \quad B = \{(x, f(x)) \mid f(x) < x\}$$

If we assume that there is no fixed point, then the graph is the disjoint union of the two open sets

$$G(f) = A \sqcup B$$

and hence the graph is disconnected, which is a contradiction.



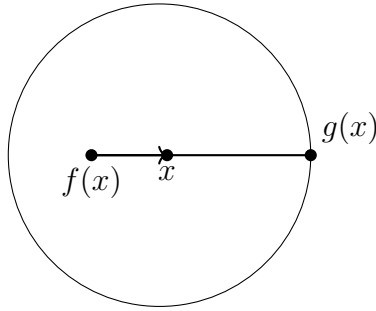
However, the general case is not easy to prove, and the connectness will be a hard condition to use, the dilemma is that we do not have a good property like intermediate value theorem for any n , in another word, the topology of real line is special.

Proof. Here is a proof by homology theory, which is a pure algebraic topology method, we sketch the outline here:

- (1) Assume that there exists a continuous function f without fixed point when $n \geq 2$, then we can construct a retraction $r : D^n \rightarrow \mathbb{S}^{n-1}$ by method of analytic geometry.
- (2) We claim that there is no retraction from D^n to $\mathbb{S}^{n-1}$ by homology theory, and prove it.
- (3) conclude the result.

Here are the details:

- (1) For any $x \in D^n$, we define that $r(x)$ is the intersection of $\mathbb{S}^{n-1}$ and the ray line from $f(x)$ to x :



By vector we can write

$$r(x) = x + \lambda(x - f(x)), \quad \lambda \neq 0 \text{ iff } \|x\| \neq 1$$

by $\|r(x)\| = 1$, we can get a quadratic equation of λ

$$\|x - f(x)\|^2 \lambda^2 + 2\langle x, x - f(x) \rangle \lambda + \|x\|^2 - 1 = 0$$

with the positive determination, we can get the exact formula of λ , which is dependent of x continuously

$$\lambda(x) = \frac{-2\langle x, x - f(x) \rangle + 2\sqrt{\langle x, x - f(x) \rangle^2 - \|x - f(x)\|^2(\|x\|^2 - 1)}}{2\|x - f(x)\|^2}$$

in particular, when $x \in \mathbb{S}^1$ i.e. $\|x\| = 1$, we can get

$$\lambda(x) = \frac{-\langle x, f(x) \rangle + \sqrt{\langle x, f(x) \rangle^2}}{\|x - f(x)\|^2} = 0$$

hence $r(x) = x$, which means that r is a retraction from D^n to $\mathbb{S}^{n-1}$.

(2) We prove the claim by homology theory, notice that we have the following commutative diagram:

$$\begin{array}{ccc} & D^n & \\ i \nearrow & & \searrow r \\ \mathbb{S}^{n-1} & \xrightarrow{id} & \mathbb{S}^{n-1} \end{array}$$

then by the functoriality of homology, we have the commutative diagram:

$$\begin{array}{ccc} & H_k(D^n) & \\ i_* \nearrow & & \searrow r_* \\ H_k(\mathbb{S}^{n-1}) & \xrightarrow{id_*} & H_k(\mathbb{S}^{n-1}) \end{array}$$

notice that $H_{n-1}(D^n) = 0$ and $H_{n-1}(\mathbb{S}^{n-1}) \cong \mathbb{Z}$, so i_* is the trivial map, and $id_* = r_* \circ i_*$ is also the trivial map, which leads to a contradiction (functor keeps identity map).

□

There are other proofs by different methods, but generally the fixed point theorem can be generalized to Banach space by the projective theory.

Theorem 4.3 (Schauder). Let X be a Banach space and let $K \subset X$ be a nonempty, closed, convex set. If $f : K \rightarrow K$ is continuous and compact (i.e. $f(K)$ is relatively compact), then f has a fixed point in K .

5 Complex Analysis

5.1 Holomorphe function

As a vector space $\mathbb{C}$, it is dimension 2 over $\mathbb{R}$ but dimension 1 over $\mathbb{C}$, although sometimes complex analysis is seen as a special case of real analysis in $\mathbb{R}^2$, but the differential structure is quite different.

Lemma 5.1. Define two vector space as following:

$$\mathcal{L}_{\mathbb{C}}(\mathbb{C}) = \{u : \mathbb{C} \rightarrow \mathbb{C} \mid \forall x, y \in \mathbb{C}, \forall c \in \mathbb{C}, u(x + cy) = u(x) + cu(y)\}$$

and

$$\mathcal{L}_{\mathbb{R}}(\mathbb{C}) = \{v : \mathbb{C} \rightarrow \mathbb{C} \mid \forall x, y \in \mathbb{C}, \forall r \in \mathbb{R}, v(x + ry) = v(x) + rv(y)\}$$

where $\mathcal{L}_{\mathbb{C}}(\mathbb{C})$ is the set of $\mathbb{C}$ -linear maps from $\mathbb{C}$ to $\mathbb{C}$, and $\mathcal{L}_{\mathbb{R}}(\mathbb{C})$ is the set of $\mathbb{R}$ -linear maps from $\mathbb{C}$ to $\mathbb{C}$. Then

- $\mathcal{L}_{\mathbb{R}}(\mathbb{C})$ is a vector space over $\mathbb{R}$ with dimension 2, and a basis is given by

$$dz : z \mapsto z, \quad d\bar{z} : z \mapsto \bar{z}$$

- $\mathcal{L}_{\mathbb{R}}(\mathbb{C})$ has another basis

$$dx : z \mapsto \Re(z), \quad dy : z \mapsto \Im(z)$$

and the relation between two basis is given by

$$dz = dx + idy, \quad d\bar{z} = dx - idy$$

- $\mathcal{L}_{\mathbb{C}}(\mathbb{C})$ is a vector space over $\mathbb{C}$ with dimension 1, and it is the subspace of $\mathcal{L}_{\mathbb{R}}(\mathbb{C})$ with $\{dz\}$ as a basis.

Proof.

□

A complex function f can be naturally seen as a real function

$$F : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad (x, y) \mapsto (\Re[f(x + iy)], \Im[f(x + iy)])$$

Hence we can consider the differentiability of F in the sense of real function, and its differentiability reflects the property of f but not equivalent, It is the difference between complex analysis and real analysis.

Definition 5.1. Let $U \subset \mathbb{C}$ be an open set, a function $f : U \rightarrow \mathbb{C}$ is called **holomorphe** at $z_0 \in U$ if there exists a $\mathbb{C}$ -linear map $L \in \mathcal{L}_{\mathbb{C}}(\mathbb{C})$ such that

$$f(z_0 + h) = f(a) + L(h) + o(|h|)$$

If f is holomorphe at any point of U , then we say that f is holomorphe on U , and we denote the set of all holomorphe functions on U by $\mathcal{O}(U)$.

Equivalently, we also call "holomorphe" by " **$\mathbb{C}$ -differentiable**", immediately a complex function is called " **$\mathbb{R}$ -differentiable**" if L is $\mathbb{C}$ -linear map in the definition above, but it actually means that the corresponded real function F is differentiable in the sense of Banach space $\mathbb{R}^2$.

In particular $\mathbb{C}$ -differentiable is $\mathbb{R}$ -differentiable, but conversely it is not true. We consider the example $f(z) = \bar{z}$. The corresponding real function is

$$F : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad (x, y) \mapsto (x, -y)$$

which is clearly $\mathbb{R}$ -differentiable at any point with the invertible Jacobian matrix

$$J_F(x, y) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

However, it is not holomorphe ($\mathbb{C}$ -differentiable) at any point: Assuming that it is holomorphe at z_0 , then there exists a $\mathbb{C}$ -linear map L such that

$$\overline{z_0 + h} = \bar{z}_0 + L(h) + o(|h|)$$

By the structure of $\mathbb{C}$ -linear map, then $L(h) = ch$ for some $c \in \mathbb{C}$, hence

$$\frac{\bar{h}}{h} = c + \frac{o(|h|)}{h}$$

as h tends to zero, The right side tends to c while the left side does not have a limit, which is absurd. It is necessary to avoid the confusion between the two type of differentiable case, to get rid of the notation of dx and dy , sometimes it is terrible.

If $f : U \rightarrow \mathbb{C}$ is $\mathbb{R}$ -differentiable, in the sense of real function, its differential at $a \in U$, as a $\mathbb{R}$ -linear map, can be written as

$$(Df)_a = \frac{\partial f}{\partial x}(a)dx + \frac{\partial f}{\partial y}(a)dy$$

with two $\mathbb{R}$ -linear maps dx and dy , then we can change the basis $\{dx, dy\}$ to $\{dz, d\bar{z}\}$:

$$(Df)_a = \frac{\partial f}{\partial z}(a)dz + \frac{\partial f}{\partial \bar{z}}(a)d\bar{z}$$

Definition 5.2. The **Wirtinger operators** are defined by

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$$

Hence under the differential operator with respect to z and $\bar{z}$, the complex function f can be independent of x and y completely, with the following properties of differential:

1. For any $\mathbb{R}$ -differentiable function $f : U \rightarrow \mathbb{C}$

$$\overline{\frac{\partial f}{\partial z}} = \frac{\partial \bar{f}}{\partial \bar{z}}$$

2. For any $\mathbb{R}$ -differentiable function $f, g : U \rightarrow \mathbb{C}$,

$$\frac{\partial(fg)}{\partial z} = f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} \quad \frac{\partial(fg)}{\partial \bar{z}} = f \frac{\partial g}{\partial \bar{z}} + g \frac{\partial f}{\partial \bar{z}}$$

3. For any $\mathbb{R}$ -differentiable $f : U \rightarrow \mathbb{C}$ and $g : f(U) \subset V \rightarrow \mathbb{C}$

$$\frac{\partial(h \circ g)}{\partial z} = \left(\frac{\partial h}{\partial w} \circ g\right) \cdot \frac{\partial g}{\partial z} + \left(\frac{\partial h}{\partial \bar{w}} \circ g\right) \cdot \frac{\partial \bar{g}}{\partial z}$$

and

$$\frac{\partial(h \circ g)}{\partial \bar{z}} = \left(\frac{\partial h}{\partial w} \circ g\right) \cdot \frac{\partial g}{\partial \bar{z}} + \left(\frac{\partial h}{\partial \bar{w}} \circ g\right) \cdot \frac{\partial \bar{g}}{\partial \bar{z}}$$

Although the holomorphic function is different from real differentiable function, but we can not omit the relation between them, and in many cases the correspondent real function is easier to study, hence the following proposition is important:

Proposition 5.2. If $U \subset \mathbb{C}$ is open and $f : U \rightarrow \mathbb{C}$ is a complex function, then the following conditions are equivalent:

(1) f is holomorphic on U .

(2) f is $\mathbb{R}$ -differentiable on U and $\frac{\partial f}{\partial \bar{z}} = 0$ on U .

(3) Writing $f = u + iv$ with real functions $u, v : U \rightarrow \mathbb{R}$, then u and v are $\mathbb{R}$ -differentiable on U and satisfy the **Cauchy-Riemann equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

In particular, if f is holomorphic at $a = p + iq \in U$, then its $\mathbb{C}$ -derivative at a is given by

$$f'(a) = \frac{\partial f}{\partial z}(a) = \frac{\partial f}{\partial x}(a) = 2 \frac{\partial u}{\partial z}(a)$$

and write F as the correspondent real function of f , the Jacobian of F at (p, q) is

$$\det J_F(p, q) = |f'(a)|^2$$

Proof.

□

In the sense of complex analysis, the constant function are a very strong condition by comparison with real analysis, hence local constant condition can imply global constant condition in some requirement of connectness.

Corollary 5.3. Let $D \subset \mathbb{C}$ be a connected open set, then the following conditions are equivalent:

- (1) f is constant. (2) $f' = 0$. (3) $|f|$ is constant.
(4) $\Re(f)$ is constant. (5) $\Im(f)$ is constant.

5.2 Cauchy's Integral Formula

By the view of manifold, $\mathbb{C}$ can be seen as a **2-dimensional real manifold** with a complex structure, hence it is natural to consider the differential form on it. Let $f : U \rightarrow \mathbb{C}$ a differentiable function on an open set $U \subset \mathbb{C}$, then it induces a differential

$$df : U \rightarrow \mathcal{L}_{\mathbb{R}}(\mathbb{C}), \quad a \mapsto (Df)_a = \frac{\partial f}{\partial z}(a)dz + \frac{\partial f}{\partial \bar{z}}(a)d\bar{z}$$

about the $d(df) = \dots$

Definition 5.3. Let U be an open set in $\mathbb{C}$, and then we define

$$\begin{aligned} \text{0-form : } \quad \Omega^0(U) &= C^\infty(U, \mathbb{C}) \\ \text{1-form : } \quad \Omega^1(U) &= \{f dz + g d\bar{z} \mid f, g \in C^\infty(U, \mathbb{C})\} \\ \text{2-form : } \quad \Omega^2(U) &= \{h d\bar{z} \wedge dz \mid h \in C^\infty(U, \mathbb{C})\} \end{aligned}$$

and generally we define differential forms on U by

$$\Omega(U) = \bigoplus_{k=0}^2 \Omega^k(U)$$

As the one of the most part in complex analysis, Cauchy's integral formula shows the deep relation between the integral and holomorphic function, in this section we will give a general statement with the help of differential form, and we will see that it is an application of Stokes's theorem in complex plane, i.e. Green-Riemann theorem.

Theorem 5.4 (Green-Riemann).

Let K be a compact with smooth boundary ∂K in $\mathbb{C}$, and $w \in \Omega^1(\overset{\circ}{K})$ be a 1-form of class C^1 , then

$$\int_K dw = \int_{\partial K} w$$

Proof.

□

The classic statement of Cauchy's integral formula is that if f is holomorphic on an open set U , then for any point $a \in U$ and an open ball $B(a, r) \subsetneq U$, then

$$\int_{C(a, r)} f dz = 0$$

where $C(a, r)$ is the circle centre at a with radius r and positive orientation, it can be seen as the boundary of the $B(a, r)$, and $f dz$ is always a closed 1-form of class C^1 , hence by Green-Riemann formula, the formula holds. But we can do more with the Green-Riemann theorem, it gives a more general statement about the region of integration.

Theorem 5.5 (1912, Cauchy-Pompeiu).

Let U be an open subset of $\mathbb{C}$ and K is a compact of U with smooth boundary ∂K , if $f \in C^1(U)$, then we have

$$\frac{1}{2\pi i} \int_{\partial K} \frac{f(z)}{z-a} dz + \int_K \frac{\partial f}{\partial \bar{z}}(z) \frac{1}{z-a} d\bar{z} \wedge dz = \begin{cases} f(a) & a \in \mathring{K} \\ 0 & a \in \mathbb{C} - K \end{cases}$$

in particular, if $f \in \mathcal{O}(U)$ holomorphe,

$$\int_{\partial K} f dz = 0$$

and for any $a \in \mathring{K}$,

$$f^{(k)}(a) = \frac{k!}{2\pi i} \int_{\partial K} \frac{f(z)}{(z-a)^{k+1}} dz$$

5.3 Analytic function

previous section, it is an important properties, it claims that the local value of the power series determines the global value. In general, two analytic function f and g , if they are equal in a local voinsinage V , then for any $x \in V$, there exists a open ball $B(x, r)$ such that their power series expanasions are equal. The most important things is that V is not compact, so our open cover $B(x, r)$ will bring the properties by extending, so f and g are actually equal!

The statement above is a little abstract, it is better to draw a picture from a voinsinage V , then see that if we want to let $f = g$ we just need to consider the point nearly the ∂V ...

In this section we will prove the analytic continuation, and three other important theorems in complex anaylsis, although they have various methods to porve, but here our inspiration is continuation:

Analytic continuation $\implies$ Isolated zero $\implies$ Open mapping $\implies$ Maximum modulus

Theorem 5.6 (analytic continuation).

Given an **doamin** set U in complex plane, and $f, g : U \rightarrow \mathbb{C}$ are two holomorphic (analytic) functions. If they are equal in a voinsinage of U , then $f = g$ in whole U .

Proof. We let $S = \{z \in U | f^{(n)}(z) = g^{(n)}(z) \text{ for any } n \in \mathbb{N}\}$, and clearly $S = \bigcup_{n \in \mathbb{N}} \{z \in U | f^{(n)}(z) = g^{(n)}(z)\}$. It is easy to verify that S is the intersection of infinite closed sets, so S is closed.

S is also open. We take any $z_0 \in S$, we notice that S is a subset of the voinsinage where f and g are equal, then by corollary ??, it implies that they have the same seires in a voinsinage of z_0 contained in S , then clearly $f^{(n)}(z) = a_n = b_n = g^{(n)}(z)$ in the voinsinage, so that S must be open. What's more S is not empty, and U is connect, so $U = S$.

□

Remark 5.1. We notice that the proof is based on the topological property of the domain U , so the properties of holomorphic is special.

One directly result of analytic continuation is the isolated zero theorem, it claims that the zero of a holomorphic function must be discrete:

Theorem 5.7 (isolated zero). Suppose that U is a **domain** in complex plane, and $f \in H(U)$ is a non-zero holomorphic function, then

- If $z \in U$ such that $f(z_0) = 0$, then f can be expanded in a neighborhood $\Omega \subset U$:

$$f(z) = (z - z_0)^n g(z)$$

where g is a non-vanishing holomorphic function in Ω .

- Any zero z_0 of f , there exists a neighborhood such that z_0 is the unique zero in the neighborhood.

- The set of all zeros of f is discrete.

Proof. If z_0 is a zero, then there exists a expansion of power series near the z_0 with

$$f(z) = \sum_{k \in \mathbb{N}} a_k (z - z_0)^k, \quad a_0 = 0$$

So we just take $n = \min\{i \in \mathbb{N} | a_i \neq 0\}$, then

$$f(z) = (z - z_0)^n \sum_{k \in \mathbb{N}} a_{n+k} z^k$$

So we just take $g(z) = \sum_{k \in \mathbb{N}} a_{n+k} z^k$. It is well-defined since in a small reigon near z_0 and does not contain z_0 we have surely $g(z) = \frac{f(z)}{(z - z_0)^n}$, then by analytic continuation g is holomorphic on a samll neighborhood with z_0 in it, and $g(z_0) = a_n \neq 0$. Hence by continuity, there exists a positive $r > 0$ such that

$$||f(z_0)| - |a_n|| \leq |f(z_0) - a_n| < |a_n|/2$$

for any $z \in B(z_0, r)$, which implies $|f(z_0)| > |a_n|/2 > 0$ in the reigon, so it is non-vanishing. Just take the intersection of the previous neighborhood and $B(z_0, r)$, then we get the Ω we hope.

The second statement is immediate from the first statement. For the third, if there exists a limit point a in the set of zeros, then for any small neighborhood with a in it, it will conatins infinite other zeros, which contradicts with the second statement. $\square$

With the help of the isolated zero theorem, we can generalize the analytic continuation, locally two holomorphic function are just equal in a non-discrete set:

Theorem 5.8 (strong analytic continuation). Given an **domain** set U in complex plane, and $f, g : U \rightarrow \mathbb{C}$ are two holomorphic (analytic) functions. If they are equal in a set $S \subset U$, and U admet an accumulation point, then $f = g$ in U .

Proof. Just take $f - g$ as the new holomorphic function in U , notice it will be zero in S and S is not discrete, so $f - g$ must be zero by isolated zero. $\square$

Remark 5.2. An direct result if two holomorphic functions are equal in a non-constant convergent sequence, then they will behave equal globally.

The other two result is the maximum modulus principle and open mapping theorem.

Theorem 5.9 (open mapping).

A non-constant holomorphic function defined on a **domain** must be an open map.

Proof. Take any $z_0 \in U$, and we will prove that $f(z_0) \in f(U)$ must be an interior point. By isolated zero theorem, there exists $B = B(z_0, r) \subset U$ such that in it we have

$$f(z) - f(z_0) = (z - z_0)^n g(z)$$

so f can be written as the composition of the following holomorphic functions:

- $u : z \mapsto (z - z_0)g^{1/n}(z)$ from B to $\mathbb{C}$.
- $v : z \mapsto z^n$ from $\mathbb{C}$ to $\mathbb{C}$.
- $t : z \mapsto z + f(z_0)$ from $\mathbb{C}$ to $\mathbb{C}$.

so $f = t \circ v \circ u$, we just need to verify the three maps are open, it can be proved via **the local inversion theorem: if the derivative is not zero at some point, then locally there exists a diffeomorphism, so the image of the point will be an interior.**

For u we can compute $u'(z) = g^{1/n}(z) + \frac{(z-z_0)}{n} g'(z) g^{\frac{1-n}{n}}(z)$, clearly $u'(z_0) = g^{1/n}(z_0) \neq 0$, and u is holomorphic so class of C^1 , so locally there exists a diffeomorphism between neighborhoods U_{z_0} and $V_{u(z_0)}$, so $u(z_0)$ is a interior point of $f(B)$.

For v the proof is similar if we take $z_0 \neq 0$ since $v'(z) = nz^{n-1}$. In the case of $z_0 = 0$, we take a enough small positive $r > 0$, then $|z| < r$ implies $|z|^n < r^n$, so $B(v(0), r) \subset B(0, r^n) = v(B(0, r))$, so v is open. Finally, t is clearly open. Thus after composition, f is open. $\square$

Theorem 5.10 (Maximum modulus).

If f in a non-constant holomorphic function on a **domain** U , then $|f|$ can not attain a maximum in U .

-Weakly, if $K \subset U$ is a compact set, then $|f|_K$ attain the maximum in the boundary:

$$\sup_{z \in K} |f(z)| \leq \sup_{\partial K} |f(z)|$$

Proof. Assume that f did attain a maximum in $z_0 \in U$, then we take a small voinsinage of x_0 , called V , by open mapping theorem $f(V)$ must be open with $f(z_0)$ in it, so there exists $r > 0$ such that $B(f(z_0), r) \subset f(D)$. We take $\delta = \text{sign}(\text{Re}(f(z_0)))$, then $f(z_0) + \delta \frac{r}{2} \in f(D)$, we denote it $f(w_0)$, then $|f(w_0)| > |f(z_0)|$ clearly, which contradicts with the assumption.

The weakly statement is immediate since $|f|$ is continuous, so $f(K)$ will be compact in $\mathbb{R}$, so $|f|$ must attain a maximum in K , with the strong statement we know that the maximum can not be in $\text{int}(K)$, so in boundary. $\square$

5.4 Infinite Product

The form of infinite products is important in complex analysis to study the poles and zeros of functions, here is the definition:

Definition 5.4. Let $(a_n)_{n=1}^{\infty}$ be a sequence of complex numbers. The infinite product $\prod_{n=1}^{\infty} a_n$ is said to converge if there exists $N \in \mathbb{N}$ such that:

- (1) $a_n \neq 0$ for all $n \geq N$
 - (2) the sequence of partial products $(\prod_{k=n}^N a_k)_{n \geq N}$ converges to a non-zero limit as $n \rightarrow \infty$.
- If the limit is zero, we say the product **diverges to zero**. If the limit does not exist, we say the product diverges.

we avoid the case that some terms are zero, because it will make the product zero, which is not interesting; and we notice that if the sequence (a_n) converges to zero, then the product will tend to zero trivially, so we need to avoid this case too. Anyway, the definition here is delicate enough to avoid any trivial case.

Proposition 5.11 (The properties of convergence). Let $(a_n)_{n=1}^{\infty}$ be a sequence of complex numbers such that the infinite product $\prod_{n=1}^{\infty} a_n$ converges, then:

- (1) $\lim_{n \rightarrow \infty} a_n = 1$
- (2-LOG) If $a_n \in \mathbb{C} - \mathbb{R}_-$ for sufficient large n , then we have equivalent condition for the convergence of the product: the series $\sum_{n=1}^{\infty} \text{Log } a_n$ converges.
- (3-CVA) If $\sum_{n \in \mathbb{N}} |a_n| < \infty$, then the product $\prod_{n=1}^{\infty} 1 + a_n$ converges or diverges to zero.

Proof. (1) is immediate from $a_n = (\prod_{k=1}^n a_k / \prod_{k=1}^{n-1} a_k)$ for $n \geq 2$.

(2) By the convergence of (1), for sufficient large n , a_n stays near 1, hence we can choose the principal branch such that for sufficient large N

$$\text{Log } \prod_{n \geq N} a_n = \sum_{n \geq N} \text{Log } a_n$$

(3) If $\sum_{n \in \mathbb{N}} |a_n| < \infty$, so a_n converges to zero so we can expand

$$\text{Log}(1 + a_n) = a_n - \frac{a_n^2}{2} + o(a_n^3)$$

Then by (2), the series $\sum_{n \in \mathbb{N}} \text{Log}(1 + a_n)$ converges absolutely, hence the product converges or diverges to zero. $\square$

With the basic technics, we can study the products of functions, that's the core to construct some important functions like gamma function.

Definition 5.5 (Convergence of product of functions).

Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of continuous function on a open set U

- The product $\prod_{n \in \mathbb{N}} f_n$ converges pointwise to F on U if for each $z \in U$, the product $\prod_{n \in \mathbb{N}} f_n(z)$ converges to $F(z)$.
- The product $\prod_{n \in \mathbb{N}} f_n$ converges uniformly to F on U if there exists $N \in \mathbb{N}$ such that for all $n \geq N$, f_n does not vanish on U and the sequence of partial products $(\prod_{k=n}^N f_k)_{n \geq N}$ converges uniformly to F on U .

Here is the definition following above definition, the zero of the functions in sequence may cause some problems, so there are some difficult in definition. Another clear definition is from **Henri Cartan's book**, here we give as a lemma:

Lemma 5.12. In above definition, let $K \subset U$ as a subset, then the product $\prod_{n \in \mathbb{N}} f_n$ converges uniformly on K if f_n satisfies the following condition:

- (1) $(f_n)_{n \in \mathbb{N}}$ converges uniformly to 1 on K .
- (2) The series $\sum_{n \in \mathbb{N}} \text{Log } f_n$ converges normally on K .

Proof. By (1), for sufficient large N , we have $|f_N - 1| < 1/2$, so for all $n \geq N$, $\text{Log } f_n$ is well defined and f_n does not vanish on K . so for any $z \in K$

$$\text{Log } \prod_{k=n}^N |f_k| = \sum_{k=n}^N |\text{Log } f_k| \leq \sum_{k=n}^N \|\text{Log } f_k\|_K$$

it implies that the sequence of partial products converges uniformly on K by weierstrass M-test. $\square$

Infinite products is a strong tool to construct function with certain zeros, for example we can construct a function with zeros at all integers like following:

$$z \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)$$

It is not difficult to verify that the product converges locally uniformly to a holomorphic function, and one holomorphic function sharing the properties is **sine function**, actually we can prove that they are equally up to a constant factor, which is a famous result called **Euler's sine product formula**. With the following proposition, we can completely prove the formula:

$$\sin(\pi z) = \pi z \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)$$

The proof can be found in [Stein 2, page 142].

Proposition 5.13. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of holomorphic functions on an open set $\Omega \subset \mathbb{C}$, and suppose that the product $\prod_{n \in \mathbb{N}} f_n$ converges Ω to a function f locally uniformly, then

- (1) f is holomorphic on Ω
- (2) The zeros and multiplicities of f follows from the sequence:

$$Z(f) = \bigcup_{n \in \mathbb{N}} Z(f_n) \quad m_f(z) = \sum_{n \in \mathbb{N}} m_{f_n}(z)$$

(3) If f_n does not vanish on Ω for any n , then the series of meromorphic functions $\sum_{n \in \mathbb{N}} f'_n/f_n$ converges locally uniformly on $\Omega - Z(f)$ to f'/f .

Proof. It's the result of locally uniformly convergence on the infinite product, notice that the proof of (3) is referred to a identity:

$$\frac{(\prod_{k=1}^n f_k)'}{\prod_{k=1}^n f_k} = \sum_{k=1}^n \frac{f'_k}{f_k}$$

which can be proved by induction. Another point is that the proof of (2) needs that the multiplicities of zeros are finite, i.e. for any $a \in \Omega$ we have

$$\#\{n \in \mathbb{N} | f_n(a) = 0\} < \infty$$

Which is ensured by the definition of convergence of infinite product. □

There are two natural questions arising from Euler's formula, one is that if we find another entire function with zeros at all integers, whether the function is same with sine function up to coefficients or anything else? The other is that if there exists a general method to construct entire function with given finite or infinite zeros? The answer is from **Weierstrass's** construction.

Firstly, the zeros of holomorphic is isolated or discrete, so the given zeros can not have any limit point in $\mathbb{C}$, hence the problem is limited to discrete set of points. **For finite zeros, we can use the fundamental theorem of algebra to construct a polynomial with given zeros**, so the problem is limited to infinite zeros. By Borel-Weierstrass theorem, we can deduce that the set of zeros must be unbounded.

Another point about zero is that the multiplicity of zeros must be finite. Suppose that f is a non-zero entire function, then by property of analytic function, f can be expanded at any point z_0 as a Taylor series

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

If z_0 is a zero with infinite multiplicity, then $f^{(n)}(z_0) = 0$ for all $n \in \mathbb{N}$, hence $f \equiv 0$ in the neighborhood of z_0 , which implies $f \equiv 0$ in $\mathbb{C}$ by analytic continuation, it is absurd. Together with the above statement, if $(a_n)_{n \in \mathbb{N}}$ is the sequence of zeros, then the zeros must satisfy $\lim_{n \rightarrow \infty} |a_n| = \infty$.

With the basic analysis of zeros, now we can construct the entire function by given zeros. Notice that we can not simply combine the zeros together by linear terms

$$\prod_{n \in \mathbb{N}} (z - a_n)$$

The product diverges for infinite unbounded zeros, so we can copy the idea of Euler's formula, i.e.

$$\prod_{n \in \mathbb{N}} (1 - \frac{z}{a_n})$$

However, we can not ensure the convergence of the product, for example we take zeros as all positive integers. To ensure the convergence we need to add some extra terms in each factor, that is called **Weierstrass's canonical factors**.

Lemma 5.14. We define the weierstrass's canonical factor of the degree p as

$$E_p(z) := \begin{cases} (1 - z) & p = 0 \\ (1 - z)e^{(z + \frac{z^2}{2} + \dots + \frac{z^p}{p})} & p \in \mathbb{N} \end{cases}$$

then the factor has the following propoerties:

- (1) It is an entire function with a simple zero at $z = 1$ and no other zeros.
- (2) It is a function of finite order $p + 1$.
- (3) For $|z| \leq r < 1$, we have the estimate

$$|E_p(z) - 1| \leq C_r |z|^{p+1}$$

for some constant $C_r > 0$. That means the larger the p is, the convergence of $E_p(z)$ to 1 is faster.

Proof. (1) and (2) is clear, we prove (3).

$$\begin{aligned} \log(E_p(z)) &= \log(1 - z) + z + \frac{z^2}{2} + \dots + \frac{z^p}{p} \\ &= -\sum_{n=1}^{\infty} \frac{z^n}{n} + (z + \frac{z^2}{2} + \dots + \frac{z^p}{p}) \\ &= -\sum_{n=p+1}^{\infty} \frac{z^n}{n} = O(|z|^{p+1}) \end{aligned}$$

the expansion here is ensured by $|z| < 1$, hence

$$E_p(z) = \exp(\log E_p(z)) = 1 + O(|z|^{p+1})$$

by expansion of exponential function $e^z = 1 + z + o(z^2)$, so we can conlude the result. $\square$

So we can response the original question by collecting the above ideas:

Theorem 5.15 (Weierstrass's Factorization Theorem).

Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of complex numbers such that $\lim_{n \rightarrow \infty} |a_n| = \infty$, and we make convention that $E_n(\frac{z}{a_n}) := z$, then the infinite product

$$f(z) = \prod_{n=1}^{\infty} E_n\left(\frac{z}{a_n}\right)$$

converges locally uniformly in $\mathbb{C}$ to an entire function f whose zeros are precisely the points a_n , with multiplicities by counting. Moreover, if g is any other entire function with the same zeros and multiplicities, then there exists an entire function h such that

$$g(z) = f(z)e^{h(z)}$$

for all $z \in \mathbb{C}$.

Another better theorem is given by **Hadamard**, it shows the growth of the entire function will influence the construction of the function by given zeros.

Theorem 5.16 (Hadamard's Factorization Theorem).

Suppose that f is an entire function of finite order ρ with zeros $(a_n)_{n \in \mathbb{N}}$ (counting multiplicities), then there exists a polynomial P of degree at most $k = \lfloor \rho \rfloor$ such that

$$f(z) = e^{P(z)} \prod_{n=1}^{\infty} E_n\left(\frac{z}{a_n}\right)$$

5.5 Mellin transformation for analytic continuation

The Mellin transformation is a integral transformation which is useful in complex analysis to study the analytic continuation of functions. For example, the gamma function is well-defined for $\Re(s) > 0$ by the integral

$$\Gamma(s) = \int_0^{\infty} e^{-t} t^{s-1} dt$$

Moreover, zeta function is well-defined for $\Re(s) > 1$ by the integral (**Ex15, Chapter6, Stein 2, a easy exercise**)

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{t^{s-1}}{e^t - 1} dt$$

and similarly Beta function, a function has a deep relationship with Gamma function, is well-defined for $\Re(a), \Re(b) > 0$ by the integral (**Ex7, chapter 6, Stein 2**)

$$B(a, b) = \int_0^{\infty} \frac{t^{a-1}}{(1+t)^{a+b}} dt$$

Hence we can pay attention to the form of the integration, and then we can a form of integral transformation:

$$\int_0^{\infty} f(t) t^{z-1} dt$$

which is called **Mellin transformation** of the function f , and we usually denote it by $\mathcal{M}[f; z]$.

5.6 Gamma function

As we have mentioned above, the gamma function is defined by two equivalent forms, this section is to study the specific properties of gamma function.

Definition 5.6. Let γ be the Euler constant, $z \in \mathbb{C} - \mathbb{Z}_-$

-infinite product definition:

$$\Gamma(z) = \frac{1}{ze^{\gamma z} \prod_{n \geq 1} (1 + \frac{z}{n}) e^{-z/n}}$$

-integral definition:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$$

as the mellin transformation of $f(t) = e^{-t}$.

The Gamma function is a **meromorphic** function on $\mathbb{C}$ with poles at all $\mathbb{Z}_- = \{0, -1, -2, \dots\}$, and its residues are

$$\text{Res}(\Gamma, -n) = \frac{(-1)^n}{n!}, \quad -n \in \mathbb{Z}_-$$

The existence of poles is clear by the infinite product, but the calculation of residues is not clear, it depends on the functional equation to give a simple proof.

Proposition 5.17. For any $z \in \mathbb{C} - \mathbb{Z}_-$

$$\Gamma(z+1) = z\Gamma(z)$$

In particular, for any $n \in \mathbb{N}$, $\Gamma(n+1) = n!$.

Proof.

□

Remark 5.3. With the formula, we can verify the residues at poles easily. For any $-n \in \mathbb{Z}_-$ and $\Re(z) > -n-1$

$$\Gamma(z)(z+n) = \frac{\Gamma(z+n+1)}{z(z+1)\cdots(z+n-1)}$$

by the functional equation, hence calculate the limit of right term at $-n$, we can get the residue at $-n$.

Guass integration is one of the most famous integration

$$I = \int_0^\infty e^{-x^2} dx$$

we can set $x^2 = t$, then substitute

$$I = \frac{1}{2} \int_0^\infty t^{-1/2} e^{-t} dt = \frac{\Gamma(1/2)}{2}$$

hence the gamma function can give the integration result, which motivates the following functional equation:

Proposition 5.18. For any $z \in \mathbb{C} - \mathbb{Z}_-$

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$$

In particular, $\Gamma(1/2) = \sqrt{\pi}$.

Proof.

□

Proposition 5.19 (Legendre's Formula). For any $z \notin \mathbb{Z}_{-1} \cap (\mathbb{Z}_{-1} - \frac{1}{2})$

$$\Gamma(z)\Gamma(z + \frac{1}{2}) = 2^{1-2z}\sqrt{\pi}\Gamma(2z)$$

In particular for any $n \in \mathbb{N}$

$$\Gamma(n + \frac{1}{2}) = \frac{(2n)!}{4^n n!} \sqrt{\pi}$$

Proof.

□

In many case, we need to study the asymptotic behavior of gamma function when $z \rightarrow \infty$, and gamma function has a deep relation with factoria, a useful equivalent formula help us to do that:

Proposition 5.20 (Euler-Gauss's Formula). For any $z \in \mathbb{C} - \mathbb{Z}_-$

$$\Gamma(z) = \lim_{n \rightarrow \infty} \frac{n!n^z}{z(z+1)(z+2) \cdots (z+n)}$$

In particular, we have the asymptotic formula:

$$\Gamma(z + n + 1) \sim_{n \rightarrow \infty} n!n^z$$

5.7 Harmonic function

For the function of class C^2 , the laplace operator is defined as

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

By the Wirtinger operators, we can rewrite the laplace operator as

$$\Delta = 4 \frac{\partial^2}{\partial z \partial \bar{z}} = 4 \frac{\partial^2}{\partial \bar{z} \partial z}$$

Definition 5.7. Let $U \subset \mathbb{C}$ be an open set and $K = \mathbb{C}$ or $\mathbb{R}$, a function $u : U \rightarrow K$ of class C^2 is called **harmonic** if it satisfies the laplace equation:

$$\Delta u = 0$$

The set of complex harmonic function on U is denoted by $\mathcal{H}(U, \mathbb{C})$, the set of real hamonic function on U is denoted by $\mathcal{H}(U, \mathbb{R})$.

The holomorphe function is naturally related to harmonic function in two senses:

1. If f is holomorphe sur U , then $\frac{\partial f}{\partial \bar{z}} = 0$ implies that $\Delta = 0$.

2. write f as $f = u + iv$, then cauchy Riemann equation gives that

$$u_x = v_y, \quad u_y = -v_x$$

holomorphic ensures that u and v are of class C^∞ as real-differentiable functions, hence we can apply the lemma of schwartz

$$\Delta = u_{xx} + u_{yy} = v_{xy} + (-v_{yx}) = 0$$

and similarly $\Delta v = 0$, hence **both real part and imaginary part of holomorphic function are harmonic functions.**

We can see that holomorphic function is a strong condition for harmonic function.

Proposition 5.21. Let $U \subset \mathbb{C}$ be an open set and $f \in \mathcal{O}(U)$, if $f(z) \neq 0$ for any $z \in U$, then $\ln |f| \in \mathcal{H}(U, \mathbb{R})$.

Proof.

□

A classic question is to find the holomorphic function by a known real part or imaginary part, it refers to the solution of Cauchy-Riemann equation, formally

Definition 5.8. Let $u, v : U \rightarrow \mathbb{R}$ be two function of class C^2 , then (u, v) is called **a pari of harmonic conjugates** on U if they are all harmonic functions and satisfy the Cauchy-Riemann equation.

Harmonic conjugates allows us to denote a harmonic function as the real (imaginary) part of a holomorphic function.

Proposition 5.22. Let $U \subset \mathbb{C}$ be an open set and $u : U \rightarrow \mathbb{R}$ be a real harmonic function, then for any $a \in U$ there exists a neighborhood $V \subset U$ of a and a holomorphic function $f \in \mathcal{O}(V)$ such that

$$u = \Re(f) \text{ on } V$$

and locally on V , $\Im(f)$ is a harmonic conjugate of u .

The global existence does not hold in general, for example, we take $u(z) = \ln |z|$ on $U = \mathbb{C} - \{0\}$

Proposition 5.23 (Simply-connectness).

Let $U \subset \mathbb{C}$ be a simply-connected open set and $u : U \rightarrow \mathbb{R}$ be a real harmonic function, then there exists a holomorphic function $f \in \mathcal{O}(U)$ such that

$$u = \Re(f) \text{ on } U$$

generally, $f \mapsto \Re(f)$ induces an isomorphism between function spaces:

$$\mathcal{O}(U)/i\mathbb{R} \xrightarrow{\sim} \mathcal{H}(U, \mathbb{R})$$

Proposition 5.24 (Mean value property).

Let $u : U \rightarrow \mathbb{C}$ be a harmonic function on an open set $U \subset \mathbb{C}$, then for any $a \in U$ with $B(a, r) \subset U$ for some positive $r > 0$, we have

$$u(a) = \frac{1}{2\pi r} \int_{C(a,r)} u(z) |dz| = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}) d\theta$$

where $C(a, r) = \{z \in \mathbb{C} \mid |z - a| = r\}$ is the circle centered at a with radius r , and $|dz|$ is the length element on the circle.

Proof.

□

Mean value property shows that the value at a point is determined by the integration along a circle centered at the point, exactly the harmonic function is characterized by this property, that means given the information of the function along the boundary of a disk, we can get all the values inside the disk.

Theorem 5.25 (Dirichlet problem on $\mathbb{D}$).

For any continuous function $\varphi : \mathbb{T} \rightarrow \mathbb{C}$, then there exists a unique continuous function $u : \mathbb{D} \rightarrow \mathbb{C}$ such that u is harmonic on $\mathbb{D}$ and $u|_{\mathbb{T}} = \varphi$.

Proof.

□

Corollary 5.26. Let $f : U \rightarrow \mathbb{C}$ be a continuous function on an open set $U \subset \mathbb{C}$, then f is harmonic on U if and only if f has the mean value property on U .

One of the most important properties of holomorphic function is the principle of maximum modulus, although we can get the property from analytic continuation, it is actually the properties of harmonic function:

Maximum modulus $\iff$ Mean value property

Proposition 5.27 (principle of maximum).

Let $u : U \rightarrow \mathbb{R}$ be a real harmonic function on an **connected** open set $U \subset \mathbb{C}$, then u can not attain a maximum or minimum in U unless u is constant. In particular, for any compact set $K \subset U$, we have

$$\max_{z \in K} u(z) = \max_{z \in \partial K} u(z) \quad \min_{z \in K} u(z) = \min_{z \in \partial K} u(z)$$

Corollary 5.28 (Principle of maximum modulus).

Let $f : U \rightarrow \mathbb{C}$ be a harmonic function on an **connected** open set $U \subset \mathbb{C}$, then $|f|$ can not attain a maximum in U unless f is constant. In particular, for any compact set $K \subset U$, we have

$$\|f\|_K = \|f\|_{\partial K}$$

6 Commutative Algebra

The ring talked about in this section is always commutative ring with unity.

6.1 Modules

Definition 6.1. Let R be a ring, an R -module M is an abelian group $(M, +)$ together with an operation of R on M :

$$\cdot : R \times M \rightarrow M$$

satisfying the following axioms for all $r, s \in R$ and $m, n \in M$:

- (1) $r \cdot (m + n) = r \cdot m + r \cdot n$
- (2) $(r + s) \cdot m = r \cdot m + s \cdot m$
- (3) $(rs) \cdot m = r \cdot (s \cdot m)$
- (4) $1_R \cdot m = m$

- A R -module induces a natural ring homomorphism

$$\phi : R \rightarrow \text{End}(M), \quad r \mapsto \phi_r$$

with $\phi_r(m) = r \cdot m$. Here axiom (1) ensures ϕ is well-defined (i.e. ϕ_r is an endomorphism of the group), axiom (2)(3)(4) ensures ϕ is a ring homomorphism.

Conversely, any ring homomorphism $\phi : R \rightarrow \text{End}(M)$ induces a R -module structure on M by defining

$$r \cdot m := \phi_r(m)$$

Hence we can conclude a correspondence:

$$\{R\text{-module structures on } M\} \leftrightarrow \{\text{Ring homomorphisms } R \rightarrow \text{End}(M)\}$$

- Similarly we can define the morphism between modules, if M, N are two R -modules, then a group homomorphism $f : M \rightarrow N$ is a **R -module homomorphism (or R -linear)** if two conditions are satisfied:
 - (1) $f(m_1 + m_2) = f(m_1) + f(m_2)$ for all $m_1, m_2 \in M$
 - (2) $f(r \cdot m) = r \cdot f(m)$ for all $r \in R$ and $m \in M$

Example 6.1. Here are some important examples of modules:

(1) Any vector space can be seen as a k -module, here k is a field.

(2) Any abelian group can be seen as a $\mathbb{Z}$ -module.

(3) Any ideal I of a ring R can be seen as a R -module.

(4) Any $k[x]$ -module can be seen as a pair (V, T) where V is a k -vector space and $T : V \rightarrow V$ is a linear transformation. The module structure is given by

$$\left(\sum_{i=0}^n a_i x^i\right) \cdot v := \sum_{i=0}^n a_i T^i(v)$$

In the another view, it refers a morphism of polynomial rings:

$$\phi : k[x] \rightarrow \text{End}(V), \quad x \mapsto T$$

Another important example is algebra, it always appears in the theory of field extension.

Definition 6.2. Let R and A be two rings with a ring homomorphism $\varphi : R \rightarrow A$, then we say (A, φ) is a R -algebra, and we can define a R -module structure on A by

$$r \cdot a := \varphi(r)a$$

Remark 6.1. In particular, an R -module is not necessary a R -algebra, because the multiplication in A may not be defined in M . For example, an ideal $I \subset R$ is a R -module, but it is not a R -algebra unless $I = R$.

There are some properties about algebras:

(1) If M is a A -module with (A, φ) a R -algebra, then M is also a R -module. It is clear by

$$R \longrightarrow A \longrightarrow \text{End}(M)$$

(2) If M and N are two A -modules with (A, φ) a R -algebra, then any A -module homomorphism $f : M \rightarrow N$ is also a R -module homomorphism, i.e.

$$\text{Hom}_A(M, N) \subset \text{Hom}_R(M, N)$$

In particular, we can take equality if φ is surjective.

Similar with vector spaces, we can define the submodule and quotient module as following:

Definition 6.3. Let M be a R -module, a subset $N \subset M$.

- N is called a submodule of M if N is also a R -module with the induced operations from M . Equivalently, N satisfies the following conditions:
 - (1) N is an additive subgroup of M .
 - (2) For any $r \in R$ and any $n \in N$, we have $r \cdot n \in N$.
- Let N be a submodule of M , then there exists a natural quotient group homomorphism $\pi : M \rightarrow M/N$, if we add the condition that π is a R -module homomorphism, then we can naturally get the multiplication on M/N by

$$r \cdot \bar{m} := r \cdot \pi(m) = \pi(rm) = r\bar{m}$$

Then M/N is called the quotient module of M by N .

- the R -module homomorphism $\pi : M \rightarrow M/N$ is called the quotient map or **canonical projection**. It induced a correspondence:

$$\{\text{submodules of } M \text{ containing } N\} \leftrightarrow \{\text{submodules of } M/N\}$$

Now we conclude the isomorphism theorems for modules, the results are similar with groups and vector spaces, so we just give the statements here without proof.

Theorem 6.1 (UPQ-Module).

Let $f : M \rightarrow M'$ be a R -module homomorphism, and N be a submodule of M such that $N \subset \ker f$, then there exists a unique R -module homomorphism

$$\bar{f} : M/N \rightarrow M'$$

such that the following diagram commutes:

$$\begin{array}{ccc} M & \xrightarrow{f} & M' \\ \pi_N \downarrow & \nearrow \bar{f} & \\ M/N & & \end{array}$$

The results of the universal properties are three important isomorphism for modules:

Theorem 6.2 (Isomorphism Theorems for Modules).

Let M be a R -module, and let N, P be two submodules of M , then:

(1) (First Isomorphism Theorem) If $f : M \rightarrow N$ is a R -module homomorphism, then we have the isomorphism:

$$M/\ker f \cong f(M)$$

(2) (Second Isomorphism Theorem) We have the isomorphism:

$$(N + P)/P \cong N/(N \cap P)$$

(3) (Third Isomorphism Theorem) If $P \subset N$, then we have the isomorphism:

$$(M/P)/(N/P) \cong M/N$$

Proof. Proof for (1) is directly from UPQ-Module when $N = \ker f$. For (2), we consider diagram:

$$\begin{array}{ccc} N & \xrightarrow{i} & N + P \\ \pi_{N \cap P} \downarrow & & \downarrow \pi_P \\ N/N \cap P & \xrightarrow{\exists! f} & (N + P)/P \end{array}$$

Here we just need to verify that $\ker(\pi_P \circ i) = N \cap P$ and $\pi_P \circ i$ is surjective, then by UPQ-Module we can get the isomorphism. For (3), we consider diagram:

$$\begin{array}{ccc} M/N & \xrightarrow{\bar{\pi}_P} & M/P \\ \pi_{P/N} \downarrow & \nearrow \exists! f & \\ (M/N)/(P/N) & & \end{array}$$

□

where $\bar{\pi}_P$ is induced by the natural quotient map $\pi_P : M \rightarrow M/P$. We just need to verify that $\ker \bar{\pi}_P = P/N$ and $\bar{\pi}_P$ is surjective, then by (1) we can get the isomorphism.

Like vector spaces, we also hope to find some elements to generate the whole module, here is some vocabulary:

Definition 6.4. Let M be a R -module, let $(m_i)_{i \in I}$ be a family of elements in M , then:

(1) The family is a **generate set** of M if any element of M can be written as a **finite** linear combination of elements in the family, i.e. for any $m \in M$, there exists a finite subset $J \subset I$ and $c_j \in R, j \in J$, such that

$$m = \sum_{j \in J} c_j m_j$$

(2) The family is **linearly independent** if for any finite subset $J \subset I$ and $c_j \in R, j \in J$, the condition $\sum_{j \in J} c_j m_j = 0$ implies that $c_j = 0$ for all $j \in J$.

(3) The family is a **basis** of M if it is a generate set and linearly independent.

(4) The module M is called **finitely generated** if there exists a finite generate set of M .

(5) The module M is called **free** if it has a basis.

Remark 6.2. Remember that "a free module is a lucky accident", not all modules are free, and even finitely generated modules are not necessarily free. Here are some examples for clarification:

(a) Any vector space is a free module, but the proof is not trivial, it is something about **Zorn's lemma (Axiom of choice.)**. In particular, we have implication

$$\left\{ \begin{array}{c} \text{finite generated} \\ k\text{-module} \end{array} \right\} = \{\text{finite-dimensional vector space}\} \implies \text{free}$$

(b) An example that a finitely generated module is not free: let $R = \mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$ as a $\mathbb{Z}$ -module, then M is finitely generated by $\{1 + 2\mathbb{Z}\}$, but it is not free since it is not a base by

$$2 \cdot \bar{1} = \bar{2} = \bar{0}$$

here 2 is non-zero.

(c) In a R -module M , we define the **torsion** to be the element $m \in M$ such that

$$r \cdot m = 0 \quad \text{for some } r \in R - \{0\}$$

The torsion elements will obstruct the module to be free, in example (2) we can see that $\bar{1}$ is a torsion element.

(d) An submodule of a finitely generated module is not necessarily finitely generated, for example, let $R = k[x_1, x_2, \dots]$ be the polynomial ring of infinite variables over a field k , then R is a finitely generated module over itself, but the ideal $I = \langle x_1, x_2, \dots \rangle$ is not finitely generated.

With the basic knowledge of modules, we can define the the category of modules

$$\mathbf{Mod}_R \left| \begin{array}{l} \text{objects: } R\text{-modules} \\ \text{morphisms: } R\text{-linear maps} \end{array} \right.$$

We define the product of modules as following:

Definition 6.5. Let $(M_i)_{i \in I}$ be a family of R -modules, then the **product module** is defined by

$$\prod_{i \in I} M_i := \{(m_i)_{i \in I} \mid m_i \in M_i\}$$

with the operations defined by

$$(m_i)_{i \in I} + (n_i)_{i \in I} := (m_i + n_i)_{i \in I}$$

and

$$r \cdot (m_i)_{i \in I} := (r \cdot m_i)_{i \in I}$$

for any $r \in R$.

There exists a natural projection (a R -module homomorphism)

$$\pi : \prod_{i \in I} M_i \rightarrow M_j, \quad \pi((m_i)_{i \in I}) = m_j \text{ for some fixed } j \in I$$

We can conclude that the product of modules satisfies the universal property of product in category $\mathbf{Mod}_R$.

Proposition 6.3. Let $f : N \rightarrow \prod_{i \in I} M_i$ be a R -module homomorphism, then there exists a unique R -module homomorphism $\tilde{f} : N \rightarrow M_i$ such that $\pi_i \circ f = \tilde{f}$ for all $i \in I$. Equivalently, we have a natural isomorphism

$$\text{Hom}_R(N, \prod_{i \in I} M_i) \cong \prod_{i \in I} \text{Hom}_R(N, M_i)$$

Proof. □

Moreover, we can define the dual structure of product, that is the direct sum of modules.

Definition 6.6. Let $(M_i)_{i \in I}$ be a family of R -modules, then the **direct sum module** is defined by

$$\bigoplus_{i \in I} M_i := \{(m_i)_{i \in I} \mid m_i \in M_i, \text{ and } m_i \neq 0 \text{ for finitely many } i\}$$

with the operations defined by

$$(m_i)_{i \in I} + (n_i)_{i \in I} := (m_i + n_i)_{i \in I}$$

and

$$r \cdot (m_i)_{i \in I} := (r \cdot m_i)_{i \in I}$$

for any $r \in R$.

Remark 6.3.

(1) $\bigoplus_{i \in I} M_i$ is a submodule of $\prod_{i \in I} M_i$. In particular, if the index set is finite, then they are same.

(2) There exists a projection (a R -module homomorphism)

$$\pi : \bigoplus_{i \in I} M_i \rightarrow M_i, \quad \pi((m_i)_{i \in I}) = m_j \text{ for some fixed } j \in I$$

However, the projection is not natural in general, because it does not satisfy the universal property of product. For example we consider $I = \mathbb{N}$ and $M_i = M = \mathbb{Z}$, if we decide the application in component by $f_i : \mathbb{Z} \rightarrow \mathbb{Z}$, $z \mapsto z$, then by universal property we can get a application f such that $\pi_i \circ f = f_i$ for all i . But f is not well-defined in direct sum since the image of $1 \in \mathbb{Z}$ is $(1, 1, 1, \dots)$ which is not in $\bigoplus_{i \in I} M_i$.

(3) There exists a natural injection (a R -module homomorphism)

$$\iota : M_j \rightarrow \bigoplus_{i \in I} M_i, \quad \iota(m) = (0, \dots, 0, m, 0, \dots) \text{ for some fixed } j \in I$$

We can conclude that the direct sum of modules satisfies the universal property of **co-product** in category $\mathbf{Mod}_R$.

Proposition 6.4. Let $f_i : M_i \rightarrow N$ be a R -module homomorphism for all $i \in I$, then there exists a unique R -module homomorphism $\tilde{f} : \bigoplus_{i \in I} M_i \rightarrow N$ such that $\tilde{f} \circ \iota_i = f_i$ for all $i \in I$. Equivalently, we have a natural isomorphism

$$\text{Hom}_R\left(\bigoplus_{i \in I} M_i, N\right) \cong \prod_{i \in I} \text{Hom}_R(M_i, N)$$

Proof. We can define $\tilde{f}$ by

$$\tilde{f}((m_i)_{i \in I}) = \sum_{i \in I} f_i(m_i)$$

for all $(m_i)_{i \in I} \in \bigoplus_{i \in I} M_i$. It is easy to see that $\tilde{f}$ is well-defined (direct sum supports on finitely many non-zero components) and R -linear. Moreover, we have

$$\tilde{f} \circ \iota_i(m) = \tilde{f}((0, \dots, 0, m, 0, \dots)) = f_i(m)$$

for all $m \in M_i$, which shows $\tilde{f}$ is a desired homomorphism. For the uniqueness, if there exists another solution f , then $(\tilde{f} - f) \circ \iota_i = 0$ for each $i \in I$, hence for any $(m_i)_{i \in I} \in \bigoplus_{i \in I} M_i$, we have

$$(\tilde{f} - f)((m_i)_{i \in I}) = (\tilde{f} - f)\left(\sum_{i \in I} \iota_i(m_i)\right) = \sum_{i \in I} (\tilde{f} - f) \circ \iota_i(m_i) = 0$$

it implies $\tilde{f} = f$. □

The properties of direct sum and product can be concludes by the diagram as following:
–product:

$$\begin{array}{ccccc} & & N & & \\ & \swarrow \exists! f_k & \downarrow f & \searrow \exists! f_j & \\ M_k & \xleftarrow{\pi_k} & \prod_i M_i & \xrightarrow{\pi_j} & M_j \end{array}$$

and

–direct sum (coproduct):

$$\begin{array}{ccccc} & & N & & \\ & \nearrow f_k & \uparrow \exists! f & \nwarrow f_j & \\ M_k & \xrightarrow{\iota_k} & \bigoplus_i M_i & \xleftarrow{\iota_j} & M_j \end{array}$$

Sum of modules

Let M be a R -module, and let $E \subset M$ be a subset, then we can define the module generated by E as following:

$$M(E) := \{a_1 e_1 + \dots + a_n e_n \mid n \in \mathbb{N}, a_i \in R, e_i \in E\}$$

in particular, if $E = \{a\}$, then we denote $Ra := M(E)$ to be the **cyclic module** generated by a . Hence $M(E)$ is actually the finite sum of linear combinations of elements in E , so we can define the sum of modules as following:

Definition 6.7. Let $(M_i)_{i \in I}$ be a family of submodules of a R -module M , then the **sum** of modules is defined as the generated module by the union of the submodules $(M_i)_{i \in I}$, i.e.

$$\sum_{i \in I} M_i := \left\{ \sum_{i \in J} m_i \mid J \subset I \text{ finite}, m_i \in M_i \right\}$$

Naturally, we consider the inclusion map $\phi : M_i \hookrightarrow M$ for each $i \in I$, then by the universal properties of direct sum, there exists a unique R -module homomorphism

$$\Phi : \bigoplus_{i \in I} M_i \rightarrow M, \quad (m_i)_{i \in I} \mapsto \sum_{i \in I} m_i$$

Then we can conclude that the image of Φ is exactly the sum of modules $\sum_{i \in I} M_i$:

$$\sum_{i \in I} M_i = \text{Im } \Phi$$

Definition 6.8. A sum of modules $\sum_{i \in I} M_i$ is called a **(internal) direct sum**, if the homomorphism $\Phi : \bigoplus_{i \in I} M_i \rightarrow M$ above is injective. In this case, an isomorphism is induced:

$$\sum_{i \in I} M_i \cong \bigoplus_{i \in I} M_i$$

Remark 6.4. Equivalently, the family of submodules $(M_i)_{i \in I}$ is a direct sum if for any finite subset $J \subset I$, the following condition holds:

$$\sum_{i \in J} m_i = 0 \implies m_i = 0 \text{ for all } i \in J, m_i \in M_i$$

In particular, if I is **finite**, then the sum $\sum_{i=1}^n M_i$ is a direct sum if and only if

$$M_i \cap \sum_{j \neq i} M_j = \{0\} \quad \text{for all } i = 1, 2, \dots, n$$

The definition of external direct sum is actually the direct sum defined at first, i.e. let $(M_i)_{i \in I}$ be a family of R -modules, then the direct sum $\bigoplus_{i \in I} M_i$ is the **(external) direct sum** of the submodules. If we choose exactly the submodules of the same modules, then the external direct sum will be reduced to the internal direct sum.

A question raised here is that: for a module M , if we take a submodule N , can we find another submodule P such that $M = N \oplus P$ such that M has a good decomposition? The answer is not always true, and we define that the submodule N is **direct summand** if such P exists.

Example 6.2.

(1) $\mathbb{Z}$ is a module over itself, and let $2\mathbb{Z}$ be the submodule of $\mathbb{Z}$, then we can find that $2\mathbb{Z}$

is not direct summand. (Reason: any submodule of $\mathbb{Z}$ is of the form $n\mathbb{Z}$ for some $n \in \mathbb{N}$, then $2\mathbb{Z} \cap n\mathbb{Z} = \{0\}$ iff $n = 0$.)

(2) Any subspace of a **vector space** is direct summand, because any vector space has a basis. If V is a vector space with a basis $\{v_i\}_{i \in I}$, and let $W \subset V$ be a subspace with $W = \text{Span}(\{v_i\}_{i \in J})$ with $J \subset I$, then we can define the complement subspace by $U = \text{Span}(\{v_i\}_{i \in I-J})$.

(3) Similarly with (2), any submodule of a **free module** is direct summand.

An condition can be given here to determine whether a submodule is direct summand or not.

Proposition 6.5. Let M be a R -module, and let $N \subset M$ be a submodule, then the following conditions are equivalent:

- (1) N is direct summand of M
- (2) M/N is isomorphic to a submodule of M .
- (3) There exists a homomorphism (**section**) $s : M/N \rightarrow M$ such that $\pi \circ s = \text{id}_{M/N}$, where $\pi : M \rightarrow M/N$ is the canonical projection.
- (4) There exists a homomorphism (**retraction**) $\rho : M \rightarrow N$ such that $\rho(x) = x$ for all $x \in N$.

Proof. (1) $\implies$ (2): Let P be a submodule such that $M = N \oplus P$, then by the second isomorphism theorem we have

$$M/N = (P + M)/N \cong P/(P \cap N) \cong P$$

since direct sum ensures that $P \cap N = \{0\}$.

(2) $\implies$ (3): Let $P \subset M$ be a submodule such that $M/N \cong P$, then the canonical projection $\pi : M \rightarrow M/N$ restricts to an isomorphism $\pi|_P : P \rightarrow M/N$, hence we can define the section $s : M/N \rightarrow M$ by $s^{-1} = (\pi|_P)^{-1}$, and immediately we have $\pi \circ s = \text{id}_{M/N}$.

(3) $\implies$ (1): Let $s : M/N \rightarrow M$ be a section, then we can define a homomorphism $\rho : M \rightarrow M$ by $\rho = \text{id}_M - s \circ \pi$, then we will show that it is a retraction onto N . For any $m \in M$, we have

$$\pi \circ \rho(m) = \pi(m) - \pi \circ s \circ \pi(m) = \pi(m) - \pi(m) = \bar{0}$$

which implies that $\text{Im } \rho \subset N$. Moreover, for any $x \in N$, we have $\rho(x) = x - s \circ \pi(x) = x - s(\bar{0}) = x$ hence we finish the proof.

(4) $\implies$ (1): Let $\rho : M \rightarrow N$ be a retraction, then we will prove that $M = N \oplus \ker \rho$. For any $m \in M$, we have

$$m = m - \rho(m) + \rho(m)$$

where $m - \rho(m) \in \ker \rho$ and $\rho(m) \in N$, hence $M = N + \ker \rho$. Moreover, we can calculate the intersection to prove the directness:

$$N \oplus \ker \rho = \{x \in N \mid \rho(x) = 0\} = \{0\}$$

□

Remark 6.5. It is a motivation to consider the decomposition of the exact sequence. Let

$$0 \longrightarrow N \xrightarrow{f} M \xrightarrow{g} P \longrightarrow 0$$

be a short exact sequence of R -modules, then the following conditions are equivalent:

- (1) There exists an isomorphism $M \cong N \oplus P$.
- (2) There exists a section $s : P \rightarrow M$ such that $g \circ s = \text{id}_P$.
- (3) There exists a retraction $r : M \rightarrow N$ such that $r \circ f = \text{id}_N$.

If one of the condition holds, then the s.e.q is called **split**. The proof is similar with the previous proposition, we just need to notice that f is injective and g is surjective in a s.e.q.

Free module

For the convenience of notation, we define the product of copies of a commutative ring A as following:

$$A^I := \prod_{i \in I} A \quad \text{and} \quad A^{(I)} := \bigoplus_{i \in I} A$$

for any index set I . In particular, if $I = \{1, 2, \dots, n\}$ is finite, then $A^I = A^{(I)} = A^n$.

Remark 6.6. Recall that a module is free if it has a basis, i.e. a linearly independent generate set.

- (1) $A^{(I)}$ is a free A -module with the standard basis as following:

$$e_k := (\delta_{k,i})_{i \in I} = (\dots, 0, \underset{k\text{-th}}{1}, 0, \dots)$$

for all $(a_i)_{i \in I} \in A^{(I)}$, we have

$$(a_i)_{i \in I} = \sum_{i \in I} a_i e_i = \sum_{i \in J} a_i e_i$$

where J is the finite subset of I such that $a_i \neq 0$, so the family is a set of generators. Moreover, if $\sum_{i \in J} a_i e_i = 0$, then $a_i = 0$ for all $i \in J$, so the family is linearly independent.

- (2) A^I is not a free A -module if I is infinite. For example, let $I = \mathbb{N}$, then consider the element $(1, 1, 1, \dots) \in A^I$, it can not be expressed as a finite combination of the standard basis elements above. However, the proof is not trivial here, we will do it later.

As we have constructed, $A^{(I)}$ is a **standard model** as a free module, it refers to the following universal property:

Proposition 6.6 (UP-Free module).

Let M be a free A -module with a family of elements $\{m_i\}_{i \in I}$, there exists a unique morphism of A -modules $\Phi : A^{(I)} \rightarrow M$ such that $\Phi(e_i) = m_i$ for all $i \in I$.

$$\begin{array}{ccc} A & \xrightarrow{\iota_i} & A^{(I)} \\ & \searrow \phi_i & \downarrow \Phi \\ & & M \end{array}$$

Equivalently, it induces a natural isomorphism by $\Phi \mapsto (\phi(e_i))_{i \in I}$

$$\text{Hom}_A(A^{(I)}, M) \cong \prod_{i \in I} M$$

Proof. The proof is similar to the properties of linear map (**the choice of images of basis determines a unique linear map**). For any $(a_i)_{i \in I} \in A^{(I)}$, we can define

$$\Phi((a_i)_{i \in I}) = \sum_{i \in I} a_i m_i$$

It is easy to verify that Φ is well-defined (direct sum supports on finitely many non-zero components) and A -linear. For the uniqueness, if there exists another morphism $\Psi : A^{(I)} \rightarrow M$ such that $\Psi(e_i) = m_i$ for all $i \in I$, then for any $(a_i)_{i \in I} \in A^{(I)}$, we have

$$\Psi((a_i)_{i \in I}) = \Psi\left(\sum_{i \in I} a_i e_i\right) = \sum_{i \in I} a_i \Psi(e_i) = \sum_{i \in I} a_i m_i = \Phi((a_i)_{i \in I})$$

hence $\Psi = \Phi$. Hence we can conclude that the two A -linear maps are the same if and only if they have the same images of the basis elements, which implies the natural isomorphism. $\square$

Hence we can generalize the statement of the generating set and basis by the language of morphisms.

Corollary 6.7. Let M be an A -module, and let I be an index set, then

- (1) M is **generated** by $\{m_i\}_{i \in I}$ if and only if the morphism $\Phi : A^{(I)} \rightarrow M$ defined by $\Phi(e_i) = m_i$ for all $i \in I$ is surjective.
- (2) $\{m_i\}_{i \in I}$ is **linearly independent** set of M if and only if the morphism $\Phi : A^{(I)} \rightarrow M$ defined by $\Phi(e_i) = m_i$ for all $i \in I$ is an injective.
- (3) $\{m_i\}_{i \in I}$ is a **basis** of M if and only if the morphism $\Phi : A^{(I)} \rightarrow M$ defined by $\Phi(e_i) = m_i$ for all $i \in I$ is an isomorphism.
- (4) M is a **finitely generated** A -module if and only if there exists a surjective morphism $\Phi : A^n \rightarrow M$ for some $n \in \mathbb{N}$.

If M is a free A -module and finitely generated, then we can define "dimension" of M by proving the following statement:

Lemma 6.8. Let M be a free A -module, and let $\{m_1, \dots, m_n\}$ and $\{n_1, \dots, n_k\}$ be two bases of M , then $n = k$.

Proof. □

6.2 Polynomial and Series

As we know that a polynomial can be defined as following:

$$\mathbb{Z}[X] = \{a_n x^n + \dots + a_1 x + a_0 \mid a_i \in \mathbb{Z}\}$$

It is clearly a free $\mathbb{Z}$ -module with basis $\{x^n \mid n \in \mathbb{N}\}$, that means $\mathbb{Z}^{(\mathbb{N})} \cong \mathbb{Z}[X]$ as modules. Moreover, we notice that the polynomial is furthermore a ring, so it is a $\mathbb{Z}$ -algebra, hence the operations should be extended here to generalize the definition of polynomial ring.

Definition 6.9. Let $A^{(\mathbb{N})}$ be a free A -module with basis $(e_i)_{i \in \mathbb{N}} = (\delta_j, i)_{i \in \mathbb{N}}$, then we can define a multiplication on it by make conservation:

$$e_n \cdot e_m = e_{n+m} \quad \text{for all } n, m \in \mathbb{N}$$

and develop it for any elements

$$(a_i)_{i \in \mathbb{N}} \cdot (b_i)_{i \in \mathbb{N}} = (c_i)_{i \in \mathbb{N}} \quad \text{with} \quad c_i = \sum_{l+k=i} a_l b_k$$

Then $A^{(\mathbb{N})}$ is a A -algebra with ring homomorphism

$$\varphi : A \rightarrow A^{(\mathbb{N})}, \quad a \mapsto a e_0 = (a, 0, 0, \dots)$$

Remark 6.7.

(1) It is easy to verify that the multiplication is well-defined, i.e. the result is still in $A^{(\mathbb{N})}$ since only finitely many a_i and b_i are non-zero.

(2) The multiplicative identity is e_0 , since for any $(a_i)_{i \in \mathbb{N}} \in A^{(\mathbb{N})}$, we have

$$e_0 \cdot \sum_{i \in \mathbb{N}} a_i e_i = \sum_{i \in \mathbb{N}} a_i (e_0 \cdot e_i) = \sum_{i \in \mathbb{N}} a_i e_i = (a_i)_{i \in \mathbb{N}}$$

(3) The multiplication is associative, commutative and distributive with respect to addition, which can be verified by direct calculation.

with the work of (1)(2)(3), we can conclude that $A^{(\mathbb{N})}$ is a commutative ring with unity, hence it is a A -algebra.

Then we can construct the polynomial ring as following:

Definition 6.10. Let $A^{(\mathbb{N})}$ be the A -algebra defined above, we identify it as the **polynomial ring** $A[X]$ over A by making the following notation:

- basis: $X^n := e_n$ for all $n \in \mathbb{N}$ with $1_A = X^0 := e_0$.
- elements: $f(X) = \sum_{i \in \mathbb{N}} a_i X^i := (a_i)_{i \in \mathbb{N}}$ for all $(a_i)_{i \in \mathbb{N}} \in A^{(\mathbb{N})}$.
- addition: $\sum_n a_n X^n + \sum_n b_n X^n = \sum_n (a_n + b_n) X^n$.
- multiplication: $(\sum_n a_n X^n) \cdot (\sum_n b_n X^n) = \sum_n c_n X^n$ with $c_n = \sum_{l+k=n} a_l b_k$.

Or equivalently, if we define the polynomial ring as what we usually do, then we can get the isomorphism $A[X] \cong A^{(\mathbb{N})}$ by sending X^n to e_n for all $n \in \mathbb{N}$.

More generally, we can define the polynomial ring with multiple variables, it is similar to what we have done just now, we need to extend the multiplication of free-module on any reasonable index set.

Definition 6.11. Let $(N, +, 0)$ be an associative monoid with identity, we can define a multiplication on the free A -module $A^{(N)}$ with basis $(e_n)_{n \in N} = (\delta_{k,n})_{n \in N}$ by

$$e_u \cdot e_v = e_{u+v} \quad \text{for all } u, v \in N$$

and develop it for any elements

$$(a_n)_{n \in N} \cdot (b_n)_{n \in N} = (c_n)_{n \in N} \quad \text{with} \quad c_n = \sum_{u+v=n} a_u b_v$$

Then $A^{(N)}$ is a A -algebra with ring homomorphism

$$\varphi : A \rightarrow A^{(N)}, \quad a \mapsto a e_0$$

and we call it the **polynomial ring** over A with index monoid N , denoted by $A[N]$.

In particular, if we take $N = \mathbb{N}$ we can get the polynomial ring in one variable $A[X]$. If we take $N = \mathbb{N}^n$ with the component-wise addition, then we can get the polynomial ring in n variables $A[X_1, \dots, X_n]$ by identifying the basis element $e_{(k_1, \dots, k_n)}$ with $X_1^{k_1} X_2^{k_2} \dots X_n^{k_n}$ for all $(k_1, \dots, k_n) \in \mathbb{N}^n$, then we can get the algebra of polynomial as following:

- elements:

$$f(X_1, \dots, X_n) = \sum_{(k_1, \dots, k_n) \in \mathbb{N}^n} a_{(k_1, \dots, k_n)} X_1^{k_1} X_2^{k_2} \dots X_n^{k_n} = \sum_{\kappa \in \mathbb{N}} a_\kappa X^\kappa$$

where only finitely many a_κ are non-zero.

- addition:

$$\left(\sum_{\kappa \in \mathbb{N}} a_\kappa X^\kappa \right) + \left(\sum_{\kappa \in \mathbb{N}} b_\kappa X^\kappa \right) = \sum_{\kappa \in \mathbb{N}} (a_\kappa + b_\kappa) X^\kappa$$

- multiplication:

$$\left(\sum_{\kappa \in \mathbb{N}} a_\kappa X^\kappa \right) \cdot \left(\sum_{\kappa \in \mathbb{N}} b_\kappa X^\kappa \right) = \sum_{\kappa \in \mathbb{N}} c_\kappa X^\kappa \quad \text{with} \quad c_\kappa = \sum_{\lambda + \mu = \kappa} a_\lambda b_\mu$$

where the addition $\lambda + \mu$ is defined by component-wise addition in $\mathbb{N}^n$. The verification here is omitted for brevity, the process is similar to the one-variable case.

Polynomial is a natural object in algebra, in a certain vocabulary, we say that polynomial ring $A[X]$ is a free commutative A -algebra, it refers to the following universal properties:

Proposition 6.9 (UP-Poly).

For any A -algebra B with elements $b_1, \dots, b_n$, there exists a unique morphism of A -algebras $\Phi : A[X_1, \dots, X_n] \rightarrow B$ such that $\Phi(X_i) = b_i$ for all $i = 1, 2, \dots, n$.

Or equivalently, it induces a natural bijection by $\Phi \mapsto (\Phi(X_1), \dots, \Phi(X_n))$

$$\text{Hom}_{A\text{-alg}}(A[X_1, \dots, X_n], B) \cong B^n$$

The inverse is given by evaluation map $(b_1, \dots, b_n) \mapsto \text{ev}_{(b_1, \dots, b_n)}$, where $\text{ev}_{(b_1, \dots, b_n)} : A[X_1, \dots, X_n] \rightarrow B$ is defined by $\text{ev}_{(b_1, \dots, b_n)}(f) = f(b_1, \dots, b_n)$.

Proof. For any $f(X_1, \dots, X_n) = \sum_{\kappa \in \mathbb{N}^n} a_\kappa X^\kappa \in A[X_1, \dots, X_n]$, we can define

$$\Phi(f) = \sum_{\kappa \in \mathbb{N}^n} a_\kappa b^\kappa \in B$$

where $b^\kappa = b_1^{k_1} b_2^{k_2} \dots b_n^{k_n}$ for $\kappa = (k_1, \dots, k_n) \in \mathbb{N}^n$. It is easy to verify that Φ is well-defined (only finitely many a_κ are non-zero) and a morphism of A -algebras. Moreover, we have

$$\Phi(X_i) = \Phi(e_{(0, \dots, 1, \dots, 0)}) = b_i$$

for all $i = 1, 2, \dots, n$, which shows that Φ is a desired morphism. For the uniqueness, if there exists another solution Ψ , then for any $f(X_1, \dots, X_n) = \sum_{\kappa \in \mathbb{N}^n} a_\kappa X^\kappa \in A[X_1, \dots, X_n]$, we have

$$\Psi(f) = \Psi\left(\sum_{\kappa \in \mathbb{N}^n} a_\kappa X^\kappa\right) = \sum_{\kappa \in \mathbb{N}^n} a_\kappa \Psi(X^\kappa) = \sum_{\kappa \in \mathbb{N}^n} a_\kappa b^\kappa = \Phi(f)$$

hence we finish the proof. □

6.3 Tensor product

Definition 6.12. Let M and N be two A -modules, the **tensor product** of M and N over A is an A -module $M \otimes_A N$ together with a bilinear map

$$t : M \times N \rightarrow M \otimes_A N, \quad (m, n) \mapsto m \otimes n$$

such that the following Universal properties (UPQ-TENSOR) holds:

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & P \\ \downarrow t & \nearrow \bar{f} & \\ M \otimes_A N & & \end{array}$$

for any A -module P and any bilinear map $f : M \times N \rightarrow P$, there exists a unique A -linear $\bar{f} : M \otimes_A N \rightarrow P$ such that $f = \bar{f} \circ t$, i.e. $\bar{f}(a \otimes b) = f(a, b)$.

The definition is concluded by some work to give a good construction

(1) The existence of tensor product can be constructed by taking the free module $A^{(M \times N)}$ and its submodule R generated by the following elements:

$$\begin{cases} e_{m+m',n} - e_{m,n} - e_{m',n} \\ e_{m,n+n'} - e_{m,n} - e_{m,n'} \\ e_{am,n} - ae_{m,n}, \\ e_{m,an} - ae_{m,n} \end{cases}$$

where $m, m' \in M, n, n' \in N, a \in A$, and $(e_{m,n})_{(m,n) \in M \times N}$ is the standard basis of $A^{(M \times N)}$. Then we can define

$$M \otimes_A N := A^{(M \times N)} / R$$

with $m \otimes n := \overline{e_{m,n}}$. then some verification needs here to show that the construction satisfies the universal property.

(2) The tensor product is unique up to isomorphism, i.e. if (T, t) and (T', t') are two tensor products of M and N , then the universal properties induce two morphism $j : T \rightarrow T'$ and $j' : T' \rightarrow T$ with the following diagram:

$$\begin{array}{ccccc} & & & & T \\ & & \nearrow & & \downarrow j \\ M \times N & \xrightarrow{b'} & T' & & \\ & \searrow & & & \downarrow j' \\ & & & & T \end{array}$$

then $j \circ j' : T \rightarrow T$ a morphism satisfying $b = (j \circ j') \circ b$, by the uniqueness of universal property we have $j \circ j' = \text{id}_T$. Similarly, we can get $j' \circ j = \text{id}_{T'}$, hence $T \cong T'$.

(3) The concret construction of tensor product is not important, the nice universal property allows us to treat a multilinear map as a linear map, usually we call $M \otimes_A N$ as ||tensor|| , and $m \otimes n \in M \otimes_A N$ as $\text{||tensor element||}$.

Some basic algebraic properties of tensor product can be concluded as following, isomorphism here should be understood to avoid the abuse of equality:

Lemma 6.10. The letter we use are A -modules, then we have the following isomorphisms of A -modules:

- (1) $A \otimes_A M \cong M$ with the isomorphism $a \otimes m \mapsto am$.
- (2) $M \otimes_A N \cong N \otimes_A M$ with the isomorphism $m \otimes n \mapsto n \otimes m$.
- (3) $(M \otimes_A N) \otimes_A P \cong M \otimes_A (N \otimes_A P)$ with the isomorphism $(m \otimes n) \otimes p \mapsto m \otimes (n \otimes p)$.
- (4) $M \otimes_A (N \oplus P) \cong (M \otimes_A N) \oplus (M \otimes_A P)$ with the isomorphism $m \otimes (n, p) \mapsto (m \otimes n, m \otimes p)$.

Remark 6.8. Pay attention to the abuse of the equality, sometimes we write $=$ instead of $\cong$ to simplify the notation.

- (1) the first result can be generalized to $A^n \otimes_A M \cong M^n$.
- (2) the last result can be generalized to

$$M \otimes_A \left(\bigoplus_{i \in I} M_i \right) \cong \bigoplus_{i \in I} (M \otimes_A M_i)$$

where I is any index set, with the isomorphism we can conclude that tensor product can preserve the free module structure by isomorphism

$$A^{(I)} \otimes_A M \cong M^{(I)}$$

which covers the first remark. Hence we can conclude: **Mod_A** with tensor product $\otimes_A$ forms a **monoidal category**.

Definition. A monoidal category contains following data:

- (1) a category $\mathcal{C}$.
- (2) a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$.
- (3) an unit object $I \in \mathcal{C}$ having two natural isomorphisms.

$$I \otimes X \cong X \cong X \otimes I$$

- (4) a natural isomorphism (associator)

$$X \otimes Y \otimes Z \cong X \otimes (Y \otimes Z)$$

- (5) These isomorphism satisfy the MacLane axiom and triangle axiom.

We can generalize the universal property to construct the correspondence between multilinear maps and linear maps from the tensor product as following:

Proposition 6.11 (UPQ-MULTI-TENSOR).

Let $M_1, \dots, M_k$ be A -modules, then for any A -module P , there exists a natural isomorphism

$$\text{Hom}_A(M_1 \otimes_A \cdots \otimes_A M_k, P) \cong \text{Mult}_A(M_1 \times \cdots \times M_k, P)$$

with the bijection $f \mapsto f(- \otimes - \cdots - \otimes -)$

The tensor product of two module is a larger module containing both information of two object in parallel way, hence the morphism of two tensor product is similarly a "product" of morphisms of two modules, which is the tensor product of morphisms:

Proposition 6.12. Let $f : M \rightarrow M'$ and $g : N \rightarrow N'$ be two morphisms of A -modules, then there exists a unique morphism of A -modules

$$f \otimes g : M \otimes_A N \rightarrow M' \otimes_A N', \quad m \otimes n \mapsto f(m) \otimes g(n)$$

Review that if $f : A \rightarrow B$ be a ring homomorphism, and M is a B -module, then it is reasonable to view M as an A -module by

$$a \cdot m := f(a)m, \quad \forall a \in A$$

and we say that A -module M is obtained from B -module M by **restriction of scalars** along f . Conversely, if M is an A -module, then we can construct a B -module by tensor product:

Proposition 6.13 (Extension of scalars).

Let M be an A -module, and N be a (A, B) -bimodule, then $N \otimes_A M$ is a B -module with the unique structure defined by

$$b \cdot (n \otimes m) := (bn) \otimes m, \quad \forall b \in B, n \in N, m \in M$$

In particular, if B is a A -algebra, then $B \otimes_A M$ is a B -module with a natural map

$$\tau : M \rightarrow B \otimes_A M, \quad m \mapsto 1_B \otimes m$$

Hence for any A -algebra B and any A -module M , we define the module obtained by **extension of scalar from A to B** as following

$$M_B := B \otimes_A M$$

Sometimes we need to calculate tensor product under different ring, it is useful to do some identification:

Lemma 6.14 (Mixed associativity of tensor product).

Let M be a A -module, N be a B -module and P be a (A, B) -bimodule, then there exists a natural isomorphism

$$M \otimes_A (P \otimes_B N) \cong (M \otimes_A P) \otimes_B N$$

In particular, if B is an A -algebra, then there exists a natural isomorphism

$$M \otimes_A N \cong M_B \otimes N$$

And we can talk more about morphism after extension of scalars, than restriction of scalars:

Proposition 6.15 (UPA-Extension of scalars).

Let B be an A -algebra, M be a A -module and N be a B -module, then there exists a natural isomorphism by $f \mapsto \varphi_f$, $\varphi_f(m) = f(1 \otimes m)$

$$\mathrm{Hom}_B(M_B, N) \xrightarrow{\sim} \mathrm{Hom}_A(M, N)$$

with the inverse given by $g \mapsto \psi_g$, $\psi_g(b \otimes m) = b \cdot g(m)$.

Remark 6.9. It is the adjoint properties in category of module, we consider a ring homomorphism $f : A \rightarrow B$, then the restriction of scalar can be described as a functor by

$$\mathrm{Res}_A^B : \mathbf{Mod}_B \rightarrow \mathbf{Mod}_A \quad \left| \begin{array}{l} \text{objects: } B\text{-module } M \mapsto A\text{-module } M \\ \text{morphisms: } \mathrm{Hom}_B(M, N) \subset \mathrm{Hom}_A(M, N) \end{array} \right.$$

It is a forgetful functor, since it forgets the extra structure of B -module. Conversely, the extension of scalars can be described as a functor by

$$B \otimes_A - : \mathbf{Mod}_A \rightarrow \mathbf{Mod}_B \quad \left| \begin{array}{l} \text{objects: } M \mapsto M_B \\ \text{morphisms: } f \mapsto \mathrm{id}_B \otimes f \end{array} \right.$$

Hence by the above proposition, the isomorphism can be rewritten as

$$\mathrm{Hom}_B(B \otimes_A M, N) \cong \mathrm{Hom}_A(M, \mathrm{Res}_A^B(N))$$

which shows that extension of scalar can be seen as the **left adjoint** functor of restriction of scalar in the sense of category:

$$B \otimes_A - \dashv \mathrm{Res}_A^B$$

We have seen that extension of scalars is a left adjoint functor, hence it is more meaningful to discuss its functoriality than a forgetful functor, and indeed it behaves well in exactness:

Lemma 6.16. Let

$$M \xrightarrow{f} M' \xrightarrow{g} M'' \longrightarrow 0$$

be an exact sequence of A -modules, and let N be another A -module, it induces an exact sequence of A -modules

$$M \otimes_A N \xrightarrow{f \otimes \mathrm{id}_N} M' \otimes_A N \xrightarrow{g \otimes \mathrm{id}_N} M'' \otimes_A N \longrightarrow 0$$

Equivalently, the functor $- \otimes_A N : \mathbf{Mod}_A \rightarrow \mathbf{Mod}_A$ is a right exact functor.

Proof.

□

Hence which motivates us to classify the modules like projective modules or injective modules to guarantee the exactness after tensor product.

Definition 6.13. An A -module N is called a **flat module** if the functor $- \otimes_A N : \mathbf{Mod}_A \rightarrow \mathbf{Mod}_A$ is an exact functor, i.e. it preserves the exactness of any exact sequence of A -modules.

- In particular, if $N = B$ is a A -algebra, then we say that B is a **flat algebra** over A if B is flat as an A -module.

Flat algebra is a good concept for extension of scalars, for example let M be an A -module, then the natural map

$$\tau : M \rightarrow M_B = B \otimes_A M$$

is not necessary a inclusion. For example we consider $\mathbb{Z}$ -module $M = \mathbb{Z}/(3)$ and $B = \mathbb{Z}/(2)$, then $M_B = 0$ shows a collapse as a case of bad extension. However, **faithful flatness** can avoid this problem.

Category of Algebra

As we have discussed before, for a commutative ring A , all commutative A -algebras forms a category

$$\mathbf{CAlg}_A \mid \begin{array}{l} \text{objects: } (B, f) \text{ or } A \xrightarrow{f} B \\ \text{morphisms: } h : (B, f) \rightarrow (C, g) \text{ s.t. } h \circ f = g \end{array}$$

We will see that tensor product can be described as the product with universal property in this category.

Proposition 6.17 (UP-cproduct-tensor). we have natural isomorphism

$$\mathrm{Hom}_{A\text{-alg}}(B \otimes_A C, D) \cong \mathrm{Hom}_{A\text{-alg}}(B, D) \times \mathrm{Hom}_{A\text{-alg}}(C, D)$$

With the universal property, we can conclude some basic identifications

Claim 6.18.

$$A/I \otimes_A A/J \cong A/(I + J)$$

Proof. By the universal property

$$\mathrm{Hom}_{A\text{-alg}}(A/I \otimes_A A/J, R) \cong \mathrm{Hom}_{A\text{-alg}}(A/I, R) \times \mathrm{Hom}_{A\text{-alg}}(A/J, R)$$

and morphisms $f : A/I \rightarrow R$ and $g : A/J \rightarrow R$ are uniquely (**verify**) determined by a morphism $T : A \rightarrow R$ with $I \subset \ker T$ and $J \subset \ker T$, and it is equivalent to $I + J \subset \ker T$, since $I + J$ is the smallest ideal containing both I and J , hence it uniquely determined a morphism $\bar{T} : A/I + J \rightarrow R$. $\square$

Claim 6.19.

$$A[X, Y] \cong A[X] \otimes_A A[Y]$$

Proof. Review the universal property of polynomial gives

$$\text{Hom}_{A\text{-alg}}(A[X_1, \dots, X_n], R) \cong R^n$$

In particular $n = 1$ gives $\text{Hom}_{A\text{-alg}}(A[X], R) \cong R$, hence

$$\text{Hom}_{A\text{-alg}}(A[X_1, \dots, X_n], R) \cong \prod_{i=1}^n \text{Hom}_{A\text{-alg}}(A[X_i], R)$$

hence we can conclude the result. □

Claim 6.20.

$$A[X] = A \otimes_{\mathbb{Z}} \mathbb{Z}[X]$$

which shows that any polynomial ring is the result of extension of scalars.

6.4 Ideal

Ideal is something like normal subgroups in group theory, it is important to reflect the structure and relationship of rings. Here is the definition:

Definition 6.14. Let R be a ring, a subset $I \subset R$ is called a ideal of R if the following conditions are satisfied:

- (1) I is an additive subgroup of R , that is for any $a, b \in I$, we have $a - b \in I$.
- (2) For any $r \in R$ and any $a \in I$, we have $ra \in I$ and $ar \in I$.

Here is something to add as the properties, proof is not difficult:

1. Ideal $I \subset R$ can be seen as an R -submodule of R , roughly speaking, it can be seen as a "vector subspace". (The union of ideals is not necessary an ideal)
2. An ideal generated by a subset $S \subset R$ is defined by

$$\langle S \rangle := \bigcap_{I \text{ ideal}, S \subset I} I$$

it is the smallest ideal containing S , and we can verify that

$$\langle S \rangle = \left\{ \sum_{\text{finite}} r_i s_i \mid r_i \in R, s_i \in S \right\}$$

In particular, a **principal ideal** is an ideal of the form $(a) = \langle \{a\} \rangle$, i.e a ideal generated by a single element (or it is a cyclic R -module).

3. A useful statement: If a unit $r \in I$, then $I = R$.
4. The operation of ideal: $IJ \subset I \cap J \subset I \subset \langle I \cup J \rangle$

In this section, many different ideals will be introduced.

Lemma 6.21. Let $I \subset R$ be an ideal of a ring R , then

$$I = \sqrt{I} = \{a \in R \mid \exists n \in \mathbb{N}^*, a^n \in I\}$$

defines an ideal of R , called the **radical** of I .

Definition 6.15. Let $I \subset R$ be an ideal of a ring R , we say that

$-I$ is a **maximal ideal** if it is a proper ideal such that for any ideal J of R satisfying $I \subset J$, we have either $J = I$ or $J = R$.

$-I$ is a **prime ideal** if it is a proper ideal such that for any $a, b \in R$ satisfying $ab \in I$, we have either $a \in I$ or $b \in I$.

$-I$ is called a **radical ideal** if $I = \sqrt{I}$.

The set of all prime ideals of R is called the **spectrum** of R , denoted by $\text{Spec}(R)$; The set of all maximal ideals of R is denoted by $\text{Max}(R)$.

Here are some important remarks:

1. Any ring has at least one maximal ideal, the proof will depend on Zorn's lemma.
2. It is necessary to verify that radical ideal is indeed an ideal of R , and it satisfy $I \subset \sqrt{I}$.
3. The nilpotent elements of a ring R is an element $a \in R$ such that there exists $n \in \mathbb{N}^*$ satisfying $a^n = 0$, and a ring is called **reduced** if its nilpotent elements is only 0. It is easy to verify that the set of nilpotent element is a radical ideal, and we call it the **nilradical** of R , denoted by $\text{Nil}(R)$:

$$\text{Nil}(A) = \sqrt{(0)}$$

4. A exercise: $I \subsetneq \sqrt{I} \iff \text{Nil}(A/I) \neq \{0\}$
5. A exercise: $\text{Nil}(A/I) = \sqrt{I}/I$
6. prime ideals or maximal ideals are radical with:

$$\sqrt{\mathfrak{J}} = \mathfrak{J}, \quad \forall \mathfrak{J} \in \text{Max}(R) \cup \text{Spec}(A)$$

Quotient rings gives us a good way to describe this ideals via the characterizations of rings:

Proposition 6.22. Let $I \subset R$ be an ideal of a ring R , then

(a) I is a maximal ideal if and only if the quotient ring R/I is a field.

(b) I is a prime ideal if and only if the quotient ring R/I is an integral domain.

(c) I is a radical ideal if and only if the quotient ring R/I is a reduced ring.

The radical ideal has a deep relation with prime ideals, one can characterize the radical ideal by the intersection of prime ideals:

Lemma 6.23. Let $I \subseteq R$ be an ideal of a ring R , then

$$\sqrt{I} = \bigcap_{\substack{\mathfrak{p} \in \text{Spec}(R) \\ I \subset \mathfrak{p}}} \mathfrak{p}$$

In particular, if $I = 0$ we can conclude that nilradical is the intersection of all prime ideals:

$$\text{Nil}(R) = \bigcap_{\mathfrak{p} \in \text{Spec}(R)} \mathfrak{p}$$

Proof.

□

The intersection of ideals gives an equivalent definition of nilradical ideal, similarly we can define **Jacobson radical** by the intersection of maximal ideals:

$$\text{Jac}(R) = \bigcap_{\mathfrak{m} \in \text{Max}(R)} \mathfrak{m}$$

An important property of Jacobson radical is the following:

Lemma 6.24. Let R be a ring, then for any $a \in R$, we have

$$a \in \text{Jac}(R) \iff \forall b \in R, 1 - ab \text{ is a unit in } R$$

Proof.

□

with the Jacobson radical, we can use the condition of finitely generated module, which is a certain technical lemma in commutative algebra but useful in many situations:

Theorem 6.25 (Nakayama). Let M be a finitely generated A -module, and $I \subset \text{Jac}(A)$ be an ideal of A . If $IM = M$, then $M = 0$.

Proof.

□

Notice that the condition $IM = M$ is equivalent to the quotient module $M/IM = 0$, a better understanding of Nakayama lemma can be based on the tensor product:

Lemma 6.26. Let M be a A -module, and $I \subset A$ be an ideal of A . Then there exists a natural isomorphism $\bar{m} \mapsto 1 \otimes m$

$$M/IM \cong (A/I) \otimes_A M$$

In particular, Nakayama lemma gives a useful support to show that the module over local ring has many good properties like field.

6.5 Noetherian ring

Some basic properties about finitely generated modules is useful, because we always need to claim whether a module is finitely generated or not.

Proposition 6.27. Let M be a R -module, and $N \subset M$ be a submodule:

(1) If M is finitely generated, then quotient module M/N is also finitely generated.

(2) If N and M/N are finitely generated, then M is also finitely generated.

(3) If M_i is finitely generated R -modules for all $i = 1, 2, \dots, n$, then the product module $\prod_{i=1}^n M_i$ is also finitely generated.

(4) Let M be A -module, and (A, ϕ) be a R -algebra, then if M is finitely generated as a R -module, then M is also finitely generated as a A -module.

Proof. (1) Let $(m_1, m_2, \dots, m_n)$ be a finite generate set of M , then we claim that $(\bar{m}_1, \bar{m}_2, \dots, \bar{m}_n)$ is a finite generate set of M/N . For any $\bar{m} \in M/N$, there exists $c_i \in R$ such that

$$m = \sum_{i=1}^n c_i m_i$$

hence

$$\bar{m} = \sum_{i=1}^n c_i \bar{m}_i$$

it implies the result.

(2) Let $(n_1, n_2, \dots, n_k)$ be a finite generate set of N , and let $(\bar{m}_1, \bar{m}_2, \dots, \bar{m}_l)$ be a finite generate set of M/N . We claim that $(n_1, n_2, \dots, n_k, m_1, m_2, \dots, m_l)$ is a finite generate set of M . For any $m \in M$, there exists $c_j \in R$ such that

$$\bar{m} = \sum_{j=1}^l c_j \bar{m}_j$$

hence

$$m - \sum_{j=1}^l c_j m_j \in N$$

so there exists $d_i \in R$ such that

$$m - \sum_{j=1}^l c_j m_j = \sum_{i=1}^k d_i n_i$$

it implies that

$$m = \sum_{i=1}^k d_i n_i + \sum_{j=1}^l c_j m_j$$

hence the result.

(3) Let $(m_{i1}, m_{i2}, \dots, m_{in_i})$ be a finite generate set of M_i for each $i = 1, 2, \dots, r$, then the finite set (order is $n_1 + \dots + n_r$)

$$\{(0, \dots, 0, m_{ij}, 0, \dots, 0) \mid i = 1, 2, \dots, r; j = 1, 2, \dots, n_i\}$$

is a generate set of $\prod_{i=1}^r M_i$.

(4) Let $(m_1, m_2, \dots, m_n)$ be a finite generate set of M as a R -module, then for any $m \in M$, there exists $c_i \in R$ such that

$$m = \sum_{i=1}^n c_i \cdot m_i = \sum_{i=1}^n \phi(c_i) m_i$$

□

The statement (2) can be generalized to a exact sequence as following:

Corollary 6.28. Let $0 \rightarrow N \rightarrow M \rightarrow P \rightarrow 0$ be an exact sequence of R -modules, then

$$\{N, P\} \text{ finitely generated} \implies M \text{ finitely generated}$$

However, we can not say more about the implication because the submodule of a finitely generated module is not necessarily finitely generated. We should make some restriction on the module such that some good properties can be ensured, that is called **Noetherian module**.

Definition 6.16. A R -module M is called a **Noetherian module** if it satisfies the following equivalent conditions:

- (1) Any submodule of M is finitely generated.
- (2) Any ascending chain of submodules of M

$$N_1 \subset N_2 \subset N_3 \subset \dots$$

stabilizes, i.e. there exists $k \in \mathbb{N}$ such that for all $n \geq k$, $N_n = N_k$.

- (3) Any non-empty set of submodules of M has a maximal element with respect to inclusion.

The equivalent of the conditions need a proof here for clarification.

Proof. □

Immediate with the definition, we can get some properties of Noetherian modules like proposition 6.27:

- Let $0 \rightarrow N \rightarrow M \rightarrow P \rightarrow 0$ be an exact sequence of R -modules, then

$$\{N, P\} \text{ Noetherian} \iff M \text{ Noetherian}$$

in particular, let N be a submodule of M , then

$$M \text{ Noetherian} \iff \{N, M/N\} \text{ Noetherian}$$

- Let M_i be Noetherian R -modules for all $i = 1, 2, \dots, n$, then the product module $\prod_{i=1}^n M_i$ is also Noetherian.
- Let M be a A -module, and (A, ϕ) be a R -algebra, then M is Noetherian as a A -module if it is Noetherian as a R -module.

Noetherian condition is very important in commutative algebra, because it ensures that many bad situations will not happen. For example, in a locally ring (R, m) , we only have one chain of ideals

$$\cdots I_2 \subset I_1 \subset m$$

so which ensures the maximal ideal m is finitely generated, and other ideals will be generated by part of the generators of m .

Definition 6.17. Let R be a commutative ring, R is called a **Noetherian ring** if it is Noetherian as a R -module, i.e. it satisfies one of the following equivalent conditions:

- (1) Any ideal of R is finitely generated.
- (2) A.C.C condition on ideals.
- (3) Any non-empty set of ideals has a maximal element with respect to inclusion.

Remark 6.10. Many rings we meet are Noetherian rings:

- (a) Any field k is a Noetherian ring, because the only ideals are $\{0\}$ and k .
- (b) Any principal ideal domain (PID) is a Noetherian ring, because any ideal is generated by a single element.
- (c) The extension of $\mathbb{Z}$ is a Noetherian ring, for example $\mathbb{Z}[\sqrt{2}]$ and $\mathbb{Z}[i]$. The result can be generalized to the following proposition.

Proposition 6.29. Let R be a Noetherian ring, and M is a finitely generated R -algebra, then M is also a Noetherian ring.

Proof. Let $\{m_1, \dots, m_n\}$ be a set of generators of M as a R -module, then define $R^n := \prod_{i=1}^n R$ to be the product R -module. It is clear that R^n is a Noetherian module since R is a Noetherian module as a R -module, then we can define a natural surjection of R -modules by

$$\varphi : R^n \rightarrow M, \quad (a_1, \dots, a_n) \mapsto \sum_{i=1}^n a_i m_i$$

It is actually a R -module homomorphism, so by the first isomorphism can conclude that $R^n / \ker \varphi \cong M$, then the quotient module of the Noetherian module R^n is also Noetherian, hence M is a Noetherian ring. $\square$

The idea of the proof is important, In the proof we try to construct a module isomorphic to M , so we define a natural object R^n and a natural morphism φ to M , we can find that the property of R^n is so nice that we can transfer the property to M by the isomorphism theorem, i.e. R^n is a free object.

The statement without the finiteness condition is not correct, for example we consider rational number $\mathbb{Q}$, it is not finitely generated as a $\mathbb{Z}$ -module, and it is not a noetherian ring since we can find a chain of ideals:

$$\mathbb{Z} \subset \frac{1}{2}\mathbb{Z} \subset \dots \subset \frac{1}{2^n}\mathbb{Z} \subset \dots$$

it will not stop, so $\mathbb{Q}$ is not a Noetherian ring.

Theorem 6.30 (Hilbert's Basis Theorem). If A is a Noetherian ring, then the polynomial ring $A[X]$ is also Noetherian.

6.6 UFD

In this section we will study the commutative ring with nice factorization properties, which allows us to define arithmetic structure as what we have done in $\mathbb{Z}$.

Definition 6.18. An integral domain A is called **unique factorization domain (UFD)** if it satisfies the following conditions:

(Ex): A is a factorization ring: any non-unit element $a \in A - \{0\}$ can be written as a finite product of irreducible elements.

(Un): The factorization is unique up to order and unit factors, i.e. if

$$x = p_1 p_2 \dots p_r = p'_1 p'_2 \dots p'_{r'}$$

Then $r = r'$ and there exists a permutation $\sigma \in S_r$ such that p_i and $p'_{\sigma(i)}$ are **associate** (equal up to a unit) for all $i = 1, 2, \dots, r$.

Remark 6.11. Here is some example of UFD:

(1) $\mathbb{Z}$ is a UFD by the fundamental theorem of arithmetic.

(2) $\mathbb{Z}[\sqrt{-5}]$ is not a UFD since

$$6 = (1 - \sqrt{-5}) \times (1 + \sqrt{-5}) = 2 \times 3$$

where $2, 3, 1 \pm \sqrt{-5}$ are irreducible elements. pay attention that many algebraic number rings are not UFDs.

(3) The elementary number theory is based on the division with remainder, with that we can develop the fundamental theorem of arithmetic in $\mathbb{Z}$, similarly we can define the division with remainder in the polynomial ring over a field $K[X]$, which allows us to conclude that $K[X]$ is a UFD. This is an informal proof, we will talk about it later in Euclidean domains.

(4) An interesting example is $\mathcal{O}(\mathbb{C})$, the ring of all entire function on $\mathbb{C}$, in complex analysis we have Euler's formula

$$\sin(\pi z) = \pi z \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)$$

where $\sin(\pi z)$ is not a unit but it can write as an infinite product of irreducible elements, hence $\mathcal{O}(\mathbb{C})$ is not a UFD.

Review the example (2) above, we notice here is a difference between prime elements and irreducible elements:

$$2 \mid 6 \nRightarrow 2 \mid 1 + \sqrt{-5}$$

Hence 2 is not a prime elements. However, in $\mathbb{Z}$ all irreducible elements are essentially prime elements, which leads to the proof of FTA according to the properties of prime numbers (Euclid's lemma). This motivates us to redefine UFD in another way:

Proposition 6.31. If A is a FD (factorization domain), then the following conditions are equivalent:

- (1) A is a UFD.
- (2) Euclid's lemma holds in A : irreducible elements are prime elements.
- (3) Gauss's lemma holds in A : $(a \mid bc \quad \wedge \quad \gcd(a, b) = 1) \implies (a \mid c)$

Valuation

Valuation is a useful tool to study the local properties of commutative rings, in $\mathbb{Z}$ it reflects the divisibility of integers by prime numbers, which is a fundamental tool in number theory,

we define it on UFD, but the definition can be generalized to FD although some good properties may not hold.

Definition 6.19. Let A be a UFD, and let $p \in A$ be an irreducible (prime) element, then we define the p -adic valuation a function $v_p : A - \{0\} \rightarrow \mathbb{N}$ by

$$v_p(a) = \max\{n : p^n \mid a\}$$

and we make convention that $v_p(0) = +\infty$.

Remark 6.12. The following statement can be seen as a lemma for the definition, we suppose that A is just a FD here, and p is a prime element:

(1) For any $a \in A - \{0\}$, the set $E = \{n : p^n \mid a\} \subset \mathbb{N}$ is finite, ensure the existence of the maximum such that $v_p(a)$ is well-defined.

(2) If $v_p(a) = n$ the maximum of E above, then there exists a unique element $b \in A - \{0\}$ such that $a = p^n b$ and $p \nmid b$.

The valuation has some nice properties as following, pay attention that the proof is independent of the UFD property:

Proposition 6.32. Let v_p be a p -adic valuation on a UFD A , then for any $a, b \in A$, the following conditions hold:

- (1) $v_p(ab) = v_p(a) + v_p(b)$
- (2) $v_p(a + b) \geq \min\{v_p(a), v_p(b)\}$, with equality if $v_p(a) \neq v_p(b)$

with the help of valuation, we can give some characterizations of elements in UFD, sometimes it rewrites the definitions of some concepts we meet in elementary number theory.

Proposition 6.33. Let A be a UFD

(1) an element $a \in A$ is a **unit** if and only if $v_p(a) = 0$ for all irreducible elements $p \in A$.

(2) an element $a \in A$ is an **irreducible** element if and only if there exists an irreducible element $p \in A$ such that $v_p(a) = 1$ and $v_q(a) = 0$ for all irreducible elements $q \neq p$.

(3) two elements $a, b \in A$ are **associate** if and only if $v_p(a) = v_p(b)$ for all irreducible elements $p \in A$.

(4) two elements $a, b \in A$, the **divisibility** $a \mid b$ if and only if $v_p(a) \leq v_p(b)$ for all irreducible elements $p \in A$.

(5) an element $a \in A$ with factorization $a = p_1 p_2 \dots p_r$ into irreducible elements, then $v_p(a)$ is the number of factors p_i that are associate to p .

(6) Let P be the set of representatives of the associate classes of irreducible elements in A , then for any $a \in A - \{0\}$ there exists a unique unit $u \in A$ such that

$$a = u \prod_{p \in P} p^{v_p(a)}$$

(7) The **greatest common divisor** of two elements $a, b \in A$ is given by

$$\gcd(a, b) = \prod_{p \in P} p^{\min(v_p(a), v_p(b))}$$

(8) The **least common multiple** of two elements $a, b \in A$ is given by

$$\text{lcm}(a, b) = \prod_{p \in P} p^{\max(v_p(a), v_p(b))}$$

Proof.

□

Remark 6.13. In a noetherian intger or a FD, none of the above properties may hold true, that is the reason why we restrict our discussion about valuation technic on UFD. For example we consider the ring $\mathbb{Z}[\sqrt{-5}]$ again, then some counterexample will happenn:

- take $a = 2(1 + \sqrt{-5})$ and $b = 6$, then (4) fails since $v_2(a) = 1$ but $v_2(b) = 2$, however $a \nmid b$.

- take $6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, then (5) fails since $v_2(6) = 1$ but there is only one factor 2 in the factorization of 6.

- take $a = 6$, then (6) fails so that (7) and (8) also fail, since

$$\prod_{p \in P} p^{v_p(a)} = 2 \times 3 \times (1 + \sqrt{-5}) \times (1 - \sqrt{-5}) = 36 \neq 6$$

As the consequence of the valuation tool, we prove the transfert theorem of UFD, the traditional proof is based on the study of primitive polynomial, and it is a bit complicated.

Theorem 6.34 (Guass's Theorem).

If A is a UFD, then the polynomial ring $A[X]$ is also a UFD.

Proof.

□

6.7 Localization

Localization is a technique to extend a commutative ring by inverting some elements, such that we can study the local properties of the original ring. For example, we want to view 2 as a unit in some algebraic ring, it is natural to consider $\mathbb{Z}[\frac{1}{2}]$ such that $2^{-1} = \frac{1}{2}$.

Definition 6.20. Let A be a ring, a subset $S \subset A - \{0\}$ is called a **multiplicative subset** if $1 \in S$ and for any $s, t \in S$, we have $st \in S$.

with the multiplicative subset, we choose the elements we want to invert, then we can construct the localization of the ring as following:

Lemma 6.35. Let A be a ring and S be a multiplicative subset of A .

- We define a equivalent relation $\sim$ on $A \times S$ by

$$(a, s) \sim (a', s') \iff \exists t \in S, t(as' - a's) = 0$$

it induces a quotient set denoted by $S^{-1}A = (A \times S) / \sim$.

(1-addition) we can define an addition on $S^{-1}A$ by

$$\frac{a}{s} + \frac{a'}{s'} := \frac{as' + a's}{ss'}$$

with a additive identity $\frac{0}{1}$.

(2-multiplication) we can define a multiplication on $S^{-1}A$ by

$$\frac{a}{s} \cdot \frac{a'}{s'} := \frac{aa'}{ss'}$$

with a multiplicative identity $\frac{1}{1}$.

Then $S^{-1}A$ is a ring with the operations above, called the **localization** of A at S .

Proof. □

Remark 6.14. There exists a natural ring homomorphism onto any localization of A :

$$\ell : A \rightarrow S^{-1}A, \quad a \mapsto \frac{a}{1}$$

which shows that any localization of A is a A -algebra. And we can verify the kernel of ℓ is given by

$$\ker \ell = \{a \in A \mid \exists s \in S, sa = 0\}$$

it is actually some zero divisors of A "killed" by some elements in S . Hence if A is an **integral domain**, then ℓ is injective, and $\frac{a}{s} = 0$ if and only if $a = 0$.

As a A -algebra object, the localization satisfies the universal properties a little like "analytic continuation" in complex analysis, it refers to extend the morphism to a larger initial object:

Proposition 6.36 (UP-Localization).

Let $S^{-1}A$ be the localization of a ring A , then for any A -algebra (B, f) such that $f(S) \subset B^\times$, then there exist a unique morphism of A -algebras $\bar{f} : S^{-1}A \rightarrow B$ such that the $\bar{f} \circ \ell = f$.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \ell \downarrow & \nearrow \bar{f} & \\ S^{-1}A & & \end{array}$$

Proof.

□

Remark 6.15. There are some consequences of the universal property above:

(1) For a A -module M , if we consider $B = \text{End}(M)$ and f is the structure morphism of M , then UP induces a natural morphism $\bar{f}$ if any element in S acts as an automorphism on M , i.e. for any $s \in S$, the map $f(s) : M \rightarrow M$ is bijective. Hence M can be viewed as a $S^{-1}A$ -module by defining

$$\frac{a}{s} \cdot m := f(a) \circ f(s)^{-1}(m)$$

for any $\frac{a}{s} \in S^{-1}A$ and $m \in M$.

(2) Conversely, any $S^{-1}A$ -module can be trivially viewed as a A -module since $S^{-1}A$ is a A -algebra. Hence a correspondence about morphisms of module can be concluded here if the condition in (1) holds:

$$\text{Hom}_A(M, N) = \text{Hom}_{S^{-1}A}(M, N)$$

(3) If we set $B = S^{-1}A$ itself, then the UP implies that the uniqueness of the endomorphism of localization, i.e.

$$\text{End}_{A\text{-alg}}(S^{-1}A) = \{\text{id}\}$$

which shows that the localization is a canonical construction.

Another important respects of localization is about correspondence of ideals, which allows us to identify the ideals in the localization, it is the base of the study of local rings.

Let $S^{-1}A$ be the localization of a ring A at a multiplicative subset S , then there exists two maps

$$\begin{aligned}\{I \subset A \text{ ideal} \mid I \cap S = \emptyset\} &\longleftrightarrow \{J \subset S^{-1}A \text{ ideal}\} \\ I &\longmapsto S^{-1}I := \left\{ \frac{a}{s} \mid a \in I, s \in S \right\} \\ \ell^{-1}(J) &\longleftarrow J\end{aligned}$$

Proof. □

However, the correspondence above may not be a bijection, so it is not nice to do some identificaion:

$$\ell^{-1}(S^{-1}I) = \{a \in A \mid \frac{a}{1} \in S^{-1}I\} = \{a \in A \mid \exists t \in S, ta \in I\}$$

If the ideal here is prime, then we can get a better result that $\ell^{-1}(S^{-1}I) = I$, hence we can conclude a good corresponence as following:

Proposition 6.37. Let $S^{-1}A$ be the localization of A , and define

$$D(S) := \{p \in \text{Spec}(A) \mid p \cap S = \emptyset\}$$

then there exists a bijection

$$\text{Spec}(S^{-1}A) \rightarrow D(S), \quad \mathfrak{p} \mapsto \ell^{-1}(\mathfrak{p})$$

Remark 6.16.

(1) $D(S)$ is not empty if $S^{-1}A \neq 0$ (which map happen if $o \in S$, but we aviod it by definition), the reason is that any ring has at least one maximal ideal by Zorn's lemma, hence by bijection we can at least find one element in $D(S)$.

(2) The statement (1) can be concluded to be a techinic lemma to prove the existence of prime ideals in any ring, which is a fundamental result in commutative algebra:

Lemma. Let S be a multiplicative subset of a ring A , if $I \subsetneq A$ is an ideal such that $I \cap S = \emptyset$, then there exists a prime ideal $p \subset A$ such that $I \subset p$ and $p \cap S = \emptyset$.

The quotient of localization is another basic question as a object of ring. Here we make some remarks to avoid talk too much in the proof of main propoition below, and it gives a clear motivation to the construction:

Let I be a ideal of A such that $I \cap S = \emptyset$ disjoint, then by above correspondence we can know that $S^{-1}I$ is actually an ideal of localization $S^{-1}A$, so it makes sense to define the quotient ring $S^{-1}A/S^{-1}I$.

On the other hand, we consider natural morphism $\pi : A \rightarrow A/I$, then $\bar{S} = \pi(S)$ is a multiplicative subset (verify) of A/I if $I \cap S = \emptyset$, then we can do localization on A/I at $\bar{S}$ to get the ring $\bar{S}^{-1}(A/I)$.

So a immediate question is that the two objects above are isomorphic or not? It is essentially that the quotient of localization is again a localization, and it is the localization of the quotient of initial ring.

Proposition 6.38 (UPQ-Localization). Let $S^{-1}A$ be a localization of a ring A , and I is a ideal of A such that $I \cap S = \emptyset$, then there exists a natural isomorphism of rings

$$\bar{S}^{-1}(A/I) \cong S^{-1}A/S^{-1}I$$

Proof. □

The localization can be extended to modules in a natural way, which allows us to study the local properties of modules over commutative rings.

Definition 6.21. Let M be a A -module, with $S^{-1}A$ the localization of A , then we define the **localization** of M at S to be a $S^{-1}A$ -module

$$S^{-1}M := M \times S / \sim$$

with the equivalence

$$(m, s) \sim (m', s') \iff \exists t \in S, \quad t \cdot (s'm - sm') = 0_M$$

and similarly we denote the class of (m, s) by $\frac{m}{s}$.

the process of localization is actullay a type of extension of scalar when we see M as a subset of $S^{-1}M$, hence we can apply the tensor product here to give isomorphic construction of localization of modules:

Proposition 6.39 (localization-module).

The inclusion $M \hookrightarrow S^{-1}M$ induces a natural isomorphism of $S^{-1}A$ -modules

$$S^{-1}M \cong S^{-1}A \otimes_A M$$

Proof. □

The structure shows the essential difference of localization of modules and rings here, the localization of **intergal domian** is process of extension, it adds some new elements to the ring such that it can be the inverse of some elements; while the localization of modules is a process of extension of scalar, or by the proposition above it is a tensor product, which means that

localization of modules is not necessarily larger than the initial module!

here is some examples:

Lemma 6.40. Let M be a A -module, and S be a multiplicative subset of A , then the following conditions are equivalent:

- (1) $S^{-1}M = 0$
- (2) for any $m \in M$, there exists $s \in S$ such that $s \cdot m = 0_M$

Proof. (1) $\implies$ (2): for any $m \in M$, we have $\frac{m}{1} = 0$ in $S^{-1}M$, by the definition of equivalence, there exists $s \in S$ such that $s \cdot (1 \cdot m - 1 \cdot 0_M) = 0_M$, hence $s \cdot m = 0_M$.

(2) $\implies$ (1): for any $\frac{m}{s} \in S^{-1}M$, by the condition there exists $t \in S$ such that $t \cdot m = 0_M$, then by the definition of equivalence we have

$$\frac{m}{s} = \frac{t \cdot m}{t \cdot s} = \frac{0_M}{t \cdot s} = 0$$

hence $S^{-1}M = 0$. □

(1) If G is an finite abelian group, then any element has finite order, for example $g \in G$ with order n , then $n \cdot g = g^n = 0$ which implies that

$$\mathbb{Q} \otimes_{\mathbb{Z}} G = S^{-1}G = 0$$

which is a example that the localization of module is even vanishing.

(2) In the previous example, the choice $\mathbb{Q}$ is a little large such that we can not get any interesting result, here we consider $G = \mathbb{Z}/15\mathbb{Z}$ and we consider the localization $\mathbb{Z}_{(3)}$, i.e. take multiplicative subset $S = \mathbb{Z} - 3\mathbb{Z}$, then

$$\mathbb{Z}_{(3)} \otimes_{\mathbb{Z}} G \cong S^{-1}\mathbb{Z}/3\mathbb{Z} \oplus S^{-1}\mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z}$$

It is easy to the result by the lemma above. The result can be generalized to any finite abelian group G

$$\mathbb{Z}_{(p)} \otimes_{\mathbb{Z}} G \cong G_p$$

where G_p is the p -primary component of G , which is an example that the localization module is smaller.

(3) A A -module M induce a map $\psi : A \rightarrow \text{End}(M)$ by

$$\psi_a : (m \mapsto a \cdot m) \in \text{End}(M)$$

then the localization $S^{-1}M = M$ if and only if ψ_s is bijective for any $s \in S$, a simple example is that

$$\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}$$

Localization is a strong tool to study the local properties of commutative rings and modules without changing the global structure too much, and conversely we can study the global structure by gluing the local structures together, in number theory it is called the **local-global principle**, which is a fundamental idea in modern number theory.

Proposition 6.41. The **localization of ring** preserves the following properties:

- (1) reduced ring (2) integral domain (3) Noetherian ring
 (4) UFD (5) integral closed

Proof. □

Finally, the examples of localization will be introduced as the technic base.

Example 6.3. Let A be a ring, and f be a non-nilpotent element of A , then $S = \{f^n, n \in \mathbb{N}\}$ is a multiplicative subset of A , and the localization is called the **localization of A at f** , denoted by $A[f^{-1}]$. Here are some examples:

(1) $\mathbb{Z}[\frac{1}{2}]$ is the localization of $\mathbb{Z}$ at 2.

(2) Laurent polynomial ring $A[X, X^{-1}]$ is the localization of polynomial ring $A[X]$ at X .

(3) **The localization at a point is just to omit the information at this point.** Let M be a $\mathbb{Z}$ -module with 2 is a torsion element (for example $M = \mathbb{Z}/(6)$), then the localization of module M at point 2 allows us to view 2 as a unit, hence in $M[2^{-1}]$

$$2x = 0 \implies x = 0$$

we loose the information about 2 as a torsion element after localization, which is actually the important characterizations of 2 in $\mathbb{Z}$ -module M .

The requiremenet that f is non-nilpotent ensures that $0 \notin S$, hence it is necessary to avoid the form of $\frac{a}{0}$. tHE structure OF localization at a single elemet is easier to study, and it is clearly that f^{-1} as the element in $A[f^{-1}]$ is always not intergal, but it can be write as a solution of a linear equation:

Lemma 6.42. Let A be a ring, and $f \in A$ be a non-nilpotent element

$$A[f^{-1}] \cong A[X]/(fX - 1)$$

Example 6.4. Let $\mathfrak{p}$ be a prime ideal of a ring A , then $S = A - \mathfrak{p}$ is a multiplicative subset of A (**verify!**), and the localization is called the **localization of A at $\mathfrak{p}$** , denoted by $A_{\mathfrak{p}}$. Here are some examples:

(1) For any integral domian A , the localization $A_{\{0\}}$ is the field of fractions of A .

(2) Notice $2\mathbb{Z}$ is a prime ideal of $\mathbb{Z}$, then the localization is

$$\mathbb{Z}_{(2)} = \left\{ \frac{a}{b} \in \mathbb{Q} \mid a, b \in \mathbb{Z}, b \notin 2\mathbb{Z} \right\}$$

In the view of valuation it is

$$\mathbb{Z}_{(2)} = \{x \in \mathbb{Q} \mid v_2(x) \geq 0\}$$

while

$$\mathbb{Z}[2^{-1}] = \{x \in \mathbb{Q} \mid v_2(x) < 0\}$$

In a PID, the localization at a prime ideal (p) is opposite to the localization at a point, i.e. it will keep all the information about p to give an extension.

(3) for any ring A and any maximal ideal $\mathfrak{m}$, the localization $A_{\mathfrak{m}}$ is a local ring with unique maximal ideal $\mathfrak{m}A_{\mathfrak{m}}$.

In particular, the prime ideals of localization at a prime ideal have a nice description:

Lemma 6.43. Let A be a ring and $\mathfrak{p}$ be a prime ideal of A , then there exists a bijective correspondence

$$\text{Spec}(A_{\mathfrak{p}}) \rightarrow \{\mathfrak{q} \in \text{Spec}(A) \mid \mathfrak{q} \subset \mathfrak{p}\}$$

In particular, $\mathfrak{p}A_{\mathfrak{p}}$ is the unique maximal ideal of $A_{\mathfrak{p}}$, i.e. $A_{\mathfrak{p}}$ is a **local ring**.

Proof.

□

The fraction field of an integral domain A is the localization of A at $S = A - \{0\}$, and trivially the set contains all zero-divisors of A , hence it motivates a general case

Example 6.5. For any ring A , we define A_{reg} to be the set of all regular elements (non-zero divisors) of A , then A_{reg} is a multiplicative subset of A (**verify!**), the localization is called the **total ring of fractions** of A , and it is denoted by $Q(A)$. Here are some examples:

(1) If A is an integral domain, then the total ring of fractions is just the fraction field of A .

(2) Let $A = \mathbb{Z}/6\mathbb{Z}$, then the regular elements are $\{1, 5\}$, hence the total ring of fractions is

$$Q(A) = \left\{ \frac{a}{1}, \frac{a}{5} \mid a \in \mathbb{Z}/6\mathbb{Z} \right\}$$

notice that $5^2 = 1 \pmod{6}$, then $\frac{a}{5} = \frac{5a}{1}$, which shows that $Q(A) = A$.

Total ring of fractions can be seen as the "biggest localization" of a ring A such that $A \rightarrow Q(A)$ is an inclusion.

6.8 PID and modules over it

Module is the generalization of vector space, but module is not as nice as vector space in many respects, many properties of vector space can not hold in module.

- A submodule of a finitely generated module may not be finitely generated, which needs the noetherian condition, and in particular PID is a noetherian ring; a submodule of a free module may not be free, the noetherian condition is not sufficient here: For example in $k[X, Y]$ -module $K[X, Y]$, the submodule (ideal) (X, Y) is not free since $YX - XY = 0$ gives a non-trivial relation.
- A submodule of a module may not have a complement, even in free module a linear independent lists can not be extended to a basis: for example in $\mathbb{Z}$ -module $\mathbb{Z}$, the only basis is $\{1\}$, and we can not extend $\{2\}$ to be a basis.
- Similarly with the above point, any generated family can not be reduced to a basis: For example in $\{z\}$, the family $\{2, 3\}$ generates the module by Bezout's theorem, but it can not be reduced to a basis.
- Although the torsion element will prevent the module to be free, but a torsion-free module may not be free: Like the first example a $k[X, Y]$ -module (X, Y) , it has no torsion element, but it is not free.
- Even more terrible, a module can not have any linearly independent family: for example $\mathbb{Z}$ -module $\mathbb{Q}$, any two elements $\frac{a}{b}$ and $\frac{c}{d}$ are linearly dependent since

$$ad \cdot \frac{a}{b} - bc \cdot \frac{c}{d} = 0$$

That is the reason why we talk about principal ideal domain, the module over it will show the similar properties as vector space although they are not same, but it keeps many important structures and conversely we can explain the good behavior of vector space from a higher view. The module over PID can be divided into two parts: torsion part and free part, so we review the notation here:

Definition 6.22. Let M be a A -module and A is a integral domain

- An element $m \in M$ is called a **torsion element** if there exists a non-zero element $a \in A$ such that $a \cdot m = 0_M$, and we define the **annihilator** of m be the set of elements in A that kills m :

$$\text{Ann}(m) := \{a \in A \mid a \cdot m = 0_M\}$$

- We denote the set of all torsion elements of M by $T(M)$ or M_{tor} , and module M is called a **torsion module** if $M = T(M)$, and it is called a **torsion-free module** if $T(M) = \{0_M\}$.

By the definition, we can say that a is torsion element if and only if $\text{Ann}(m) \neq \{0\}$. Any free module over a integral domain is torsion-free, and any finite abelian group can be seen as a torsion module over $\mathbb{Z}$.

Proposition 6.44. Let M be a module over a integral domian A .

(a) $\text{Ann}(m)$ is a ideal of A and $T(M)$ is a submodule of M .

(b) $M/T(M)$ is torsion-free as a A -module.

Proof. (a) is just the verification. For (b), we suppose that $\bar{m} \in M/T(M)$ is a torsion element, and we will show that $m \in T(M)$, that means the torsion element in $M/T(M)$ is $\bar{0}$, which finishes the proof. By the assumption, there exists non-zero $a \in A$ such that

$$a \cdot \bar{m} = \overline{a \cdot m} = 0$$

which shows that $am \in T(M)$, and there exists non-zero $b \in A$ such that $bam = 0$. A is integral domain, then $ba \neq 0$, so at least $ba \in \text{Ann}(m)$, which shows that $m \in T(M)$. $\square$

Later we will see that the decomposition of module over PID is analogue to the decomposition of finite abelian group, so cyclic module is important to study here as cyclic group in finite abelian group. As we have mentioned before, a A -module M is **cyclic** if there exists $m \in M$ such that

$$M = (m) = Am = \{a \cdot m \mid a \in A\}$$

Here is some properties of cyclic module over PID or not:

Proposition 6.45. Let M be a module over A .

(a) If M is cyclic, then there exists an ideal $I \subset A$ such that $M \cong A/I$, in particular $I = \text{Ann}(m)$ with m a generator of M , so $Am \cong A/\text{Ann}(m)$.

(b) If A is a PID, then a cyclic module M is torsion-free if and only if $M \cong A$, i.e. M is free of rank 1.

Proof. (a) Consider the surjective morphism of A -modules

$$\varphi : A \rightarrow M, \quad a \mapsto a \cdot m$$

then by the first isomorphism theorem we can conclude the result.

(b) If $M \cong A$, then it is free hence torsion-free. Conversely if M is cyclic then by (a) there exists $a \in A$ such that $M \cong A/(a)$, it must be principal since A is PID. If $a \neq 0$, then for the class $\bar{1} \in A/(a)$ we have $a \cdot \bar{1} = \bar{0}$, which shows that $\bar{1}$ is a torsion element, contradicting the assumption. Hence $a = 0$ and $M \cong A$. $\square$

Theorem 6.46. Any finitely generated torsion-free module over a PID is free.

Proof. Let M is a module over a PID A , and here is the outline of the proof:

- (1) If $x, y \in M$ and $a, b \in A - \{0\}$ such that $ax = by$, then there exists $z \in M$ such that $x = bz$ and $y = az$.
- (2) For $m \in M - \{0\}$, we define **saturated set**

$$S(m) = \{x \in M \mid \exists a \in A - \{0\}, ax \in Am\}$$

then prove that $S(m)$ is a cyclic submodule of M and $M/S(M)$ is torsion-free.

- (3) Prove by induction.

Deatils:

(1) We assume that $(a, b) = 1$, i.e. they have no common divisors, then by PID there exists $u, v \in A$ such that $ua + vb = 1$, we can define $z = uy + vx$, then we can verify that $x = bz$ and $y = az$. If $(a, b) \neq 1$, we can divide them by their gcd to reduce to the above case.

(2) For any $x, y \in S(m)$, there exists non-zero $a, b \in A$ such that $ax \in Am$ and $by \in Am$, then $abx, aby \in Am$, hence $x + y \in S(m)$. For any $c \in A$ and $x \in S(m)$, there exists non-zero $a \in A$ such that $ax \in Am$, then $a(cx) = c(ax) \in Am$, hence $cx \in S(m)$. So $S(m)$ is a submodule of M .

To show that it is cyclic, for any $x, y \in S(M)$, there exists non-zero $a, b \in A$ such that $ax = rm$ and $by = sm$ for some $r, s \in A$, then $(sa)x = (rb)y$, and we can apply (1) to conclude that there exists $z \in M$ such that $x = bz$ and $y = az$, hence $z \in S(m)$ and $S(m) = Az$ is cyclic.

Finally we show that $M/S(m)$ is torsion-free, for any $\bar{x} \in M/S(m)$ and non-zero $a \in A$ such that $ax \in S(m)$, which implies $b \in A - \{0\}$ and $\alpha \in M$ such that $(ba)x = \alpha m$, which shows that $x \in S(m)$, hence the unique torsion element in $M/S(m)$ is $\bar{0}$.

(3) Let $\{v_1, \dots, v_n\}$ be the generated family of M , we will prove by induction on n . If $n = 1$, then M is cyclic and so it is free by proposition 6.45. If $n > 1$, we assume that the statement holds for $n - 1$, and we will prove it for n :

Let $v \in \{v_1, \dots, v_m\}$ and then saturated set $S(v)$ is a torsion-free cyclic submodule of M generated by some $z \in M$ by (2), and notice that $v \in S(v)$, which shows that $M/S(v)$ is generated by $n - 1$ elements, so by induction $M/S(v)$ is free with basis $\bar{e}_1, \dots, \bar{e}_{n-1}$, and then we can lift them to prove that M is free with basis $z, e_1, \dots, e_{n-1}$. $\square$

Remark 6.17. Notice that in the final parts of the proof, if we allow to use the properties of projective module, then the result is immediate: Since $M/S(v)$ is free so projective, then the exact sequence

$$0 \rightarrow S(v) \rightarrow M \rightarrow M/S(v) \rightarrow 0$$

splits, which shows that M is the direct sum immediately.

With the theorem, we can get the rough decomposition of module over PID, for any **finitely generated** module M over PID A , we have the short exact sequence

$$0 \rightarrow T(M) \rightarrow M \rightarrow M/T(M) \rightarrow 0$$

as we have proved before, $M/T(M)$ is torsion-free, and it is finitely generated as the quotient of finitely generated module, hence by the theorem $M/T(M)$ is free, which allows us to decompose any finitely generated module over PID into **torsion part** and **free part**:

$$M = M_{tor} \oplus A^r$$

where r is the rank of the free part. Finite abelian group invites the following work we will do, it is easy to prove that any finite abelian group is a torsion module over $\mathbb{Z}$, so as a finitely generated PID module it has no free part, hence the structure theorem of finite abelian group is indeed the decomposition of finitely generated torsion module over PID, that is main motivation to do the further decomposition.

Definition 6.23. Let M be a module over a integral domain A and $p \in A$ is a prime element, we define the **p -primary component** of M to be the submodule

$$M_p := \{m \in M \mid \exists k \in \mathbb{N}, p^k \cdot m = 0\}$$

The p -primary component of module is analogue to the p -primary component of finite abelian group, more generally we can define the primary component with respect to any irreducible element in a integral domain, it refers to the primary decomposition in commutative algebra, here we just focus on the specific case here.

In particular, if we rewrite the definition in the context of abelian group, we can find that the p -primary component is just the **p -group**, the following proposition shows the analogue between them:

Proposition 6.47. Let M be a module over PID A ,

- (a) If p and q are two distinct prime elements of A , then $M_p \cap M_q = \{0_M\}$.
- (b) A p -primary component M_p is cyclic if and only if $M_p \cong A/(p^k)$ for some $k \in \mathbb{N}$.
- (c) A finitely-generated p -primary component M_p has a decomposition of the form

$$M_p = \bigoplus_{i=1}^l A/(p^{e_i})$$

for some $e_i \in \mathbb{N}$, the decomposition is unique up to the order of the factors.

Proof. (a) For any non-zero element $x \in M_p \cap M_q$, there exists $k, l \in \mathbb{N}$ such that $p^k \cdot x = 0$ and $q^l \cdot x = 0$, since p and q are distinct prime elements in PID, they are coprime, hence there exists $u, v \in A$ such that $up^k + vq^l = 1$, then

$$x = (up^k + vq^l)x = u(p^k x) + v(q^l x) = 0$$

which is a contradiction, hence $M_p \cap M_q = \{0_M\}$.

(b) By proposition 6.45, if M_p is cyclic, then $M_p \cong A/\text{Ann}(x)$ with x a generator of M_p , and since $x \in M_p$, there exists $t \in \mathbb{N}$ such that $p^t \in \text{Ann}(x)$, so $(p^t) \subset \text{Ann}(x)$. A is PID, then $\text{Ann}(x)$ must be a principal ideal contains p^k , which shows that $\text{Ann}(x)$ is generated by a divisor of p^t , hence $\text{Ann}(x) = (p^k)$ for some $k \in \mathbb{N}$, which shows that $M_p \cong A/(p^k)$. The converse is trivial.

(c) Let $m_1, \dots, m_n$ be a minimal generated family of M_p , then we can do induction on n . If $n = 1$, it can be done by (b). If $n > 1$ we assume that the statement holds for $n - 1$, and we will prove it for n :

Let x be one of family satisfies that $\min\{n \mid p^n m_i = 0\}$ is the largest, and we set $\text{Ann}(x) = (p^k)$, then $p^k y = 0$ for any $y \in M_p$. Then we set $N = M_p/Ax$ and consider the exact sequence

$$0 \rightarrow Ax \rightarrow M_p \rightarrow N \rightarrow 0$$

we extend the scalar to $K = A/(p)$ -vector space by tensoring $A/(p)$:

$$Ax \otimes_A A/(p) \rightarrow M_p \otimes_A A/(p) \rightarrow N \otimes_A A/(p) \rightarrow 0$$

using the formula of tensor product $M/IM \cong M \otimes_A A/I$ we can verify that

$$Ax \otimes_A A/(p) \cong Ax/pAx \cong (A/(p^k))/p(A/p^k) \cong A/(p)$$

so we can conclude that

$$A/(p) \rightarrow M_p/pM_p \rightarrow N/pN \rightarrow 0$$

The left map is in fact injective, since the map can be written as

$$Ax/pAx \rightarrow M_p/pM_p, ax + pAx \mapsto ax + pM_p$$

If $ax \in pM_p$, then there exists $y \in M_p$ such that $ax = py$. Because $\text{Ann}(x) = (p^k)$, then $p^{k-1}ax = p^k y = 0$, which shows that $p^{k-1}a \in (p^k)$ and which implies that $a \in (p)$, and we can conclude that the kernel is trivial. Hence the sequence is exact, and it splits since the modules are vector space over K , which shows that N/pN is a K -vector space with dimension $\leq n - 1$, so we can lift the basis to get a generated family of N with size $l - 1 \leq n - 1$, hence by induction N there exists the decomposition

$$N \cong \bigoplus_{i=1}^{l-1} A/(p^{e_i})$$

to finish the proof it is hard to use the original exact sequence to get the decomposition, we consider the localization of the module at (p) , then A_p is the local ring with unique maximal ideal pA_p , and M_p will not change after localization since it is a p -primary component, **hence we get the decomposition by the technique of localization.** $\square$

Above results can be viewed as the structure theorem of finite p -groups, with the basic preparation done, we can finally give the structure theorem of finitely generated module over PID, which is a strong result in module theory that appears in many branches.

Theorem 6.48 (Primary Decomposition).

Let M be a finitely generated module over a PID A , then there exists a unique decomposition up to the order of the factors:

$$M \cong A^r \oplus M_{p_1} \oplus \cdots \oplus M_{p_n}$$

where $r \in \mathbb{N}$ and p_i are distinct prime elements of A , and M_{p_i} is the p_i -primary component of M . Moreover each M_{p_i} has a decomposition

$$M_{p_i} \cong \bigoplus_{j=1}^{l_i} A/(p_i^{e_{ij}})$$

for some $e_{ij} \in \mathbb{N}$, the decomposition is unique up to the order of the factors.

Proof.

□

7 Affine Algebraic Geometry

The object of this section is to give a basic **geometry-algebra correspondence** between affine algebraic varieties and a polynomial ring over a field, which can be concluded by the following graph:

$$\begin{array}{ccc}
 k^n & \xrightarrow{\mathcal{O}} & \mathcal{O}(k^n) \\
 \uparrow \text{Nullstellensatz} & & \uparrow \text{infinite field} \\
 \mathbb{A}_k^n & \xleftarrow{\text{Spec}} & k[X_1, \dots, X_n]
 \end{array}$$

Firstly we should clarify the isomorphism of polynomial functions and the formal polynomial in certain condition, which allows us to freely use to two objects without confusion.

Definition 7.1. A formal polynomial is a element of the polynomial ring $k[X_1, \dots, X_n]$ over a field k , while a polynomial function is a function with a formal polynomial expression, i.e. a map

$$f : k^n \rightarrow k, \quad (x_1, \dots, x_n) \mapsto \sum_{i=0}^n a_i x_i$$

and we denote the set of all polynomial functions on k^n by $\mathcal{O}(k^n)$.

It is easy to verify that $\mathcal{O}(k^n)$ is a k -algebra type fini, hence by the universal property of polynomial, there exists a unique surjective k -linear map

$$\text{ev} : k[X_1, \dots, X_n] \rightarrow \mathcal{O}(k^n), \quad X_i \mapsto \pi_i : (x_1, \dots, x_n) \mapsto x_i$$

In particular, when k is an **infinite field (verify and find counterexample)**, the map ev is also injective, hence it is an isomorphism of k -algebras, which allows us to abuse the notation of polynomial functions and polynomials.

A classic question in old algebraic geometry is to study the solution of the polynomial equations over some field k , for example the famous Fermat's last theorem is to study the integer solutions of the equation

$$x^n + y^n = z^n$$

if we see it as a polynomial equation over the field $\mathbb{Q}$. generally, if we have a set of polynomial equations

$$\begin{cases} f_1(x_1, \dots, x_n) = 0 \\ f_2(x_1, \dots, x_n) = 0 \\ \vdots \\ f_m(x_1, \dots, x_n) = 0 \end{cases} \quad (3)$$

then it induces a subset of k^n , called algebraic subsets:

Definition 7.2. Let k be a field, and I is a ideal (subset) of the polynomial ring $k[X_1, \dots, X_n]$, then we define the **algebraic subset** of k^n defined by I to be

$$V(I) := \{(a_1, \dots, a_n) \in k^n \mid f(a_1, \dots, a_n) = 0, \forall f \in I\}$$

it is also called the **variety** defined by I .

Remark 7.1. we should remark that the following facts(**verify!**):

$$V(S) = V((S)), \quad (S) \text{ is the ideal generated by } S$$

then the common zeros of $f_1, \dots, f_m$ is just the algebraic subsets defined by the ideal $(f_1, \dots, f_m)$.

Moreover, some algebraic facts should be known(**verify**):

- (1) $V(1) = \emptyset$ and $V(0) = k^n$ as two trivial cases.
- (2) $V(I) \cup V(J) = V(IJ) = V(I \cap J)$ for any two ideals.
- (3) $\cap_k V(I_k) = V(\sum_k I_k)$ for any index.

Conversly, for any subset $V \subset k^n$, we can try to find the largest sets of polynomials vanishing on V , which leads to the definition of **vanishing ideal**:

$$I(V) = \{f \in k[X_1, \dots, X_n] \mid f(a) = 0, \forall a \in V\}$$

equivalently, consider the ideal of polynomial functions, it makes sense to consider the restriction:

$$I(V) = \{f \in \mathcal{O}(k^n) \mid f|_V = 0\}$$

Remark 7.2. Here are some facts:

- (1) $I(k^n) = \{0\}$ and $I(\emptyset) = k[X_1, \dots, X_n]$ as two trivial case.

- (2) vanishing ideal $I(V)$ is a radical ideal, **should know**

$$\text{maximal ideal} \subset \text{prime ideal} \subset \text{radical ideal}$$

- (3) **antitone Galois connection:**

- (4) $V = V(I(V))$ for any subset $V \subset k^n$:

It is clear that $V \subset V(I(V))$, for avoid confusion of notation in traditional proof, we define a set $\mathcal{J} = \{I \subset K[X_1, \dots, X_n] \text{ ideal} \mid f|_V = 0, \forall f \in I\}$, then the set is a partial order set under inclusion, it is easy to verify that $I(V)$ is a maximal element in it, i.e. each chain in $\mathcal{J}$ will stop at $V(I(V))$:

$$I_1 \subset I_2 \subset \dots \subset I(V)$$

then immediately apply $V(-)$, we can get a decreasing chain of algebraic subsets:

$$V \subset V(I(V)) \subset \dots \subset V(I_2) \subset V(I_1)$$

$V(I(V))$ must be the minimal element in the chain, hence $V(I(V)) = V$.

(5) $I \subset \sqrt{I} \subset I(V(I))$ for any ideal $I \subset k[X_1, \dots, X_n]$:

For example we take $I = (X^2)$ in $k[X]$, it is clearly that the common zeros is 0, but the vanishing ideal of $\{0\}$ is (X) , strictly larger than (X^2) .

Here we need a strong condition of field such that we can get a correspondence between algebraic subsets and vanishing ideals, which motivates the Hilbert's zero theorem (Nullstellensatz), we review the original statement here:

Theorem 7.1 (1893). Let $f_1, \dots, f_m$ be the polynomials of $\mathcal{O}(\mathbb{C}^n)$, and V is the set of common zeros of $f_1, \dots, f_m$ in $\mathbb{C}^n$, and it is non-empty, then for any other polynomial $g \in \mathcal{O}(k^n)$

$$g(x) = 0, \forall x \in V \iff g^k \in (f_1, \dots, f_m) \text{ for some } k \in \mathbb{N}$$

We will give a general proof later, but firstly we can see how it makes sense. In above we have shown that $\sqrt{I} \subset I(V(I))$, the other side of inclusion is to ask: when we take a polynomial g vanishing on $V(I)$, i.e. the common zeros of some polynomials, what can we say about g ? Clearly, $I = (f_1, \dots, f_m)$ since we know

$$V(\{f_1, \dots, f_m\}) = V((f_1, \dots, f_m))$$

then the theorem gives the answer when $k = \mathbb{C}$, it shows that $g \in \sqrt{I}$, hence we have

$$I(V(I)) = \sqrt{I} = I?$$

the first equality holds by the theorem, but the second equality holds under the condition that I is a radical ideal, which shows that the zeros of polynomials has a closed relationship with the radical ideals in certain condition, i.e. algebraic closed fields. Here we give the general statement of zero theorem:

Theorem 7.2 (weak Nullstellensatz). Let K be the k -algebra of type fini, if K is a field extension of k , then K is a finite extension of k .

Proof. Let K be generated by n elements over k , nothing needs to prove, for $n \geq 1$ we can prove it by induction:

(1) We suppose that for $n - 1$ the statement holds, then for n , there exists $a_1, \dots, a_n$ in K such that $K = k[a_1, \dots, a_n] = k(a_1)[a_2, \dots, a_n]$, then by induction K is a finite extension of $k(a_1)$ and it suffices to show that $k(a_1)$ is a finite extension of k , then

$$[K : k] = [K/k(a_1)] \cdot [k(a_1) : k]$$

(2) K is finite over $k(a_1)$, which allows us to find a finite basis $\{e_1, \dots, e_m\}$ for K , then for any generator a_i , there exists $d_{i,k} \in k(a_1)$ such that

$$a_i = \sum_{k=1}^m d_{i,k} \cdot e_k$$

and algebra allows us to get $c_{rs,k}$ such that

$$e_r \cdot e_s = \sum_{t=1}^m c_{rs,t} \cdot e_t$$

Hence we can define a finite generated subalgebra $A = k[d_{i,k}, c_{rs,t}]_{i,r,s,k,t} \subset k(a_1)$, then It is not difficult to deduce

$$K = k[a_1, \dots, a_n] \subset \bigoplus_{j=1}^m A e_j$$

since each element of K is a $k(a_1)$ -linear combination of e_j 's, and each a_i can be expanded as an A -linear combination of e_j .

(2) We finish the proof by absurd: if $k(a_1)$ is not finite over k , then it is not algebraic over k , i.e. $k[a_1] \cong k[X]$, then any generator of $A \subset k(a_1)$ can be written as the form of f_i/g_i , and we can construct a polynomial coprime with all g_i by

$$P = 1 + \prod_i g_i$$

then $1/P \notin A$, and it shows that $1/P \notin K$, and then K is not a field. In fact, if $1/P \in K$, then there exists $h_j \in A$ such that $1/P = \sum_{j=1}^m h_j \cdot e_j$, then

$$1 = \sum_{j=1}^m (P \cdot h_j) \cdot e_j$$

by direct sum the representation is unique by $1 = 1 + 0 \dots + 0$, so it shows that $1/P = h_1 \in A$, which is a contraction.

□

In particular, if $\mathfrak{m}$ is a maximal ideal of $k[x_1, \dots, x_n]$, then the residue field $k[x_1, \dots, x_n]/\mathfrak{m}$ is a finite extension of k , hence if k is algebraically closed, then the residue field is just k itself, which shows that the maximal ideals is just the "closed point":

$$\mathfrak{m} = (X_1 - a_1, \dots, X_n - a_n)$$

for some $(a_1, \dots, a_n) \in k^n$, which allows us to reformulate the nullstellensatz in the language of geometry:

Corollary 7.3. If k is algebraically closed, and then

(1) For any ideals $I, J \subset K[X_1, \dots, X_n]$, $V(I) = V(J) \iff \sqrt{I} = \sqrt{J}$.

(2) application $I \mapsto V(I)$ and $V \mapsto I(V)$ induces a bijection

$$\{\text{radical ideal of } K[X_1, \dots, X_n]\} \longleftrightarrow \{\text{algebraic variety of } k^n\}$$

and

$$\{\text{maximal ideal of } K[X_1, \dots, X_n]\} \longleftrightarrow \{\text{one-point variety of } k^n\}$$

Proof. (1) $\Leftarrow$ is clear. Conversely, a radical ideal I can be written as the intersection of some maximal ideals (**verify**),

$$\sqrt{I} = \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$$

then by weak nullstellensatz $V(\mathfrak{m}) = \{a\}$ for some point of k^n , hence $I \subset \mathfrak{m}$ if and only if $a \in V(I)$, i.e.

$$\sqrt{I} = \bigcup_{a \in V(I)} \mathfrak{m}_a$$

where notice that $a = (a_1, \dots, a_n) \mapsto (X - a_1, \dots, X - a_n) = \mathfrak{m}_a$ gives a bijection by nullstellensatz. Hence $V(I) = V(J)$ gives the same radical ideal.

(2) For any algebraic subset $V \subset k^n$, by definition $I(V)$ is a radical ideal, and $V(I(V)) = V$ holds without any condition. Conversely, if $I = K[X_1, \dots, X_n]$, then $V(I) = \emptyset$ and $I(V(I)) = I(\emptyset) = I$. If $I \subsetneq K[X_1, \dots, X_n]$, then there exists at least one maximal ideal $\mathfrak{m}$ such that $I \subset \mathfrak{m}$ by Zorn's lemma, hence $V(I) \neq \emptyset$, hence we can apply (1) to $V(I(V(I))) = V(I)$, then $\sqrt{I(V(I))} = \sqrt{I}$, both are radical ideals, hence we can get the result. $\square$

Remark 7.3. Actually, we can say more about the original zero theorem, If $I = (f_1, \dots, f_m)$ forms a proper ideal, then there exists a maximal ideal $\mathfrak{m}$ containing it, similarly techniques as above shows that there exists a common zeros of $f_1, \dots, f_m$ in k^n if k is algebraically closed. Hence the original zero theorem holds without assumption of non-empty common zeros, generally it is called the **strong Nullstellensatz**:

$$V(I) = \emptyset \iff I = A$$

Indeed it reflects the fact about zeros: if a set of polynomials has common zeros if and only if the polynomials forms a proper ideal, i.e. there does not exist some coefficients $c_i \in k$ such that

$$\sum_i c_i f_i = 1$$

A immediate question is that how to describe the prime ideals in terms of algebraic variety, it is more meaningful when consider the Zariski topology, here we suppose that $I(V)$ is a prime ideal of $K[X_1, \dots, X_n]$, then for any $f, g \in K[X_1, \dots, X_n]$,

$$\begin{aligned} fg \in I(V) &\iff f \in I(V) \text{ or } g \in I(V) \\ &\iff V \subset V(f) \text{ or } V \subset V(g) \\ &\implies V \subset V(f) \cup V(g) \end{aligned}$$

If we want complete above implication, some other condition is needed here.

Definition 7.3. An algebraic subset $V \subset k^n$ is called **irreducible** if for any algebraic subsets $V_1, V_2 \subset k^n$ such that

$$V = V_1 \cup V_2$$

then $V = V_1$ or $V = V_2$.

The condition is a little like connectness in topology, hence it is more natural to consider a irreducible space in Zariski topology, here we just finish the correspondence via the condition:

Proposition 7.4. Let k be an algebraically closed field, then there exists a bijection

$$\{\text{prime ideal of } K[X_1, \dots, X_n]\} \longleftrightarrow \{\text{irreducible algebraic variety of } k^n\}$$

Proof. Let I be a prime ideal of $K[X_1, \dots, X_n]$, then we need to show that $V(I)$ is irreducible. If $V(J)$ and $V(L)$ are two algebraic subsets such that

$$V(I) = V(J) \cup V(L) = V(JL)$$

then by the correspondence of radical ideals, we have $\sqrt{I} = \sqrt{JL}$, I is prime so it is radical, hence $I = JL$, then by the definition of prime ideal, $J \subset I$ or $L \subset I$, hence $V(I) \subset V(J)$ or $V(I) \subset V(L)$, which shows that $V(I)$ is irreducible.

Let V be an irreducible algebraic subset of k^n , then we need to show that $I(V)$ is a prime ideal. For any $f, g \in K[X_1, \dots, X_n]$ such that

$$fg \in I(V) \iff V \subset V(fg) = V(f) \cup V(g)$$

then by irreducibility of V , we have $V \subset V(f)$ or $V \subset V(g)$, hence $f \in I(V)$ or $g \in I(V)$, which shows that $I(V)$ is prime. $\square$

So far we have built the basic correspondence between variety and ideals, to say something geomtry, we need to consider the topology of the affine space k^n under the condition of algebraically closed, Review the properties of algebraic varieties in remark 7.1, we can see that the algebraic varieties naturally defined closed sets of k^n , hence it induces a topology on k^n , and we call it the **Zariski topology**. Now we will conclude some data of the topological properties:

- (a) It is difficult to describe the open sets in Zariski topology exactly, but we can define a basis of open sets as following:

$$D(f) := k^n \setminus V(f) = \{x \in k^n \mid f(x) \neq 0\}$$

for any polynomials $f \in K[X_1, \dots, X_n]$, it is called the principal open subset defined by f , and **the set of all principal open subsets forms a basis of Zariski**

topology(verify). It is easy to verify some properties of principal open subsets:

- (1) $D(0) = \emptyset$ and $D(1) = k^n$.
- (2) $D(f) \cap D(g) = D(fg)$ for any polynomials f, g .
- (3) $D(f) = \emptyset \iff f$ is nilpotent.

- (b) When $n=1$, the Zariski topology on k is just the cofinite topology: for any finite subset $F = \{c_1, \dots, c_n\} \subset k$, we can find that $f(X) = (X - c_1) \cdots (X - c_n)$ vanishes exactly on F , hence $V(f) = F$, which shows that any finite subset is closed, and there does not exist a polynomial vanishing at infinitely many points in $k[X]$, hence all possible closed sets are

$$\mathcal{F} = \{k, \emptyset, \text{finite subsets}\}$$

which is exactly the cofinite topology, it is a natural example to understand the Zariski topology. **Pay attention** that when $n \geq 2$, the Zariski topology is not the cofinite topology, a example is that

$$V(X_1) = \{(0, a_2, \dots, a_n) \mid a_i \in k\}$$

it is an infinite closed subset of k^n , hence

$$\text{cofinite topology} < \text{Zariski topology} < \text{Euclidean topology}$$

when “ $\mid$ ” means “is coarser than”.

- (c) **The Zariski topology is a T_1 -space**, i.e. any single point is closed. Indeed, and point $a \in k^n$ can be seen uniquely as a maximal ideal $\mathfrak{m}_a = (X_1 - a_1, \dots, X_n - a_n)$ by nullstellensatz. However, **the Zariski topology is not T_2 (Hausdorff)**, consider any two points $a, b \in k^n$, and two non-empty set principal open subsets $D(f), D(g)$ containing them respectively, then

$$D(f) \cap D(g) = \emptyset \iff D(fg) = \emptyset$$

which implies that fg is nilpotent, then f or g is nilpotent, which contradicts the assumption that $D(f), D(g)$ are non-empty. Hence any two non-empty open subsets of Zariski topology always intersect, which shows that it is not Hausdorff.

- (d) **The Zariski topology is quasi-compact**, i.e. any open cover has a finite sub-cover(easy to verify).
- (e) **The Zariski topology is connected**: Suppose that $k^n = U \cup V$ are two non-empty disjoint open subsets, then the open basis allows to choose non-empty open sets $D(f)$ and $D(g)$ such that $D(f) \cap D(g) = \emptyset$, then disjoint condition shows that $D(fg) = \emptyset$, hence fg is nilpotent, then $D(f) = \emptyset$ or $D(g) = \emptyset$, which leads to contradiction.

We can do more with the help of nullstellensatz, we consider a non-empty variety $V \subset k^n$, we can consider the quotient ring $K[X_1, \dots, X_n]/I(V)$, it means we kill all polynomial functions vanishing on V , hence we can view it as the set of polynomial functions on V :

Definition 7.4. Let $V \subset k^n$ be an algebraic subset, then a function $f : V \rightarrow k$ is called a **polynomial function** on V if there exists a polynomial function $\tilde{f} \in \mathcal{O}(k^n)$ such that $f = \tilde{f}|_V$. We denote the set of all polynomial functions on V by $\mathcal{O}(V)$.

The restriction induces a natural surjective morphism of k -algebras

$$r : \mathcal{O}(k^n) \rightarrow \mathcal{O}(V), \quad f \mapsto f|_V$$

with the kernel $\ker r = I(V)$, hence by the isomorphism of quotient we have

$$K[X_1, \dots, X_n]/I(V) \cong \mathcal{O}(k^n)/I(V) \cong \mathcal{O}(V)$$

and we call $\mathcal{O}(V)$ the **coordinate ring** of the variety V , the following corollary reflects the name of "coordinate", it reflects all the information of the variety:

Proposition 7.5 (UPQ-Coordinate ring).

Let $V \subset k^n$ be an algebraic subset with k algebraically closed, then for any k -algebra A , there exists a natural bijection

$$V \longleftrightarrow \text{Hom}_{k\text{-alg}}(\mathcal{O}(V), k)$$

by mapping a point $a \in V$ to the evaluation map ev_a :

$$\text{ev}_a : \mathcal{O}(V) \rightarrow k, \quad f \mapsto f(a)$$

Proof. In particular, if $V = k^n$, then $k^n \cong \text{Hom}(\mathcal{O}(k^n), k)$ is just the dual property of polynomial ring. $\square$

Remark 7.4. In particular, if $V = \{a\}$ is a single point, then the coordinate ring $\mathcal{O}(V) \cong k$ itself, and then **the endomorphism of a k -algebra is unique (verify!)**, i.e.

$$\text{Hom}_{k\text{-alg}}(A, A) = \{id\}$$

which satisfy the fact here, furthermore we notice that

$$\mathcal{O}(k^n)/I(V) \cong \mathcal{O}(V) \cong k$$

hence $I(V)$ must be a maximal ideal, without the assumption of algebraically closed field, we can not get bijection between points and maximal ideals.

We should notice that the coordinate ring actually reflects the geometric information of variety in the way of algebra, for example we consider two algebraic subsets

$$V_1 = \{(x, 0) \in k^2 \mid x \in k\} \quad V_2 = \{(x, x, x) \in k^3 \mid x \in k\}$$

clearly that they are all stright lines in affine spaces with different dimensions, and we can calculate their coordiante rings:

$$\mathcal{O}(V_1) \cong k[X, Y]/(Y) \cong k[X], \quad \mathcal{O}(V_2) \cong k[X, Y, Z]/(X - Y, Y - Z) \cong k[X]$$

hence the coordiante rings reflects the the functoriality of $\mathcal{O}$, from a geometric object to an algebraic object. To complete the functoriality of morphism, we naturally define the morphism between varieties:

Definition 7.5. Let $V \subset k^n$ and $W \subset k^m$ be two algebraic subsets, then a map $\varphi : V \rightarrow W$ is called a **polynomials** if it is a restriction of a polynomials map $k^n \rightarrow k^m$

$$(x_1, \dots, x_n) \mapsto (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))$$

where $f_i \in \mathcal{O}(k^n)$.

Indeed such a map is continous under Zariski topology (**verify!**), which allows us to construct a sub-category of **Top**, called **the category of affine algebraic varieties over k** :

$$\mathbf{Aff}_k \left| \begin{array}{l} \text{objects: algebraic variety } V \subset k^n \\ \text{morphisms: polynomial functions } f : V \subset k^n \rightarrow W \subset k^m \end{array} \right.$$

with this definition, then as a functor $\mathcal{O}$ maps each object to a coordinate ring, exactly a k -algebra of type fini, here we denote it by $\mathbf{Alg}_k^{\text{fini}}$.

On the level of morphisms, each polynoimial function $\varphi : k^n \rightarrow k^m$ induces a morphism of k -algebra like dual map:

$$\varphi^* : \mathcal{O}(k^m) \rightarrow \mathcal{O}(k^n), \quad f \mapsto f \circ \varphi$$

hence furthermore let $\varphi|_V : V \rightarrow W$ be the restriction, then it induces a morphism of coordiante rings by quotient:

$$\begin{array}{ccc} \mathcal{O}(k^m) & \xrightarrow{\varphi^*} & \mathcal{O}(k^n) \\ \pi_W \downarrow & & \downarrow \pi_V \\ \mathcal{O}(W) & \dashrightarrow & \mathcal{O}(V) \end{array}$$

It is easy to check:

$$\begin{aligned} \varphi^*(f) \in I(V) &\iff f \circ \varphi \in I(V) \\ &\iff f(\varphi(a)) = 0, \forall a \in V \\ &\iff f(b) = 0, \forall b \in W \\ &\iff f \in I(W) \end{aligned}$$

hence it induces a unique morphism of k -algebras

$$\varphi^* : \mathcal{O}(W) \rightarrow \mathcal{O}(V), \quad f + I(W) \mapsto f \circ \varphi + I(V)$$

Hence we finish the construction of functor completely

$$\mathcal{O} : \mathbf{Aff}_k \rightarrow \mathbf{Alg}_k^{\text{fini}}$$

without the assumption that k is algebraically closed, the functor will not be an equivalence, actually we need another condition to ensure that, which makes us can give a one-side correspondence completely:

Theorem 7.6. Let k be an algebraically closed field, then the functor $\mathcal{O}$ gives a contravariant equivalence to the reduced k -algebras of type fini:

$$\mathbf{Aff}_k \cong (\mathbf{Alg}_k^{\text{fini, red}})^{\text{op}}$$

Part II

M1 Notes II

8 Some group and linear algebra

This section can be seen as a brief review of linear algebra and group theory, it mainly focus on something related to lie theory and representation theory.

8.1 Semi-direct product

Definition 8.1. Let G and H be groups, and let $\varphi : H \rightarrow \text{Aut}(G)$ be a group homomorphism. The **semi-direct product** $G \rtimes_{\varphi} H$ is the set $G \times H$ with the multiplication defined by

$$(g_1, h_1)(g_2, h_2) = (g_1\varphi(h_1)(g_2), h_1h_2)$$

for all $g_1, g_2 \in G$ and $h_1, h_2 \in H$.

Remark 8.1. If φ is trivial, i.e. $\varphi(h) = \text{id}_G$ for all $h \in H$, then the semi-direct product $G \rtimes_{\varphi} H$ is just the **direct product** $G \times H$.

Conversely, if $G \rtimes H$ is abelian, then φ must be trivial by calculating $(1, h)(g, 1) = (g, 1)(1, h)$, since it will be $\varphi(h)(g) = g$ for any h and g .

Lemma 8.1. Let $N \triangleleft G$ be a normal subgroup, and $H < G$ be a subgroup, then $\langle H \cup N \rangle = NH$.

Proof. □

Definition 8.2. A group G is an inner semi-direct product of N and H if $N \triangleleft G$, $H < G$ and $G = NH$ with $N \cap H = \{e\}$.

Remark 8.2. The definition implies that the group has semi-direct product structure, and the automorphism comes from the inner automorphism:

$$G \cong N \rtimes_{\varphi} H,$$

where $\varphi : H \rightarrow \text{Aut}(N)$ is defined by $\varphi(h)(n) = hnh^{-1}$, it is well-defined since N is normal. To prove the isomorphism, we set $f : N \rtimes H \rightarrow G$ by $f(n, h) = nh$, then it is surjective since $NH = G$, and it is indeed a morphism

$$\begin{aligned} f((n_1, h_1)(n_2, h_2)) &= f(n_1\varphi(h_1)(n_2), h_1h_2) \\ &= n_1h_1n_2h_1^{-1}h_1h_2 \\ &= n_1h_1n_2h_2 \\ &= f(n_1, h_1)f(n_2, h_2) \end{aligned}$$

So it is enough to check the injectivity: if $f(n, h) = e$, then $n = h^{-1} \in N \cap H$, thus $n = e$ and $h = e$.

Proposition 8.2. Let $G \rtimes H$ be a semi-direct product with respect to $\varphi : H \rightarrow \text{Aut}(G)$, then G can be identified as a normal subgroup of $G \rtimes H$ while H can be identified as a subgroup of $G \rtimes H$, and we have a splitting exact sequence

$$0 \rightarrow G \rightarrow G \rtimes H \rightarrow H \rightarrow 0$$

Proof. We consider the inclusion $i : G \rightarrow G \rtimes H$ defined by $i(g) = (g, 1)$ and the projection $\pi : G \rtimes H \rightarrow H$ defined by $\pi(g, h) = h$, then we can verify that i and π are homomorphisms, and $\ker \pi = i(G) \cong G$, so it shows that G can be identified as a normal subgroup of $G \rtimes H$ and the sequence is exact. Finally, to prove that the sequence splits, we define a section $s : H \rightarrow G \rtimes H$ by $s(h) = (1, h)$, then it is easy to check that s is an injective homomorphism such that $\pi \circ s = id_H$, it shows that the sequence splits, and H can be identified as $s(H)$. $\square$

8.2 Minimal polynomial

Lemma 8.3. Let $f : V \rightarrow V$ be a linear endomorphism with V a finite-dimensional vector space over k , and $\mathcal{V} = \{v_1, \dots, v_n\}$ be a basis of V such that A is the matrix of f with respect to $\mathcal{V}$, then the minimal polynomial of the matrix A is exactly the minimal polynomial of the endomorphism f .

Proof. We consider the twist map $\varphi : k^n \rightarrow V$ defined by $\varphi(e_i) = v_i$, where e_i is the standard basis of k^n , then we have the following commutative diagram:

$$\begin{array}{ccc} k^n & \xrightarrow{\varphi} & V \\ A \downarrow & & \downarrow f \\ k^n & \xrightarrow{\varphi} & V \end{array}$$

thus for any polynomial $p \in k[X]$, we have $p(f) \circ \varphi = \varphi \circ p(A)$, hence f will have the same minimal polynomial as A . $\square$

Remark 8.3. We can consider two representations of $k[X]$ -module by

$$\rho : k[X] \rightarrow \text{End}(V), \quad X \mapsto f$$

and

$$\rho_A : k[X] \rightarrow \text{End}(k^n), \quad X \mapsto A$$

then φ gives an isomorphism of two representations.

8.3 Decomposition of endomorphisms

Theorem 8.4 (Real-Polar decomposition).

For any real endomorphism $A \in \text{GL}_n(\mathbb{R})$, there exists a unique orthogonal endomorphism $O \in O(n)$ and a unique symmetric positive definite endomorphism $P \in S_n^{++}(\mathbb{R})$ such that

$$A = OP$$

Proof. Here is the outline of the proof:

- (1) A **lemma** about the existence of square root are needed here: If $A \in S_n^+(\mathbb{R})$, then there exists a (unique) $B \in S_n^+(\mathbb{R})$ such that $B^2 = A$.
- (2) Prove that $A^t A$ is (semi-definite) positive, thus by the lemma there exist a (unique) squar root $P = \sqrt{A^t A}$, and we prove that P is positive definite.
- (3) Define $O = AP^{-1}$, and prove that O is orthogonal.
- (4) Prove the uniqueness of the decompoition.

Here is the details:

- (1) By real-spectral theorem, there exists $O \in O(n)$ such that

$$A = ODO^T, \quad D = \text{dig}(\lambda_1, \dots, \lambda_n)$$

where $\lambda_i \geq 0$ are the eigenvalues of A , and we set $\Sigma = \text{dig}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$ such that

$$A = (O\Sigma O^T)^2$$

which shows that $B = O\Sigma O^T$ is a (unique) square root of A and it is positive since $\sqrt{\lambda_i} \geq 0$.

(2) $A^T A$ is clearly symmetric, and it is positive since

$$x^T (A^T A) x = \|Ax\|^2 \geq 0$$

for any $x \in \mathbb{R}^n$ under the common inner product, and A is invertible such that $\|Ax\| = 0$ if and only if $x = 0$, thus $A^T A$ is positive definite, so does its square root (the eigenvalues are strictly positive).

(3) $P^{-1} = (P^{-1})^T$ since P is symmetric, so we have

$$O^T O = (P^{-1})^T A^T A P^{-1} = (P^{-1})^T P^2 P^{-1} = I$$

(4) Suppose that there exists another decomposition $A = O_0 P_0$, then we have

$$A^T A = P_0^T O_0^T O_0 P_0 = P_0^2$$

so P_0 is the positive square root of $A^T A$, if the uniqueness of (1) is proved, we can finish the proof, but here we give another proof of the uniqueness only under the existence of the square root: we have $O_0^{-1} O = P_0 P^{-1}$, so it is easy to check that $P_0 P^{-1}$ is **orthogonal**; notice that P_0 and $A^T A$ commute, and there exists a polynomial f such that

$$f(A^T A) = P$$

it can be done by using lagrange interpolation, thus P_0 and P commute as two symmetric matrix, then P and P_0 can be diagonalized simultaneously by one orthogonal H such that

$$P = H D H^T \quad P_0 = H D_0 H^T$$

where $D = \text{dig}(\lambda_1, \dots, \lambda_n)$ and $D_0 = \text{dig}(\mu_1, \dots, \mu_n)$, thus

$$P_0 P^{-1} = H D_0^{-1} D H^T$$

it is **symmetric and positive definite** since $D_0^{-1} D$ is diagonal with positive entries, thus $P_0 P^{-1} \in O(n) \cap S_n^{++}(\mathbb{R}) = \{I\}$, which implies that $P_0 = P$ and $O_0 = O$. $\square$

Theorem 8.5 (Complex-Polar decomposition).

For any complex endomorphism $A \in \text{GL}_n(\mathbb{C})$, there exists a unique unitary endomorphism $U \in U(n)$ and a unique hermitian positive definite endomorphism $P \in H_n^{++}(\mathbb{C})$ such that

$$A = UP$$

The proof is similar to the real case, we set $P = \sqrt{A^* A}$ and $U = AP^{-1}$, and the lemma is based on the complex-spectral theorem for normal operator: The positive Hermitian endomorphism has a unique positive Hermitian square root.

Remark 8.4. In particular, if $n = 1$, we have the polar decomposition in $GL_1(\mathbb{C}) = \mathbb{C}^*$ by

$$z = |z|e^{i\text{Arg}(z)}$$

it is easy to see that $U(1) \cong \mathbb{S}^1$, it is the original ideal of the decomposition. Similarly, in the real case, we can embed $\mathbb{C}$ into $GL_2(\mathbb{R})$ by

$$a + ib \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

and we have the polar decomposition for the matrix form of complex numbers:

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} \sqrt{a^2 + b^2} & 0 \\ 0 & \sqrt{a^2 + b^2} \end{pmatrix} \begin{pmatrix} \frac{a}{\sqrt{a^2 + b^2}} & -\frac{b}{\sqrt{a^2 + b^2}} \\ \frac{b}{\sqrt{a^2 + b^2}} & \frac{a}{\sqrt{a^2 + b^2}} \end{pmatrix}$$

Another useful decomposition is QR decomposition, which is also called as Gram-Schmidt decomposition. Suppose that

8.4 Lattice

Definition 8.3. Let V be a $\mathbb{R}$ -vector space of dimension n

Definition 8.4. Let V be a $\mathbb{R}$ -vector space of dimension n , a parallelotope of V with respect to a basis $\{v_1, \dots, v_n\}$ is the set of points

$$P = \left\{ \sum_{i=1}^n c_i v_i \mid c_i \in [0, 1) \right\}$$

The volume of P corresponds to the an inner product $\langle \cdot, \cdot \rangle$ is defined by

$$\text{Vol}(P) = \sqrt{\det G(v_1, \dots, v_n)}$$

where $G(v_1, \dots, v_n)$ is the Gram matrix with entries $G_{ij} = \langle v_i, v_j \rangle$.

The definition is resonable with the following reasons:

1. Let P be of dimension 3 with respect to the euclidean inner product, then the volume of P is given by the triple product

$$\text{Vol}(P) = |v_1 \cdot (v_2 \times v_3)| = |\det[v_1, v_2, v_3]|$$

with respect to the standard basis.

2. Let $\{v_i\}_i$ be an orthogonal basis, then the Gram matrix $G = I_n$ exactly, hence the volume of a hypercube is 1, which same as what we do in plane geometry and solid geometry.

3.

with the convention above, it allows us to construct the invariant measure on any real inner product space.

Proposition 8.6. For any real inner product space V of dimension n , there exists a natural isometry ($\mathbb{R}^n$ equipped with the standard inner product)

$$\Phi : \mathbb{R}^n \rightarrow V, \quad (x_1, \dots, x_n) \mapsto \sum_{i=1}^n x_i e_i$$

where $\{e_1, \dots, e_n\}$ is an orthonormal basis of V , then $\mu = \lambda \circ \Phi^{-1}$ defines a Haar measure on V such that for any parallelotope P we have $\mu(P) = \text{Vol}(P)$ exactly.

Proof. μ is exactly a pushforward measure, to verify that it is a Haar measure, we need to check the translation invariance: for any $v \in V$ and any measurable set $S \subset V$, we have

$$\mu(v + S) = \lambda(\Phi^{-1}(v + S)) = \lambda(\Phi^{-1}(v) + \Phi^{-1}(S)) = \lambda(\Phi^{-1}(S)) = \mu(S)$$

where the second equality is due to the linearity of Φ and the third equality is due to the translation invariance of the Lebesgue measure λ . Now we consider the parallelotope P with respect to a basis $\{v_1, \dots, v_n\}$ and set $\Phi(a_i) = v_i$, then we have

$$\mu(P) = \lambda(\Phi^{-1}(P)) = |\det A| = \sqrt{\det A^t A}$$

where A is the matrix with columns a_i , and isometry shows that $A^t A$ is exactly the Gram matrix $G(v_1, \dots, v_n)$, which finishes the proof. $\square$

9 Representation of groups

The representation theory appears in almost all branches of mathematics, but the basic ideas and techniques are quite similar, among them the representation theory of finite groups is the most basic one to start with.

9.1 Category and Schur's lemma

Definition 9.1. Let G be a finite group and k be a field, $\mathbf{Rep}_k(G)$ is a category defined as following:

- The objects is of the form (V, ρ) , where V is a k -vector space such that ρ a group homomorphism

$$\rho : G \rightarrow \mathrm{GL}(V)$$

such a objects is called a **k -representation of G** .

- A morphism from (V, ρ) to (W, τ) is a k -linear map $T \in \mathcal{L}(V, W)$ such that for any $g \in G$ we have the following commutative diagram:

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \rho(g) \downarrow & & \downarrow \tau(g) \\ V & \xrightarrow{T} & W \end{array}$$

such a morphism is called an **intertwiner** or a **G -equivariant**. The isomorphisms in the category is just a linear isomorphism such that above diagram commutes. Usually, we will denote the set of morphisms from (V, ρ) to (W, τ) by $\mathrm{Hom}_G(V, W)$, and the set of endomorphisms of (V, ρ) by $\mathrm{End}_G(V)$.

- In fact under the composition of map, the morphisms is well-defined and associative, and the identity inside a representation (V, ρ) is the identity map $T(x) = x$ for any $x \in V$, thus it is indeed a category.
- A representation is called faithful if ρ is injective, and it is called trivial if $\rho(g) = id_V$ for any $g \in G$.
- If V is finite-dimensional over k , then a representation (V, ρ) is called finite-dimensional, and the dimension of V is called the dimension of the representation.

It is the basic definition of representation of groups, so firstly we need to describe more structures of the objects and morphisms such that we can know more about what the category exactly looks like.

Definition 9.2. Let $W \subset V$ as a subspace, we say that W is invariant under the representation (V, ρ) if for any $g \in G$ we have

$$\rho(g)W \subset W$$

and we define $(W, \rho|_W)$ as a subrepresentation of (V, ρ) . There always exists two trivial subrepresentations: $(\{0\}, \rho|_{\{0\}})$ and (V, ρ) itself.

Remark 9.1. The definition is exactly natural, in the sense of morphism, subrepresentation is just a representation can be embedded into a larger representation by a injective intertwiner, i.e. if (W, τ) a representation with $W \subset V$ as a subspace, then the inclusion $i : W \rightarrow V$ will be an intertwiner such that

$$\tau(w) = \tau(g)(i(w)) = i(\rho(w)) = \rho(w)$$

by the definition of τ , then we can recover the definition of invariant.

In linear algebra, we always consider the invariant subspace of a linear operator, in the context of representation, the invariant subspace is the invariant under a collection of linear operators, so there are some analogue.

Lemma 9.1. Let (V, ρ) and (W, π) be two representation of G , then

1. If $T : V \rightarrow W$ is an interwiner, then $\ker T$ is an invariant subspace of V and $\text{im} T$ is an invariant subspace of W .
2. If $T \in \text{End}_G(V)$, then the eigenspaces of T are invariant subspaces of V .

Proof. □

As we study groups by classifying their subgroups, we can also study representation by considering their subrepresentations, so it motivates us to consider a special case, which is analogous to simple groups.

Definition 9.3. A representation (V, ρ) is called irreducible if it has no non-trivial subrepresentation, i.e. the only subrepresentations are $(\{0\}, \rho|_{\{0\}})$ and (V, ρ) itself.

Theorem 9.2 (Schur).

Let (V, ρ) and (W, π) be two irreducible representations of G , and $T : V \rightarrow W$ be an interwiner, then T is either an isomorphism or the zero map.

Proof. By above lemma, $\ker T$ is an invariant subspace of V , so we can conclude that $\ker T$ is either $\{0\}$ or V since (V, ρ) is irreducible. If $\ker T = V$, then T is the zero map; If $\ker T = \{0\}$, then T is injective, notice that $\text{im} T$ is an invariant subspace of W and (W, π) is irreducible, in this case we have $\text{im} T = W$ since T is injective, thus T is an isomorphism. □

Remark 9.2. If two representations are not isomorphic, then interwiner must be zero, i.e. $\text{Hom}_G(V, W) = \{0\}$, and we can define an equivalence relation on **the irreducible representations** of G as following:

$$(V, \rho) \sim (W, \pi) \iff \text{they are isomorphic}$$

and we use $\hat{G}$ to denote the set of equivalence classes.

Corollary 9.3. $\text{End}_G(V)$ is a division k -algebra for any irreducible representation (V, ρ) of G , **in particular if $k = \mathbb{C}$ and V is finite-dimensional**, then $\text{End}_G(V) \cong \mathbb{C}$ is the set of scalar multiplications.

Proof. $\text{End}_G(V)$ contains zero map and isomorphisms by schur's lemma, thus it is a division algebra over k , and the unique finite-dimensional division algebra over $\mathbb{C}$ is $\mathbb{C}$ itself, so we have $\text{End}_G(V) \cong \mathbb{C}$. $\square$

9.2 Group algebra and convolution

Definition 9.4. Let G be a group and k be a field, then the group algebra $k[G]$ is a k -algebra defined as following:

$$k[G] := \left\{ \sum_{g \in G} c_g e_g \mid c_g \in k \text{ and } c_g = 0 \text{ for almost all } g \right\}$$

where e_g is a formal component with respect to g , which satisfies

$$\forall g, h \in G, \quad e_g \times e_h = e_{gh}$$

and e_1 is the multiplicative identity of $k[G]$, the operation is defined as following:

- Addition:

$$\sum_{g \in G} a_g e_g + \sum_{g \in G} b_g e_g = \sum_{g \in G} (a_g + b_g) e_g$$

- Scalar multiplication:

$$c \cdot \sum_{g \in G} a_g e_g = \sum_{g \in G} (ca_g) e_g$$

- Multiplication:

$$\left(\sum_{g \in G} a_g e_g \right) \times \left(\sum_{h \in G} b_h e_h \right) = \sum_{g \in G} c_g e_g$$

with $c_g = \sum_{uv=g} a_u b_v$.

Remark 9.3. Generally, the definition can be extended to a monoid and a ring instead of a group and a field. If G is a finite group, then $k[G]$ is a finite-dimensional k -vector space with basis $\{e_g\}_{g \in G}$. Moreover, the group algebra is a commutative algebra if and only if G is an abelian group.

Proposition 9.4. There exists a equivalence of categories

$$\mathbf{Rep}_k(G) \cong {}_{k[G]}\mathbf{Mod}$$

Proof. Let (V, ρ) be a representation of G , the definition induces a group action of G on V by ρ , i.e. a map

$$G \times V \rightarrow V, \quad (g, v) \mapsto \rho(g)(v)$$

defines a group action, thus we can extend it to be an $k[G]$ -module. Conversely, for an $k[G]$ -module M , the definition of scalar multiplication induces a group action of $k[G]$ on M , and by restriction we can get a representation of G .

For morphisms,...

□

Remark 9.4. Similar argument can be applied to the right $k[G]$ -module, but we are used to use left $k[G]$ -module.

Definition 9.5. A **(left) regular representation** of G is the representation of G on $k[G]$ defined as following:

$$\rho : G \rightarrow \text{GL}(k[G]), \quad g \mapsto \rho(g)$$

where $\rho(g)$ is the left multiplication by e_g :

$$\rho(g)\left(\sum_{h \in G} c_h e_h\right) = \sum_{h \in G} c_h e_{gh}$$

Remark 9.5. By above equivalence of categories, a representation of G is (left) regular if and only if the corresponding $k[G]$ -module is $k[G]$ itself, or equivalently a free $k[G]$ -module of rank 1.

More generally, any group action can be extended linearly to be a representation, we construct as following

1. Let $G \curvearrowright X$ be a group action with a well-defined operation $g * x$, firstly we can extend the set X to be a vector space over k

$$k[X] := \left\{ \sum_{x \in X} c_x e_x \mid c_x \in k \text{ and } c_x = 0 \text{ for almost all } x \in X \right\}$$

where e_x is a formal component with respect to x , and together forms a basis of $k[X]$, it is easy to check that it is a vector space.

2. The group action can be equivalently defined as a group homomorphism $\sigma : G \rightarrow S_X$ with $\sigma(g)(x) = g * x$, then we can extend the morphism to

$$\rho : G \mapsto \text{GL}(k[X]), \quad g \mapsto \rho(g)$$

where $\rho(g)$ is the linear extension of $\sigma(g)$:

$$\rho(g)\left(\sum_{x \in X} c_x e_x\right) = \sum_{x \in X} c_x e_{g*x}$$

It is easy to check that ρ is a group homomorphism, and we consider a natural inclusion $\iota : X \rightarrow K[X]$ by $\iota(x) = e_x$, then we have the following commutative diagram:

$$\begin{array}{ccc} X & \xrightarrow{\sigma(g)} & X \\ \downarrow \iota & & \downarrow \iota \\ k[X] & \xrightarrow{\rho(g)} & k[X] \end{array}$$

which shows that ρ is exactly the linear extension of σ as a representation of G .

3. The extension can be applied to morphism of G -sets. if G is a group acting on two sets X and Y with operation $*$ and $\star$ respectively, then we consider a G -map $f : X \rightarrow Y$ which satisfies

$$\forall g \in G, x \in X, \quad f(g * x) = g \star f(x)$$

then we can extend f to be a linear map $\tilde{f} : k[X] \rightarrow k[Y]$ by

$$\tilde{f}\left(\sum_{x \in X} c_x e_x\right) = \sum_{x \in X} c_x e_{f(x)}$$

Let ρ_X and ρ_Y be representations defined as 2, then we can check

$$\begin{aligned} \rho_Y(g) \circ \tilde{f}\left(\sum_{x \in X} c_x e_x\right) &= \rho_Y(g)\left(\sum_{x \in X} c_x e_{f(x)}\right) \\ &= \sum_{x \in X} c_x e_{g \star f(x)} \\ &= \sum_{x \in X} c_x e_{f(g * x)} \\ &= \tilde{f}\left(\sum_{x \in X} c_x e_{g * x}\right) \\ &= \tilde{f} \circ \rho_X(g)\left(\sum_{x \in X} c_x e_x\right) \end{aligned}$$

which shows that $\tilde{f}$ is an interwiner between ρ_X and ρ_Y .

The constuction shows that representation thoery is a powerful tool, as a core tool in group theory, group action can be contained in represetation theory, which can be concluded by the language of category as following:

Proposition 9.5. Let $L : G\text{-Set} \rightarrow \mathbf{Rep}_k(G)$ be a functor defined by the construction above:

- On objects: $L(X) = (k[X], \rho)$ where ρ is the linear extension of the group action of G on X .

- On morphisms: $L(f) = \tilde{f}$ where $\tilde{f}$ is the linear extension of the G -map f .

Then L is a well-defined functor which staifies

- (a) L is faithful but not full, and for any G -set X , the image $L(X)$ is called a **permutation representation** of G .

(b) L is a free functor.

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & U(k[X]) \\ & \searrow f & \downarrow U(\tilde{f}) \\ & & U(V) \end{array}$$

Now we turn our attention to the finite-dimensional representation of finite groups

Definition 9.6. Let G be a finite group and k be a field, we set the space of functions on G by $\mathcal{F}(G)$, and we define the operation such that it is a k -alegebra:

- Addition:

$$(f + g)(x) = f(x) + g(x)$$

- Scalar multiplication:

$$(c \cdot f)(x) = cf(x)$$

- Multiplication:

$$(fg)(x) = f(x) \cdot g(x)$$

where $f, g \in \mathcal{F}(G)$ and $c \in k$.

As a vector space, $\mathcal{F}(G)$ is finite-dimensional with basis $\{\delta_g\}_{g \in G}$ where δ_g is the delta function defined by

$$\delta_g(x) = \begin{cases} 1 & x = g \\ 0 & x \neq g \end{cases}$$

hence there exists a natural isomorphism $\mathcal{F}(G) \cong k[G]$ by sending δ_g to e_g .

However, as k -alegebra the group algebra $k[G]$ is not isomorphic to $\mathcal{F}(G)$, since $k[G]$ can be non-commutative if G while $\mathcal{F}(G)$ is always commutative, but a different multiplication can be defined on $\mathcal{F}(G)$ such that we can get an isomorphism.

Lemma 9.6. For any $f_1, f_2 \in \mathcal{F}(G)$, we define the **convolution** of f_1 and f_2 by

$$(f_1 * f_2)(g) = \frac{1}{|G|} \sum_{t \in G} f_1(t) f_2(t^{-1}g)$$

it has the following properties:

- (a) The convolution is associative.
- (b) The convolution is commutative if and only if G is an abelian group.
- (c) $\delta_g * \delta_h = \delta_{gh}$ for any $g, h \in G$.
- (d) $(\mathcal{F}(G), +, *)$ is isomorphic to group algebra $k[G]$ as k -alegebra.
- (e) $f(g_1^{-1}xg_2) = (\delta_{g_1} * f * \delta_{g_2^{-1}})(x)$

Proof.

□

Proposition 9.7. For any $g \in G$, we define the left translation on $\mathcal{F}(G)$ by

$$L_g : \mathcal{F}(G) \rightarrow \mathcal{F}(G), \quad L_g(f)(x) = f(g^{-1}x)$$

then L_g is an algebra automorphism of $(\mathcal{F}(G), +, *)$ and the map $g \mapsto L_g$ defines a representation $(\mathcal{F}(G), L)$, which is equivalent to the left regular representation of G .

Simiarly, the right translation on $\mathcal{F}(G)$ defined by

$$R_g : \mathcal{F}(G) \rightarrow \mathcal{F}(G), \quad R_g(f)(x) = f(xg)$$

and $g \mapsto R_g$ induces a representation $(\mathcal{F}(G), R)$ of G , it is equivalent to the right regular representation of G .

Proof. □

Remark 9.6. This proposition explicitly constructs a left and right regular representation via the convolution on algebra of functions on a finite group G , more generally, if G is finite then there exists a equivalence of categories

$$\mathbf{Rep}_k(G) \cong_{\mathcal{F}(G)} \mathbf{Mod}$$

where $\mathcal{F}(G)$ is the **convolution algebra**.

Finally, we will write a representation of finite group G as a $\mathcal{F}(G)$ -module, we set a representation $\rho : G \rightarrow \mathrm{GL}(V)$, we should extend it such that the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\rho} & \mathrm{GL}(V) \\ g \mapsto \delta_g \downarrow & & \downarrow \\ \mathcal{F}(G) & \xrightarrow{\tilde{\rho}} & \mathrm{End}(V) \end{array}$$

we define the extension as following:

$$\tilde{\rho}(f)(v) = \frac{1}{|G|} \sum_{g \in G} f(g) \cdot \rho(g)(v)$$

for any $f \in \mathcal{F}(G)$ and $v \in V$, it needs to check that $\tilde{\rho}$ is a ring homomorphism, then it induces a $\mathcal{F}(G)$ -module structure on V . Moreover, when $f = \delta_h$ for some $h \in G$, we have

$$\tilde{\rho}(\delta_h) = \rho(h)$$

so exactly $\tilde{\rho}$ gives a extension exactly.

9.3 Character and orthogonality relations

Definition 9.7. Suppose that (V, ρ) is a finite-dimensional representation of G , we define the character of (V, ρ) as a function $\chi_\rho : G \rightarrow k$ by

$$\chi_\rho(g) = \mathrm{tr}(\rho(g))$$

for any $g \in G$.

Notice that if $\dim V = 1$, then $\text{GL}(V) \cong k^\times$, which shows that the character of the representation will be the representation itself: a group homomorphism from G to $k^\times$.
[Generally, the character of a representation is not necessary a group homomorphism.](#)

Definition 9.8. Let G be a group and k be a field, we denote $\text{Cl}(G) = G / \sim_c$ be the quotient of G by the conjugacy relation

$$x \sim y \iff \exists g \in G, y = gxg^{-1}$$

a function $f : \text{Cl}(G) \rightarrow k$ is called a **class function**, or equivalently, a function $f : G \rightarrow k$ such that

$$f(gxg^{-1}) = f(x)$$

for any $g, x \in G$.

Remark 9.7. The algebraic properties of trace shows that [the character of a representation is always a class function](#), thus it can be regarded as a function on $\text{Cl}(G)$.

The conjugacy classes can be described by group action $G \curvearrowright G$ with operation $g * x = gxg^{-1}$, then $\text{Cl}(G)$ is exactly the set of orbits with

$$C_G(x) = \{g \in G \mid gxg^{-1} = x\}$$

the centralizer (stablizer) of x in G . When G is finite, $[G : C_G(x)]$ is exactly the size of orbit of x .

Proposition 9.8. Let (V, ρ) be a finite-dimensional representation of G , then

(a) $\chi_\rho(e) = \dim V$.

(b) $\chi_{\rho^*}(g) = \chi_\rho(g^{-1}) = \text{tr}(\rho(g)^{-1})$ for any $g \in G$.

Let (W, π) be another finite-dimensional representation of G , then

(c) $\chi_{\rho \oplus \pi} = \chi_\rho + \chi_\pi$

(d) $\chi_{\rho \otimes \pi} = \chi_\rho \cdot \chi_\pi$

10 Topological Groups

10.1 Basic properties

Group and Topological space is two different objects, group is algebraic structure, topological space is geometric structure, but it is not strange to combine them together.

Definition 10.1. A **topological group** is a object $(G, \mathcal{T}, \cdot)$ such that

- $(G, \cdot)$ is a group
- $(G, \mathcal{T})$ is a topological space
- The group structure is compatible with topological structure, i.e. multiplication

$$m : G \times G \rightarrow G, \quad (x, y) \mapsto x \cdot y$$

and inverse

$$i : G \rightarrow G, \quad x \mapsto x^{-1}$$

are continuous maps.

With the basic definition, we can talk about the properties of topological groups, here is a useful lemma, it shows that the topological group is **homogeneous**, i.e. the local topology is same at any point.

Lemma 10.1. Let G be a topological group, then for any $g \in G$, we define **left translation** and **right translation** by

$$L_g : G \rightarrow G, \quad x \mapsto gx$$

$$R_g : G \rightarrow G, \quad x \mapsto xg$$

Then both L_g and R_g are homeomorphisms on G .

Proof. It is easy to verify that $L_{gh} = L_g \circ L_h$, and $R_{gh} = R_h \circ R_g$, then we have

$$L_g^{-1} = L_{g^{-1}}, \quad R_g^{-1} = R_{g^{-1}}$$

and we can check the continuity of L_g and R_g by composition. □

It is an important in the following proofs of properties, we first talk about the basic properties of topological groups.

Proposition 10.2. Let G be a topological group, then

- (1) each open subgroup $H < G$ is also closed.

(2) each closed subgroup $H < G$ with finite index is also open.

Proof. The subgroup H gives a partition of G by left cosets:

$$G = \bigsqcup_{g \in \mathcal{S}} gH$$

with $\mathcal{S} \subset G$ the set of representatives in G/H , and we fix the representatives for H be e , then

$$G - H = \bigsqcup_{g \in \mathcal{S} - \{e\}} gH$$

If H is open, then by translation each gH is also open, thus the union of open sets $G - H$ is also open, hence H is closed, which finishes that proof of (1). For the statement (2), the finite index shows that $\mathcal{S}$ is finite, and we suppose that the index is n . If $n = 1$, then $H = G$ is open clearly; otherwise, we can deduce that $G - H$ is the finite union of closed sets gH with $g \in \mathcal{S} - \{e\}$, thus it is closed, it can be done similarly by translation, hence H is open since $G - H$ is closed. $\square$

Then it is the **separation** of topological groups.

Proposition 10.3. Let G be a topological groups, then the following statements are equivalent:

- (1) G is Hausdorff (T2).
- (2) The singleton $\{e\}$ is closed in G .
- (3) G is T1.

Proof. (3) implies (2) is clear, since T1 space makes all singletons closed; (1) implies (3) is just the implication from strong to weak separation axiom.

(2) implies (1): G is Hausdorff if and only if the diagonal set

$$\Delta = \{(g, g) \in G \times G \mid g \in G\}$$

is closed under the product topology, and topolgal group induces a continous map $f(x, y) = xy^{-1}$ for any $(x, y) \in G \times G$, then $\Delta = f^{-1}(\{e\})$ is closed by continuity. $\square$

Corollary 10.4. Let G be a topological group and $H < G$ be a subgroup, then the quotient space G/H is **Hausdorff** if and only if H is **closed** in G .

Another topological properties is about **connectness**:

Proposition 10.5. Let G be a topological group, then

- (1) If G is connected, then any open set containing e generates G .
- (2) The connected component G_0 of identity $e \in G$ is a **closed normal subgroup** of G , and each connected component of G is homeomorphic to G_0 by translation.

Proof. (1) We denote $\langle A \rangle$ be the subgroup generated by $A \subset G$, and we take A be an open set containing e , we will prove that $\langle A \rangle$ is open and closed, by the connectness of G , which implies that $\langle A \rangle = G$. It is clear that

$$\bigcup_{x \in \langle A \rangle} xA \subset \langle A \rangle$$

conversely, for any $a \in \langle A \rangle$, we have $a = ae$ with $e \in A$, so a is in the union such that we can replace the inclusion by equality. Notice that $xA = L_x(A)$ is open since A is open, so $\langle A \rangle$ is also open as the union of open sets; By the proposition 10.2, $\langle A \rangle$ is also closed since it is open subgroup, hence we finish the proof of (1).

(2) Any connected component is closed in general topology, which can be proved by contradiction to $G_0 \subsetneq \overline{G_0}$. To prove that G_0 is a subgroup, we take a continuous map $f(x, y) = xy^{-1}$ on $G \times G$, then $f(G_0 \times G_0)$ is connected since the finite product of connected spaces is connected, and we notice that $a = f(a, e)$ for any $a \in G_0$, so $G_0 \subset f(G_0 \times G_0)$ and then we can take equality by the maximality of connected component, and it shows that G_0 is exactly a subgroup.

To prove the normality, we take the conjugation map for any $g \in G$:

$$\sigma_g : G \rightarrow G, \quad x \mapsto gxg^{-1}$$

it is homeomorphism by composition $\sigma_g = L_g \circ R_g^{-1}$. Here $\sigma_g(e) = e$ and the image is also connected, so $\sigma_g(G_0) \subset G_0$ by the maximality of connected component, so it must be normal. $\square$

Notice that if $H < G$ is a subgroup, then it induces a natural equivalence relation on G by left cosets:

$$x \sim y \iff x = yh, \text{ for some } h \in H$$

similarly we can define right cosets here, and then it gives a quotient space $G/\sim$ and we denote it by G/H , and it is naturally endowed with the quotient topology with a (continuous) quotient map:

$$\pi : G \rightarrow G/H, \quad g \mapsto gH$$

In particular, if H is a normal subgroup, then G/H is exactly a topological group by UPQ:

$$\begin{array}{ccc} G \times G & \xrightarrow{f} & G \\ \pi \times \pi \downarrow & & \downarrow \pi \\ G/N \times G/N & \xrightarrow{f_{G/N}} & G/N \end{array}$$

with $f(g, h) = gh^{-1}$, so it gives the continuity of group operation. However, it is not so easy to talk about the topology of quotient space G/H directly, it is better to see the quotient space as a **homogeneous space** under the action of G , hence we need to talk about action of the topological group:

Definition 10.2. Let G be a topological group and X be a topological space, a continuous action of G on X is a well-defined action in the sense of group together with a continuous operation, that means there exists a continuous map

$$\phi : G \times X \rightarrow X, \quad (g, x) \mapsto g \cdot x$$

such that for any $g, h \in G$ and $x \in X$, we have

$$(1) \quad e \cdot x = x$$

$$(2) \quad g \cdot (h \cdot x) = (gh) \cdot x$$

In the context of topological groups or Lie groups, we always omit "continuous" when it comes to actions, it is a convention sometimes in the literature.

It is similar to the algebraic action, we can define **stabilizer** of a point $x \in G$

$$I_x = \{g \in G \mid g \cdot x = x\}$$

It is sometimes called **isotropy group** at x , and it can be written as the preimage

$$I_x = \phi_x^{-1}(\{x\}), \quad \phi_x(g) = g \cdot x$$

with ϕ_x a continuous map by fixing variable, hence the isotropy group is always a **closed subgroup** of G if X is Hausdorff (T1 is enough, but manifold makes sense). Similarly, we can define **orbit** of x by

$$Gx = O_x = \{g \cdot x \mid g \in G\}$$

which is the image of ϕ_x on the contrary, hence it gives a natural bijection

Lemma 10.6 (stabilizer-orbit).

Let G be a compact group and X be a Hausdorff space, If G acts continuously on X , then for any $x \in X$, there exists a homeomorphism $G/I_x \cong Gx$.

Proof. The proof is just based on the universal property of quotient space, and the stabilizer-orbit theorem in group theory, we just need to check the natural bijection induced by ϕ_x is indeed a homeomorphism. $\square$

In particular, the orbit gives an equivalence relation on X by

$$x \sim y \iff y = g \cdot x, \text{ for some } g \in G$$

hence it gives a quotient space denoted by X/G and we call it **orbit space**, it is naturally endowed with the quotient topology with a (continuous) quotient map:

$$\pi : X \rightarrow X/G, x \mapsto Gx$$

the following lemma is useful and important, it shows that the topology of group action is well-behaved from the view of technique:

Lemma 10.7. The natural quotient map $\pi : X \rightarrow G/X$ is open.

Proof. Suppose that U is an open set of X , then we can prove

$$\pi^{-1}(\pi(U)) = \bigsqcup_{g \in G} gU$$

and then by translation each gU is open, so we can finish the proof by the definition of quotient topology that $\pi(U)$ is open if its preimage is open. $\square$

Then we can conclude

Proposition 10.8. Let G be a topological group and H be a subgroup, then

- (1) H is closed if and only if G/H is Hausdorff.
- (2) If G is connected, then G/H is connected.
- (3) If G/H and H are both connected, then G is connected.

Proof.

$\square$

Neighborhood system

As we have mentioned before, topological group is a homogeneous space, so the local topology at one point can be translated to generate the global topology, for some convenience, we always study the local topology at identity of the group.

Formally, for any group G we define the **neighborhood system** (at x) to be the collection of subsets containing x , denoted by $\mathcal{V}_x$:

$$A \in \mathcal{V}_x \iff x \in A \wedge A \subset G$$

Hence we can induce a topology on G by neighborhood system (i.e. determine the open voisinage of identity).

Proposition 10.9. Let G be a group with $\mathcal{V}_e$ as the neighborhood system at identity e , if $\mathcal{V}_e$ satisfies the following conditions:

1. For any $U, V \in \mathcal{V}_e$, there exists $W \in \mathcal{V}_e$ such that

$$W \subset U \cap V.$$

2. If $a \in U \in \mathcal{V}_e$, then there exists $V \in \mathcal{V}_e$ such that

$$Va \subset U.$$

3. For each $U \in \mathcal{V}_e$, there exists $V \in \mathcal{V}_e$ such that

$$V^{-1}V \subset U.$$

4. For each $U \in \mathcal{V}_e$ and each $x \in G$, there exists $V \in \mathcal{V}_e$ such that

$$x^{-1}Vx \subset U.$$

then there exist a unique topology $\mathcal{T}$ such that $\mathcal{V}_e$ is the set of all open voisinage of e , and $(G, \mathcal{T})$ is a topological group.

Proof. The topology $\mathcal{T}$ is uniquely defined by translation:

$$\mathcal{B} = \{gU \mid g \in G, U \in \mathcal{V}_e\}$$

we just need to prove that $\mathcal{B}$ is the topology basis. Notice the condition (1) and (2) ensure that $\mathcal{B}$ generates a topology, the condition (3) is the requirement of continuity of multiplication and inverse; finally, the condition (4) is obligator for non-abelian group to ensure the continuity. $\square$

In partiucular, we can naturally choose subgroup as the element in the system, the condition can be simplified as following:

Corollary 10.10. Let G be a group with $\mathcal{U}$ as a collection of subgroups of G , if $\mathcal{U}$ satisfies the following conditions:

1. For any $U, V \in \mathcal{U}$, there exists $W \in \mathcal{U}$ such that

$$W \subset U \cap V.$$

2. For each $U \in \mathcal{U}$ and each $x \in G$, there exists $V \in \mathcal{U}$ such that

$$x^{-1}Vx \subset U.$$

then there exist a unique topology $\mathcal{T}$ such that $\mathcal{U}$ is a basis of open voisinage of e , and $(G, \mathcal{T})$ is a topological group.

Here are some classic examples of topological groups constructed by neighborhood system:

Example 10.1. We consider the usual topology on $\mathbb{Q}$, then the completion of $\mathbb{Q}$ is just $\mathbb{R}$, it is another method to construct real number (see in Bourkabi's book). We review that we always write the viosinage of 0 as the form of $(-\varepsilon, \varepsilon)$, which motivates us to consider the system of subgroup:

$$\mathcal{U} = \left\{ \frac{1}{n}\mathbb{Z} \mid n \in \mathbb{N}^* \right\}$$

pay attention to the choices here! We can not take $n\mathbb{Z}$ to be an open set (consider it in $\mathbb{R}$, it is closed and not open). Then we can verify the topology induced by $\mathcal{U}$ is just the subspace topology induced by $\mathbb{R}$, i.e. it induces a usual absolute value on field $\mathbb{Q}$:

$$|x|_\infty = \begin{cases} x & x \geq 0 \\ -x & x \leq 0 \end{cases}$$

It is natural to ask that can we define other topological stucture on $\mathbb{Q}$ such that we can get other different completion? The answer is yes, and a classic example is the *p-adic number field*.

Example 10.2. We fix a prime number p and then we can define a subgroup

$$\mathbb{Z}_{(p)} = \left\{ \frac{m}{n} \mid m, n \in \mathbb{Z}, p \nmid n \right\}$$

and we let $\mathcal{U} = \{p^t \mathbb{Z}_{(p)} \mid t \in \mathbb{Z}\}$ to be the collection of subgroups, which generates an unique topology (verify!). Notice that we have a unique chain in the system:

$$\dots p^2 \mathbb{Z}_{(p)} \subset p \mathbb{Z}_{(p)} \subset \mathbb{Z}_{(p)} \subset p^{-1} \mathbb{Z}_{(p)} \subset p^{-2} \mathbb{Z}_{(p)} \subset \dots$$

motivates from the above example, we consider each subgroup gives a method to measure the size of an elemnt is some sense, for example we consider the following number when $p = 3$:

$$1, \quad 4, \quad \frac{1}{2}, \quad \frac{2}{5}$$

they are all in $\mathbb{Z}_{(p)}$, and we consider the distance of them to 0 can be bounded by α (consider $(-\varepsilon, \varepsilon)$ gives the set of all elements with distance to 0 less than ε), and then we consider some other numbers:

$$\frac{1}{3}, \quad \frac{2}{3}, \quad \frac{7}{6}$$

they are all in $\frac{1}{3}\mathbb{Z}_{(p)}$ but not in $\mathbb{Z}_{(3)}$, so we can think that their distance to 0 is bouned by β , and it is natural to give a realtion $\alpha < \beta$, which motivates us to give a new method to consider the metric on $\mathbb{Q}$ by conisder the local factorization of p .

Hence we can define something to denote α and β just now:

$$v_p : \mathbb{Z} - \{0\} \rightarrow \mathbb{N}, \quad m \mapsto \max\{k : p^k \mid m\}$$

for example $v_3(6) = 2$, $v_3(-2) = 0$. Then the method of measuring can be extended to $\mathbb{Q}$ as following:

$$v_p : \mathbb{Z} - \{0\} \rightarrow \mathbb{Z}, \quad \frac{m}{n} \mapsto v_p(m) - v_p(n)$$

And we make a convention that $v_p(0) = +\infty$, it is called **p-adic valustion**. for example $v_3(\frac{1}{3}) = -1$, $v_3(\frac{9}{2}) = 2$. Then we can get a measure of number

$$p^k \mathbb{Z}_{(p)} \iff \{x \in \mathbb{Q} \mid v_p(x) \geq p^k\}$$

but a question occurs here is that $v_3(1) = 0$ and $v_3(\frac{1}{3}) = -1$, as what we discussed just now, actually we hope the measure of 1 should be less than the measure of $\frac{1}{3}$ (here $0 > -1$), hence the valuation v_p can not be treated as the absolute value directly, hence we modify it as following:

$$| - |_p : \mathbb{Q} \rightarrow \mathbb{R}_+, \quad x \mapsto \frac{1}{e^{v_p(x)}}$$

the convention just now forces $|0|_p = 0$, and again we can get a clear measure of numbers:

$$x \in p^k \mathbb{Z}_{(p)} \iff |x|_p \leq e^{-k}$$

hence $|1|_3 < |\frac{1}{3}|_3$, which admits our original idea.

Remark 10.1. it is just a short introduction of p-adic number from the view of topological group, and here are something to supplement if to go deeper:

(1) The metric of is induced by the absolute value:

$$d(x, y) = |x - y|_p$$

we can verify that is a **ultrametric (non-archimedean metric)**, i.e. a metric satifying the strong triangle inequality:

$$d(x, z) \leq \max d(x, y), d(y, z), \quad \forall x, y, z \in \mathbb{Q}$$

which induces different analysis structure on $\mathbb{Q}$.

(2) Same situation with $\mathbb{Q}$ equipped by the usual absolute value, $(\mathbb{Q}, | - |_p)$ is not complete as a metric space, we can complete it to get the **p-adic number field** $\mathbb{Q}_p$. (The completion process is same with what we do for the real number, and we can prove that the completion is unique up to isometry.)

(3) By **Ostrowski's theorem**, any non-trivial absolute value on $\mathbb{Q}$ is equivalent to either the usual absolute value or some p -adic absolute value. In detail, if $| - |$ is a non-trivial absolute value on $\mathbb{Q}$, then either there is some $c > 0$ such that

$$|x| = |x|_\infty^c, \quad \forall x \in \mathbb{Q}$$

or there is some prime p and some $c > 0$ such that

$$|x| = |x|_p^c, \quad \forall x \in \mathbb{Q}$$

10.2 Homogeneous Spaces

Review that a group action $G \curvearrowright X$ is **transitive** if for any $x, y \in X$, there exists $g \in G$ such that $g \cdot x = y$, i.e. there is only one orbit.

Definition 10.3. A **homogeneous space** is a Hausdorff topological space X with a transitive continuous action of a group G , in particular, we call it **G-homogeneous space** to emphasize the acting group G .

Before talking about the properties of homogeneous space, we consider a basic example but important example: rotation action on sphere $SO(n) \curvearrowright \mathbb{S}^{n-1}$

$$SO(n) \times \mathbb{S}^{n-1}, \quad (A, v) \mapsto Av$$

the operation is defined by matrix multiplication, so it defines a continuous action, then we can verify the following results:

1. **The action is transitive.** When $n = 2$, it is clear since $SO(2)$ is just the set of rotations of plane, so we prove the case $n \geq 3$ by induction. We assume that the action is transitive for $n - 1, \dots$
2. **The isotropy group at e_1 is isomorphic to $SO(n-1)$.** Suppose that $A \in SO(n)$ is matrix such that $Ae_1 = e_1$, then by calculation we can verify that A is of the form

$$M = \begin{pmatrix} 1 & 0_{1 \times n} \\ 0_{n \times 1} & A' \end{pmatrix}, \quad A' \in SO(n-1).$$

Then it is easy to establish an isomorphism and a homeomorphism.

3. **There exists a homeomorphism $SO(n)/SO(n-1) \cong \mathbb{S}^{n-1}$.** By the stabilizer-orbit lemma, we can deduce a continuous bijection $SO(n)/SO(n-1) \rightarrow \mathbb{S}^{n-1}$, so it is enough to verify that $SO(n)$ is compact.
4. **$SO(n)$ is connected.** When $n = 2$, a classic homeomorphism $SO(2) \cong \mathbb{S}^1$ shows that $SO(2)$ is connected, then we can prove the case $n \geq 3$ by induction.

From this example, we can see here sphere $\mathbb{S}^n$ is a homogeneous space under the action of $SO(n+1)$, we are very familiar with the topology of sphere, but we are not familiar with the topology of $SO(n)$, so homogeneous space gives us a method to understand the topology of group by the homogeneous space. Conversely, if we are familiar with the structure of the topological group, we can understand a strange space by this method, anyway homogeneous space gives a deep relation

$$\{\text{topological group}\} \sim \{\text{space with same local structure}\}$$

Theorem 10.11. Let G be a Lie group and X be a G -homogeneous space, if I_x

the isotropy group at $x \in X$, then there exists a homeomorphism

$$G/I_x \cong X$$

10.3 The Pontryagin Duality

We consider the category of locally compact abelian groups:

$$\mathbf{LCA} \left| \begin{array}{l} \text{objects: locally compact abelian groups } (G, +) \\ \text{morphisms: continuous group homomorphisms} \end{array} \right.$$

without talking about the topology of the group, we can similarly consider the pure algebraic structure:

$$\mathbf{Ab} \left| \begin{array}{l} \text{objects: abelian group } (G, +) \\ \text{morphisms: group homomorphisms} \end{array} \right.$$

Clearly it is same as category of $\mathbb{Z}$ -modules, hence it is an abelian category with tools of homological algebra, so we can choose a certain objects of abelian groups to construct a duality.

$$\hat{G} := \text{Hom}_{\mathbb{Z}}(G, \mathbb{R}/\mathbb{Z})$$

Clearly, $\mathbb{R}/\mathbb{Z}$ is an injective $\mathbb{Z}$ -module, hence the Hom-functor will keep exactness. In particular, it will have a good behavior when group is finite.

Proposition 10.12. If G is a finite abelian group, then $\hat{G}$ is isomorphic to G .

Proof.

□

However, the duality fails when G is infinite. For example, if $G = \mathbb{Z}$, then $\hat{\mathbb{Z}} \cong \mathbb{R}/\mathbb{Z}$, but if we take dual again

$$\widehat{\mathbb{R}/\mathbb{Z}} = \text{End}_{\mathbb{Z}}(\mathbb{R}/\mathbb{Z}) \cong \prod_p \mathbb{Z}_p$$

clearly it is larger than $\mathbb{Z}$, hence it actually fails to be a duality. What we have done is to give a exact restriction on the category of abelian groups to make the duality work: It depends on the topology of the group, so we need to consider the topology of the dual group firstly. Formally, we define dual group as following:

Definition 10.4. Let $G \in \mathbf{LCA}$, we define its **dual group** as G^* or $\hat{G}$ by

$$G^* := \text{Hom}_{\mathbf{LCA}}(G, \mathbb{S}^1)$$

i.e. the group of continous group homomorphism from G to the circle group ($\mathbb{S}^1 \cong \mathbb{R}/\mathbb{Z}$). Among G^* , the element are called **characters** of G , and we usually denote it by

$$\chi : G \rightarrow \mathbb{S}^1$$

with operation defined by pointwise multiplication:

$$(\chi_1 + \chi_2)(g) := \chi_1(g) \cdot \chi_2(g)$$

Generally a Hom-functor is not necessary to keep category itself, it will maps to a category of sets, so we should verify that LCA is a good condition to make the functor closed.

Lemma 10.13. Let $G \in \mathbf{LCA}$, then its dual group G^* endowed with the compact-open topology is a locally compact abelian group, in particular

$$V(K, \varepsilon) = \{\chi \in G^* : \chi(K) \subset U_\varepsilon\}$$

for any compact subset $K \subset G$ and $U_\varepsilon = \{e^{it} \mid t \in (-\varepsilon, \varepsilon)\}$, it forms a neighborhood basis of $0 \in G^*$.

Proof. It is easy to verify that G^* is indeed a abelian group under defined operation, to verify it is a topological group, we need to verify the family $\mathcal{B}$ of all $V(K, \text{varepsilon})$ forms a neighborhood basis of identity of G^* : (1) zero character $g \mapsto 1$ is in any set clearly; (2) $V(K, r) \cap V(L, t) = V(K \cup L, \min(r, t))$, so it is stable under finite intersection; (3) $-V(K, \varepsilon) = V(K, \varepsilon)$ since $U_\varepsilon^{-1} = U_\varepsilon$; (4) $V(K, r) + V(L, t)$ is the subset of $V(K \cup L, r + t)$, so it is stable under operation. Hence it forms a neighborhood system of identity by (1)-(4).

Finally, we need to verify that G^* is locally compact by Ascoli's theorem... □

Remark 10.2. In particular, we denote evaluation map by

$$\text{ev} : G \times G^* \rightarrow \mathbb{S}^1, \quad (g, \chi) \mapsto \chi(g)$$

the compact-open topology is **the coarsest topology** such that the evaluation map is continuous (page 43 [1]).

A convention is that G^* always means the dual group with compact-open topology, so we finish the level of objects in the category, and then it is naturally to consider the functoriality, or the level of morphisms.

Lemma 10.14. Let $G, H \in \mathbf{LCA}$, and $f : G \rightarrow H$ be a continuous group homomorphism, then it induces a natural map

$$f^* : H^* \rightarrow G^*, \quad \chi \mapsto \chi \circ f$$

which is a continuous (under the compact-open topology) group homomorphism.

Proof. For any two characters $\chi_1, \chi_2 \in H^*$, we have

$$f^*(\chi_1 + \chi_2)(g) = (\chi_1 + \chi_2)(f(g)) = \chi_1(f(g)) \cdot \chi_2(f(g)) = f^*(\chi_1)(g) \cdot f^*(\chi_2)(g)$$

for any $g \in G$, so $f^*(\chi_1 + \chi_2) = f^*(\chi_1) + f^*(\chi_2)$ it is a group homomorphism. To prove the continuity, we take a basic open set $V_G(K, \varepsilon)$, then

$$\begin{aligned} (f^*)^{-1}(V_G(K, \varepsilon)) &= \{\chi \in H^* \mid \chi \circ f \in V_G(K, \varepsilon)\} \\ &= \{\chi \in H^* \mid \chi(f(K)) \subset U_\varepsilon\} \\ &= V_H(f(K), \varepsilon) \end{aligned}$$

here $f(K)$ is compact since f is continuous, so the preimage of any neighborhood of identity is open, hence f^* is continuous by translation. □

It is easy to verify that $(f \circ g)^* = f^* \circ g^*$, hence we have constructed a contravariant functor inside the category **LCA**.

Lemma 10.15. If $f : G \rightarrow H$ is a surjective continuous group homomorphism, then $f^* : H^* \rightarrow G^*$ is injective; If $f : G \rightarrow H$ is both injective and open, then $f^* : H^* \rightarrow G^*$ is surjective.

Proof. If f is surjective, we take any $\chi_1, \chi_2 \in H^*$ such that $f^*(\chi_1) = f^*(\chi_2)$, we assume that $\chi_1 \neq \chi_2$, then there exists $h \in H$ such that $\chi_1(h) \neq \chi_2(h)$, since f is surjective, there exists $g \in G$ such that $f(g) = h$, hence $f^*(\chi_1)(g) \neq f^*(\chi_2)(g)$, so it is absurd.

If f is injective, then injective module $\mathbb{S}^1$ allows us to extend a character $\chi : G \rightarrow \mathbb{S}^1$ to $\tilde{\chi} : H \rightarrow \mathbb{S}^1$ such that $\tilde{\chi} \circ f = \chi$. We need to verify the continuity of $\tilde{\chi}$: notice that f is open, so $f(G)$ is the open subgroup of H , hence we can prove that $\tilde{\chi}$ is continuous on $f(G)$, then we can conclude that $\tilde{\chi}$ is continuous on H by translation (a group homomorphism is continuous on some open subgroup, then the morphism is just continuous). $\square$

Theorem 10.16 (Pontryagin-Van Kampen, 1934).

Let $G \in \mathbf{LCA}$ and G^* be its dual group, then evaluation map gives a natural isomorphism in **LCA** by fixing one variable:

$$\text{ev} : G \rightarrow (G^*)^*, \quad g \mapsto \text{ev}_g$$

where $\text{ev}_g : G^* \rightarrow \mathbb{S}^1$ is defined by $\text{ev}_g(\chi) = \chi(g)$.

Proof.

$\square$

Here is the slogan of the duality:

every LCA-group is the dual group of its dual group.

The conclusion is similar with the finite-dimension vector space, hence it will be a strong tool to study the structure of LCA-groups and the analysis on it. In the sense of category theory, the Pontryagin duality is a contravariant equivalence with inverse itself:

$$(-)^* : \mathbf{LCA} \rightarrow (\mathbf{LCA})^{op}$$

Consequences

It is wonderful to see what we can get from the duality, we make a convention that G in this section always means a LCA-group.

Proposition 10.17. In the sense of topological isomorphism

$$G_1 \oplus G_2 \cong G_1 \times G_2$$

and

$$(G_1 \times G_2)^* \cong G_1^* \times G_2^*$$

Proof. In the sense of category of abelian groups, we have

$$(G_1 \times G_2)^* \cong G_1^* \oplus G_2^* \cong G_1^* \times G_2^*$$

So we just need to verify the isomorphism is homomorphism under the compact-open topology. $\square$

Proposition 10.18. If G is compact, then G^* is discrete; If G is discrete, then G^* is compact.

Proof. $\square$

Proposition 10.19. If G is compact group, then

- G is metrizable if and only if G^* is countable.
- G is connected if and only if G^* is torsion-free.

11 Classical Groups and Lie algebras

Matrix groups are the most important examples of Lie groups and Lie algebras, this section is some preparation for the general case.

11.1 Exponential map

Remember that $M_n(K) \cong K^{n^2}$ for any field (even any ring), in particular we consider $K = \mathbb{C}$ or $\mathbb{R}$, then the space of endomorphisms or matrices is just a finite-dimensional vector space over the field, so we can define the typical norm on it:

$$\begin{aligned}\|A\| &= \sqrt{\sum_{i,j} |a_{ij}|^2} \quad (\text{euclidean norm}) \\ \|A\|_\infty &= \max_{i,j} |a_{ij}| \quad (\text{Max norm}) \\ \|A\|_1 &= \sum_{i,j} |a_{ij}| \quad (\text{Sum norm})\end{aligned}$$

Moreover, we can consider the norm of $M_n(K)$ from another view, we view it as the linear continuous operator inside K^n , hence in particular the norm of K^n can induce the norm of operator:

$$\|A\|_{op} = \sup_{x \in K^n - \{0\}} \frac{\|Ax\|}{\|x\|}$$

Anyway, the space of endomorphisms on finite-dimensional vector space or the matrix space is still finite-dimensional, as a Banach space, all the norms on $M_n(K)$ is equivalent, with the basic knowledge of topology on it, we can talk about some whether some series in analysis makes sense or not, in particular, exponential map:

Definition 11.1. For a matrix $A \in M_n(K)$, the exponential of A is defined by

$$\exp(A) = \sum_{k \geq 0} \frac{A^k}{k!}$$

or sometimes we denote it by e^A .

The definition is analogous to the exponential map in $\mathbb{C}$, and we can verify that it is well-defined for any A by

$$\|\exp(A)\| \leq \sum_{k \geq 0} \frac{\|A\|^k}{k!} = \exp(\|A\|) < +\infty$$

so exponential map still makes sense for linear algebra! Some algebraic properties can be conclude to handle different calculation, and they are important.

Proposition 11.1.

Let $K = \mathbb{C}$ or $\mathbb{R}$, then

(1) $ge^Xg^{-1} = e^{gXg^{-1}}$ for any $g \in \text{GL}_n(K)$ and $X \in M_n(K)$.

(2) If $XY = YX$, then $e^{X+Y} = e^Xe^Y$.

(3) e^X is invertible with inverse e^{-X} .

Proof. (1) It is easy to check that

$$(gXg^{-1})^k = gX^kg^{-1}$$

thus by definition we have the equality. For (2), we can expand e^{X+Y} by definition and binomial formula, and the commutativity of X and Y allows us to rearrange the terms. For (3), we see the X as the complex matrix and then by Jordan form, we have

$$gXg^{-1} = D(I + N)$$

under the base change of g , where D is diagonal and N is nilpotent such that $DN = ND$, thus

$$ge^Xg^{-1} = e^De^{I+D}$$

with e^D is diagonal and e^{I+D} an upper-triangular matrix with non-zero diagonal entries, so we can conclude that e^X is invertible with inverse e^{-X} by the property (2). $\square$

There are many other proof of the proposition, and each of them gives a new view about the matrix group and Lie group, for example, we can consider the exponential map as the solution of ODE:

Theorem 11.2. For any $X \in M_n(K)$, the one-parameter

$$\Phi_X : \mathbb{R} \rightarrow M_n(K), \quad t \mapsto e^{tX}$$

is the unique **global** C^1 **solution** of the initial value problem of ODE:

$$\frac{d}{dt}Y(t) = X\Phi_Y(t), \quad \Phi_X(0) = I$$

with Y the matrix-valued function of t .

Proof. The uniqueness of the solution can be proved by Cauchy-Lipschitz theorem, for the existence, we should verify that Φ_X is exactly the solution: For any $t \in \mathbb{R}$

$$\frac{\Phi_X(t+h) - \Phi_X(t)}{h} = \frac{e^{(t+h)X} - e^{tX}}{h} = e^{tX} \frac{(e^{hX} - I)}{h}$$

The convergence of the exponential series is uniform, which allows us to exchange the limit and the sum, thus we can deduce the derivative of Φ_X is exactly $X\Phi_X$, hence Φ_X is the solution of the ODE (surely it is C^1 , Φ_X is derivative so it is continuous, that implies that $X\Phi_X$ is continuous too). $\square$

Remark 11.1. For property (2) in proposition 11.1, here we can consider the ODE

$$Z'(t) = (X + Y)Z(t), \quad Z(0) = I$$

we can verify that $t \mapsto e^{X+Y}$ and $t \mapsto e^X e^Y$ are both solutions when X and Y commute, thus they are equal by the uniqueness of the solution.

Remark 11.2. Actually, the theorem contains a simple information is that

$$\frac{d}{dt}(e^{tX}) = Xe^{tX}$$

so we notice the amazing similarity with the case in euclidean space, which allows us to consider the one-parameter subgroup as a dynamic system, and the exponential map is the flow of the system, this is the original ideal of the exponential map in Lie theory.

Another method to prove that e^X is invertible is to consider its determinant, which is natural to motivate the following result:

Theorem 11.3 (Jacobi's formula). For any $A \in M_n(K)$, we have

$$\det(e^A) = e^{\text{tr}(A)}$$

Proof. By Schur's lemma, any $X \in M_n(\mathbb{C})$ can be upper-triangularized by some $g \in GL_n(\mathbb{C})$ such that $X = gTg^{-1}$ the upper-triangular matrix T with diagonal entries $\lambda_1, \dots, \lambda_n$, thus

$$\det(e^X) = e^{\lambda_1} \dots e^{\lambda_n} = e^{\lambda_1 + \dots + \lambda_n} = e^{\text{tr}(X)}$$

$\square$

Another analytic aspects of the exponential map is asymptotic expansion, that means we can manipulate the exponential map as what we do in analysis, but first of all we prove that the exponential map is similarly a smooth map by the requirement of the expansion:

Proposition 11.4. For any $X \in M_n(K)$, the one-parameter

$$\Phi_X : \mathbb{R} \rightarrow GL_n(K), \quad t \mapsto e^{tX}$$

defines a real-analytic homomorphism, and we have the following expansion

$$(1) \exp(tX) \exp(tY) = \exp(t(X + Y) + \frac{1}{2}[X, Y] + O(t^3))$$

$$(2) \exp(tX) \exp(tY) \exp(-tX) \exp(-tY) = \exp(t^2[X, Y] + O(t^3))$$

for any $X, Y \in M_n(K)$, with bracket $[X, Y] = XY - YX$ the commutator of X and Y .

Proof. For any $t_0 \in \mathbb{R}$, we can expand Φ_X at t_0 by definition:

$$e^{tX} = e^{t_0X} e^{(t-t_0)X} = e^{t_0X} \sum_{k=0}^{\infty} \frac{(t-t_0)^k}{k!} X^k$$

The series converges uniformly on any compact interval, so the map is real-analytic. To prove the formula, we just need to expand the left-hand side by definition and rearrange the terms. $\square$

Remark 11.3. If $XY = YX$, then the commutator $[X, Y] = 0$, thus the expansion in (1) is exact, which is just the property (2) in proposition 11.1, hence we get the same result by using the expansion.

Above properties of one-parameter allows us to consider the general case in matrix group, later we will see that the matrix group is indeed a smooth manifold, i.e. a Lie group.

Theorem 11.5. Let $\gamma : \mathbb{R} \rightarrow GL_n(\mathbb{R})$ be a one-parameter subgroup, then there exists a unique $X \in M_n(\mathbb{R})$ such that $\gamma(t) = e^{tX}$ for any $t \in \mathbb{R}$, so any one-parameter subgroup of $GL_n(\mathbb{R})$ is exactly real-analytic.

Proof. convolution... $\square$

11.2 Correspondence between lie group and lie algebra

Formally, we define the real-general linear group as following:

$$GL_n(\mathbb{R}) := \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}$$

it is the subset of $M_n(\mathbb{R})$, while the topology of $GL_n(\mathbb{R})$ is just the subspace topology induced by $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, hence we can conclude something about the topology of the group:

- $GL_n(\mathbb{R})$ is a open subset of $M_n(\mathbb{R})$, it is immediately by

$$GL_n(\mathbb{R}) = \det^{-1}(\mathbb{R} - \{0\})$$

since determinant is a continuous map, so it can be viewed as a open subset of euclidean space $\mathbb{R}^{n^2}$, so it is a smooth manifold.

- Following the above result, since it can be viewed as an open subset of euclidean space, so it is a "bon espace" in the sense of topology, i.e. semi-locally simply connected, locally path-connected, locally compact, and Hausdorff.

- Consider the operation of the group

$$m(A, B) = AB^{-1} = \frac{1}{\det(B)} A \cdot \text{adj}(B)$$

where $\text{adj}(B)$ is the adjoint matrix of B , which is a polynomial of entries of B , hence this map is smooth as polynomial map, which shows that $GL_n(\mathbb{R})$ is a topological group, and in particular, with the good smooth structure, it is a Lie group.

- As the topological group, the local structure is same, so we can just consider the point $I \in GL_n(\mathbb{R})$. As a differential manifold, we can consider the tangent space at I , so we can identify the tangent space

$$T_I(GL_n(\mathbb{R})) \cong \mathbb{R}^{n^2} \cong M_n(\mathbb{R})$$

Actually, we can view a strong result: For any $X \in M_n(\mathbb{R})$, we define a curve on manifold by

$$\gamma_X : \mathbb{R} \rightarrow GL_n(\mathbb{R}), \quad t \mapsto e^{tX}$$

by exponential map, this is smooth natural curve with derivative $\gamma'_X(0) = X$, hence we prove that $M_n(\mathbb{R})$ is exactly the tangent space of $GL_n(\mathbb{R})$ at I , and we realize the connection between the local structure of the Lie group and its tangent space by one-parameter subgroup, that is the basic ideal of Lie theory.

We conclude above discussion by the following definition, or exactly the **vocabulary**, because they are just the new notation of manifold and vector space, but they are the basic object of Lie theory, so we need to give them a name:

Definition 11.2. A **lie group** is a smooth manifold G with a group structure such that the group operation $m : G \times G \rightarrow G$ defined by $m(g, h) = gh^{-1}$ is smooth.

A **real lie algebra** is a $\mathbb{R}$ -vector space $\mathfrak{g}$ with a bilinear map $[-, -] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ such that for any $x, y, z \in \mathfrak{g}$ and $a, b \in \mathbb{R}$, we have

$$(1) [x, y] = -[y, x] \text{ (antisymmetry)}$$

$$(2) [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0 \text{ (Jacobi identity)}$$

with above discuss we can see that $GL_n(\mathbb{R})$ is exactly a Lie group, while its tangent space at identity $M_n(\mathbb{R})$ is a vector space, actually it is a Lie algebra by defining the bracket as following:

$$[X, Y] = XY - YX$$

and we say $\mathfrak{h}$ is a **sub(-lie) algebra** of $\mathfrak{g}$ if $\mathfrak{h}$ is a subspace of $\mathfrak{g}$ and is closed under the bracket, i.e.

$$\forall x, y \in \mathfrak{h}, [x, y] \in \mathfrak{h}$$

To study other matrix groups, we can just consider the submanifold of $GL_n(\mathbb{R})$, i.e. the subgroup of $GL_n(\mathbb{R})$ with the subspace topology, hence we can get the following result:

Theorem 11.6. Let G be a closed subgroup of $GL_n(\mathbb{R})$, then G is a submanifold of $GL_n(\mathbb{R})$, and the tangent space of G at identity is a Lie algebra with

$$T_I(G) = \{M \in M_n(\mathbb{R}) \mid e^{tM} \in G, \forall t \in \mathbb{R}\}$$

it is a **subalgebra** of $M_n(\mathbb{R})$ with the bracket defined by commutator.

Proof. □

which motivates the following definition:

Definition 11.3. A **linear lie group** is a closed subgroup of $GL_n(\mathbb{R})$ for some n , and its lie algebra is defined as its tangent space

$$\text{Lie}(G) = \{M \in M_n(\mathbb{R}) \mid e^{tM} \in G, \forall t \in \mathbb{R}\}$$

and we usually denote it by the character $\mathfrak{g}$.

We invite the Lie algebra of $GL_n(\mathbb{R})$ by consider its tangent space at identity, conversely, exponential map allows us to know the local structure of the group by its Lie algebra, which can be done beyond the general manifold

Theorem 11.7. The exponential map $\exp : M_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ is a local diffeomorphism at 0, in particular, it is a diffeomorphism from some neighborhood of 0 to the image.

Proof. □

Let us consider some examples related to the general linear group:

Example 11.1. We view $\mathbb{C}$ as a real vector space, then there exists a natural embedding

$$\iota : M_n(\mathbb{C}) \rightarrow M_{2n}(\mathbb{R}), \quad A + iB \mapsto \begin{pmatrix} A & -B \\ B & A \end{pmatrix}$$

it is a injective continous linear map, and we can verify that

$$(A + iB)(x + iy) = 0 \iff \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

which shows that restriction of ι on $GL_n(\mathbb{C})$ takes image in $GL_{2n}(\mathbb{R})$, and image $\iota(GL_n(\mathbb{C}))$ is indeed a closed subgroup of $GL_{2n}(\mathbb{R})$, hence it is a linear lie group, and its lie algebra is exactly the image of ι on $M_n(\mathbb{C})$, which motivates the following diagram

$$\begin{array}{ccc} M_n(\mathbb{C}) & \xrightarrow{\iota} & M_{2n}(\mathbb{R}) \\ \exp \downarrow & & \downarrow \exp \\ GL_n(\mathbb{C}) & \xrightarrow{\iota} & GL_{2n}(\mathbb{R}) \end{array}$$

similarly with general linear group, the closed subgroup of $GL_n(\mathbb{C})$ is also a submanifold of $GL_n(\mathbb{C})$, and the tangent space at identity is the subalgebra of $M_n(\mathbb{C})$ with the bracket defined by commutator, hence we can get a large class of linear lie groups and lie algebras by this method, it is a big case of linear lie groups.

Example 11.2. (Affine group) We consider a closed subgroup of $GL_{n+1}(\mathbb{R})$ defined by

$$G = \left\{ \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \mid A \in GL_n(\mathbb{R}), b \in \mathbb{R}^n \right\}$$

and we consider the group action $G \curvearrowright \mathbb{R}^n$ defined by

$$\begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \cdot x = \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} = Ax + b$$

This action is transitive so $\mathbb{R}^n$ is a homogeneous space under G , and we can establish a homeomorphism

$$G/GL_n(\mathbb{R}) \cong \mathbb{R}^n$$

It is a classic example about Erlangen programme, here G is a important geometric group, we usually call it the **affine group** $\text{Aff}_n(\mathbb{R})$.

11.3 Adjoint map

Review a bsic fact about the finite-dimensional vector space: any norm on a finite-dimensional Banach space is equivalent, hence it is natural to consider the abstract exponential map on linear opeators

Definition 11.4. Let V be a finite-dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$, we define the exponential map $\text{Exp} : \text{End}(V) \rightarrow GL(V)$ by

$$\text{Exp}(T) = \sum_{k \geq 0} \frac{T^k}{k!}$$

where $T^k = T \circ \cdots \circ T$ is the k -th composition of T .

The definition is well-defined when we consider $\text{End}(V)$ as a Banach space, and we can verify that the properties in proposition 11.1 still hold for the abstract exponential map. In particular, if we choose a basis of V and denote it by $\mathcal{B}$, then we can see their connection by the following commutative diagram:

$$\begin{array}{ccc} \text{End}(V) & \xrightarrow{\text{Exp}} & GL(V) \\ \downarrow [\cdot]_{\mathcal{B}} & & \downarrow [\cdot]_{\mathcal{B}} \\ M_n(K) & \xrightarrow{\text{exp}} & GL_n(K) \end{array}$$

or simply the equality

$$[\text{Exp}(T)]_{\mathcal{B}} = \text{exp}([T]_{\mathcal{B}})$$

Our first aim is to construct the lie alegbra from exactly a lie group.

We consider the conjugation action of a linear lie group G on itself, and then for any $g \in G$, there exists a diffeomorphism by the smooth structure of G

$$c_g : G \rightarrow G, \quad h \mapsto ghg^{-1}$$

with $c_g(I) = I$, hence it allows us to consider the differential of c_g at identity,

$$(Dc_g)_I : T_I(G) \rightarrow T_I(G), \quad X \mapsto gXg^{-1}$$

the reason is as following: for linear lie group, the tangent space at identity is just $\text{Lie}(G)$ by theorem 11.6, so for any $X \in \text{Lie}(G)$, there exists a one-parameter subgroup $\gamma_X : \mathbb{R} \rightarrow G$ defined by $\gamma_X(t) = e^{tX}$ with $X = \left. \frac{d}{dt} \right|_{t=0} \gamma_X(t)$, hence we consider the parameterziation of the conjugation

$$\left. \frac{d}{dt} \right|_{t=0} c_g(\gamma_X(t)) = \left. \frac{d}{dt} \right|_{t=0} g\gamma_X(t)g^{-1} = gXg^{-1}$$

which shows that $(Dc_g)_I(X) = gXg^{-1}$ exactly, hence by the construction we induce a representation

$$\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g}), \quad g \mapsto \text{Ad}_g = (Dc_g)_I$$

and we call the representation Ad the **adjoint representation** of G .

Notice that $\mathfrak{g}$ is finite-dimensional, so $\text{GL}(\mathfrak{g})$ is also a linear Lie group, with its Lie algebra $\text{End}(\mathfrak{g})$, hence we can consider the differential of Ad at identity

Proposition 11.8. Let G be a linear lie group with lie algebra $\mathfrak{g}$, then the direction derivative of the adjoint representation at identity is

$$(D \text{Ad})_I(X) = \left. \frac{d}{dt} \right|_{t=0} \text{Ad}_{\exp(tX)} = \text{ad}_X$$

with $\text{ad}(X) : \mathfrak{g} \rightarrow \mathfrak{g}$ defined by $\text{ad}_X(Y) = XY - YX$ the commutator of X and Y .

Proof. caculation... □

Hence we get the basic idea of the adjoint

Definition 11.5. Let G be a linear lie group with its lie algebra $\mathfrak{g}$, then

- The adjoint representation of G is defined by

$$\text{Ad} : G \rightarrow \text{Aut}(\mathfrak{g}), \quad g \mapsto \text{Ad}_g$$

with $\text{Ad}_g(X) = gXg^{-1}$ for any $X \in \mathfrak{g}$.

- The adjoint representation of Lie algebra $\mathfrak{g}$ is defined by

$$\text{ad} : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g}), \quad X \mapsto \text{ad}_X$$

with $\text{ad}_X(Y) = [X, Y]$ for any $Y \in \mathfrak{g}$.

Remark 11.4. An application between two lie algebras $f : \mathfrak{g} \rightarrow \mathfrak{h}$ is a **morphism of Lie algebra** if it is a linear map and preserves the bracket

$$f([X, Y]_{\mathfrak{g}}) = [f(X), f(Y)]_{\mathfrak{h}}$$

In particular, the isomorphism in the category of lie algebras is called **automorphism**, and we use $\text{Aut}(\mathfrak{g})$ to denote the group of automorphisms of $\mathfrak{g}$, which is subspace of $\text{GL}(\mathfrak{g})$, the latest one is the set of all bijective linear maps on $\mathfrak{g}$.

with above discussion, we can describe a core properties of the adjoint representation, which is the connection between the local structure of the group and its lie algebra:

Theorem 11.9. Let G be a linear lie group with its lie algebra $\mathfrak{g}$, then for any $X \in \mathfrak{g}$, we have

$$\text{Ad}_{\exp(X)} = \text{Exp}(\text{ad}_X)$$

i.e. the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\text{ad}} & \text{End}(\mathfrak{g}) \\ \exp \downarrow & & \downarrow \text{Exp} \\ G & \xrightarrow{\text{Ad}} & \text{GL}(\mathfrak{g}) \end{array}$$

Proof.

□

12 Lie Algebra

In this section all vector space will be finite-dimensional.

12.1 Ideal

Definition 12.1. Let $\mathfrak{g}$ be a lie algebra with bracket, then $\mathfrak{h} \subset \mathfrak{g}$ is an ideal if $\mathfrak{h}$ is a linear subalgebra and $[g, h] \in \mathfrak{h}$ for any $g \in \mathfrak{g}$ and $h \in \mathfrak{h}$.

Ideal is like the ideal in ring, it allows us to talk about the quotient of Lie algebra, i.e. we consider the quotient of linear space firstly

$$\pi : \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h}, \quad X \mapsto X + \mathfrak{h}$$

and then π is a lie algebra morphism if and only if $\mathfrak{g}/\mathfrak{h}$ is a well-defined lie algebra, that allows us to write

$$[x + \mathfrak{h}, y + \mathfrak{h}] = [\pi(x), \pi(y)] = \pi([x, y]) = [x, y] + \mathfrak{h}$$

which ensures us to define the bracket on the quotient linear space such that $\mathfrak{g}/\mathfrak{h}$ is a lie algebra, and similarly we have the isomorphism theorem.

Theorem 12.1. If $f : \mathfrak{g} \rightarrow \mathfrak{h}$ is a lie algebra morphism, then $\ker f$ is an ideal of $\mathfrak{g}$ and $\text{Im } f$ is a subalgebra of $\mathfrak{h}$, and we have the isomorphism of lie algebra

$$\mathfrak{g}/\ker f \cong \text{Im } f$$

Remark 12.1. Here are some quick notes about the ideal:

1. The intersection of ideals and sum of ideals are still ideals.
2. The intersection of ideal I and subalgebra $\mathfrak{g}$ will be the ideal of subalgebra, and it is necessary to be the ideal of the original algebra.
3. Two important examples of ideal, the first one is the **centre of a lie algebra**

$$Z(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0, \forall Y \in \mathfrak{g}\}$$

and the second one is the **derived algebra** of a lie algebra

$$[\mathfrak{g}, \mathfrak{g}] = \left\{ \sum_{i=1}^n c_i [x_i, y_i] \in \mathfrak{g} \mid x_i, y_i \in \mathfrak{g}, c_i \in k, n \in \mathbb{N} \right\}$$

4. We can define the ideal generated by two subsets A and B by

$$[A, B] = \text{span}\{[a, b] \mid a \in A, b \in B\}$$

A notation should be clarified is that sometimes we write $\mathfrak{a}\mathfrak{b}$ to instead of $[\mathfrak{a}, \mathfrak{b}]$, the reason is that the bracket is just a non-associative product defined on a module.

Definition 12.2. Let $\mathfrak{g}$ be a lie algebra, and we define

- The **central series** of $\mathfrak{g}$ is defined by

$$\mathcal{C}(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}], \quad \mathcal{C}^{n+1}(\mathfrak{g}) = [\mathfrak{g}, \mathcal{C}^n(\mathfrak{g})]$$

for any $n \geq 1$.

- The **derived series** of $\mathfrak{g}$ is defined by

$$\mathcal{D}(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}], \quad \mathcal{D}^{n+1}(\mathfrak{g}) = [\mathcal{D}^n(\mathfrak{g}), \mathcal{D}^n(\mathfrak{g})]$$

for any $n \geq 1$.

This is two types of decreasing chains of ideals, sometimes we swap the notation as following:

$$\mathcal{C}^n(\mathfrak{g}) = \mathfrak{g}^n, \quad \mathcal{D}^n(\mathfrak{g}) = \mathfrak{g}^{(n)}$$

for any $n \geq 1$, the reason is same as the notation of derived algebra. Another useful notation is for quotient algebra, we can write

$$\mathcal{C}^n(\mathfrak{g}/\mathfrak{h}) = \mathcal{C}^n(\mathfrak{g}) + \mathfrak{h}/\mathfrak{h}, \quad \mathcal{D}^n(\mathfrak{g}/\mathfrak{h}) = \mathcal{D}^n(\mathfrak{g}) + \mathfrak{h}/\mathfrak{h}$$

Definition 12.3. Let $\mathfrak{g}$ be a lie algebra, then

- $\mathfrak{g}$ is **nilpotent** if there exists some n such that $\mathcal{C}^n(\mathfrak{g}) = 0$.
- $\mathfrak{g}$ is **solvable** if there exists some n such that $\mathcal{D}^n(\mathfrak{g}) = 0$.

Here are some results immediately by the definition:

1. nilpotent implies solvable since $\mathcal{D}^n(\mathfrak{g}) \subset \mathcal{C}^n(\mathfrak{g})$ for any n .
2. If $\mathfrak{g}$ is nilpotent and non-zero, then $Z(\mathfrak{g})$ is non-zero.
3. If $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent, then $\mathfrak{g}$ is solvable.
(Hint: by definition we can prove $\mathcal{D}^{n+1}(\mathfrak{g}) \subset \mathcal{C}^n([\mathfrak{g}, \mathfrak{g}])$)

The two properties will keep stable under the morphism of lie algebra, that is the following result:

Proposition 12.2. Let $\mathfrak{g}$, $\mathfrak{h}$ and $\mathfrak{f}$ be lie algebras satisfying the short exact sequence

$$0 \rightarrow \mathfrak{f} \rightarrow \mathfrak{g} \rightarrow \mathfrak{h} \rightarrow 0$$

then $\mathfrak{g}$ is nilpotent (resp. solvable) if and only if $\mathfrak{f}$ and $\mathfrak{h}$ are nilpotent (resp. solvable), equivalently it is enough to prove the three properties:

- (1) The subalgebra of a nilpotent (resp. solvable) lie algebra is still nilpotent (resp. solvable).
- (2) The quotient of a nilpotent (resp. solvable) lie algebra is still nilpotent (resp. solvable).
- (3) If $\mathfrak{g}/\mathfrak{h}$ and $\mathfrak{h}$ are nilpotent (resp. solvable), then $\mathfrak{g}$ is nilpotent (resp. solvable).

Proof. □

upper-triangular matrix

We have seen that many matrix ring is exactly a lie algebra, and each linear lie group matches a lie algebra, which motivates to consider the representation of an abstract lie algebra.

Definition 12.4.

Lemma 12.3. For any representation $\rho : \mathfrak{g} \rightarrow \text{End}(V)$ with V non-zero, if $\rho(x)$ is nilpotent for any x , then there exists some non-zero vector $v \in V$ such that $\rho(x)(v) = 0$ for any x .

Theorem 12.4 (Engel's theorem). Let $\mathfrak{g}$ be a lie algebra, then $\mathfrak{g}$ is nilpotent if and only if ad_x is nilpotent for any $x \in \mathfrak{g}$.

Proof. It is clearly all adjoint map is nilpotent in $\text{End}(\mathfrak{g})$ if $\mathfrak{g}$ is nilpotent, so it is enough to prove the converse. If $\mathfrak{g} = 0$ or $\dim \mathfrak{g} = 1$, then $\mathfrak{g}$ is nilpotent, so we can assume for any $\dim \mathfrak{g} = n \geq 2$, if ad_x is nilpotent for any x , then $\mathfrak{g}$ is nilpotent.

By lemma 12.3 there exists $y \in \mathfrak{g}$ such that $\text{ad}_x(y) = [x, y] = 0$ for any $x \in \mathfrak{g}$, so it means $Z(\mathfrak{g}) \neq 0$. Then we can consider the quotient algebra $\mathfrak{g}/Z(\mathfrak{g})$ with dimension less than n , and any adjoint map of $\mathfrak{g}/Z(\mathfrak{g})$ is nilpotent if any adjoint map of $\mathfrak{g}$ is nilpotent, hence by induction $\mathfrak{g}/Z(\mathfrak{g})$ is nilpotent, so $\mathfrak{g}$ is nilpotent since $Z(\mathfrak{g})$ is always nilpotent. □

Theorem 12.5 (Lie's theorem). Let $\mathfrak{g}$ be a **solvable** lie algebra over an **algebraically closed field with characteristic zero**, then for any finite-dimensional representation $\rho : \mathfrak{g} \rightarrow \text{End}(V)$, there exists $v \in V - 0$ such that

$$\rho(x)(v) = \lambda(x) \cdot v$$

for some scalar function $\lambda : \mathfrak{g} \rightarrow K$ and any $x \in \mathfrak{g}$.

Proof. □

12.2 Killing form

Definition 12.5. A **non-abelian** lie algebra $\mathfrak{g}$ is simple if $\mathfrak{g} \neq 0$ and the only ideals are 0 and $\mathfrak{g}$.

Remark 12.2. The condition of non-abelian is necessary, otherwise the definition of ideal will be reduced to the linear subspace, and the unique case of "simple" algebras is one-dimensional lie algebra.

Proposition 12.6. The lie algebra $\mathfrak{g}$ is simple if and only if the adjoint representation $\text{ad} : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g})$ is irreducible.

Proof. Suppose that the adjoint representation is not irreducible, then there exists a proper non-zero subspace $V \subset \mathfrak{g}$ such that $\text{ad}_X(V) \subset V$ for any $X \in \mathfrak{g}$, i.e.

$$\forall X \in \mathfrak{g}, \forall Y \in V, \quad [X, Y] \in V$$

it shows that V is an proper non-zero ideal of $\mathfrak{g}$, hence $\mathfrak{g}$ is not simple. Conversely, notice that if $\mathfrak{h}$ is an ideal of $\mathfrak{g}$, then

$$\forall X \in \mathfrak{g}, \forall Y \in \mathfrak{h}, \quad [X, Y] \in \mathfrak{h}$$

i.e. which shows that $\mathfrak{h}$ is an invariant subspace of the adjoint representation, hence if $\mathfrak{g}$ is simple, then the adjoint representation is irreducible. $\square$

Definition 12.6. The radical of a lie algebra $\mathfrak{g}$ is defined as the maximal solvable ideal of $\mathfrak{g}$, and we denote it by $\text{rad}(\mathfrak{g})$.

Remark 12.3. The definition is well-defined since **the sum of two solvable ideals is still a solvable ideal**, alternatively, we can define the radical ideal as

$$\text{rad}(\mathfrak{g}) = \sum \mathfrak{h}$$

where the sum runs over all solvable ideals $\mathfrak{h}$ of $\mathfrak{g}$, and the finite-dimensional condition ensures the choice of ideals is finite, hence the sum is exactly an ideal.

Definition 12.7. The killing form of a lie algebra $\mathfrak{g}$ is the trace form of its adjoint representation, i.e. it is the bilinear form defined by

$$\kappa(x, y) = \text{tr}(\text{ad}_x \circ \text{ad}_y)$$

Remark 12.4. Here are some properties of the killing form:

1. The killing form is symmetric since the trace is symmetric.
2. The killing form is invariant under the adjoint representation

$$\kappa([x, y], z) + \kappa(y, [x, z]) = 0$$

Proposition 12.7. Let $\mathfrak{g}$ be a lie algebra, then the following statements are equivalent:

- (a) The only abelian ideal is $\{0\}$.
- (b) $\text{rad}(\mathfrak{g}) = \{0\}$

If the base field is of characteristic 0

- (c) The killing form of $\mathfrak{g}$ is non-degenerate.
- (d) $\mathfrak{g}$ is a direct sum of simple lie algebras.

Proof.

□

13 Field and Galois Theory

To study the structure of the field, consider a natural map: Let K be a field

$$\phi : \mathbb{Z} \rightarrow K, \quad 1 \mapsto 1_K$$

It is a well-defined homomorphism, which invites a ideal of $\mathbb{Z}$

$$\ker \phi = \{n \in \mathbb{Z} | n \cdot 1_K = 0_K\}$$

By the structure of $\mathbb{Z}$, so there exists a integer $p \in \mathbb{Z}$ such that $\ker \phi = p\mathbb{Z}$, which gives the definition of the characteristic of the field.

Definition 13.1. For any field K , we define $\text{char}(K)$ be the characteristic of the field: either 0 or the smallest integer $n \in \mathbb{N}$ such that $n \cdot 1_K = 0$, by the natural map equivalently

$$\begin{cases} \text{char}(K) = 0 \iff \text{im} \phi \cong \mathbb{Z} \\ \text{char}(K) = p \iff \text{im} \phi \cong \mathbb{Z}/p\mathbb{Z} \end{cases}$$

The characteristic of a field is either zero or a prime number. Suppose that $\text{char}(K) = p \neq 0$, if $p = ab$ is not a prime, then we will get two zero divisor $a \cdot 1_K$ and $b \cdot 1_K$, but a field can not contain any zero divisor, so p must be prime. An amazing thing in a field (or ring) with non-zero characteristic p is that we can write a equation

$$(x + y)^p = x^p + y^p$$

it has an interesting name: **freshman's dream**, this is an equality that would be written by someone who has studied very little mathematics or is just beginning to learn it.

Another natural map is **Frobenius endomorphism**, let K be a field with non-zero characteristic p , then we define

$$\sigma : K \rightarrow K, \quad x \mapsto x^p$$

It is a well-defined injective field homomorphism, so it is a endomorphism, but it is not necessary surjective.

Example 13.1. Consider $K = \mathbb{F}_p(x)$ a field of rational functions. $\text{char}(K) = p$ since $\text{im} \phi = \mathbb{F}_p$, and it is easy to observe that there exists no $f \in \mathbb{F}_p(x)$ such that $f^p(x) = x$, so the Frobenius map here is not surjective.

A good case is finite field, in that case frobenius endomorphism is furthermore a automorphism, so the frobenius map will be in galois group and it reflects the structure of the finite field.

13.1 Finite extension

For the beginning, notice a classic isomorphism:

$$\mathbb{R}[X]/(X - a) \cong \mathbb{R}$$

For any $a \in \mathbb{R}$, and we review the isomorphism given in complex number field:

$$\mathbb{R}[X]/(X^2 + 1) \cong \mathbb{C}$$

Clearly, $\mathbb{C}$ is a larger field containing $\mathbb{R}$, but we wonder how we can get it from the algebra structure. Here it motivates us to consider the question from the polynomial structure defined on a field. Firstly we need some lemmas, it is very clear after commutative ring:

Lemma 13.1. Let L be a field containing K as a subfield, then L is a vector space over K .

This lemma allows us to give a rough classification of field extension:

Definition 13.2. If L is a field containing K as a subfield.

- L/K or $K \subset L$ is denoted a **field extension** of K .
- L/K is a **finite extension** if L is a finite-dimensional vector space over K . The **degree** of the finite extension is defined as $[L : K] := \dim_K L$.
- $a \in L$ is **algebraic** over K if there exists a non-zero polynomial $f \in K[X]$ such that $f(a) = 0$, L/K is an **algebraic extension** if every element in L is algebraic over K .
- a is **transcendental** over K if it is not algebraic over K .

Another lemma is the key to construction of field, the reason is that in a PID ring a ideal is maximal if and only if it is generated by an irreducible element.

Lemma 13.2. If K a field and $p \in K[X]$, then p is irreducible if and only if $K[X]/(p)$ is a field.

Hence the a contrsuction of field extension can be given by the quotient of polynomial ring over a field by a good choice of ideal:

Proposition 13.3. Let K be a field and I be a principiapl ideal generated by a monic irreducible polynomial $p \in K[X]$ of degree d , Let $L = K[X]/I$, then

- (1) L is a field and K can be embedded in L , so K can be identified as a subfield of L .
- (2) p has a root β in L , exactly $\beta = X + I \in L$
- (3) If $g \in K[X]$ and β is a root of g in L , then $p|g$.

- (4) p is the unique monic irreducible polynomial in $K[X]$ having β as a root in L .
- (5) L is a finite extension of K of degree d with the basis $\{1, \beta, \dots, \beta^{d-1}\}$.

Proof. (1) L is a field by above lemma, and we consider an embedding $i(a) = a + I$ for any $a \in K$, clearly it is injective and actually it is $\pi|_K$, where π is the canonical map from $K[x]$ to $K[x]/I$.

(2) Suppose $p(x) = a_0 + \dots + a_d x^d$, then in L , we have

$$\begin{aligned} p(\beta) &= a_0 + a_1\beta + \dots + a_d\beta^d \\ &= a_0 + a_1(X + I) + \dots + a_d(X + I)^d \\ &= a_0 + a_1(X + I) + \dots + a_d(X^d + I) \\ &= p(X) + I = I \end{aligned}$$

Here $p(X) \in I$ and I is the zero element in L .

(3) If p does not divide g , then $\gcd(p, g) = 1$ since p is irreducible, so PID ring $K[X]$ implies the existence of r, t in $K[X]$ such that

$$1 = s(x)p(x) + t(x)g(x)$$

put $x = \beta$ in L , then $1 = 0$ leads to a contradiction.

(4) immediately from (3). For (5) we use euclidean division for polynomial, any $f \in K[X]$, there exists $q, r \in K[X]$ with $\deg r < d$ such that $f(x) = q(x)p(x) + r(x)$, then $f + I = r + I$ in L , and if $r(x) = b_0 + \dots + b_k x^k$, $k < d$, by operations of ideal

$$r + I = b_0 + b_1\beta + \dots + b_k\beta^k$$

so $\{1, \beta, \dots, \beta^{d-1}\}$ spans L . They are linearly independent because if we assume they are linearly dependent, that means there exists a polynomial $h \in K[X]$ of degree $< d$ such that h has β as the root, but by (3) we know that $p|h$, which leads to a contradiction since $d = \deg p \leq \deg h$. $\square$

Review the example in the beginning, the complex number field can also be constructed by adjoining element i to $\mathbb{R}$ such that $i^2 + 1 = 0$, and we always write a complex number of the form $a + ib$ with a, b real numbers, which is another construction of field extension, and more generally we have the following proposition:

Proposition 13.4. Let K be a field and $f \in K[X]$ irreducible, adjoin element $c \notin K$ such that c is a root of f , then $K[c]$ is a field containing K as a subfield, and it can be written as $K(c)$.

$$K(c) = \{a_0 + a_1c + \dots + a_k c^k \mid a_1, \dots, a_k \in K, k = \deg f - 1\}$$

Proof. $K[c]$ naturally is a ring with same identity with K , so we just need to find the inverse. For any polynomial $p \in K[X]$ with degree less than $\deg f$, it is co-prime with

f since f is irreducible, so by Bezout's theorem for polynomials, there exists $r, t \in K[X]$ such that

$$p(x)r(x) + t(x)f(x) = 1$$

hence we take $x = c$ then immediately $p(c)r(c) = 1$, so in $K[c]$ we find the inverse of $p(c)$. and notice that $K[c]$ is a field guarantee any rational polynomial can have the form of polynomial, so $K[c] = K(c)$ evidently. $\square$

Here we find two methods to extend the field, the first method is consider the quotient of polynomial ring, and we can embed K into it; the second is more direct, we just add some element in the field.

Theorem 13.5 (Structure of field extension).

Let L/K be field extension of K and $a \in L$ is algebraic, then

(1) There exists a unique **monic** irreducible polynomial in $K[X]$ having a as a root, formally we call it **minimal polynomial** of a over K , and denote it by Π_a .

(2) If $I = (\Pi_a)$, then there exists an isomorphism

$$\phi : K[X]/I \rightarrow K(a)$$

with $X + I \mapsto a$ and $c + I \mapsto c$ for all $c \in K$.

(3) Let L'/K is another field extension and $a' \in L'$ is also a root of Π_a , then there exists an isomorphism $\psi : K(a) \rightarrow K(a')$ such that $\psi|_K = \text{id}_K$ and $\psi(a) = a'$.

Proof. It needs to consider the **valuation map**, we define $h : K[X] \rightarrow L$ by $p \mapsto p(a)$, then clearly $\ker h$ contains all polynomials having a as a root, then notice that $K[X]$ is a principal ideal ring, so $\ker h$ must be the form (p) with monic $p \in K[X]$. By first isomorphism theorem, we know that $K[X]/I \cong K(a)$, then by Lemma 13.2 p must be irreducible, hence we prove (1) and (2), and we can draw commutative diagram to finish (3):

$$\begin{array}{ccc} K[X] & \xrightarrow{h} & K[a] \\ \downarrow h' & \searrow \pi & \uparrow \simeq \\ K[a'] & \xleftarrow{\simeq} & K[X]/I \end{array}$$

where $h'(p) = p(a')$, and π is the canonical map. $\square$

Remark 13.1. The theorem shows an induction

$$\{\text{algebraic elements in } L/K\} \rightsquigarrow \{\text{monic irreducible polynomial over } K\}$$

conversely it is not right since the field extension L may be not enough large, which refers the "**closure**" in the sense of algebraic element. For statement (3), a simple example is $\mathbb{Q}(\sqrt{2}) = \mathbb{Q}(-\sqrt{2})$, and monic irreducible polynomial $X^2 - 2 \in \mathbb{Q}[X]$.

13.2 Algebraic Closed field

13.3 Normal extension and Automorphism

Example 13.2. We consider the field extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$, here the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is $X^3 - 2$, we can solve it in $\mathbb{C}$ with roots:

$$\sqrt[3]{2}, \quad \sqrt[3]{2}\omega, \quad \sqrt[3]{2}\omega^2 \quad \text{with } \omega = e^{2\pi i/3}$$

by above theorem we can conclude $\mathbb{Q}(\sqrt[3]{2}) \cong \mathbb{Q}(\sqrt[3]{2}\omega) \cong \mathbb{Q}(\sqrt[3]{2}\omega^2)$, and it also implies a **question**: quotient ring such as $\mathbb{Q}[X]/(X^3 - 2)$ can not construct a field containing all roots of irreducible polynomial such as $X^3 - 2$.

However, we can construct a larger field since $\mathbb{Q}(\sqrt[3]{2})$ has been a field: equivalently, we can adjoin ω to $\mathbb{Q}(\sqrt[3]{2})$, or consider the irreducible polynomial $X^2 + X + 1$ over $\mathbb{Q}(\sqrt[3]{2})$, then we can get a larger field $\mathbb{Q}(\sqrt[3]{2}, \omega)$ containing all roots of $X^3 - 2$.

Above is a typical example to motivate the definition of splitting field, and we can give a more general definition:

Suppose L/K is a extension and a polynomial $f \in K[X]$, we say f **splits** over K if there exists $c, a_1, \dots, a_n \in K$ such that f can be factorized as linear terms:

$$f(x) = c(x - a_1)(x - a_2) \cdots (x - a_n)$$

above examples shows that a extension L/K can not always contain all roots of a polynomial, and we can construct a larger field.

Theorem 13.6 (splitting field).

Let K be a field and $f \in K[X]$ with degree $n \geq 1$, then there exists a smallest field extension L/K such that f splits over L , formally such a field extension is called **splitting field** of f over K , and it satisfies:

- $L \cong K[r_1, \dots, r_n]$ with $r_1, \dots, r_n$ are the roots of f in L , it shows that the splitting field is unique up to isomorphism.

- As a finite extension, $[L : K] \leq n!$

Proof.

□

Remark 13.2. L.Kronecker proved that for any field K and $f \in K[X]$, there exists a extension L/K such that f splits over L . The smallest of such extension is the splitting field of f , and an extremely larger field is the algebraic closed field of K , although the result is a little old in nowadays.

For the example at the beginning, the experience of $\mathbb{C}$ gives us a hint that $\mathbb{Q}(\sqrt[3]{2})$ can not

contain the complex number ω , which reminds us to consider the roots of the polynomial in algebraic closure.

Definition 13.3. Let L/K be an algebraic extension, we define

$$\text{Hom}_{K\text{-alg}}(L, \bar{K}) = \{f : L \hookrightarrow \bar{K} \mid f|_K = \text{id}_K\}$$

to be **K -embedding of L** .

Remark 13.3. It is natural to consider the category of field extensions of K , we denote it by $\mathbf{Ext}_K$. Its objects are the extension $K \subset L$, or we can see it as a K -algebra by an embedding $\tau : K \hookrightarrow L$. Hence its morphism is just the morphism of K -algebra, for example if we consider $f : L \rightarrow L'$, then f must be injective since it is a morphism of field, and $f(r) = rf(1) = r$ for any $r \in K$, in particular $L' = \bar{K}$ we get the definition of K -embedding of L . Hence here we get a (non-full) subcategory of $\mathbf{CAlg}_K$.

For any $L/\mathbb{Q}$ algebraic extension, we consider f is a embedding, then we have

$$f(X^3 - 2) = f(X)^3 - 2 \quad f(X^2 + X + 1) = f(X)^2 + f(X) + 1$$

for any $X \in L$. which allows us to calculate that

$$\text{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{Q}(\sqrt[3]{2}), \bar{\mathbb{Q}}) = \{f_1, f_2, f_3\}$$

with $f_i(\sqrt[3]{2}) = \sqrt[3]{2}\omega^{i-1}$ for $i = 1, 2, 3$.

$$\text{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{Q}(\sqrt[3]{2}, \omega), \bar{\mathbb{Q}}) = \{f_{1,1}, f_{1,2}, f_{1,3}, f_{2,1}, f_{2,2}, f_{2,3}\}$$

with $f_{i,j}(\sqrt[3]{2}) = f_i(\sqrt[3]{2})$ and $f_{i,j}(\omega) = \omega^j$ for $i = 1, 2, 3$ and $j = 1, 2$. We can observe that the second case forms exactly a group since the image of $f_{i,j}$ is $\mathbb{Q}(\sqrt[3]{2}, \omega)$ itself, and in particular it can be identified as S_3 (notice $f_{2,1}$ is of order 3 and $f_{1,2}$ is of order 2, they are not commute).

Definition 13.4. Let L/K be a field extension, we define

$$\text{Gal}(L/K) = \text{Hom}_{K\text{-alg}}(L, L) = \{\sigma \in \text{Aut}(L) \mid \sigma|_K = \text{id}_K\}$$

to be the **Galois group** (or **automorphism group**) of L/K , it is a subgroup of $\text{Aut}(L)$ which fixes K .

Notice a clear inclusion

$$\text{Gal}(L/K) \subset \text{Hom}_{K\text{-alg}}(L, \bar{K})$$

which is sufficient to prove that when $L = \mathbb{Q}(\sqrt[3]{2}, \omega)$, the inclusion turns to be an equality. while $L = \mathbb{Q}(\sqrt[3]{2})$, we can only get $\text{Gal}(L/\mathbb{Q}) = \{f_1\}$, hence the difference between Galois group and the embeddings is the key to distinguish the extensions.

Theorem 13.7. Let L/K be a algebraic extension, then the following conditions are equivalent:

1. L is a splitting field of a set of polynomials in $K[X]$.
2. $\text{Gal}(L/K) = \text{Hom}_K(L, \bar{K})$
3. For any $a \in L$, the minimal polynomial $\Pi_a \in K[X]$ splits over L .

If one of the above conditions is satisfied, we say L/K is a **normal extension**.

Remark 13.4. Let $a \in L$ and Π_a is the minimal polynomial of a , then K -embedding induces a bijection to the roots of Π_a in $\bar{K}$, denote the set by R_a

$$\text{Hom}_K(L, \bar{K}) \rightarrow R_a, \quad f \mapsto f(a)$$

exactly

13.4 Separable extension

Example 13.3. Let us consider the function field $K = \mathbb{F}_p(t)$

- $X^p - t$ is irreducible in $K[X]$ since K is the fraction field of $\mathbb{F}_p[t]$, and

$$\mathbb{F}_p[t][X] \cong \mathbb{F}_p[t, X] \cong \mathbb{F}_p[X, t]$$

allow us to prove the irreducibility in $\mathbb{F}_p[t][X]$, so it holds in fraction field as well.

- consider $L = K[X]/(X^p - t)$, then we can get a extension $L \cong \mathbb{F}_p(t^{1/p})$, which allows to factorize the minimal polynomial

$$X^p - t = (X - t^{1/p})^p$$

That shows L is a splitting field of $X^p - t$ over K , so L/K is a normal extension of degree p .

- We exactly construct a extension L/K such that the minimal polynomial has repeated root, in this case the $\bar{K}$ -embedding fails to distinguish the roots and $\text{Gal}(L/K) = \text{Hom}_K(L, \bar{K}) = \{id\}$.

Review the case in $\mathbb{R}[X]$, although we do not prove the FTA exactly, but the our experience in $\mathbb{C}$ enough to conclude that any irreducible polynomial in $\mathbb{R}[X]$ can not have multiple roots in $\mathbb{C}$, since for the given distinct roots $R = \{r_1, \dots, r_k\}$ with multiplicity $\{m_1, \dots, m_k\}$, we can uniquely determine the monic polynomial of the form

$$\prod_{r_i \in \mathbb{R} \cap R} (X - r_i)^{m_i} \cdot \prod_{\text{otherwise}} (X^2 + b_j x + c_j)^{m_j}$$

since the complex roots always appear in pairs, so we can roughly conclude the result. Even more, we can not find a counterexample in $\mathbb{Q}[X]$ as well, this section is to distinguish such a field.

Definition 13.5. Let A be a k -algebra, an endomorphism $\partial : A \rightarrow A$ is called a **derivation** if it satisfies the Leibniz rule:

$$\forall f, g \in A, \quad \partial(fg) = \partial(f)g + f\partial(g)$$

It is easy to verify that

$$\begin{aligned} \partial(\lambda) &= 0, \quad \forall \lambda \in k \\ \partial(f^n) &= n f^{n-1} \partial(f), \quad \forall f \in A, n \in \mathbb{N} \end{aligned}$$

We can use the derivation to define the formal derivative of a polynomial, which is a good view that does not depend on the the exactly form of the polynomial.

$$\Omega_{L/K} = 0 \iff L/K \text{ is separable}$$

This is the general explanation of the separable extension by Kähler differentials. Exactly in the definition we take $A = k[X]$, and then endomorphism will be determined by the image of X by the universal property of polynomial ring, then we can get the formal derivative of any polynomial

Definition 13.6. Let $\partial(X) = 1$, then for any $f = \sum_n a_n X^n \in k[X]$, we can define the formal derivative of f by

$$f'(X) := \partial(f) = \sum_n n a_n X^{n-1}$$

and a polynomial f is called **separable** if f is coprime with its formal derivative f' in $k[X]$.

Notice that $k[X]$ is a PID, so the coprime condition is equivalent to the existence of $r, s \in k[X]$ such that

$$1 = r(X)f(X) + s(X)f'(X)$$

we describe the separable polynomial without talking about roots of polynomial, and similarly we can give a definition of roots of a polynomial via arithmetic of polynomial:

Definition 13.7. Let $f \in k[X]$ and $a \in k$, then a is a **root** of f if there exists $k \in \mathbb{N}^*$ such that

$$(X - a)^k \mid f(X)$$

and the **order of the root** is defined as the largest integer m such that

$$(X - a)^m \mid f(X) \wedge (X - a)^{m+1} \nmid f(X)$$

in particular, a is a **simple root** if the order of the root is 1, and it is a multiple root if the order of the root is greater than 1.

Remark 13.5. Following this definition, $f(a) = 0$ if a is a root of f , the converse is not right in general, the ring should at least be an integral domain (**sure for field**) or the polynomial should be monic, the proof depends on the division of polynomial.

Proposition 13.8. For polynomial $f \in k[X]$, the following conditions are equivalent:

1. f is separable.
2. f has no multiple root in $\bar{k}$.
3. f and f' has no common root in a algebraic closure $\bar{k}$ of k .
4. f has exactly $\deg f$ distinct roots in $\bar{k}$.
5. $\bar{k}[X]/(f)$ is reduced and isomorphic to $\bar{k}^{\deg f}$.

Proof. (1) $\Rightarrow$ (2): If f has a multiple root a , then $(X - a)^2 \mid f(X)$, so $f(X) = (X - a)^2 g(X)$ for some $g \in k[X]$, then we can calculate the formal derivative of f :

$$f'(X) = 2(X - a)g(X) + (X - a)^2 g'(X)$$

hence we get $(X - a) \mid f(X)$ and $(X - a) \mid f'(X)$, which contradicts the coprime condition.

(2) $\Rightarrow$ (3): in $\bar{k}$, we can factorized f as linear terms, and the formal derivative of f can be caculated of the form

$$f'(X) = \sum_a \frac{f(X)}{X - a}$$

where a runs over all roots of f , we suppose that b is a common root of f' , then $f'(b) = f(X)/(X - b)|_{b=0} = 0$ implies that $X - b$ divides $f(X)/ - b$, which gives a contradiction.

(3) $\Rightarrow$ (4): similar argument as above.

(4) $\Rightarrow$ (5): The isomorphism can be proved by the **chinese remainder theorem**, and the product of fields is reduced surely.

(5) $\Rightarrow$ (4): Notice that $\bar{k}[X]/(X - a)^2$ is not reduced with $\overline{X - a}^2 = 0$ in the quotient, and for any f we can factorized it to get a isomorphism

$$\bar{k}[X]/(f) \cong \prod_a \bar{k}[X]/(X - a)^{m_a}$$

with a distinct roots of f and m_a the multiplicity, so surely m_a should be all 1.

(4) $\Rightarrow$ (1): caculation. □

actually we can characterize the irreducible polynomial if it is separable, it depends on the characteristic of the field.

Proposition 13.9. Let f be an irreducible polynomial in $k[X]$, then f is inseparable if and only if $f' = 0$, which shows that

- If $\text{Char}(k) = 0$, any irreducible polynomial is separable.
- If $\text{Char}(k) = p$, any irreducible polynomial is of the form $g(X^{p^r})$ with g a separable irreducible polynomial and $r \in \mathbb{N}$, and the representation is unique.

Proof. In $k[X]$ (a UFD), f and f' coprime means their gcd associates to 1, or equivalently the gcd is a unit. When f is irreducible, then f is a prime element in $k[X]$, so the gcd is either f or a unit. If f is inseparable, then the gcd is f , so $f|f'$ implies $f' = 0$ since $\deg f' < \deg f$.

□

Definition 13.8. Let L/K be a algebraic extension, then L/K is called a **separable extension** if for any $a \in L$, the minimal polynomial Π_a of a over K is separable.

Remark 13.6. Any subextension of a separable extension is separable: Let $K \subset E \subset L$ be a tower of algebraic extension, and L/K is separable. If $a \in E$, then $\Pi_{a,E}$ divides $\Pi_{a,L}$, and factor of separable polynomial is still separable (verify).

By proposition above, we can conclude that **any algebraic extension of a field with characteristic zero is separable**, in particular the extension of $\mathbb{Q}$, it is the core object in number theory. Hence, it is hopeless to find a example of inseparable extension of $\mathbb{Q}$ or $\mathbb{R}$, when we change our attention to field of non-zero characteristic, we can find a such example like beginning of this section.

Theorem 13.10. Let L/K be a finite extension, then the following conditions are equivalent:

1. L/K is separable.
2. $|\text{Hom}_K(L, \bar{K})| = [L : K]$
3. $\bar{K} \otimes_K L$ is reduced and is isomorphic to $\bar{K}^{[L:K]}$.

Proof.

□

Remark 13.7. When the extension is infinite, the condition 2 makes no sense, but it is still make sense to consider the condition 3. In the example of $L = \mathbb{F}_p(t^{1/p})$, we can find that the degree is p while the number of embedding is 1, so it is another way to show that L is an inseparable extension.

Corollary 13.11 (Primitive element theorem). Any finite separable extension is a simple extension, that is, for any finite separable extension L/K , there exists $a \in L$ such that $L = K(a)$, such a element a is called a **primitive element** of the extension.

Proof.

□

Remark 13.8. If $L = K(a)$ a simple algebraic extension, we denote the roots of the

minimal polynomial Π_a in $\bar{K}$ by R , then we can get a natural bijection:

$$\text{Hom}_K(L, \bar{K}) \rightarrow R, \quad f \mapsto f(a)$$

It is well-defined since embedding sends a to another root, and the map is always injective: For any $f, g \in \text{Hom}_K(L, \bar{K})$ with $f(a) = g(a)$, take any $x \in L = K(a)$, then there exists a polynomial $p \in K[X]$ such that $x = p(a)$, so we can calculate

$$f(x) = f(p(a)) = p(f(a)) = p(g(a)) = g(p(a)) = g(x)$$

which shows that $f = g$.

Hence the simple algebraic extension allows to identify the conjugations of the primitive element with the K -embeddings of the extension, and surely simple algebraic extension is not necessary separable, for example $L = \mathbb{F}_p(t^{1/p})$.

13.5 Galois extension

Let K/k be a finite extension, with $\text{Aut}(K/k)$ the automorphism group then we can consider the group action $\text{Aut}(K/k) \curvearrowright K$ defined by

$$\sigma \cdot a := \sigma(a)$$

the fixed point of the action will be remarkable:

Lemma 13.12. Let $S \subset \text{Aut}(K/k)$ be a non-empty subset, then

$$K^S := \{x \in K \mid \sigma(x) = x \text{ for all } \sigma \in S\}$$

is a intermediate field of K/k .

Proof. $\sigma \in \text{Aut}(K/k)$ fixes k , so $k \subset K^S$. For any $x, y \in K^S$ and $\sigma \in S$, we can calculate

$$\sigma(x + y) = \sigma(x) + \sigma(y) = x + y$$

$$\sigma(xy) = \sigma(x)\sigma(y) = xy$$

it is enough to show the result. □

Theorem 13.13. Let K/k be a finite extension, then the following conditions are equivalent:

1. K/k is normal and separable.
2. For any $a \in K$, the minimal polynomial of a splits over K with

$$\Pi_a(X) = \prod_{\sigma \in \text{Aut}(K/k)} (X - \sigma(a))$$

3. K is a splitting field of a separable polynomial in $k[X]$.
4. $|\text{Aut}(K/k)| = [K : k]$
5. Fixed field $K^{\text{Aut}(K/k)} = k$.

Proof. (1) implies (2): Since K/k is normal, then $\text{Aut}(K/k)$ is just the k -embedding onto itself, that means $\sigma(a)$ runs over all roots of Π_a in $\bar{k}$ exactly, and since K/k is separable, the roots are of one-multiplicity, so Pi_a splits like the form in (2).

(2) implies (3): Π_a is a separable polynomial in $K[X]$ and splits over K , so we can consider the splitting field of all minimal polynomials.

(3) implies (1), it is immediate.

(3) implies (4): Since (1) and (3) are equivalent, so it is enough to show that (1) implies (4), and it can be proved by cardinal

$$|\text{Aut}(K/k)| = |\text{Hom}(K, \bar{k})| = [K : k]$$

The first equality holds since K/k is normal, and the second equality holds since K/k is separable.

(4) implies (5): Set $k' = K^{\text{Aut}(K/k)}$, then we can prove that $\text{Aut}(K/k') = \text{Aut}(K/k)$, and then we can prove that $k' = k$ by proving that

$$[K : k'] \leq [K : k] = |\text{Aut}(K/k)| = |\text{Aut}(K/k')| \leq [K : k']$$

the first inequality holds since $k \subset k'$, and the last inequality holds since for any finite extension K/k we have

$$\text{Aut}(K/k) \subset \text{Hom}(K, \bar{k}), \quad |\text{Hom}(K, \bar{k})| \leq [K : k]$$

(5) implies (1): For any $a \in K$, we can set consider the orbit of a , it contains the distinct roots of Π_a in K , and then we set $f(X) = \prod_{r \in \text{Orb}(a)} (X - r)$. For any $\sigma \in \text{Aut}(K/k)$, we can view that $\sigma(f) = f$ since σ just permutes the roots

$$\sigma(f) = \prod_{r \in \text{Orb}(a)} (X - \sigma(r)) = \prod_{r \in \text{Orb}(\sigma(a))} (X - r) = f$$

that means any coefficient of f will lie in $K^{\text{Aut}(K/k)} = k$, i.e. $f \in k[X]$. Hence Π_a divides f since $f(a) = 0$, and Π_a is separable since f is separable. So we can actually show (5) implies (3), and (3) implies (1) as before. $\square$

The equivalent statement motivates us to give the definition:

Definition 13.9. An algebraic extension K/k is called a **Galois extension** if it is normal and separable, and in this case we call $\text{Aut}(K/k)$ the **Galois group** of the extension, and denote it by $\text{Gal}(K/k)$.

Another **alternative definition** of galois extension is an algebraic extension K/k such that $K^{\text{Aut}(K/k)} = k$, which is equivalent to the definition above by the theorem without limitation on the finiteness of the extension.

To establish the correspondence between the intermediate fields of a galois extension and the subgroups of the galois group, we investigate an example carefully:

Example 13.4. Let $K = \mathbb{Q}(\sqrt[3]{2}, \omega)$ be the extension of $\mathbb{Q}$, it is exactly a galois extension with $\text{Gal}(K/\mathbb{Q}) = S_3$. We set $a = \sqrt[3]{2}$, then we can denote the basis by

$$1, a, a^2, \omega, a\omega, a^2\omega$$

Exactly the galois group can be generated by two elements $g = f_{1,2}$ and $h = f_{2,1}$, with

$$g(a) = a\omega, \quad g(\omega) = \omega$$

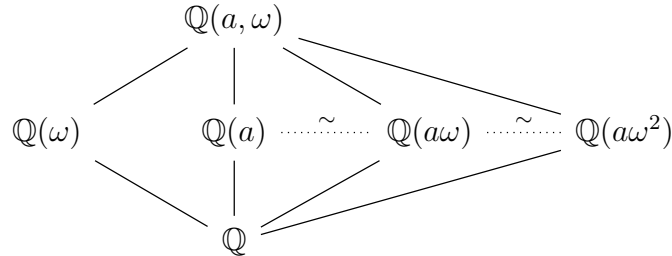
$$h(a) = a, \quad h(\omega) = \omega^2$$

so we can find that g is of order 3 and fixes $\mathbb{Q}(\omega)$, while h is of order 2 and fixes $\mathbb{Q}(\sqrt[3]{2})$, i.e. we construct two intermediate fields in according to the subgroups

$$K^G = \mathbb{Q}(\omega), \quad G = \langle g \rangle \cong \langle (1, 2, 3) \rangle \cong A_3$$

$$K^H = \mathbb{Q}(\sqrt[3]{2}), \quad H = \langle h \rangle \cong \langle (1, 2) \rangle$$

Similarly, we can recover all possible intermediate fields by considering the subgroups of S_3 , and it can be drawn as following



where the intermediate fields correspond to all non-trivial proper subgroups of S_3 :

$$A_3, \langle (1, 2) \rangle, \langle (2, 3) \rangle, \langle (1, 3) \rangle$$

Notice that A_3 is the unique normal subgroup of S_3 , while $\mathbb{Q}(\omega)$ is the unique sub-galois extension of $K/\mathbb{Q}$, so it motivates us to consider the general case.

Proposition 13.14. Let K/k be a galois extension with L a intermediate field, then

1. K/L is a galois extension.
2. $\text{Gal}(K/L)$ is a subgroup of $\text{Gal}(K/k)$, if extension K/k is finite, then the index of galois group is $[L : k]$.

3. For any $\sigma \in \text{Gal}(K/k)$, we have conjugation

$$\text{Gal}(K/\sigma(L)) = \sigma \text{Gal}(K/L) \sigma^{-1}$$

4. L/k is galois if and only if $\text{Gal}(K/L)$ is a normal subgroup of $\text{Gal}(K/k)$, and in this case we have an isomorphism

$$\text{Gal}(L/k) \cong \text{Gal}(K/k) / \text{Gal}(K/L)$$

Proof. (1) For any $a \in K$, Let $\Pi_a \in k[X]$ be the minimal polynomial of a over k , and $M_a \in L[X]$ be the minimal polynomial of a over L , then M_a divides Π_a , it implies that M_a is separable since Π_a is separable. What's more, the roots of M_a are also the roots of Π_a , and Π_a takes all roots in K , so does M_a , hence K/L is normal and separable.

(2) Since $k \subset L$, hence $\text{Gal}(K/L) \subset \text{Gal}(K/k)$, and it is not hard to verify it is exactly a subgroup, for index we can conclude that

$$[\text{Gal}(K/k) : \text{Gal}(K/L)] = [K : k] / [K : L] = [L : k]$$

The first equality holds since the two extensions are galois, and the second holds by tower law.

(3) For any σ we have

$$\begin{aligned} f \in \text{Gal}(K/\sigma(L)) &\iff f(\sigma(x)) = \sigma(x), \quad \forall x \in L \\ &\iff \sigma^{-1} \circ f \circ \sigma|_L = \text{id}_L \\ &\iff \sigma^{-1} \circ f \circ \sigma \in \text{Gal}(K/L) \\ &\iff f \in \sigma \text{Gal}(K/L) \sigma^{-1} \end{aligned}$$

(4) If L/k is galois, then $\text{Gal}(L/k) = \text{Hom}_k(L, \bar{k})$ since it is normal, hence for any $\sigma \in \text{Gal}(K/k)$, when consider $\sigma|_L$ as the restriction map, it will be contained in the k -embedding of L , so $\sigma(L) = L$, which shows that the following map is well-defined

$$\text{Gal}(K/k) \rightarrow \text{Gal}(L/k), \quad \sigma \mapsto \sigma|_L$$

the kernel of the map is exactly $\text{Gal}(K/L)$, so it is enough to prove it is surjective.

Conversely, if $\text{Gal}(K/L)$ is normal, then for any $\sigma \in \text{Gal}(K/k)$ we have

$$\sigma(L) = K^{\text{Gal}(K/\sigma(L))} = K^{\sigma \text{Gal}(K/L) \sigma^{-1}} = K^{\text{Gal}(K/L)} = L$$

so again above map is well-defined and surjective, so any $\tau \in \text{Gal}(L/k)$, there exists $\sigma \in \text{Gal}(K/k)$ such that $\tau(L) = \sigma|_L(L) = L$, which shows that L/k is normal, so it is galois as well since the sub-extension of a separable extension is separable.

□

Definition 13.10. Let $(\mathcal{F}, \leq_F)$ and $(\mathcal{G}, \leq_G)$ be two partially ordered sets, a application $r : \mathcal{F} \rightarrow \mathcal{G}$ is called **order-preserving (isotone)** if for any $x, y \in \mathcal{F}$ with $x \leq_F y$, we have $r(x) \leq_G r(y)$. Conversely, is is called **order-reversing (antitone)** if $r(x) \geq_G r(y)$.

The galois correspondence can be stated by the language of category, firstly any partial ordered set $(P, \leq)$ can be viewed as a category as following:

$$\mathcal{C}_P \mid \begin{array}{l} \text{objects: } x \in P \\ \text{morphisms: } \text{Hom}(x, y) = \{*\} \text{ if } x \leq y, \text{ otherwise } \emptyset \end{array}$$

hence an isotone (antitone) bijection between two partially ordered sets can be viewed as an (anti)equivalence between the corresponding categories, so we can restate the galois correspondence as following:

Theorem 13.15 (Galois correspondence).

Let K/k be a galois extension, let $\mathcal{I}$ be the set of intermediate fields of K/k and $\mathcal{S}$ be the set of subgroups of $\text{Gal}(K/k)$, then they are all partial ordered sets by inclusion $\subset$, and there exists an antitone bijection between them by

$$L \mapsto \text{Gal}(K/L) \quad \text{with inverse} \quad H \mapsto K^H$$

In another word, the antitone as a functor gives an equivalence

$$\mathcal{C}_{\mathcal{I}} \simeq \mathcal{C}_{\mathcal{S}}^{op}$$

Proof.

□

As we have seen in the previous example ($a = \sqrt[3]{2}$), we can consider the galois extension tower

$$\mathbb{Q} \subset \mathbb{Q}(\omega) \subset \mathbb{Q}(a, \omega)$$

with Galois groups $\text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}) \cong \mathbb{Z}/(2)$ and $\text{Gal}(\mathbb{Q}(a, \omega)/\mathbb{Q}) \cong \mathbb{Z}/(3)$, while in the theory of group the symmetric group can be viewed as a semi-direct product

$$S_3 = \mathbb{Z}/(3) \rtimes \mathbb{Z}/(2)$$

so it is natural to ask whether the structure of galois group can be reflected in a extension tower.

Definition 13.11. Let K/k and L/k be two extensions contained in an extension Ω/k , then we define the composite of K and L over k by

$$KL = \langle K \cup L \rangle = \bigcap_{K \cup L \subset E \subset \Omega} E$$

as the smallest subextension of Ω/k containing both K and L .

Remark 13.9. For any two algebraic extensions, we can always take $\Omega = \bar{k}$ to define, Furthermore if K and L are finite extensions, then we always have

$$[KL : K] \leq [K : k][L : k]$$

since if we write $K = k(a_1, \dots, a_n)$ and $L = k(b_1, \dots, b_m)$, clearly $KL \subset k(a_i b_j)$ since $K \cup L \subset k(a_i b_j)$. In the category of $\mathbf{Ext}_k$

Remark 13.10. Composite of extensions is **stable** under the separable and normal condition, hence the composite of galois extensions is still a galois extension, if the composite is exactly a pushout, we can talk something about the structure of galois group.

Proposition 13.16. Let K_1 and K_2 be the separable extensions of k , denote $K_{12} = K_1 K_2$, then the following conditions are equivalent:

1. The canonical map $K_1 \otimes_k K_2 \rightarrow K_{12}, x_1 \otimes x_2 \mapsto x_1 x_2$ is an isomorphism.
2. $K_1 \otimes_k K_2$ is a integral domain.
3. For any $a \in K_2$, the minimal polynomial of a over k is still irreducible in $K_1[X]$.

If K_1/k and K_2/k are finite

4. $[K_{12} : k] = [K_1 : k][K_2 : k]$

If K_1/k or K_2/k are galois (i.e. normal here)

5. $K_1 \cap K_2 = k$

If one of above conditions holds, we say that K_1 and K_2 are **linearly disjoint** over k , and K_{12} is the **extension étrangère**.

Remark 13.11. The normal condition here is necessary, consider the previous example $a = \sqrt[3]{2}$, we take $K_1 = \mathbb{Q}(a)$ and $K_2 = \mathbb{Q}(a\omega)$, then we can find that $K_{12} = \mathbb{Q}(a, \omega)$ exactly, while

$$9 = [K_1 : k][K_2 : k] \neq [K_{12} : k] = 6$$

although $K_1 \cap K_2 = \mathbb{Q}$ exactly.

Proof.

□

Corollary 13.17. Along above notation, if K_{12}/k and K_1/k are finite galois extensions, and K_{12} is a extension étrangère, then galois group of the composite extension is **the inner semi-direct product**

$$\text{Gal}(K_{12}/K_1) \rtimes \text{Gal}(K_{12}/K_2) \cong \text{Gal}(K_{12}/k)$$

In particular the product is reduced to direct product **if and only if** K_2/k is a galois extension.

Proof.

□

14 Theory of Algebraic Number

14.1 Norm, Trace and Discriminant

For any finite extension L/K of degree n , we can view element $a \in L$ as a K -linear map

$$m_a : L \rightarrow L, \quad x \mapsto ax$$

in particular, it is an automorphism of L if $a \neq 0$, and it satisfies several properties:

- $m_{a+b} = m_a + m_b$
- $m_{ab} = m_a \circ m_b$
- $m_{c \cdot a} = c \cdot m_a$

with $a, b \in L$ and $c \in K$. Since L is a finite-dimensional, it is natural to consider m_a as a matrix and define two invariants in linear algebra

- Trace of a over K : $\text{Tr}_{L/K}(a) = \text{Tr}(m_a)$

- Norm of a over K : $N_{L/K}(a) = \det(m_a)$

exactly what we do can be concluded as the following

$$L \hookrightarrow \text{End}_K(L) \rightarrow K$$

the first map refers to a **regular representation** of L over K , and the second map is what we are familiar with, and we can verify some properties of trace and norm:

- $\text{Tr}_{L/K}(a + b) = \text{Tr}_{L/K}(a) + \text{Tr}_{L/K}(b)$
- $\text{Tr}_{L/K}(c \cdot a) = c \cdot \text{Tr}_{L/K}(a)$
- $N_{L/K}(ab) = N_{L/K}(a) \cdot N_{L/K}(b)$
- $N_{L/K}(c \cdot a) = c^n \cdot N_{L/K}(a)$
- $\text{Tr}_{L/K}(c) = nc$ and $N_{L/K}(c) = c^n$

with $a, b \in L$ and $c \in K$. We can also give a more explicit description of trace and norm by the minimal polynomial of a over K , moreover, for any endomorphism m_a we can consider its **characteristic polynomial** $\chi_a(t) = \det(t \cdot \text{id}_L - m_a)$, and conclude the following result:

Proposition 14.1. For any L/K finite extension with degree n and $a \in L$,

(a) $\chi_a = (\Pi_a)^d$ with $d = [L : K(a)]$.

(b) Suppose $\Pi_a(t) = t^m + c_{m-1}t^{m-1} + \dots + c_0$, then

$$\mathrm{Tr}_{L/K}(a) = -dc_{m-1} \quad \mathrm{N}_{L/K}(a) = (-1)^n c_0^d$$

Proof. (a) It is clear that $\deg \Pi_a = [K(a) : K]$, we denote it by m , then

$$\Pi_a(t) = t^m + c_{m-1}t^{m-1} + \dots + c_0$$

for some $c_i \in K$, and $\{1, a, \dots, a^{m-1}\}$ is a basis for $K(a)/K$, we suppose that $\{e_1, \dots, e_d\}$ is a basis for $L/K(a)$, then we can conclude a basis of L/K is

$$e_1, \dots, e_1 a^{m-1} \mid e_2, \dots, e_2 a^{m-1} \mid \dots \mid e_d, \dots, e_d a^{m-1}$$

so we can write the matrix of m_a under this basis

$$m_a \sim \mathrm{diag}(A_1, \dots, A_d), \quad A_i = \begin{pmatrix} 0 & 0 & \dots & -c_0 \\ 1 & 0 & \dots & -c_1 \\ 0 & 1 & \dots & -c_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -c_{m-1} \end{pmatrix}$$

By calculate the characteristic polynomial of A_i we can get exactly

$$\chi_a(t) = \prod_{i=1}^d \det(t \cdot I - A_i) = (\Pi_a(t))^d$$

(b) By vieta's formula we can conclude that

$$\chi_a(t) = t^n - \mathrm{Tr}_{L/K}(a)t^{n-1} + \dots + (-1)^n \mathrm{N}_{L/K}(a)$$

then by formula of (a) we can calculate the result. □

The invariant behaves well in separable extension (especially, Galois extension), for example the number fields, the reason is that we can talk about the conjugations of elements with invariant.

Proposition 14.2. Let L/K be a separable finite extension of degree n , then

- $\mathrm{N}_{L/K}(a) = \prod_{\sigma} \sigma(a)$

- $\mathrm{Tr}_{L/K}(a) = \sum_{\sigma} \sigma(a)$

- $\chi_a(X) = \prod_{\sigma} (X - \sigma(a))$

where $a \in L$ and $\sigma \in \mathrm{Hom}_K(L, \bar{K})$ runs over all K -embeddings

Proof. We consider a relation of equivalence for $\Sigma = \text{Hom}_K(L, \bar{K})$ by

$$\sigma_1 \sim_a \sigma_2 \iff \sigma_1(a) = \sigma_2(a)$$

it holds for any finite extension, then we calculate the class of equivalence:

$$\begin{aligned} [\sigma] &= \{\tau \in \Sigma \mid \tau(a) = \sigma(a)\} \\ &= \{\tau \in \Sigma \mid \tau(p(a)) = \sigma(p(a)), \forall p \in K[X]\} \\ &= \{\tau \in \Sigma \mid \tau = \sigma \text{ on } K(a)\} \end{aligned}$$

which shows that $|\llbracket \sigma \rrbracket| = |\text{Hom}_{K(a)}(L, \bar{K})|$, in particular, if L/K is separable, then $L/K(a)$ is also separable, so exactly $|\llbracket \sigma \rrbracket| = [L : K(a)]$, it implies

$$\prod_{\sigma} (X - \sigma(a)) = \prod_{[\sigma] \in \Sigma / \sim_a} (X - \sigma(a))^{[L:K(a)]} = \Pi_a(X)^{[L:K(a)]} = \chi_a(X)$$

Here $\sigma(a)$ runs over all roots of Π_a since the restriction map

$$\text{res} : \text{Hom}_K(L, \bar{K}) \rightarrow \text{Hom}_K(K(a), \bar{K}), \quad \sigma \mapsto \sigma|_{K(a)}$$

is surjective.

□

Definition 14.1. Let L/K be a finite extension of degree n

- The **trace pairing** of the extension is a symmetric K -bilinear form

$$Q_{L/K}(x, y) = \text{Tr}_{L/K}(xy)$$

- we define the **discriminant** of a family $\{e_1, \dots, e_n\}$ in L by

$$\text{Disc}(e_1, \dots, e_n) = \det(\text{Tr}_{L/K}(e_i e_j)) = \det(Q_{L/K}(e_i, e_j))$$

It is better to consider the discriminant in a separable extension, although we always use the "**free condition**" in the extension of $\mathbb{Q}$, the reason is as following:

Proposition 14.3. Let L/K be a finite extension, then the following conditions are equivalent:

1. L/K is separable.
2. $\text{Tr}_{L/K} \neq 0$.
3. The trace pairing $Q_{L/K}$ is non-degenerate.

Proof.

□

Remark 14.1. Consider the inseparable example of $L = \mathbb{F}_p(t^{1/p})$, we can calculate $\text{Tr}_{L/K}(u^k) = 0$ with $u = t^{1/p}$ and $k = 0, 1, \dots, p-1$, which shows that the trace is identically zero, so the discriminant of any basis is zero, which provides no information about the extension.

Proposition 14.4. Let L/K be a finite separable extension of degree n , and $a_1, \dots, a_n$ be a basis of L/K , then the discriminant of the basis

$$\text{Disc}(a_1, \dots, a_n) = \det(\text{Tr}_{L/K}(\sigma_i(a_j)))^2$$

with $\sigma_i \in \text{Hom}_K(L, \bar{K})$ n distinct K -embeddings of L .

Proof. Review that for any two n order square matrices A, B , we have

$$(AB)_{i,j} = \sum_{k=1}^n A_{i,k} B_{k,j}$$

In particular

$$(A^T A)_{i,j} = \sum_{k=1}^n A_{k,i} A_{k,j}$$

so For any (i, j) -entry of trace matrix, applying proposition 14.2 we can calculate

$$\text{Tr}_{L/K}(a_i a_j) = \sum_{k=1}^n \sigma_k(a_i) \sigma_k(a_j) = \sum_{k=1}^n \sigma_k(a_i) \sigma_k(a_j) = \sum_{k=1}^n = (A^T A)_{i,j}$$

where $A = \text{Tr}_{L/K}(\sigma_i(a_j))_{1 \leq i, j \leq n}$. □

Corollary 14.5. Let L/K be a finite separable extension of degree n , then the discriminant of any basis of L/K is no-zero, in particular, if a is the primitive element such that $L = K(a)$ and $\{1, \dots, a^{n-1}\}$, then

$$\text{Disc}(1, a, \dots, a^{n-1}) = \prod_{i < j} (r_i - r_j)^2$$

where r_i are the distinct n roots of Π_a in $\bar{K}$.

Proof. Review that

$$(PAP^T)_{i,j} = \sum_{1 \leq k, l \leq n} P_{i,k} P_{j,l} A_{k,l}$$

For any two n order square matrices A, P . For any basis $\{e_1, \dots, e_n\}$, there exists a invertible matrix P such that $e_i = \sum_{j=1}^n P_{ij} a^{j-1}$, since $\{1, \dots, a^{n-1}\}$ is a basis, then

$$\begin{aligned} \text{Tr}_{L/K}(e_i e_j) &= \text{Tr}_{L/K} \left(\left(\sum_{k=1}^n P_{ik} a^{k-1} \right) \left(\sum_{l=1}^n P_{jl} a^{l-1} \right) \right) \\ &= \sum_{k, l=1}^n P_{ik} P_{jl} \text{Tr}_{L/K}(a^{k+l-2}) \\ &= \sum_{k, l=1}^n P_{ik} P_{jl} \text{Tr}_{L/K}(a^{k+l-2}) \end{aligned}$$

which shows that the trace matrix of the basis is $P^T AP$, where $A = (\text{Tr}_{L/K}(a^{i-1}a^{j-1}))_{i,j}$, which suffice to calculate that

$$\text{Disc}(e_1, \dots, e_n) = |\det(P)|^2 \text{Disc}(1, a, \dots, a^{n-1})$$

Hence just need to calculate the discriminant of the basis $\{1, \dots, a^{n-1}\}$, and prove that it is non-zero, which finishes the proof. By the above proposition, we can calculate

$$\text{Disc}(1, a, \dots, a^{n-1}) = \begin{vmatrix} 1 & \sigma_1(a) & \dots & \sigma_1(a)^{n-1} \\ 1 & \sigma_2(a) & \dots & \sigma_2(a)^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \sigma_n(a) & \dots & \sigma_n(a)^{n-1} \end{vmatrix}^2$$

where $\sigma_i(a)$ runs over all distinct roots since L/K is separable, hence we can conclude the result by the **Vandermonde determinant formula**.

Remark 14.2. A remarkable fact in the proof is the scale of the discriminant, suppose that $\{e_1, \dots, e_n\}$ and $\{a_1, \dots, a_n\}$ are two family with $[e_1, \dots, e_n] = P[a_1, \dots, a_n]$ for some endomorphism P , then

$$\text{Disc}(e_1, \dots, e_n) = |\det(P)|^2 \text{Disc}(a_1, \dots, a_n)$$

□

Example 14.1. If $P(X) = X^n + aX + b$, the the discriminant of the polynomial is

$$\text{Disc}(P) = (-1)^{n(n-1)/2} [n^n b^{n-1} + (1-n)^{n-1} a^n]$$

14.2 Number fields

To motivate the algebraic approach, we firstly consider a typical example in number theory, it will be the basis of the discussion.

- The finite extension of $K/\mathbb{Q}$ is called a **number field**, typically it an separable extension since $\text{Char}(\mathbb{Q}) = 0$, so we can apply **primitive element** theorem to conclude that $K = \mathbb{Q}(a)$ for some $a \in K$.
- The element $a \in K$ is called an **algebraic integer** if there exists a monic polynomial $f(x) \in \mathbb{Z}[x]$ such that $f(a) = 0$. Notice that the following conditions are equivalent:
 - (a) a is an algebraic integer.
 - (b) minimal polynomial $\Pi_a(X) \in \mathbb{Z}[X]$.
 - (c) $\mathbb{Z}[a]$ is a finitely generated $\mathbb{Z}$ -module.

Proof.

□

- Following the above definition, we can verify a similar result with the algebraic numbers: (a) sum of two algebraic integers is an algebraic integer, (b) product of two algebraic integers is an algebraic integer.

Proof. □

- It is trivial that algebraic integers are algebraic numbers, so similarly we can define

$$\tilde{\mathbb{Q}} = \{x \in \bar{\mathbb{Q}} : x \text{ is an algebraic integer}\}$$

It is a subring of algebraic closed field $\bar{\mathbb{Q}}$ or $\mathbb{C}$, and then we can define the ring of integers of number field K as

$$\mathcal{O}_K := \tilde{\mathbb{Q}} \cap K$$

equivalently, it is the set of all algebraic integers contained in K .

So we sketch the basic notation and definitions, the ring of integers of number field is special and important, since it is analogous to $\mathbb{Z}$ in $\mathbb{Q}$, and in particular number theory is the study of "integers", although here the integrality is not clear, so we will focus on it.

Theorem 14.6. Let $K/\mathbb{Q}$ be a number field with $\mathcal{O}_K$ its ring of integers, then

1. $(\mathcal{O}_K, +)$ is a free $\mathbb{Z}$ -module of rank $[K : \mathbb{Q}]$, hence it is a noetherian ring.
2. $\mathcal{O}_K$ is integrally closed in K (following the definition).
3. Any non-zero prime ideal in $\mathcal{O}_K$ is maximal, i.e. the krull dimension of $\mathcal{O}_K$ is 1.

Proof. □

To study the arithmetic of "integers" we should understand the factorization of elements in $\mathcal{O}_K$, unfortunately, the ring of integers are not always UFD, typical example is $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ with $K = \mathbb{Q}(\sqrt{-5})$, in it we have

$$6 = 2 \times 3 = (1 + \sqrt{-5}) \times (1 - \sqrt{-5})$$

even more terrible, for the simplest example $K = \mathbb{Q}(\sqrt{d})$ (d is square-free), we have the following known results:

- When $d < 0$, $\mathcal{O}_K$ is a UFD if and only if d is one of the following integer:

$$-1, -2, -3, -7, -11, -19, -43, -67, -163$$

It is owe to the **Heegner-Baker-Stark theorem**, and the proof is non-trivial.

- When $d > 0$, it is conjectured by Gauss that there are infinitely many d such that $\mathcal{O}_K$ is a UFD, but it is still open (till 2026...).

Hence it is almost hopeless to consider the factorization of elements in $\mathcal{O}_K$. After dedekind and Kummer introduced the concept of ideal, a more general factorization is possible, in particular it holds for any number fields.

Theorem 14.7. For any number field K with ring of integers $\mathcal{O}_K$, any non-zero proper ideal $I \subsetneq \mathcal{O}_K$ can be factored into prime ideals, i.e. there exist prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ such that

$$I = \mathfrak{p}_1 \cdots \mathfrak{p}_n$$

and the factorization is unique up to permutation of the prime ideals.

For example we consider $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$, we have the following factorization of ideals:

$$(6) = \mathfrak{p}_2^2 \mathfrak{p}_3 \mathfrak{p}'_3$$

with

$$\mathfrak{p}_2 = (2, 1 + \sqrt{-5}), \quad \mathfrak{p}_3 = (3, 1 + \sqrt{-5}), \quad \mathfrak{p}'_3 = (3, 1 - \sqrt{-5})$$

then we can find that the factorization of 6 is unique in the sense of ideals:

$$(2) = \mathfrak{p}_2^2, \quad (3) = \mathfrak{p}_3 \mathfrak{p}'_3$$

while

$$(1 + \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}_3, \quad (1 - \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}'_3$$

To prove the theorem, some preparation should be done since the result is essentially not trivial, firstly we should consider the division of ideals:

Definition 14.2. Let $\mathfrak{a}$ and $\mathfrak{b}$ be two non-zero ideals in $\mathcal{O}_K$, we define

$$\mathfrak{a} | \mathfrak{b} \iff \mathfrak{b} \subset \mathfrak{a}$$

This definition can be done for any commutative ring, but the properties of rings of integers ensures the following results, which is the key to the proof.

Lemma 14.8. $\mathfrak{p}$ is a prime ideal of $\mathcal{O}_K$ if and only if for any ideals $\mathfrak{a}$ and $\mathfrak{b}$ in $\mathcal{O}_K$, we have $\mathfrak{p} | \mathfrak{a}\mathfrak{b} \implies \mathfrak{p} | \mathfrak{a}$ or $\mathfrak{p} | \mathfrak{b}$.

Proof.

□

Lemma 14.9. For any non-zero ideal $\mathfrak{a} \in \mathcal{O}_K$, there exists non-zero prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ such that $\mathfrak{p}_1 \cdots \mathfrak{p}_n \subset \mathfrak{a}$.

Proof.

□

Lemma 14.10. For any prime ideal $\mathfrak{p}$ in $\mathcal{O}_K$, we define its inverse ideal as

$$\mathfrak{p}^{-1} = \{x \in K : x\mathfrak{p} \subset \mathcal{O}_K\}$$

then for any non-zero ideal $\mathfrak{a}$ in $\mathcal{O}_K$, we have $\mathfrak{a} \subsetneq \mathfrak{a}\mathfrak{p}^{-1}$. In particular, $\mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}_K$.

Proof. □

With the two lemma we can finish the proof:

Proof of theorem 14.7. For the existence, we set a set S of ideals which can not be factorized into prime ideals, if S is empty then we are done, **hence we assume S is non-empty**. We consider the partial ordered set $(S, \subset)$, then the noetherian condition ensures the existence of maximal element, denoted by $\mathfrak{a}$.

$\mathfrak{a}$ can not be prime (i.e. maximal), otherwise it is already factorized, hence there exists a maximal ideal (i.e prime) such that $\mathfrak{a} \subsetneq \mathfrak{p}$, which implies

$$\mathfrak{a} \subsetneq \mathfrak{a}\mathfrak{p}^{-1} \subsetneq \mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}_K$$

the chain holds true by lemma 14.10, hence we construct an element $\mathfrak{a}\mathfrak{p}^{-1} \notin S$, which allows us to factorize

$$\mathfrak{a}\mathfrak{p}^{-1} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$$

then $\mathfrak{a} = \mathfrak{a}\mathcal{O}_K = \mathfrak{a}\mathfrak{p}^{-1}\mathfrak{p}$ shows that the maximal element can be factorized, which is a contradiction.

For the uniqueness, we assume that there are two factorization for an ideal $\mathfrak{a}$

$$\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_n = \mathfrak{q}_1 \cdots \mathfrak{q}_m$$

then $\mathfrak{p}_1|\mathfrak{a}$ shows that there exists i such that $\mathfrak{p}_1|\mathfrak{q}_i$, prime ideals are maximal so $\mathfrak{p}_1 = \mathfrak{q}_i$ exactly, and we can cancel then by multiplying $\mathfrak{p}_1^{-1}$, then we can repeat the process to conclude that $n = m$ and $\mathfrak{p}_i = \mathfrak{q}_{\sigma(i)}$ for any permutation $\sigma \in S_n$. □

with unique factorization, we can extend the definition of norm from elements to ideals.

Definition 14.3. For any non-zero ideal $\mathfrak{a}$ in $\mathcal{O}_K$, we define its **(absolute) norm** by

$$N(\mathfrak{a}) = |\mathcal{O}_K/\mathfrak{a}|$$

where $\mathcal{O}_K/\mathfrak{a}$ is called the **residue ring** of $\mathfrak{a}$.

Remark 14.3. The definition should be clarified, any non-zero ideal $\mathfrak{a}$ is exactly a $\mathbb{Z}$ -submodule of $\mathcal{O}_K$, while $\mathcal{O}_K$ is free and $\mathbb{Z}$ is PID, hence $\mathfrak{a}$ is a free $\mathbb{Z}$ -module with rank $\leq n = [K : \mathbb{Q}]$. We can take any non-zero element x of $\mathfrak{a}$ such that we have a chain

$$(x) \subset \mathfrak{a} \subset \mathcal{O}_K$$

principal ideal must be a free $\mathbb{Z}$ -module of rank n by an $\mathbb{Z}$ -module isomorphism with $\mathcal{O}_K$, hence the chain forces $\mathfrak{a}$ is also a free $\mathbb{Z}$ -module of rank n , hence the residue ring is exactly finite.

Proposition 14.11. Let K be a number field with ring of integers $\mathcal{O}_K$

- (a) For non-zero ideals $\mathfrak{a}$ with factorization $\mathfrak{a} = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_n^{r_n}$

$$N(\mathfrak{a}) = N(\mathfrak{p}_1)^{r_1} \cdots N(\mathfrak{p}_n)^{r_n}$$

- (b) Norm is **multiplicative**, i.e. $N(\mathfrak{a}\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b})$ for any two non-zero ideals.

- (c) Let $\mathfrak{a}$ be a non-zero ideal with $\mathbb{Z}$ -basis $\{a_1, \dots, a_n\}$, then

$$N(\mathfrak{a}) = \sqrt{\frac{d_a}{d_K}}$$

- (d) For principal ideal, $N((a)) = |N_{K/\mathbb{Q}}(a)|$ with a non-zero in $\mathcal{O}_K$.

Proof. If the factorization of a non-zero ideal $\mathfrak{a}$ is $\mathfrak{a} = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_n^{r_n}$, then by Chinese remainder theorem, the residue ring can be decomposed as

$$\mathcal{O}_K/\mathfrak{a} \cong \mathcal{O}_K/\mathfrak{p}_1^{r_1} \times \cdots \times \mathcal{O}_K/\mathfrak{p}_n^{r_n}$$

hence it is enough to prove that for any prime ideal $\mathfrak{p}$ we have $N(\mathfrak{p}^k) = N(\mathfrak{p})^k$. For any $i \geq 1$, $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ is a $\mathcal{O}_K/\mathfrak{p}$ -vector space since $\mathfrak{p}$ is a $\mathcal{O}_K$ -module,

$$\mathfrak{p}^i/\mathfrak{p}^{i+1} \cong \mathcal{O}_K/\mathfrak{p} \otimes_{\mathcal{O}_K} \mathfrak{p}^i$$

consider $k = \mathcal{O}_K/\mathfrak{p}$, then k is the residue field of local ring $\mathcal{O}_{K,\mathfrak{p}}$, hence

$$\begin{aligned} \mathfrak{p}^i/\mathfrak{p}^{i+1} &\cong (k \otimes_{\mathcal{O}_K} \mathcal{O}_{K,\mathfrak{p}}) \otimes_{\mathcal{O}_K} \mathfrak{p}^i \\ &\cong k \otimes_{\mathcal{O}_{K,\mathfrak{p}}} \mathfrak{p}^i \mathcal{O}_{K,\mathfrak{p}} \\ &\cong k \end{aligned}$$

which shows that $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ is one-dimensional over k (**an alternative proof can be found in [ST01] Lemma 6.20 , with direct discussion isomorphism**), hence we can conclude that

$$N(\mathfrak{p}^k) = |\mathcal{O}_K/\mathfrak{p}| \cdot |\mathfrak{p}/\mathfrak{p}^2| \cdots |\mathfrak{p}^{k-1}/\mathfrak{p}^k| = N(\mathfrak{p})^k$$

which finishes the proof of (a), and (b) is a direct consequence of (a).

For (c), □

Similarly with Gauss integer, the norm of an ideal can be used to determine whether it is prime, and it will open a complicated story about the arithmetic of number fields.

Lemma 14.12. Let $\mathfrak{p}$ be a non-zero prime ideal of $\mathcal{O}_K$, then

- (a) there exists a unique prime number p such that $p \in \mathfrak{p}$.
(b) $N(\mathfrak{p}) = p^e$ for some integer $1 \leq e \leq n$.

(c) $\mathfrak{p} \cap \mathbb{Z} = (p)$.

Proof. Notice $\mathcal{O}_K/\mathfrak{p}$ is a finite field if $\mathfrak{p}$ is prime (in $\mathcal{O}_K$ it is maximal), which proves (b). For (c), notice that $\mathfrak{p} \cap \mathbb{Z}$ is pre-image of $\mathfrak{p}$ under the inclusion $\mathbb{Z} \hookrightarrow \mathcal{O}_K$, and the contraction of a prime ideal is either zero or a prime ideal. $\mathfrak{p} \cap \mathbb{Z}$ can't be zero, in fact we take any non-zero $x \in \mathfrak{p} \subset \mathcal{O}_K$, then using the minimal polynomial $\Pi_x \in \mathbb{Z}[X]$, we can deduce that $f = \Pi_x - \Pi_x(0) \in \mathbb{Z}[X]$ is a polynomial such that $f(x) = -\Pi_x(0) \in \mathbb{Z}$, which shows that $f(x) \in \mathfrak{p} \cap \mathbb{Z}$, and it proves (b). For (a), (c) shows the existence of prime number p exactly, if there are another prime number $q \in \mathfrak{p}$ then

$$N(\mathfrak{p}) \text{ divides } N(p) = p^n, \quad N(\mathfrak{p}) \text{ divides } N(q) = q^n$$

it shows that $N(\mathfrak{p}) = 1$, it is impossible. $\square$

The properties of prime ideals invite some vocabulary, for any prime number p , there exists a factorization in sense of principal ideal

$$(p) = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_k^{e_k}$$

a prime ideal $\mathfrak{p}$ is called **lying over** p if $\mathfrak{p} | (p)$, and the **ramification index** of $\mathfrak{p}$ is defined as

$$e = \max\{l \in \mathbb{N} : \mathfrak{p}^l | (p)\}$$

and the **residue index** of $\mathfrak{p}$ is defined as the extension degree

$$f = [\mathcal{O}_K/\mathfrak{p} : \mathbb{F}_p]$$

exactly such $\mathfrak{p}$ must be one of $\mathfrak{p}_i$ with e_i as its ramification index, and residue index is just the exponent f_i such that $N(\mathfrak{p}_i) = p^{f_i}$, hence we immediately have the following relation

Corollary 14.13.

$$[K : \mathbb{Q}] = \sum_{\mathfrak{p}_i} e_i f_i$$

where $\mathfrak{p}_i$ runs over all prime ideals lying over p , with e_i and f_i as its ramification index and residue index respectively.

We can talk more about the factorization of prime numbers, which is the basis of the study of arithmetic of number fields.

1. For any number field K , as a separable extension there exists $a \in K$ such that $K = \mathbb{Q}(a)$ and $\mathcal{O}_K = \mathbb{Z}[a] \cong \mathbb{Z}[X]/(\Pi_a(X))$. We want to study $(p) = p\mathcal{O}_K$ for some **integral prime** p , so it is natural to consider the quotient $\mathcal{O}_K/p\mathcal{O}_K$: if the quotient is a field or just integral domain, we can determine that (p) is prime.
2. Otherwise, we consider $\mathcal{O}_K$ as a $\mathbb{Z}$ -module

$$\mathcal{O}_K/p\mathcal{O}_K \cong (\mathbb{Z}/(p)) \otimes_{\mathbb{Z}} \mathbb{Z}[X]/(\Pi_a) \cong \mathbb{F}_p[X]/(\overline{\Pi}_a)$$

we consider the factorization of $\bar{\Pi}_a$ in $\mathbb{F}_p[X]$, it will be unique since $\mathbb{F}_p[X]$ is a PID, so by chinese remainder theorem we can decompose the quotient as a product

$$\mathcal{O}_K/p\mathcal{O}_K \cong \prod_i \mathbb{F}_p[X]/(\bar{Q}_i^{e_i})$$

notice that the structure is analogue to the factorization of ideals $(p) = \prod_i \mathfrak{p}_i^{e_i}$, so it is natural to find $\mathfrak{p}_i$ by consider the lifting of each irreducible factor $\bar{Q}_i$, so it motivates us to define

$$I_p(Q) = p\mathcal{O}_K + Q(a)\mathcal{O}_K$$

where $Q \in \mathbb{Z}[X]$ is a monic polynomial such that $\bar{Q} \in \mathbb{F}_p[X]$ is a factor of $\bar{\Pi}_a$.

3. Notice that $\bar{a} \mapsto \bar{X}$ gives the first isomorphism in 2, so if $\bar{Q}$ is a irreducible factor of $\bar{\Pi}_a$, then we can conclude

$$I_p(Q)/p\mathcal{O}_K \cong (Q)/(Q^e)$$

this is the maximal ideal in local ring $\mathbb{F}_p[X]/(Q^e)$, so by this isomorphism we can conclude that $I_p(Q)$ **is the maximal (so prime) ideal**.

4. The lifting Q is not unique, but $I_p(Q)$ **does not depend on the choice of Q** : In fact if we set $Q' \in \mathbb{Z}[X]$ as another lifting of $\bar{Q}$, then $Q(a) - Q'(a) = 0 \pmod p$, which shows that $Q(a) \in I_p(Q')$ and $Q' \in I_p(Q)$ respectively, hence $I_p(Q) = I_p(Q')$.
5. In fact as we have known, the factorization of (p) is unique, so in priori we can write the isomorphism $\mathcal{O}_K/p\mathcal{O}_K \cong \prod_i \mathcal{O}_K/\mathfrak{p}_i^{e_i}$, and $I_p(Q)$ as we have shown is a prime ideal lying over p , so we can conclude directly the following result (It is another view to prove the uniqueness of factorization of ideals)

Theorem 14.14 (Dedkind-Kummer).

Let K be a number field with primitive element a , and $p \in \mathbb{Z}$ a prime, then

- (a) The bijection $Q \mapsto I_p(Q)$ gives a correspondence

$$\{\text{monic factors of } \bar{\Pi}_a \in \mathbb{F}_p[X]\} \leftrightarrow \{\text{ideal dividing } (p)\}$$

$$\{\text{irreducible factors}\} \leftrightarrow \{\text{prime ideals}\}$$

- (b) The norm of $I_p(Q)$ is $p^{\deg(Q)}$. When $\bar{Q}$ is a irreducible factor, $\deg(Q)$ is exactly the residue index of $I_p(Q)$.
- (c) there exists integer $e_i \in \mathbb{N}$ such that

$$(p) = \prod_i I_p(Q_i)^{e_i}$$

where $\bar{Q}_i$ runs over all distinct irreducible factors of $\bar{\Pi}_a$ in $\mathbb{F}_p[X]$, and e_i is the multiplicity of $\bar{Q}_i$ in $\bar{\Pi}_a$, it is exactly the ramification index of $I_p(Q_i)$.

14.3 Minkowski's space

This theory corresponds to the geometry of numbers, the idea is to consider embedding of number fields into $\mathbb{R}$ or $\mathbb{C}$, we now consider a number field K of degree n , and review some basic notations:

1. an embedding $\sigma : K \rightarrow \mathbb{C}$ is called a **real embedding** if $\sigma(K) \subset \mathbb{R}$, otherwise it is called a **complex embedding**. Since $K/\mathbb{Q}$ is separable, hence the number of embeddings is exactly n .
2. It is easy to verify that if σ is an embedding, then $\bar{\sigma}$ is also an embedding, we say that two complex embeddings σ and σ' are conjugate if $\sigma' = \bar{\sigma}$. Hence the set of embeddings can be divided as following:

$$\Sigma = \Sigma_r \sqcup \Sigma_c \sqcup \bar{\Sigma}_c$$

where Σ_r is the set of real embeddings, and Σ_c is the set of distinct complex embeddings up to conjugation. In particular, we set

$$\Sigma_0 = \Sigma_r \sqcup \Sigma_c \cong \Sigma / \sim$$

to denote the set of distinct embeddings up to conjugation, and usually we write $n = r + 2s$ with

$$r = |\Sigma_r|, \quad s = |\Sigma_c|$$

3. For any $\mathbb{R}$ -vector space V of dimension n with an inner product $\langle -, - \rangle$, an additive subgroup $L \subset V$ is a **complete lattice** if V/L is compact (or equivalently, L is discrete of rank n), and we can define the volume and covolume

$$\text{covol}(L) = \text{Vol}(V/L) = \mu(F)$$

where μ is the induced measure, and F is the fundamental domain. If $L = \oplus_{i=1}^n \mathbb{Z}v_i$ for some basis $\{v_1, \dots, v_n\}$, then we have

$$\text{covol}(L) = \text{Vol}(V/L) = \sqrt{\det G}$$

where G is the Gram matrix with entries $G_{ij} = \langle v_i, v_j \rangle$.

Definition 14.4. For any number field K of degree $n = r + 2s$, we define the **Minkowski's space** of K as a $\mathbb{R}$ -vector space $K_\infty = \mathbb{R}^n$ together with an inner product defined by

$$\langle x, y \rangle_m = \sum_{i=1}^n a_i \cdot x_i y_i$$

with

$$a_i = \begin{cases} 1 & \text{if } i \leq r \\ 2 & \text{if } i > r \end{cases}$$

Remark 14.4. The Minkowski's space always appears together with the embedding of K by

$$\iota : K \hookrightarrow \mathbb{R}^r \times \mathbb{C}^s \cong K_\infty, \quad x \mapsto (x_\sigma)_{\sigma \in \Sigma_0}$$

where $x_\sigma = \sigma(x)$, the inner product can be pulled back to K by

$$\langle x, y \rangle_M = \langle \iota(x), \iota(y) \rangle_m = \sum_{\sigma \in \Sigma_0} a_\sigma \cdot \operatorname{Re}(x_\sigma \overline{y_\sigma})$$

with

$$a_\sigma = \begin{cases} 1 & \text{if } \sigma \text{ real} \\ 2 & \text{if } \sigma \text{ complex} \end{cases}$$

It is not hard to verify that the inner product concides with the trace pairing if K is totally real ($s = 0$).

Remark 14.5. We can view that this inner product is the restriction of the Hermitian inner product on $\mathbb{C}^n$, i.e. it is same with the pulled back from the embedding

$$K \rightarrow \mathbb{C}^n, \quad x \mapsto (x_\sigma)_{\sigma \in \Sigma}$$

the reason to consider the restriction is that $\mathbb{R}$ -vector space is exactly the place where the convex geometry works, and we always view ideal as a lattice in the Minkowski's space, such that we can talk about the volume and the density.

Theorem 14.15 (The structure of Minkowski's space).

Along the notation above

- (a) $K_\infty \cong K \otimes_{\mathbb{Q}} \mathbb{R}$.
- (b) A subset of K_∞ is measurable if and only if it is lebesgue measuable, in particular we they are distict up to a constant

$$\operatorname{Vol}(S) = 2^s \cdot \operatorname{Vol}_{\operatorname{Lebesgue}}(S)$$

- (c) If $\mathfrak{a}$ is a non-zero ideal in $\mathcal{O}_K$, then $\iota(\mathfrak{a})$ is a lattice in K_∞ with

$$\operatorname{covol}(\iota(\mathfrak{a})) = \sqrt{|d_{\mathfrak{a}}|}$$

Moreover, the aboslute norm of $\mathfrak{a}$ can be defined **equivalently** by

$$N(\mathfrak{a}) = \frac{\operatorname{covol}(\iota(\mathfrak{a}))}{\operatorname{covol}(\iota(\mathcal{O}_K))}$$

Proof. (a)

(b) The inner product is the euclidean inner product with weight, hence the norm are equivalent to the standard norm. Notice that the lebesgue measure and induced measure on Minkowski's space are both Haar measure, so they are distinct up to a constant, so it is enough to compute the hypercube $S = [0, 1]^n$ in both measures, and we have

$$\operatorname{Vol}(S) = \sqrt{\det(\langle e_i, e_j \rangle_m)_{i,j}} = 2^s$$

(c) Review that for square matrix A we have

$$(A^T \bar{A})_{i,j} = \sum_k A_{k,i} \overline{A_{k,j}}$$

by definition of the discriminant, if $a_1, \dots, a_n$ is a basis of $\mathfrak{a}$, then we have

$$d_{\mathfrak{a}} = \det[\sigma_i(a_j)]^2$$

hence we set here A is the matrix with entries $A_{i,j} = \sigma_i(a_j)$, then we have

$$(\bar{A}^T A)_{i,j} = \langle a_i, a_j \rangle_M$$

hence $\text{covol}(\iota(\mathfrak{a})) = \sqrt{\det(\bar{A}^T A)} = |\det A| = |d_{\mathfrak{a}}|$, notice here $A \in M(n, \mathbb{C})$ exactly with $\det A \in \mathbb{C}$.

□

Remark 14.6. In sense of Minkowski's space, the discriminant of a number field K can be described as a geometric invariant, it is exactly the square of the volume of the fundamental domain with respect to the translation action $\mathcal{O}_K$ on K_{∞} .

Since ι is an inclusion, so we it has no problem to identify the subset of K (numbers) as the subset of K_{∞} (vectors), from now on we will not distinguish them with a little abuse of notation, i.e.

$$\text{Vol}(S) = \text{Vol}(\iota(S))$$

for any subset $S \subset K$ (or K_{∞}).

Theorem 14.16 (Minkowski's convex theorem).

Let K be a number field of degree n with ring of integers $\mathcal{O}_K$, and let $\iota(S) \subset K_{\infty}$ be a convex symmetric measurable subset with

$$\mu(\iota(S)) > 2^n \cdot \mu(\iota(\mathcal{O}_K))$$

for any Haar measure on K_{∞} , then there exists a non-zero element $x \in \mathcal{O}_K$ such that $\iota(x) \in S$.

The proof is owe to the theory of lattice, should pay attention to the habite that we usually consider the lebesgue measure λ , and choose a proper convex body we can get some interesting results about number fields.

Theorem 14.17. For any non-zero ideal $\mathfrak{a}$ in $\mathcal{O}_K$, there exists a non-zero element $a \in \mathfrak{a}$ such that

$$|N_{K/\mathbb{Q}}(a)| \leq M_K N(\mathfrak{a})$$

where $M_K = \frac{n!}{n^n} \left(\frac{4}{\pi}\right)^s \sqrt{|d_K|}$ is defined as the **Minkowski's bound** of K .

Proof. We define a L^1 norm on K_{∞} by

$$\|(x_1, \dots, x_r, z_1, \dots, z_s)\|_1 = \sum_{i=1}^r |x_i| + 2 \sum_{i=1}^s |z_i|$$

and then under this norm we define a convex symmetric body by

$$C_R = \{x \in K_\infty : \|x\|_1 \leq R\}, \quad R > 0$$

then we can calculate the volume $\lambda(C_R) = 2^r \left(\frac{\pi}{2}\right)^s \frac{R^n}{n!}$, and then we apply Minkowski's theorem here to conclude the lower bound of R such that $\mathfrak{a} \cap \iota^{-1}(C_R) \neq \emptyset$ will be

$$R^n \geq n! \cdot \frac{2^{n-r}}{\pi^s} \cdot N(\mathfrak{a}) \sqrt{|d_K|}$$

under this condition, we set $a \in \mathfrak{a}$ to be a such element, to estimate the norm of a , we apply **the inequality of arithmetic and geometric means** to get

$$\begin{aligned} |N_{K/\mathbb{Q}}(a)| &= \prod_{\sigma \in \Sigma} |\sigma(a)| \\ &\leq \frac{1}{n^n} \|\iota(a)\|_1^n \\ &\leq \frac{R^n}{n^n} \end{aligned}$$

so it is enough to take the lower bound of R to finishe the proof. $\square$

14.4 Logarithmic space and Unit theorem

This section is the complement of the the last section, and the notation will be the same as before for the sake of consistency.

Definition 14.5. The **logarithmic space** of K is defined as the $\mathbb{R}$ -vector space $\mathbb{L} \cong \mathbb{R}^{r+s}$ together with the map

$$l : K^\times \rightarrow \mathbb{L}, \quad x \mapsto (a_\sigma \cdot \log |x_\sigma|)_{\sigma \in \Sigma_0}$$

the map is a **homomorphism onto a additive group**, i.e.

$$l(xy) = l(x) + l(y)$$

For any $x, y \in K^\times$.

Remark 14.7. The alternative definition is via the minkowski's space

$$K^\times \xrightarrow{\iota} K_\infty^\times \xrightarrow{\text{Log}} \mathbb{L}$$

with the logarithm map naturally defined by

$$\log : K_\infty^\times \rightarrow \mathbb{L}, \quad (x_1, \dots, x_r, z_1, \dots, z_s) \mapsto (\log |x_1|, \dots, \log |x_r|, 2 \log |z_1|, \dots, 2 \log |z_s|)$$

hence exactly here $l = \text{Log} \circ \iota$, based on the definition we can quickly investigate some properties:

- Define the norm map (similar with the adele norm)

$$N : K_\infty \rightarrow \mathbb{R}, \quad x \mapsto \prod_{x \text{ real}} |x| \prod_{z \text{ complex}} |z|^2$$

and trace map

$$T : \mathbb{L} \rightarrow \mathbb{R}, \quad x \mapsto \sum_{i=1}^{r+s} x_i$$

such that the following diagram commutes

$$\begin{array}{ccccc} K^\times & \hookrightarrow & K_\infty^\times & \xrightarrow{\text{Log}} & \mathbb{L} \\ \downarrow N_{K/\mathbb{Q}} & & \downarrow N & & \downarrow T \\ \mathbb{Q}^\times & \hookrightarrow & \mathbb{R}^\times & \xrightarrow{\log|\cdot|} & \mathbb{R} \end{array}$$

- The kernel of l is the group of roots of unity $\mu(K)$, hence it motivates us to restrict the map l on $\mathcal{O}_K^\times$ for the structure, and it is not difficult to verify the following equivalent condition:

$$x \in \mathcal{O}_K^\times \iff N_{K/\mathbb{Q}}(x) = \pm 1 \iff N(\iota(x)) = 1 \iff T(l(x)) = 0$$

Hence it motivates us to define the **hyperplane**

$$H := \{x \in \mathbb{L} \mid x_1 + \cdots + x_{r+s} = 0\}$$

it is a subspace of $\mathbb{L}$ of dimension $r + s - 1$, and **norm-one surface**

$$S = \{x \in K_\infty \mid N(x) = 1\}$$

it is a closed subgroup of minkowski space K_∞ .

Theorem 14.18 (Dirichlet). $l(\mathcal{O}_K^\times)$ is a complet lattice of $\mathbb{L}$, and the structure of unit group can be given by isomorphism

$$\mathcal{O}_K^\times \cong \mu(K) \times \mathbb{Z}^{r+s-1}$$

Proof.

□

The lebesgue volume of fundamental domain $\mathbb{L}/l(\mathcal{O}_K^\times)$ is called the **regulator** of number field K , and we usually denote it by d_K ; The generators of the free module $l(\mathcal{O}_K^\times)$ is called **fundamental unit** of K , and if $\{\varepsilon_1, \dots, \varepsilon_{r+s-1}\}$ is a basis, then each unit can be written as the form

$$u \cdot \varepsilon_1^{m_1} \cdots \varepsilon_{r+s-1}^{m_{r+s-1}}$$

where $u \in \mu(K)$.

14.5 Class group and Class number

It is natural to consider the group structure of the ideal under multiplication since the factorization of prime ideals is unique, similarly with $\mathbb{Z}$, it can not exactly form a group under multiplication, and it needs a "localization" to complete the elements, it is back to the Dedekind's original definition of ideal.

Definition 14.6. Let K be a number field with ring of integers $\mathcal{O}_K$, a fractional ideal of K is a **non-zero** subset $\mathfrak{a} \subset K$ such that

- (1) $\mathfrak{a}$ is an $\mathcal{O}_K$ -submodule of K .
- (2) There exists a non-zero element $x \in \mathcal{O}_K$ such that $x\mathfrak{a} \subset \mathcal{O}_K$. The set of all fractional ideals of K is denoted by J_K .

Remark 14.8. The definition is a little axiomatic, here is some descriptions:

- condition (2) is equivalent to $\mathfrak{a}$ is finitely generated with rank equal to $[K : \mathbb{Q}]$, and in particular it is also a free $\mathbb{Z}$ -module of rank $[K : \mathbb{Q}]$.
- The non-zero ideal $\mathfrak{a}$ is exactly a $\mathcal{O}_K$ -submodule of $\mathcal{O}_K$ itself, hence naturally it is a fractional ideal, and we call it **integral ideal** to distinguish with the fractional ideal.
- Notice that $x\mathfrak{a} \subset \mathcal{O}_K$ is equivalent to $x\mathfrak{a}$ is an integral ideal, hence the definition can be expressed precisely by saying that a fractional ideal is a set of the form

$$xI := \{xa \mid a \in I\} \subset K$$

where $x \in K - \{0\}$ and I is an integral ideal, since as a fractional field $K = \text{Frac}(\mathcal{O}_K)$, we can always write $x = a/b$ with $a, b \in \mathcal{O}_K$.

Lemma 14.19. For any fractional ideal $\mathfrak{a}$, there exists a unique factorization up to order

$$\mathfrak{a} = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_n^{r_n}$$

where $\mathfrak{p}_i$ are prime ideals and r_i are integers.

Proof. By the definition of fractional ideal, there exists $x \in \mathcal{O}_K - \{0\}$ such that $x\mathfrak{a} \subset \mathcal{O}_K$ is an integral ideal, hence there exists a unique factorization of $x\mathfrak{a}$ up to order

$$x\mathfrak{a} = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_k^{r_k}$$

notice $x\mathfrak{a} = (x)\mathfrak{a}$, hence there exists a unique factorization for the principal ideal

$$(x) = \mathfrak{q}_1^{s_1} \cdots \mathfrak{q}_m^{s_m}$$

then we take the inverse of the principal ideal to rewrite $\mathfrak{a} = (x)^{-1}x\mathfrak{a}$, then we can conclude the result. $\square$

Remark 14.9. There are several immediate consequences of the properties:

1. J_K forms a abelian group under multiplication, it is clear that $\mathcal{O}_K = (1)$ is the identity, so for any fractional ideal $\mathfrak{a} = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_n^{r_n}$, we can exactly find an inverse by $\mathfrak{a}^{-1} = \mathfrak{p}_1^{-r_1} \cdots \mathfrak{p}_n^{-r_n}$ since lemma 14.10 shows that $\mathfrak{a}\mathfrak{a}^{-1} = \mathcal{O}_K$. In the sense of monoid, the existence of inverse ensures uniqueness of inverse, hence J_K is exactly a group.
2. Indeed J_K is a free abelian group generated by the prime ideals, it is immediate by a natural bijection

$$v : J_K \rightarrow \mathbb{Z}^{(\mathcal{P})}, \quad \prod_{\mathfrak{p} \in \mathcal{P}} \mathfrak{p}^{a_{\mathfrak{p}}} \mapsto (a_{\mathfrak{p}})_{\mathfrak{p} \in \mathcal{P}}$$

where $\mathcal{P}$ is the set of all prime ideals of $\mathcal{O}_K$.

3. Hence we can extend the definition of **absolute norm** of ideals to J_K by

$$N : J_K \rightarrow \mathbb{R}_{>0}, \quad \mathfrak{a} \mapsto \prod_{\mathfrak{p} \in \mathcal{P}} N(\mathfrak{p})^{v_{\mathfrak{p}}(\mathfrak{a})}$$

where $v_{\mathfrak{p}} = \pi_{\mathfrak{p}} \circ v$ is exactly the projection of v at $\mathfrak{p}$ -component, here naturally we extend the absolute norm as a group homomorphism, and invite the valuation of ideals.

To consider the ideal as a number, we define **the principal fractional ideal** are the fractional ideal of the form $(x) = x\mathcal{O}_K$ for some $x \in K - \{0\}$, and P_K the set of all principal fractional ideals, then we define a relation on J_K by

$$\mathfrak{a} \sim \mathfrak{b} \iff \exists x \in K - \{0\}, \mathfrak{a} = x\mathfrak{b} = (x)\mathfrak{b}$$

then the relation is exactly equivalent, and we can get the classes

$$[\mathfrak{a}] = \{x\mathfrak{a} \mid (x) \in P_K\}$$

and it gives a partition of J_K

Lemma 14.20. In each class of, there exists an integral ideal I such that

$$N(I) \leq M_K$$

where M_K is the Minkowski's bound of K , thus J_K/P_K is exactly a finite abelian group, and we call it the **class group** of K , and its order is called the **class number** of K .

Proof. J_K is an abelian group with P_K as its subgroup, and the relation defined above is exactly the relation defined by cosets of P_K , so it is enough to show the first statement, since there exists only finitely many ideal of $\mathcal{O}_K$ with norm less than a fixed number, and each class can be represented by an integral ideal.

For any class $[\mathfrak{a}]$, let $y \in \mathcal{O}_K$ such that $y\mathfrak{a}^{-1} \subset \mathcal{O}_K$ be a non-zero ideal of $\mathcal{O}_K$, the there exists a non-zero element $x \in y\mathfrak{a}^{-1}$ such that

$$|N_{K/\mathbb{Q}}(x)| \leq M_K N(y\mathfrak{a}^{-1})$$

we set $xy^{-1}\mathfrak{a} = \mathfrak{b}$, then $\mathfrak{b}$ is an integral ideal since $xy^{-1} \in \mathfrak{a}^{-1}$, hence we multiply the inequality both sides by $N(y^{-1}\mathfrak{a})$ to conclude that $\mathfrak{b}$ is exactly the ideal we hope for the class. $\square$

15 Analytic approach to number theory

15.1 Arithmetic function

15.1.1 Dirichlet convolution and multiplicative functions

An arithmetic function is a complex-valued function defined on the positive integers $\mathbb{N}^*$, a remarkable operation on the set is defined as following:

Definition 15.1. Let f, g be two arithmetic functions, we define the **Dirichlet convolution** of f and g by

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d)$$

such a operation is commutative and associative, and the identity element is the dirac function

$$\delta : \mathbb{N}^* \rightarrow \mathbb{C}, \quad n \mapsto \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases} = \left[\frac{1}{n}\right]$$

that shows the set of arithmetic functions forms a monoid under the convolution, furthermore some restrictions can make it a group.

Proposition 15.1. If f is an arithmetic function with $f(1) \neq 0$, then there exists a unique arithmetical function g such that

$$f * g = g * f = \delta$$

and g is given by the recursion formulas

$$g(1) = f(1)^{-1}, \quad g(n) = -g(1) \sum_{d|n, d < n} f(n/d)g(d)$$

Proof.

□

Remark 15.1. We shall denote the set of all arithmetical functions by $\mathcal{A}$, and the set of all arithmetical functions with $f(1) \neq 0$ by $\mathcal{A}^*$, then $(\mathcal{A}, *)$ is a monoid, and $(\mathcal{A}^*, *)$ is a group. Notice that such a construction holds generally for any function $f : \mathbb{N}^* \rightarrow R$ with R a commutative ring, in this case $f(1) \in R^\times$ is the condition for the existence of inverse.

Example 15.1. 1-function is defined by $\mathbf{1}(n) = 1$ for any $n \in \mathbb{N}^*$, it is a function with $\mathbf{1}(1) = 1$, and we can calculate that

$$(\mathbf{1} * \mathbf{1})(n) = \sum_{d|n} \mathbf{1}(d)\mathbf{1}(n/d) = \sum_{d|n} 1 =: \tau(n)$$

τ is defined as the **divisor function** which counts the number of divisors. More generally, use some combinatorial trick we can prove that

$$\underbrace{1 * \cdots * 1}_{k \text{ times}}(n) = \tau_k(n) = \#\{(a_1, \dots, a_k) \in \mathbb{N}^k \mid a_1 \cdots a_k = n\}$$

where τ_k is the **k-th power divisor function**, with a convention that $\tau_2 = \tau$ and $\tau_1 = 1$.

Example 15.2. Möbius function μ is defined as following:

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^k & n = p_1 \cdots p_k \\ 0 & \text{otherwise} \end{cases}$$

a remarkable equality is that

$$\sum_{d|n} \mu(d) = \delta(n)$$

that shows $\mu \in \mathcal{A}^*$ with inverse $\mathbf{1}$, so $\tau_k^{-1} = \mu^{*k}$.

Example 15.3. Identity map is denoted by I with $I(n) = n$ for any $n \in \mathbb{N}^*$, it is a function with $I(1) = 1$, and we can calculate that

$$(I * I)(n) = \sum_{d|n} d \cdot \frac{n}{d} = n\tau(n)$$

which shows that $I * I = I\tau$, and more generally we have

$$\underbrace{I * \cdots * I}_{k \text{ times}}(n) = n\tau_k(n)$$

and its inverse is $\mu \cdot I$, in fact

$$(I * \mu I)(n) = \sum_{d|n} n\mu(n/d) = n\delta(n) = \delta(n)$$

Example 15.4. Euler totient function φ is defined by

$$\varphi(n) = \sum_{\substack{1 \leq k \leq n \\ (k,n)=1}} 1$$

and a famous formula is that

$$\sum_{d|n} \varphi(d) = n$$

that shows $\varphi * \mathbf{1} = I$, with I the identity function mapping n to n , so

$$\varphi(n) = (\varphi * \mathbf{1} * \mu)(n) = (I * \mu)(n) = n \sum_{d|n} \frac{\mu(d)}{d}$$

and the inverse of φ is $\mu I * \mathbf{1}$.

The example of Euler totient function can be seen as a motivation to define the convolution, since it plays an important role in number theory, and another property related to the Euler function can be concluded as following:

Definition 15.2. A arithmetic function f is called **multiplicative** if

$$f(mn) = f(m)f(n), \quad (m, n) = 1$$

and f is called **completely multiplicative** if

$$f(mn) = f(m)f(n), \quad \forall m, n \in \mathbb{N}^*$$

Remark 15.2. The multiplicative function f must be in $\mathcal{A}^*$, the reason is that $f(n) = f(1)f(n)$ for any $n \in \mathbb{N}^*$.

Example 15.5. The Euler totient function is multiplicative, which allows to deduce the formula for $n = p_1^{a_1} \cdots p_k^{a_k}$,

$$\varphi(n) = \prod_{i=1}^k (p_i - 1)p_i^{a_i-1}$$

while φ is not completely multiplicative, since $\varphi(4) = 2 \neq \varphi(2)^2 = 1$.

Example 15.6. The Möbius function is multiplicative, but not completely multiplicative. Hence in example 15.4 we notice that $d \mapsto \mu(d)/d$ is multiplicative, which allows us to deduce the another formula for Euler totient function

$$\varphi(n) = n \prod_{p|n} (1 - p^{-1})$$

Theorem 15.2. The set of multiplicative functions, denote by $\mathcal{M}$, it forms a subgroup of $\mathcal{A}^*$ under the convolution.

Proof. It □

Remark 15.3. The completely multiplicative functions is not stable, for example 1-function $\mathbf{1}$ is completely multiplicative, while divisor function $\tau = \mathbf{1} * \mathbf{1}$ is mutiplicative but not completely ($\tau(p^2) = 3 \neq 4 = \tau(p)\tau(p)$).

Proposition 15.3. Let f be a multiplicative function, then

- (a) f is completely multiplicative if and only if $f^{-1} = \mu \cdot f$.

(b) we have

$$\sum_{d|n} \mu(d)f(d) = \prod_{p|n} (1 - f(p))$$

and consequently $\varphi^{-1}(n) = \sum_{p|n} (1 - p)$.

Proof. □

Proposition 15.4 (Euler's product). Suppose f is an arithmetic function such that the series $\sum_{n \geq 1} f(n)$ converges absolutely, then $\prod_p \sum_{k \geq 0} f(p^k)$ converges absolutely, and

-If f is multiplicative, $\sum_{n \geq 1} f(n) = \prod_p \sum_{k \geq 0} f(p^k)$.

-If f is completely multiplicative, then $|f(p)| < 1$ for any prime p and

$$\sum_{n \geq 1} f(n) = \prod_p \frac{1}{1 - f(p)}$$

Proof. □

1

15.1.2 sommation formula and average

Let f be an arithmetic function, its average in many cases means a limit

$$\lim_{N \rightarrow \infty} \frac{\sum_{n \leq N} f(n)}{N} = \lim_{x \rightarrow \infty} \frac{\sum_{n \leq x} f(n)}{x}$$

and we are interested in the existence of such a limit, or the asymptotic behavior at infy.

Definition 15.3. Let f be an arithmetic function and g be a smooth function, we say that **the average of f is of order g** if

$$\frac{1}{x} \sum_{n \leq x} f(n) \sim g(x) \quad \text{as } x \rightarrow \infty$$

the order of average measures the growth of the average of arithmetic function, or behavior of the average at infy, by a comparison with a well-kown function.

Example 15.7 (distribution of prime numbers). Let $\pi(n) = \sum_{p \leq n} 1$ be the prime counting function, a remarkable result is that

$$\lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1$$

in the language of average, we define the prime characteristic function by

$$\mathbf{1}_{\mathbb{P}}(n) = \begin{cases} 1 & \text{if } n \text{ is a prime} \\ 0 & \text{otherwise} \end{cases}$$

it means that the average of $\mathbf{1}_{\mathbb{P}}$ is of order $1/\log n$, which shows that the density of prime numbers is zero.

Theorem 15.5 (Euler-Maclaurin). If f est class C^1 on $[a, b]$ with $a, b \in \mathbb{Z}$. then

$$\sum_{a \leq n \leq b} f(n) = \int_a^b f(t) + \psi(t)f'(t) dt + \frac{f(a) + f(b)}{2}$$

where $\psi(t) = t - [t] - 1/2$.

Example 15.8. If $x \geq 1$, then we have asymptotic equality:

$$(1) \sum_{n \leq x} \frac{1}{n} = \log x + C + O(1/x) \text{ with } C \text{ the euler constant.}$$

$$(2) \sum_{n \leq x} \frac{1}{n^s} = \zeta(s) + \frac{x^{1-s}}{1-s} + O(x^{-s}) \text{ if } s > 1 \text{ and } s \neq 0.$$

Proof. In (1), we take $a = 1$ and $b = [x]$, apply the formula to $f(n) = 1/n$ we have

$$\begin{aligned} \sum_{n \leq x} \frac{1}{n} &= \int_1^b \frac{dt}{t} + \int_1^b \psi(t) \frac{-1}{t^2} dt + \frac{1}{2} + \frac{1}{2b} \\ &\leq \log b + \frac{1}{2} - \underbrace{\int_1^b \frac{1}{2t^2} dt}_{\text{converges if } b \rightarrow \infty} + O(1/b) \end{aligned}$$

and notice that $b = x + O(1)$, so there exists a constant C such that (a) holds, and take the limit $x \rightarrow \infty$ we can conclude the value of C as the euler constant. $\square$

Theorem 15.6 (Abel). Let $a(n)$ be an arithmetical function and f be a function of class C^1 on $[y, x]$, then

$$\sum_{y < n \leq x} a(n)f(n) = A(x)f(x) - A(y)f(y) - \int_y^x A(t)f'(t) dt$$

where $A(x) = \sum_{n \leq x} a(n)$ with convention $A(x) = 0$ if $x < 1$.

Proof. One proof is based on Abel's sum, such a proof can be found in theorem 4.2 [Apo13]. Another proof is more direct, we notice that $x \mapsto A(x)$ is a step function and increasing, so the sum can be expressed as a Riemann-Stieltjes integral

$$\sum_{y < n \leq x} a(n)f(n) = \int_y^x f(t)dA(t)$$

then we can apply the integration by parts to get the result. $\square$

Remark 15.4. An other version of Euler-Maclaurin formula can be derived here by setting $a(n) = 1$, then we can get

$$\sum_{y < n \leq x} f(n) = \int_y^x f(t) + (t - [t])f'(t) dt + f(y)(y - [y]) - f(x)(x - [x])$$

a comparison with theorem 15.5 shows that the two versions are equivalent.

Another method of summation is **Hyperbola method**, loosely speaking, we can express the sum as following

$$\sum_{n \leq x} \sum_{d|n} f(d, n) = \sum_{qd \leq x} f(d, qd)$$

the reason can be expressed geometrically

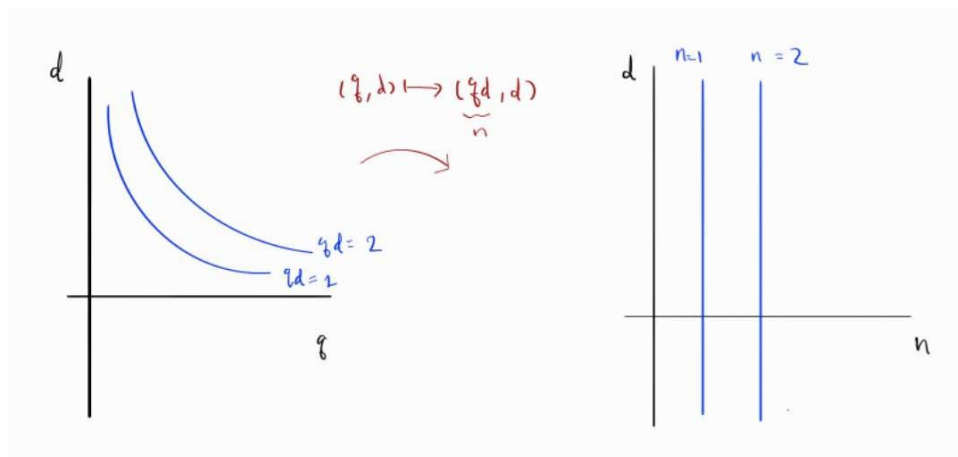


Figure 6: example $x = 2$

the sum on the left is indeed the number of integer points in a family of hyperbolas, it allows us to rewrite the sum of dirichlet convolution as following

$$\sum_{n \leq x} (a * b)(n) = \sum_{n \leq x} \sum_{d|n} a(d)b(n/d) = \sum_{qd \leq x} a(d)b(q) = \sum_{d \leq x} a(d) \sum_{q \leq x/d} b(q)$$

Example 15.9. (Dirichlet's formula) For any $x \geq 1$, we have

$$\sum_{n \leq x} \tau(n) = x \log x + (2C - 1)x + O(x^{1/2})$$

where C is the euler constant, and a rougher estimate is

$$\sum_{n \leq x} \tau(n) = x \log x + O(x)$$

Proof. By hyperbola method $\sum_{n \leq x} \tau(n) = \sum_{d \leq x} \sum_{q \leq x/d} 1 = \sum_{d \leq x} \left[\frac{x}{d} \right]$, so if we estimate $\left[\frac{x}{d} \right] \leq \frac{x}{d}$ and apply the asymptotic formula for harmonic series we can get a rough estimate. To get a refined estimate, we reconsider $\sum_{n \leq x} \tau(n) = \sum_{qd \leq x} 1$ to count the lattice, notice that the number of lattice on the line $q = d$ are $\left[\sqrt{x} \right]$, and the distribution of lattice are

symmetric with respect to the line, so we have

$$\begin{aligned}
\sum_{n \leq x} \tau(n) &= (2 \sum_{qd \leq x, d \leq \sqrt{x}} 1) - [\sqrt{x}]^2 \\
&= (2 \sum_{d \leq \sqrt{x}} [\frac{x}{d}]) - [\sqrt{x}]^2 \\
&= \{2 \sum_{d \leq \sqrt{x}} \frac{x}{d} + O(1)\} - \{\sqrt{x} + O(1)\}^2 \\
&= \{2x \sum_{d \leq \sqrt{x}} \frac{1}{d}\} + O([\sqrt{x}]) - \{x + O(\sqrt{x})\} \\
&= 2x \{\log \sqrt{x} + C + O(1/\sqrt{x})\} - x + O(\sqrt{x}) \\
&= x \log x + (2C - 1)x + O(\sqrt{x})
\end{aligned}$$

□

15.1.3 periodic arithmetic function

Definition 15.4. An arithmetic function $f : \mathbb{N} \rightarrow \mathbb{C}$ is called periodic if there exists a positive integer k such that

$$f(n + k) = f(n)$$

for all $n \in \mathbb{N}$, and we call the smallest such k the (fundamental) period of f .

Remark 15.5. The fundamental period is unique, and it divides any other period of f . In fact, if we set $\mathcal{P}$ as the set of all periods of f , as a non-empty subset of $\mathbb{N}$, it has a minimum element k . If N is another period, by the division we have $N = qk + r$ for some $q \in \mathbb{N}$ and $0 \leq r < k$, hence we have

$$f(n) = f(n + N) = f(n + r + qk) = f(n + r)$$

for any $n \in \mathbb{N}$, hence r is also a period of f if $r \neq 0$, hence $\mathcal{P} = k\mathbb{N}^*$ exactly.

Example 15.10. A natural example is exponential function, for the sake of notation,

$$e(n) = e^{2\pi i n}$$

it is a 1-periodic function.

Theorem 15.7 (finite Fourier series).

Let f be an arithmetical function with period k , then there exists a unique arithmetical function $\hat{f}$ such that

$$f(m) = \sum_{n=1}^k \hat{f}(n) e(\frac{mn}{k})$$

In fact $\hat{f}$ is the fourier transform of f on $\mathbb{Z}/k\mathbb{Z}$ by

$$\hat{f}(n) = \frac{1}{k} \sum_{m=1}^k f(m) e(-\frac{mn}{k})$$

15.2 Character

Definition 15.5. Let G be a finite abelian group, a **character** of G is a group homomorphism $\chi : G \rightarrow \mathbb{C}^\times$, and we denote the set of all characters of G as $\hat{G}$, it forms a group under function multiplication

$$\forall g \in G : (\chi_1 \cdot \chi_2)(g) := \chi_1(g)\chi_2(g)$$

And we call $\hat{G}$ the **Character group** of G .

Here are some quick observations:

- **image of character should be in $\mathbb{S}^1$:** any elements $a \in G$ is of finite order, and homomorphism of groups makes $\chi(e) = 1$, hence $\chi(a)^n = 1$ with $n = \text{ord}(a)$.
- **Inverse of character is the complex conjugate:** $|\chi(n)| = 1$, which allows us to define the inverse

$$\chi^{-1}(n) = 1/\chi(n) = \bar{\chi}(n)$$

- In particular, $\chi(-1) = \pm 1$ since $\chi(-1)^2 = \chi(1) = 1$, which allows us to classify the character into two types: **even** character with $\chi(-1) = 1$ and **odd** character with $\chi(-1) = -1$.

Remark 15.6. We consider the category of all abelian groups, denoted by **Ab**, the character group can be generalized in the category by a contravariant functor $(-)^{\wedge} : \mathbf{Ab} \rightarrow \mathbf{Ab}^{op}$: For the object $G \in \mathbf{Ab}$, we have $\hat{G} = \text{Hom}(G, \mathbb{C}^\times)$; For the morphism $f : G \rightarrow H$, it induces a morphism between dual groups $\hat{f} : \hat{H} \rightarrow \hat{G}$ by $\chi \mapsto \chi \circ f$, i.e.

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ & \searrow \hat{f}(\chi) & \downarrow \chi \\ & & \mathbb{C}^* \end{array}$$

However, the functor is not an equivalent since it is not full, that is the reason why we only consider the character group of finite abelian groups.

Remark 15.7. Another view of character is that the character is naturally a **one-dimensional complex representation** of G since $\mathbb{C}^* \cong \text{GL}_1(\mathbb{C})$, and any character as such is irreducible.

Proposition 15.8. For any finite abelian group G , we have $\hat{\hat{G}} \cong G$

Proof. For any finite abelian group G , there exists a decomposition of cyclic groups $G \cong \mathbb{Z}/n_1 \oplus \cdots \oplus \mathbb{Z}/n_r$, the universal property of direct sum allows $\widehat{G \oplus H} \cong \hat{G} \times \hat{H}$ hence we have

$$\hat{G} \cong \widehat{\mathbb{Z}/n_1} \oplus \cdots \oplus \widehat{\mathbb{Z}/n_r}$$

hence it is enough to prove the isomorphism for any cyclic group $\mathbb{Z}/n$. □

Here is a useful lemma, which appears in many proofs, from the view of category, it shows that the induced morphism of a injection is a surjection.

Lemma 15.9. Let G be a finite abelian group and H be its subgroup, then for any character $\chi \in \hat{H}$, there exists exactly $[G : H]$ characters $\hat{G}$ that extend χ to G .

Proof. Consider restriction map $r : \hat{G} \rightarrow \hat{H}$ defined by $\chi \mapsto \chi|_H$, it is a group homomorphism, and we will show that r is surjective, then we consider the equivalence relation on $\hat{G}$ by

$$\chi_1 \sim \chi_2 \iff \chi_1|_H = \chi_2|_H$$

the relation gives that the class of χ is exactly the set of characters that extend χ to G , and the relation is actually same as the relation defined by cosets of $\hat{H}$, then we can deduce the number by isomorphism of dual groups:

$$\#[\chi] = [\hat{G} : \hat{H}] = [G : H]$$

Surjective: We take $g_1 \notin H$ and construct a subgroup of G by

$$G_1 = \langle H, g_1 \rangle = \{g_1^k h \mid k \in \mathbb{N}, h \in H\}$$

then we can extend the character χ by defining $\chi_1(g_1^k h) = \chi(h)$, i.e. we set $\chi_1(g_1) = 1$ and allow $\chi_1(ab) = \chi_1(a)\chi_1(b)$, then we can verify that $\chi_1 \in \widehat{G_1}$ and $\chi_1|_H = \chi$. Similarly we can construct a chain of subgroups by $G_k = \langle G_{k-1}, g_k \rangle$ with some $g_k \notin G_{k-1}$, so we have

$$G_0 = H \subsetneq G_1 \subsetneq \cdots \subsetneq G_n = G$$

the chain will stop since G is finite (or it is finitely generated $\mathbb{Z}$ -module, so we can use noetherian condition). $\square$

Remark 15.8. Notice that $\mathbb{C}^*$ is a divisible abelian group, then for any short sequence of abelian groups

$$0 \rightarrow H \rightarrow G \rightarrow G/H \rightarrow 0$$

we have a conversion of exact sequence

$$0 \rightarrow \widehat{G/H} \rightarrow \hat{G} \rightarrow \hat{H} \rightarrow 0$$

we can talk more if we restrict our view on finite abelian groups.

Theorem 15.10. Let G be a finite abelian group, then any quotient group of G can be seen as a subgroup of G , furthermore we have

$$G \cong H \oplus G/H$$

Proof. We set $H^\perp = \{\chi \in \hat{G} \mid \chi(h) = 1, \forall h \in H\}$ as the complement of $\hat{H}$ in $\hat{G}$, it is natural in the sense of $\mathbb{Z}$ -module, then we can prove that $H^\perp \cong \widehat{G/H}$. Notice that $H^\perp$ is exactly a subgroup of $\hat{G}$, and $H^\perp \cap \hat{H} = \{1_H\}$, so we have a direct sum

$$\hat{G} = \hat{H} \oplus H^\perp \cong \hat{H} \oplus \widehat{G/H}$$

by the isomorphism of dual groups, we can deduce the result. $\square$

Proposition 15.11 (orthogonality). Let G be a finite abelian group,

1. Fix $\chi \in \bar{G}$, then

$$\sum_{g \in G} \chi(g) = \begin{cases} |G| & \chi = 1 \\ 0 & \chi \neq 1 \end{cases}$$

2. Fix $g \in G$, then

$$\sum_{\chi \in \hat{G}} \chi(g) = \begin{cases} |G| & g = e \\ 0 & g \neq e \end{cases}$$

Proof. For the first case, if $\chi = 1$ then it is trivial, otherwise we can take $h \in G$ such that $\chi(h) \neq 1$, and the translation $g \mapsto h^{-1}g$ is a bijection of G , so we have

$$\sum_{g \in G} \chi(g) = \sum_{g \in G} \chi(h)\chi(h^{-1}g) = \chi(h) \sum_{g \in G} \chi(g)$$

then we can deduce the result. For the second case, we suppose that $g \neq e$, then there exists a character ψ such that $\psi(g) \neq 1$, and the translation $\chi \mapsto \psi^{-1} \cdot \chi$ is a bijection of $\hat{G}$, so we have

$$\sum_{\chi \in \hat{G}} \chi(g) = \psi(g) \sum_{\chi \in \hat{G}} (\psi^{-1} \cdot \chi)(g) = \psi(g) \sum_{\chi \in \hat{G}} \chi(g)$$

then we can deduce the result. □

Corollary 15.12. Let $a, b \in G$, then we have

$$\sum_{\chi \in \hat{G}} \chi(a) \overline{\chi(b)} = \begin{cases} |G| & a = b \\ 0 & a \neq b \end{cases}$$

Proof. □

A special result of orthogonality is about the structure of the character, we set **FinAb** to be the category of finite abelian groups, as we have mentioned before, there exists a contravariant functor inside the category, here we set $D : \mathbf{FinAb} \rightarrow \mathbf{FinAb}^{op}$, the following result shows that the functor is actually an equivalence with $D^2 = Id$.

Theorem 15.13. Let G be a finite abelian group, there exists a natural isomorphism $\phi : G \rightarrow \hat{\hat{G}}$ defined by $g \mapsto \phi_g$ with $\phi_g(\chi) = \chi(g)$.

Proof. It is easy to verify that the map is a morphism, so it is enough to show injection since it is finite, and we consider the kernel.

$$\begin{aligned}
g \in \ker \phi &\iff \phi_g(\chi) = 1, \forall \chi \in \hat{G} \\
&\iff \chi(g) = 1, \forall \chi \in \hat{G} \\
&\iff \sum_{\chi \in \hat{G}} \chi(g) = |\hat{G}| = |G| \\
&\iff g = e
\end{aligned}$$

□

Remark 15.9. Notice that the map is natural since it does not depend on the choice of groups, and we can verify the following diagram commutes for any morphism $f : G \rightarrow H$

$$\begin{array}{ccc}
G & \xrightarrow{\phi_G} & \hat{\hat{G}} \\
f \downarrow & & \downarrow D^2(f) \\
H & \xrightarrow{\phi_H} & \hat{\hat{H}}
\end{array}$$

which ensures that the functor D is an equivalence, it is a special case of the **Pontryagin duality** for locally compact abelian groups, since every finite group can be endowed with the discrete topology.

15.2.1 Dirichlet Character and Gauss sum

Definition 15.6. Let $N \in \mathbb{N}^*$, a **Dirichlet character** modulo N is a character of the multiplicative group $(\mathbb{Z}/N\mathbb{Z})^\times$, i.e.

$$\chi : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$$

Here are some quick descriptions of this type of character:

- Let χ be a Dirichlet character modulo N , then we can extend χ to a function on $\mathbb{Z}$ by

$$\tilde{\chi}(a) = \begin{cases} \chi(\bar{a}) & \gcd(a, N) = 1 \\ 0 & \gcd(a, N) \neq 1 \end{cases}$$

It defines an arithmetic function that is **completely multiplicative** and **N -periodic**. Each character can be extended like that, so we sometimes do not distinguish the character and the arithmetic function.

- The order of $(\mathbb{Z}/N\mathbb{Z})^\times$ is $\varphi(N)$ with φ the Euler totient function, so the orthogonality of characters can be written as

$$\sum_{\chi \in (\widehat{\mathbb{Z}/N\mathbb{Z}})^*} \chi(a) \overline{\chi(b)} = \begin{cases} \varphi(N) & a \equiv b \pmod{N} \\ 0 & a \not\equiv b \pmod{N} \end{cases}$$

for any $a, b \in \mathbb{Z}$ with $(b, N) = 1$.

- We call a dirchlet character to be **principal or trivial** if $\chi(g) = 1$ for any $g \in (\mathbb{Z}/N\mathbb{Z})^\times$, and we denote the principal character by χ_0 .

Theorem 15.14. Let $f : \mathbb{N} \rightarrow \mathbb{C}$ be an arithmetic function is a Dirichlet character modulo N if it satisfies the following conditions:

- (a) completely multiplicative
- (b) N -periodic for some $N \in \mathbb{N}^*$
- (c) $|f(n)| = 1$ if and only if $\gcd(n, N) = 1$

Proof. □

Example 15.11. Notice that $(\mathbb{Z}/3\mathbb{Z})^\times$ is of order 2, so there are exactly two characters, it can be given by the following table:

n	0	1	2
χ_1	0	1	1
χ_2	0	1	-1

while $(\mathbb{Z}/6\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z})^\times \times (\mathbb{Z}/3\mathbb{Z})^\times$ is of order 2, so the dirichlet character mod 6 can be given by the product of the chracter mod 2 (trival character) and mod 3 (χ_1 and χ_2 as above), and we write down the table

n	0	1	2	3	4	5
ψ_1	0	1	0	0	0	-1
ψ_2	0	1	0	0	0	1

Example 15.12. Another example is the character mod 9, there are exactly 6 characters, while in it there are two characters can be written down by character mod 3

n	0	1	2	3	4	5	6	7	8
$\tilde{\chi}_1$	0	1	-1	0	1	-1	0	1	-1
$\tilde{\chi}_2$	0	1	-1	0	1	-1	0	1	-1

for instane,

$$\tilde{\chi}_1(n) = (\chi_1 \cdot f_0)(n) = \begin{cases} \chi_1(n) & \gcd(n, 9) = 1 \\ 0 & \gcd(n, 9) \neq 1 \end{cases}$$

where f_0 is the trival character mod 9, it is clearly that $\tilde{\chi}_1$ is a 9-periodic function and completely multiplicative, and for any n coprime with 9 ($9 = 3^2$ so also coprime with 3), we have $\tilde{\chi}_1(n) = \chi_1(n) = \pm 1$, hence it is a dirichlet character mod 9, and the same for $\tilde{\chi}_2$.

The second example shows that for a dirichlet character mod N , if N is large then it may be determined by a dirichlet charcter mod the factors of N , which induces the following definition.

Lemma 15.15. Let χ be a dirichlet character mod N , and d a positive divisor of N , then the following statements are equivalent:

- (a) χ factors through $(\mathbb{Z}/d\mathbb{Z})^\times$: there exists a character ψ_d mod d such that the following diagram commutes:

$$\begin{array}{ccc} (\mathbb{Z}/N\mathbb{Z})^\times & \xrightarrow{\chi} & \mathbb{C} \\ \pi \downarrow & \nearrow \psi_d & \\ (\mathbb{Z}/d\mathbb{Z})^\times & & \end{array}$$

where π is the natural projection sending $[x]_N$ to $[x]_d$.

- (b) $\chi(a) = 1$ for any $a \in \mathbb{N}$ such that $\gcd(a, N) = 1$ and $a \equiv 1 \pmod{d}$.
(c) there is a character ψ_d mod d such that in the sense of arithmetic function

$$\chi(n) = \psi_d(n)\chi_0(n)$$

If one of the above statements holds, we say that d is an **induced modulus** of χ .

Proof.

□

Remark 15.10. Notice that N is always an induced modulus of χ , while 1 is an induced modulus of χ if and only if χ is trivial. In above example, 3 is an induced modulus of $\tilde{\chi}_1$ and $\tilde{\chi}_2$.

Definition 15.7. Let χ be a dirichlet character mod N , the conductor of χ is the smallest induced modulus of χ .

The character χ is called **primitive** if its conductor is exactly N .

Remark 15.11. Any non-trivial charcter mod a prime number p is primitive, since the divisor of p is only 1 and p , so the conductor can only be 1 or p , it is p since the character is non-trivial.

Remark 15.12. By lemma, for any dirichlet chracter χ mod N , there exists a unique primitive character ψ mod conductor of χ , such that

$$\chi(n) = \psi(n)\chi_0(n)$$

the proof depends on the fact that conductor is the smallest induced modulus (cf. [Apo13], Theorem 8.18).

Notice that dirichlet character can be viewed as a periodic arithmetic function, which allows us to consider its fourier transform, here is the notation introduced by Gauss in the study of quadratic reciprocity.

Definition 15.8. The **Gauss sum** of a dirichlet character $\chi \bmod N$ is defined by

$$G(n, \chi) = \sum_{m=1}^N \chi(m) e\left(\frac{mn}{N}\right)$$

and we write $\tau(\chi) = G(1, \chi)$ for convenience.

Lemma 15.16. If χ is a dirichlet character mod N , then for any $\gcd(n, N) = 1$ we have

$$G(n, \chi) = \bar{\chi}(n) \tau(\chi)$$

Proof. □

χ can be viewed as a function on $\mathbb{Z}/N\mathbb{Z}$, so by theorem 15.7 we can rewrite

$$G(n, \chi) = N \cdot \hat{\chi}(-n)$$

so we can describe primitive character by gauss sum again, since we believe that fourier transform can keep the information of the original function.

Lemma 15.17. Along above notation, the following statements are equivalent:

- (a) χ is primitive.
- (b) $G(n, \chi) = \bar{\chi}(n) \tau(\chi)$ for any $n \in \mathbb{N}$.
- (c) $G(n, \chi) = 0$ for any $\gcd(n, N) \neq 1$.

If one of the above statements holds, we call the Gauss sum is **separable**.

Proof. □

Proposition 15.18. Let χ be a primitive character mod N , then

$$|G(n, \chi)| = \begin{cases} \sqrt{N} & \gcd(n, N) = 1 \\ 0 & \gcd(n, N) \neq 1 \end{cases}$$

Proof. □

Remark 15.13. If χ is not primitive, then by Ramanujan sum we can show that

$$\tau(\chi) = 0$$

Finally, when the character is primitive, then the fourier expansion of character will be simple since the gauss sum is separable, we can conclude that as following, which is important in many applications.

Theorem 15.19. Let χ be a primitive character mod N , then its fourier trnasform is

$$\hat{\chi} = \frac{\tau(\chi)\bar{\chi}(-1)}{N} \cdot \bar{\chi}$$

with the fourier expansion

$$\chi(n) = \frac{\tau(\chi)}{N} \sum_{m=1}^N \bar{\chi}(n) e(-\frac{mn}{N})$$

Proof.

□

15.2.2 Legendre Symbol and real character

A character of a finite abelian group is called **real character** if its image is contained in $\{\pm 1\}$, this parts is to introduce a tycpial example of real character.

Historically, a classical diophantine problem is to solve

$$X^2 = b \pmod{p}$$

with p a prime, and b is a given non-zero intger, equivalently, solve the quadratic equation in the finite field $\mathbb{F}_p$. For example, $X^2 - 2 = 0$ has no solution in $\mathbb{F}_5$ while it has solution in $\mathbb{F}_3$.

Lemma 15.20. Let p be an odd prime, then $\varphi : \mathbb{F}_p^* \rightarrow \mathbb{F}_p^*$ with $x \mapsto x^{\frac{p-1}{2}}$ defines a homomorphism of groups, with kernel S_p the set of all squares in $\mathbb{F}_p^*$, and the image is $\{1, -1\}$.

Proof. p is prime so $\frac{p-1}{2}$ is an integer, which is enough to show the map is a homomorphism. Notice that $S_p \subset \ker \varphi$, it is enough to show that $|S_p| = |\ker \varphi| = \frac{p-1}{2}$. After that, the cardianlity of the image is 2, and $\varphi(x)$ is exactly the roots of $X^2 - 1 = 0$, which implies the image is $\{1, -1\}$.

S_p is the image of morphism $f(x) = x^2$ inside $\mathbb{F}_p^*$, the kernel of f is exactly the solutions of $X^2 - 1 = 0$ in $\mathbb{F}_p$, they are exactly $-1, 1$ since there are at most two distinct solutions, hence $|S_p| = |\mathbb{F}_p^* / \ker f| = \frac{p-1}{2}$.

$\varphi(x) = 1$ means that x is a roots of $X^{\frac{p-1}{2}} - 1 = 0$ in $\mathbb{F}_p$, which has at most $\frac{p-1}{2}$ distinct solutions, hence $|\ker \varphi| \leq \frac{p-1}{2} = |S_p|$, so immediately $\ker \varphi = S_p$. □

Definition 15.9. A integer a is called a **quadratic residue** modulo p if a is a square in $\mathbb{F}_p$, i.e. the diophantine equation

$$x^2 \equiv a \pmod{p}$$

has a solution in $\mathbb{F}_p$. The **legendre symbol** is a function defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & a = 0 \pmod{p} \\ 1 & a \text{ is a non-zero quadratic residue modulo } p \\ -1 & \text{otherwise} \end{cases}$$

where p is an odd prime and a is an integer.

Here are some quick descriptions:

- If $p = 2$, then any elements in $\mathbb{F}_2$ is square, so legendre symbol makes no sense to distinguish the quadratic residue, hence we only consider odd prime p .
- If we fixed the prime p , then the legendre symbol is a extension of the character on $(\mathbb{Z}/p\mathbb{Z})^\times$ defined by $a \mapsto a^{\frac{p-1}{2}}$ in the above lemma, exactly it is **a real primitive character mod p** .
- **It is the unique non-trivial real character mod p** : In fact, $(\mathbb{Z}/p\mathbb{Z})^\times$ is a cyclic group of order $p-1$, so we can choose a generator g , then the character will depend only on $\chi(g) = \{\pm 1\}$, if $\chi(g) = 1$, then χ is trivial, hence along the last statement, we can conclude.

Proposition 15.21. Let a be a non-zero square free integer and $m = 4|a|$, then there exists a unique dirichlet character $\chi_a \pmod{m}$ such that

$$\forall \text{ prime } p \nmid m, \quad \chi_a(p) = \left(\frac{a}{p}\right)$$

it will be non-trivial if $a \neq 1$.

Proposition 15.22. Let p be an odd prime, then we have the following formula:

- Euler's criterion: $\left(\frac{n}{p}\right) = n^{\frac{p-1}{2}} \pmod{p}$ for any intger n .
- Gauss's lemma: Let n be an integer coprime with p , and consider the following numbers:

$$n, 2n, 3n, \dots, \frac{p-1}{2}n$$

For any above numbers, there exists a unique number $1 \leq a_k \leq p-1$ such that $a_k = kn \pmod{p}$ with $1 \leq k \leq \frac{p-1}{2}$, then we have

$$\left(\frac{n}{p}\right) = (-1)^s$$

where s is the number of a_k that is greater than $\frac{p}{2}$.

Via the lemma, Gauss proved a remarkable result about the legendre symbol, so-called quadratic reciprocity, for sake of a proof, [Apo13] Theorem 9.8 sketches as such style. Here we will see legendre symbol as a character, and consider the gauss sum of other type, such that we can get a series of results about characters.

Lemma 15.23. The **quadratic gauss sum** is defined as

$$G_n = \sum_{k=1}^n e\left(\frac{k^2}{n}\right)$$

then we have

$$G_n = \frac{1+i^{-m}}{1+i^{-1}}\sqrt{m} = \begin{cases} (1+i)\sqrt{m} & m \equiv 0 \pmod{4} \\ \sqrt{m} & m \equiv 1 \pmod{4} \\ 0 & m \equiv 2 \pmod{4} \\ i\sqrt{m} & m \equiv 3 \pmod{4} \end{cases}$$

Proof.

□

Lemma 15.24. The **generalized gauss sum** is defined as

$$G_n(a) = \sum_{k=1}^n e\left(\frac{ak^2}{n}\right)$$

with $G_n(1) = G_n$ exactly, then we conclude following properties:

(a) If $\gcd(a, n) = d$, then $G_n(a) = d \cdot G_{n/d}(a/d)$.

hence we usually consider the case a coprime with n

(b) If a is coprime with $n = n_1 n_2$ and n_1 is coprime with n_2 , then

$$G_n(a) = G_{n_1}(a n_2) G_{n_2}(a n_1)$$

(c) If n is fixed, then $a \mapsto G_n(a)$ is n -periodic and in particular if $n = p$ is an odd prime then

$$G_p(a) = \left(\frac{a}{p}\right) G_p = \left(\frac{a}{p}\right) \varepsilon_p \sqrt{p}$$

with a factor depending on p

$$\varepsilon_p = \begin{cases} 1 & p \equiv 1 \pmod{4} \\ i & p \equiv 3 \pmod{4} \end{cases}$$

(d) Let p be an odd prime, then the gauss sum of legendre symbol $\chi_p = \left(\frac{\cdot}{p}\right)$ is exactly G_p with

$$\varepsilon(\chi_p) = \varepsilon_p \sqrt{p}$$

Theorem 15.25 (Quadratic reciprocity). Let p and q be two distinct odd primes, then we have

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

Proof.

□

Theorem 15.26 (Gauss sum). Let n be a square-free odd, and χ be a real primitive character mod n , then

$$\tau(\chi) = \varepsilon_m \sqrt{m}$$

with

$$\varepsilon_m = \begin{cases} 1 & m \equiv 1 \pmod{4} \\ i & m \equiv 3 \pmod{4} \end{cases}$$

Proof.

□

15.3 Riemann Zeta-function

15.3.1 Analytical Continuation: different methods

As a core object in number theory, Riemann defines the zeta-function in his short paper [Rie59] as following

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}$$

The basic description can be given as following:

1. ζ defines a homomorphic function on the half-plane $\Re(s) > 1$, the reason is that the series is uniformly convergent on any compact subset of the half-plane, and $s \mapsto n^s$ is a entire function.
2. On the half-plane $\Re(s) > 1$, we have the Euler product expansion

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$$

which encodes the information of prime numbers into this function, that is the reason why it plays an important role in number theory.

3. On the half-plane $\Re(s) > 0$, make use of the Gamma function we have the formula

$$\frac{\Gamma(1+s)}{n^s} = \int_0^\infty e^{-nt} t^s \frac{dt}{t}$$

then some calculation can be done to arrange the formula in the form of Mellin transform:

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \int_0^\infty \frac{\Theta(t) - 1}{2} t^{s/2-1} \frac{dt}{t} \quad (4)$$

with theta function defined by

$$\Theta(t) := \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t} \quad t > 0$$

and by the poisson's formula it satisfies the functional equation

$$\Theta(t) = t^{-1/2} \Theta(1/t) \quad (5)$$

It can be proved in half plane by complex analysis with no difficult, and the analytic continuation can be done by mellin transform, the result will be important in number theory, and we state as following:

Theorem 15.27. Define $Z(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ as the **completed zeta-function**, then Z is holomorphic on half plane $\Re(s) > 1$ and it satisfies

1. Z has an meomorphic continuation the whole complex plane.
2. Z has the only simple poles at $s = 0$ and $s = 1$ with residue -1 and 1 respectively.
3. For any $s \neq 0, 1$, functional equation is $Z(1 - s) = Z(s)$.

Proof. Set the auxiliary function $\psi(t) = \frac{\Theta(t)-1}{2}$ and by equation (5) we have

$$t^{1/2}(2\psi(t) + 1) = 2\psi(1/t) + 1, \quad t > 0$$

then we can rewrite the integral representation (4)

$$\begin{aligned} Z(s) &= \int_0^1 \psi(t)t^{\frac{s}{2}-1}dt + \int_1^\infty \psi(t)t^{\frac{s}{2}-1}dt \\ &= \int_1^\infty \psi(1/t)t^{-s/2} \frac{dt}{t} + \int_1^\infty \psi(t)t^{\frac{s}{2}} \frac{dt}{t} \end{aligned}$$

plug the reflection fomrula of ψ into the first integral, we can calculate as following

$$\begin{aligned} Z(s) &= \frac{1}{2} \int_1^\infty t^{-s/2-1/2}dt - \frac{1}{2} \int_1^\infty t^{s/2-1}dt + \int_1^\infty \psi(t)(t^{s/2-1} + t^{-s/2-1/2})dt \\ &= \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty \psi(t)(t^{\frac{s}{2}} + t^{\frac{1-s}{2}}) \frac{dt}{t} \\ &= \frac{1}{s-1} - \frac{1}{s} + F(s) + F(1-s) \end{aligned}$$

with $F(s) = \int_0^\infty \psi(t)t^{s/2}dt$, the integrand can be proved to be a entire function since ψ is rapidly decreasing, so it finishes the proof by a quick verification. $\square$

The consequence is immediately about the zeta function, and such a method can be seen as a standard model to get the analytic continuation and functional equation of L-fuctions.

Corollary 15.28. Riemann zeta function ζ has an analytic continuation to the whole complex plane with only a simple pole at $s = 1$ with residue 1, and it satisfies the functional equation

$$\zeta(1-s) = 2(2\pi)^{-s} \cos\left(\frac{\pi s}{2}\right) \Gamma(s) \zeta(s) \quad s \neq 1, 0, -1, -2, \dots$$

Proof. By completed zeta function, the function

$$s \mapsto \frac{Z(s)}{\pi^{-s/2} \Gamma\left(\frac{s}{2}\right)}$$

defines an meomorphic function on the hole complex plane, which concides with ζ on the half-plane $\Re(s) > 1$, so it is the analytic continuation of ζ . Notice that Γ has a simple pole at $s = 0$, which cancels the simple pole of Z at $s = 0$, hence ζ is actually holomorphic at $s = 1$ with residue

$$\text{Res}_{z=1}\zeta = \lim_{s \rightarrow 1} \frac{(s-1)Z(s)}{\pi^{-s/2}\Gamma(s/2)} = \frac{1}{\pi^{-1/2}\pi^{1/2}} = 1$$

and the funnctional equation can be derived via the formula of gamma function. $\square$

15.3.2 The values on $\mathbb{Z}$ and critical strip

We rewrite the functional equation of zeta function here

$$\zeta(1-s) = 2(2\pi)^{-s} \cos\left(\frac{\pi s}{2}\right) \Gamma(s) \zeta(s) \quad s \neq 1, 0, -1, -2, \dots \quad (6)$$

it gives a symmetry of values, and determine when the values is zero.

Example 15.13. We set $s = 2$ in the functional equation, then we have

$$\zeta(-1) = -\frac{1}{2\pi^2} \Gamma(2) \zeta(2) = -1/12$$

since $\zeta(2) = \pi^2/6$ and $\Gamma(2) = 1$. In another case, we hope the cosine term vanish, for example we set $s = 3$ then $\zeta(-2) = 0$, it gives a zero of zeta function.

Following the example, the functional equation (6) establishes a non-vanish relation between the negative odd integers and the positive even integers, to know them all we should firstly know one sides of them. To do that, a remarkable constant is introduced.

Definition 15.10. The function $z \mapsto \frac{z}{e^z - 1}$ is holomorphic near the zeros, with the expansion of the series:

$$\frac{z}{e^z - 1} = \sum_{n \geq 0} \frac{B_n}{n!} z^n$$

the n -th **Bernoulli's number** is defined as B_n , with respect to the coefficient of the series.

The original definition of Bernoulli's number is given by Jacob Bernoulli in his book "Ars Conjectandi" (1713), he defined the number by the summation of powers:

$$\sum_{k=1}^m k^n = \frac{1}{n+1} \sum_{j=0}^n \binom{n+1}{j} B_j \cdot m^{n+1-j}$$

for any $n, m \in \mathbb{N}$, and the two definitions can be shown to be equivalent by the method of **generating function**. The definition of power series is easy for us to calculate the values of Bernoulli's number, for Example

$$B_0 = 1, \quad B_1 = \frac{-1}{2}, \quad B_2 = \frac{1}{6}, \quad B_3 = 0, \quad B_4 = \frac{-1}{30} \quad \dots$$

Lemma 15.29. The Bernoulli's number is rational, and $B_{2k+3} = 0$ for any $k \in \mathbb{N}$.

Proof. Observe that

$$\frac{ze^z}{e^z - 1} = \frac{-z}{e^{-z} - 1}$$

which allows us to write

$$z = \frac{ze^z}{e^z - 1} - \frac{z}{e^z - 1} = \sum_{n \geq 0} ((-1)^n - 1) \frac{B_n}{n!} \cdot z^n$$

at the voisinage of zero, hence by comparing the coefficients we can conclude that $B_0 = 1$ and $B_n = 0$ for any odd integer $n \geq 3$. To prove the rationality, by the properties of analytical function we know

$$B_n = n! \frac{d^n}{dz^n} \left(\frac{z}{e^z - 1} \right) \Big|_{z=0}$$

the derivatives can be proved to be a rational number by induction on n , hence the result holds true. $\square$

Proposition 15.30. Let k be a positive integer, then $\zeta(-2k) = 0$ and

$$\zeta(-n) = -\frac{B_{n+1}}{n+1}$$

by the functional equation a corollary is that

$$\zeta(2k) = (-1)^{k+1} \frac{(2\pi)^{2k} B_{2k}}{2(2k)!}$$

Proof. The first result is immediate by the functional equation (6) by setting $s = 1 + 2k$, then $\Gamma(s)$ and $\zeta(s)$ are both finite, while the cosine term vanishes, hence $\zeta(-2k) = 0$.

For the second result, a classical formula about cotangent function is

$$\pi z \cot(\pi z) = 1 + \sum_{n=1}^{\infty} \frac{2z^2}{z^2 - n^2}, \quad z \in \mathbb{C} \setminus \mathbb{Z}$$

and we can expand the formula by geometric series as following

$$\begin{aligned}
\pi z \cot(\pi z) &= 1 + 2z^2 \sum_{n \geq 1} \frac{-1}{n^2} \cdot \frac{1}{1 - z^2/n^2} \\
&= 1 - 2z^2 \sum_{n \geq 1} \frac{1}{n^2} \sum_{k \geq 0} \left(\frac{z^2}{n^2}\right)^k \\
&= 1 - 2 \sum_{k \geq 1} \zeta(2k) z^{2k}
\end{aligned}$$

here the formula holds true for $|z| < 1$, and the last equality holds by Fubini's theorem, so we get a power series expansion at zeros. To finish the proof, we write the trigonometric function in terms of exponential function, then we can calculate

$$z \cot(z) = \frac{2iz}{e^{2iz} - 1} + iz = 1 + \sum_{k \geq 1} (-1)^k \frac{2^{2k}}{(2k)!} B_{2k} \cdot z^{2k}$$

which allows us to get another power series expansion at zeros, if we replace z by πz and compare the two expansions, then we can conclude the first formula. $\square$

Remark 15.14. The absence of values on positive odd integers is a mystery, if we set $s = 1 + 2k$ then $\cos(\pi s/2) = 0$, which prevents us to get any information from the functional equation, although $\zeta(-2k)$ and $\zeta(2k+1)$ are lie on two sides, before the work of Apéry in 1978, we even can not determine whether $\zeta(3)$ is rational or not.

Remark 15.15. Another observation about zeros is via the completed zeta function

$$Z(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

as we know in theorem 15.27, Z is finite at $s = -2n$, while $s \mapsto \Gamma(s/2)$ has a simple pole at $s = -2n$, so the unique possible is that ζ has a simple zero at $s = -2n$ to cancel the pole, and we call such zeros from the Gamma factor the **trivial zeros** of zeta function.

Theorem 15.31 (Hamadard, 1896). ζ does not vanish on the line $\operatorname{Re}(s) = 1$.

Proof. $\square$

Corollary 15.32. The non-trivial zeros of zeta function lie in the **critical strip** $\{s \in \mathbb{C} : 0 < \Re(s) < 1\}$.

15.3.3 Von Mangoldt function

A remarkable arithmetic function in the study of Riemann and Tchebychev is the Von Mangoldt function, which is defined as following:

$$\Lambda(n) := \begin{cases} \log p & \text{if } n = p^k \\ 0 & \text{otherwise} \end{cases}$$

Notice that $\Lambda(1) = 0$ and it is only related to the prime numbers, and we view it in the context of arithmetic functions, it is not hard to prove the following properties:

Proposition 15.33. By dirichlet convolution,

$$\log n = (\Lambda * \mathbf{1})(n) = \sum_{d|n} \Lambda(d)$$

and by mobius inversion formula

$$\Lambda(n) = (\log * \mu)(n) = \sum_{d|n} \mu(d) \log(n/d)$$

This function appears in the Riemann's paper first time although it is not named as such.

Lemma 15.34 (Riemann,1859). For any $\Re(s) > 1$, we have

$$\log \zeta(s) = \sum_{n \geq 2} \frac{\Lambda(n)}{\log(n)} \frac{1}{n^s}$$

and

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \geq 2} \frac{\Lambda(n)}{n^s}, \quad \Re(s) > 1$$

Proof. Take the logarithm to zeta function of the Euler product expansion, we have

$$\log \zeta(s) = - \sum_p \log(1 - p^{-s})$$

notice that $\log(1-x)$ has a power series expansion if $|x| < 1$, and here $|p^{-s}| = |p|^{-\Re(s)} < 1$, so we can write

$$\log \zeta(s) = \sum_p \sum_{k \geq 1} \frac{1}{k} p^{-ks}$$

Notice that 1 is not a prime by convention, and Λ only supports on the prime powers, so the sum can be rewritten as

$$\sum_{n=p^k} \frac{\Lambda(n)}{\log n} \frac{1}{n^s}$$

since

$$\frac{\Lambda(p^k)}{\log p^k} = \frac{\log p}{k \log p}$$

□

Tchebychev consider a type of summation, which is closed to the average of the von mangoldt function, that is

$$\theta(x) := \sum_{p \leq x} \log p$$

and we call it **the first Tchebychev function**, and **the second Tchebychev function** is defined by

$$\psi(x) := \sum_{n \leq x} \Lambda(n)$$

they are equivalent in the sense of asymptotic behavior, and he estimated the growth of θ , which can be translated as following:

Lemma 15.35 (Tchebychev, 1850). $\psi(x) \asymp x$, that means there exists two positive constants c_1, c_2 such that for any x large enough, we have

$$c_1 x \leq \psi(x) \leq c_2 x$$

Proof.

□

Corollary 15.36. ψ and θ have the same asymptotic behavior.

15.3.4 Prime number theorem

Fix prime counting function

$$\pi(x) = \sum_{p \leq x} 1$$

Claim 15.37. As $x \rightarrow \infty$, the following asymptotic behavior are equivalent:

- (a) $\pi(x)/x \sim 1/\log x$
- (b) $\psi(x) \sim x$
- (c) $\psi_1(x) \sim x^2/2$

and

15.4 Dirichlet L-function

15.4.1 Analytical continuation

Definition 15.11. The dirichlet L -function associated to a dirichlet character $\chi \bmod N$ is defined by the series

$$L(s, \chi) = \sum_{n \geq 1} \frac{\chi(n)}{n^s}$$

about the analytic properties, some quick descriptions are as following:

1. $s \mapsto L(s, \chi)$ defines a holomorphic function on the half-plane $\Re(s) > 1$, and the Euler product formula holds as well on the half-plane

$$L(s, \chi) = \prod_{p \text{ prime}} \frac{1}{1 - \chi(p)p^{-s}}$$

the prove is similar to the case of zeta function.

2. If χ is a character with conductor q of the form $\chi = \psi_q \cdot \chi_0$, where χ_0 is the trivial character mod N and ψ_q is a primitive character mod q , then we have the factorization

$$L(s, \chi) = L(s, \psi_q) \prod_{p|N} (1 - \psi_q(p)p^{-s})$$

which allows us to reduce the analytic properties of $L(s, \chi)$ to the primitive case.

3. Riemann zeta function can be viewed as a L-function associated to the character mod 1, and for any trivial character χ_0 mod N , we have

$$L(s, \chi_0) = \sum_{n \geq 1, (n, N)=1} \frac{1}{n^s} = \zeta(s) \prod_{p|N} (1 - p^{-s})$$

so if χ_0 is trivial, then $L(s, \chi_0)$ has a meromorphic continuation to the whole complex plane with only a simple pole at $s = 1$ with residue $\prod_{p|N} (1 - p^{-1})$.

To do the analytical continuation, we start from the integral we use in the Riemann zeta function

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \frac{1}{n^s} = \int_0^\infty e^{-\pi n^2 t} t^{\frac{s}{2}-1} dt \quad \Re(s) > 1$$

to make use of the Gauss sum of the character, so it is natural to modify the integral a little bit **with a assumption that the character is non trivial and primitive χ mod q**

$$(q/\pi)^{s/2} \Gamma\left(\frac{s}{2}\right) \frac{\chi(n)}{n^s} = \frac{1}{\sqrt{q}} \int_0^\infty [\chi(n) e^{-\pi n^2 t/q}] t^{\frac{s}{2}-1} dt \quad (7)$$

In the case of $\chi(-1) = 1$, we can sum equation (7) over n to get

$$(q/\pi)^{s/2} \Gamma\left(\frac{s}{2}\right) L(s, \chi) = \frac{1}{2\sqrt{q}} \int_0^\infty \psi(t, \chi) t^{\frac{s}{2}-1} dt \quad (8)$$

with

$$\psi(t, \chi) = \sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2 t/q}$$

In the case of $\chi(-1) = -1$, then $\psi(t, \chi) = 0$ such that representation (8) will make no sense in this case, so we should modify the equation (7) a little bit (introduce a translation $s \mapsto s + 1$) to get

$$(q/\pi)^{(s+1)/2} \Gamma\left(\frac{s+1}{2}\right) \frac{\chi(n)}{n^s} = \frac{1}{2\sqrt{q}} \int_0^\infty \psi(t, \chi) t^{\frac{s-1}{2}} dt$$

with

$$\psi(t, \chi) = \sum_{n \in \mathbb{Z}} n \chi(n) e^{-\pi n^2 t / q}$$

above discussion can be summarized as following:

Lemma 15.38. On the half-plane $\Re(s) > 1$, the integral representation of a L-function $L(s, \chi)$ associated to a non-trivial primitive character $\chi \bmod q$ is given by

$$(q/\pi)^{\frac{s+\mathfrak{v}}{2}} \Gamma\left(\frac{s+\mathfrak{v}}{2}\right) L(s, \chi) = \frac{1}{2\sqrt{q}} \int_0^\infty \psi(t, \chi) t^{\frac{s+\mathfrak{v}}{2}-1} dt$$

with $\mathfrak{v} = \frac{1-\chi(-1)}{2}$ and auxiliary function

$$\psi(t, \chi) = \sum_{n \in \mathbb{Z}} n^{\mathfrak{v}} \chi(n) e^{-\pi n^2 t / q}$$

So we should know the functional equation of ψ firstly.

Lemma 15.39. Let $t > 0$, then ψ satisfies the equation

$$\tau(\bar{\chi}) \psi(t, \chi) = (i/t)^{\mathfrak{v}} \sqrt{q/t} \cdot \psi(\bar{\chi}, 1/t)$$

where $\tau(\chi) = G(1, \chi)$ as the Gauss sum of χ .

Proof. We consider the $f(x) = e^{-\pi x^2 t}$ for some $t > 0$, and then we can calculate

$$\hat{f}(s) = \frac{1}{\sqrt{t}} f\left(\frac{s}{t}\right) = \frac{1}{\sqrt{t}} e^{-\pi s^2 / t}$$

hence by poisson's formula we have

$$\sum_{n \in \mathbb{Z}} e^{-\pi(n+a)^2 t} = \frac{1}{\sqrt{t}} \sum_{n \in \mathbb{Z}} e^{-\pi n^2 / t} e^{2\pi i n a} \quad (9)$$

now we consider χ is a arithmetic function on $\mathbb{Z}$ with period q , then we can write

$$\chi(n) = \frac{\tau(\chi)}{q} \sum_{m=1}^q \bar{\chi}(m) e^{2\pi i m n / q}$$

by the primitive character $\overline{\tau(\chi)} = \chi(-1) \tau(\bar{\chi})$, we have

$$\tau(\bar{\chi}) \chi(n) = \sum_{m=1}^q \bar{\chi}(m) e^{2\pi i m n / q}$$

hence we can calculate

$$\begin{aligned} \tau(\bar{\chi}) \psi(t, \chi) &= \sum_{n \in \mathbb{Z}} n^{\mathfrak{v}} \tau(\bar{\chi}) \chi(n) e^{-\pi n^2 t / q} \\ &= \sum_{m=1}^q \bar{\chi}(m) \sum_{n \in \mathbb{Z}} n^{\mathfrak{v}} e^{-\pi n^2 t / q} e^{2\pi i m n / q} \end{aligned}$$

when $\chi(-1) = 1$, i.e. $\mathfrak{v} = 0$, we can apply the Poisson's formula (9) to get

$$\begin{aligned}\tau(\bar{\chi})\psi(t, \chi) &= \sqrt{q/t} \cdot \sum_{m=1}^q \bar{\chi}(m) \cdot \sum_{n \in \mathbb{Z}} e^{-\pi(qn+m)^2/qt} \\ &= \sqrt{q/t} \cdot \sum_{l \in \mathbb{Z}} \bar{\chi}(l) e^{-\pi l^2/t} \\ &= \sqrt{q/t} \cdot \psi(\bar{\chi}, 1/t)\end{aligned}$$

when $\chi(-1) = -1$, i.e. $\mathfrak{v} = 1$, we can modify the poisson's formula (9) by taking derivative of $f(x)$, then we have

$$it^{3/2} \sum_{n \in \mathbb{Z}} (n+a) e^{-\pi(n+a)^2 t} = \sum_{n \in \mathbb{Z}} n e^{-\pi n^2/t} e^{2\pi i n a}$$

then apply it we can get

$$\begin{aligned}\tau(\bar{\chi})\psi(t, \chi) &= \sum_{m=1}^q \bar{\chi}(m) \cdot i(q/t)^{3/2} \sum_{n \in \mathbb{Z}} (n + \frac{m}{q}) e^{-\pi(qn+m)^2/qt} \\ &= (i/t) \sqrt{q/t} \cdot \sum_{l \in \mathbb{Z}} l \cdot \bar{\chi}(l) e^{-\pi l^2/t} \\ &= (i/t) \sqrt{q/t} \cdot \psi(\bar{\chi}, 1/t)\end{aligned}$$

□

Finally we can finish the analytical continuation as we do in Riemann zeta function

Theorem 15.40. For any L-function associated to a non-trivial primitive character $\chi \bmod q$, the completed L-function is defined by

$$\Lambda(s, \chi) = (q/\pi)^{\frac{s+\mathfrak{v}}{2}} \Gamma(\frac{s+\mathfrak{v}}{2}) L(s, \chi)$$

then $s \mapsto \Lambda(s, \chi)$ has a **holomorphic continuation** to the whole complex plane, and it satisfies the functional equation

$$\Lambda(1-s, \bar{\chi}) = \frac{i^{\mathfrak{v}} \sqrt{q}}{\tau(\chi)} \Lambda(s, \chi)$$

Proof. when $\chi(-1) = 1$, i.e. $\mathfrak{v} = 0$, we have

$$\begin{aligned}2\sqrt{q}\Lambda(s, \chi) &= \int_0^1 \psi(t, \chi) t^{\frac{s}{2}-1} dt + \int_1^\infty \psi(t, \chi) t^{\frac{s}{2}-1} dt \\ &= \int_1^\infty \psi(1/u, \chi) u^{-\frac{s}{2}-1} du + \int_1^\infty \psi(t, \chi) t^{\frac{s}{2}-1} dt \\ &= \frac{\tau(\chi)}{\sqrt{q}} \int_1^\infty \psi(u, \bar{\chi}) u^{\frac{-s-1}{2}} du + \int_1^\infty \psi(t, \chi) t^{\frac{s}{2}-1} dt \quad (a)\end{aligned}$$

notice that in this case we have $\overline{\tau(\chi)} = \tau(\bar{\chi})$, so we can write

$$\begin{aligned} 2\sqrt{q}\Lambda(s, \chi) &= \frac{\tau(\chi)}{\sqrt{q}} \left[\int_1^\infty \psi(u, \bar{\chi}) u^{-\frac{s-1}{2}} du + \frac{\tau(\bar{\chi})}{\sqrt{q}} \int_1^\infty \psi(t, \bar{\chi}) t^{-\frac{s-1}{2}-1} dt \right] \\ &= \frac{\tau(\chi)}{\sqrt{q}} \cdot 2\sqrt{q}\Lambda(1-s, \bar{\chi}) \end{aligned}$$

Finally, when $\chi(-1) = -1$, i.e. $\mathfrak{v} = 1$, we have

$$\begin{aligned} 2\sqrt{q}\Lambda(s, \chi) &= \int_0^1 \psi(t, \chi) t^{\frac{s-1}{2}} dt + \int_1^\infty \psi(t, \chi) t^{\frac{s-1}{2}} dt \\ &= \int_1^\infty \psi(1/u, \chi) u^{-\frac{s+3}{2}} du + \int_1^\infty \psi(t, \chi) t^{\frac{s-1}{2}} dt \\ &= \frac{\tau(\chi)}{i\sqrt{q}} \int_1^\infty \psi(u, \bar{\chi}) u^{-\frac{s}{2}} du + \int_1^\infty \psi(t, \chi) t^{\frac{s-1}{2}} dt \quad (b) \end{aligned}$$

and notice that in this case we have $\overline{\tau(\chi)} = -\tau(\bar{\chi})$, so we can write

$$\begin{aligned} 2\sqrt{q}\Lambda(s, \chi) &= \frac{\tau(\chi)}{i\sqrt{q}} \left[\int_1^\infty \psi(u, \bar{\chi}) u^{-\frac{s}{2}} du + \frac{\tau(\bar{\chi})}{i\sqrt{q}} \int_1^\infty \psi(t, \bar{\chi}) t^{\frac{s-1}{2}} dt \right] \\ &= \frac{\tau(\chi)}{i\sqrt{q}} \cdot 2\sqrt{q}\Lambda(1-s, \bar{\chi}) \end{aligned}$$

□

Remark 15.16. Notice that in above proof, (a) and (b) shows that the functions $\mapsto 2\sqrt{q}\Lambda(s, \chi)$ defines an entire function, since the integrand nicely converges uniformly on $(1, \infty)$, so it indicates the holomoprhic continuation of Λ .

15.4.2 The arithmetic progressions

15.5 Dedekind L-function

i want to change something here

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